Module 1
CHEMICAL BONDING

Modern Approach to structure of atom

1. Wave nature of material objects. *Einstein* had suggested that light can behave as a *wave* as well as a *particle*, i.e., it has a *dual character*. In 1924, *de-Broglie* suggested that all material objects and therefore, electrons also have a dual character. *They behave as a particle as well as a wave*. This means that an electron which has been regarded as a particle also behaves like a wave. According to *de-Broglie*, the wavelength associated with a particle of mass m, moving with velocity v is given by the relation,

$$\lambda = \frac{h}{mv}$$ where *h* is *Planck's constant.*

This equation is known as ***de-Broglie equation***. It was experimentally verified by *Davisson* and *Germer*.

It may be noted that *matter wave concept is significant for microscopic bodies only*. For large bodies, the wave length of the associated waves are very small and cannot be measured by any of the available methods. Thus, practically all these bodies are said to have no wavelengths. For example, the wavelength of an electron with mass 9.11×10^{-31} kg and moving with the velocity of 10^6 m s^{-1} is 7.27×10^{-10} m as shown below :

$$\lambda = \frac{6.626 \times 10^{-34}\ kg\ m^2\ s^{-1}}{(9.11 \times 10^{-31}\ kg) \times (10^6\ m\ s^{-1})} = 7.27 \times 10^{-10}\ m$$

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This wavelength associated with the moving electron is of the order of magnitude as of X-rays which can be easily measured.

On the other hand, the wavelength associated with a ball, for example, weighing 10 g and moving with the same velocity as that of electron (10^6 m s^{-1}) is only 6.626×10^{-38} m :

$$\lambda = \frac{6.626 \times 10^{-34}\ \text{kg m}^2\ \text{s}^{-1}}{(10 \times 10^{-3}\ \text{kg}) \times (10^6\ \text{m s}^{-1})} = 6.626 \times 10^{-38}\ \text{m}$$

This wavelength is so small in magnitude that it is difficult to grasp the idea of wave associated with the ball. Therefore, it does not have any significance.

2. Uncertainty principle. So far, we have been believing that everything can be specified to any desired degree of accuracy. However, *Heisenberg* pointed out that it *is not possible to measure simultaneously both the position and momentum (or velocity) of a microscopic particle, with absolute accuracy.* Since all observations are made by the impact of light, these microscopic objects suffer a change in position or velocity as a result of the impact of light.

It should be borne in mind that the uncertainty is not due to lack of refined techniques but it is due to the fact that we cannot observe microscopic bodies without disturbing them.

However, in our daily life, this principle has no significance. This is because, we come across only large objects. The position and velocity of these objects can be determined accurately because in such cases the changes that occur due to the impact of light are negligible.

Quantum Mechanical Model of Atom

On the basis of *de Broglie's idea of matter waves* and the *uncertainty principle*, a new model of atom was developed. This model is known as *quantum mechanical model*. In this model, behaviour of the electron in an atom is described by an equation known as *Schrodinger wave equation*. It can be represented as,

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0$$

where x,y and z are three cartesian coordinates which are used to designate the electron within the atom with nucleus as the origin, h is the

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Planck's constant, m and V are the mass and potential energy of the electron in the atom respectively, E is its total energy and ψ is the amplitude of the electron wave called *wave function*. $\frac{\partial^2\psi}{\partial x^2}$ refers to the second derivative of ψ with respect to x, $\frac{\partial^2\psi}{\partial y^2}$ refers to the second derivative with respect to y and $\frac{\partial^2\psi}{\partial z^2}$ refers to the second derivative with respect to z.

Concept of Orbitals

When the *Schrodinger equation* for the electron in a hydrogen atom is solved, several solutions are obtained. These solutions are called *wave functions*. Only some of these wave functions are physically acceptable. The acceptable wave functions are known as ***eigen functions***. Each eigen function represents a possible state of the electron in the atom and is called an *orbital wave function* or simply *orbital* . An orbital may be defined as:

the region of space around the nucleus where the probability of finding the electron is maximum.

Each orbital has a definite amount of energy. So each orbital is said to be quantised.

Orbital wave Functions and Shapes of Orbitals

Orbitals are described by wave functions known as orbital wave functions. These orbital wave functions depend upon three variables. One of these variables is a distance variable (r), while the other two are angle variables (θ and ϕ)*. The part of the wave function which depends upon the distance (r) from the nucleus is called *radial wave function* $\psi(r)$. The other part which depends upon the angles (θ and ϕ) is called *angular wave function* $\psi(\theta, \phi)$. Thus, an orbital wave function may be written as a product of two functions, the radial function and angular function, i.e., $\psi = \psi_{(r)}\, \psi_{(\theta, \phi)}$. The complete wave function represents an orbital in which the *radial part* $\psi_{(r)}$ *gives the size of the orbital and the angular part gives its shape. The radial part* $\psi_{(r)}$ depends on the quantum numbers 'n' and '*l*' where as the angular part $\psi_{(\theta,\phi)}$ depends on the quantum numbers '*l*' and 'm'. Variation of radial part of an orbital can be depicted by plotting a graph between ψ and the distance r from the nucleus. Fig. 1.1(a) and (b) show such graphs for 1s and 2s orbitals respectively.

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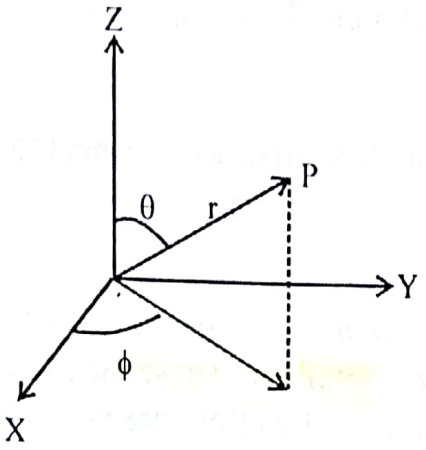
***Note:** Schrodinger equation for the hydrogen atom may be written in terms of cartesian coordinates (x, y, z) or in terms of spherical polar coordinates (r, θ, ϕ) of the electron with respect to the nucleus. As the atom is having spherical symmetry, it is more convenient to write Schrodinger equation in terms of polar coordinates. The location of a point P in spherical coordinate system relative to cartesian coordinates is shown in the following figure.



r is the length of the line between origin and the point p. θ is the angle between this line and z-axis. ϕ is the angle between the projection of this line in the xy plane and the x axis. The coordinates x, y and z of the electrons with respect to nucleus in terms of polar coordinates are : $x=r\sin\theta\cos\phi$; $y=r\sin\theta\sin\phi$; $z=r\cos\theta$, $x^2+y^2+z^2=r^2$

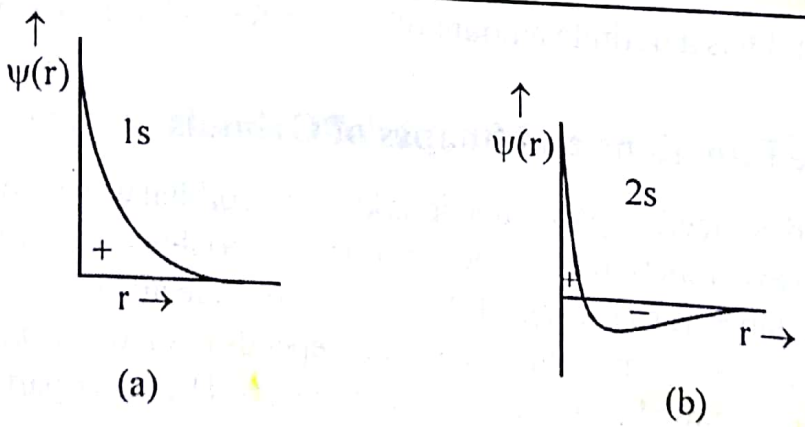


Fig. 1.1 *Plots of ψ versus r for (a) 1s and (b) 2s orbitals.*

It may be seen from the above diagrams that:

(i) The wave function has maximum value for r = 0 (i.e., at the nucleus itself).

(ii) The wave function for 1s orbital is positive for all values of r. But for 2s orbital the wave function may be positive or negative.

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Since ψ has no significance, the above plots are meaningless except that they give the sign of wave function.

The angular dependence of 1s, 2s and 2p orbital wave functions is given in Fig.1.2.

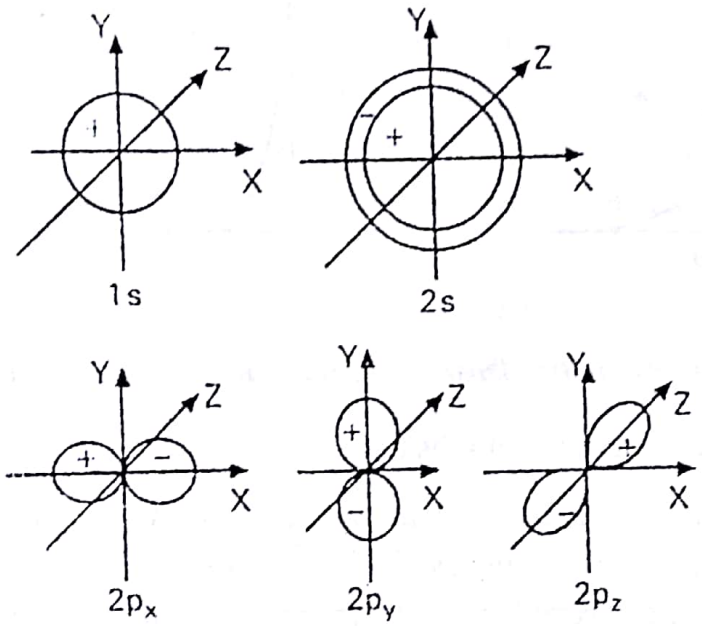

Fig. 1.2 *Angular plots of 1s, 2s and 2p orbitals.*

The following points should be noted:

1) s-orbitals do not have any angular dependance. Therefore, ***all s-orbitals are spherically symmetrical.*** However, p and d-orbital functions show angular dependance, i.e., they are directional in nature.

2) The positive and negative signs in Fig. 1.1 and Fig. 1.2 indicate whether the value of an orbital wave function is positive or negative in a particular region. These signs do not represent the positive or negative charge. Whereas the 1s orbital wave function has the same sign everywhere, the 2s orbital wave function can be positive or negative depending on the distance.

Probability Density (ψ^2) : Although the orbital wave function has no physical significance, its square ψ^2 *is a measure of the probability of finding an electron in unit volume of space around the nucleus of an atom.* Therefore, ψ^2 is called *probability density.* This suggests that to interpret an orbital, we must construct diagrams which show the variation of ψ^2 in space. *A graph of*

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ψ^2 as a function of distance from the nucleus is called ***probability density curve***. The probability density curves for 1s and 2s orbitals are shown in Fig. 1.3 (a) and (b) respectively.

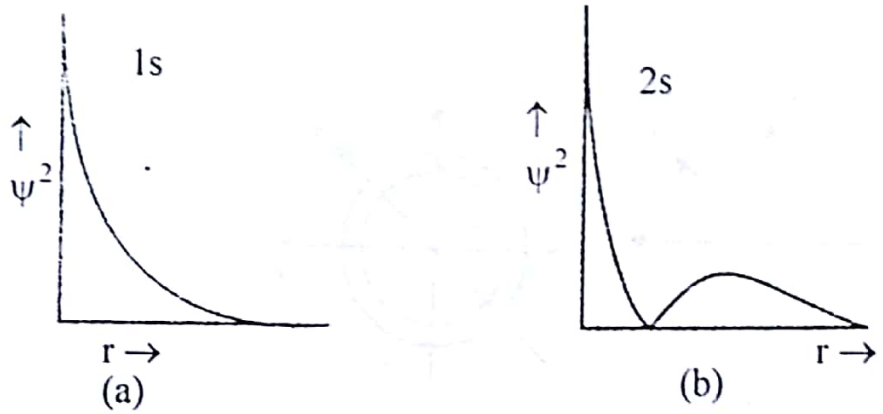

Fig. 1.3 *(a) Probability density curves of (a) 1s orbital and (b) 2s orbital.*

It can be seen from the graphs that:

(i) *ψ^2 is maximum at the nucleus and decreases with increase in distance from the nucleus and (ii) the probability of finding the electron does not become zero even at very large distances from the nucleus*. This means that an atom does not have a rigid size. Therefore, it is not possible to draw a boundary which will enclose regions of 100% probability. However, for the sake of clarity of representation we draw boundary surfaces which enclose regions of maximum probability (say 99%) of locating an electron. Since these diagrams (Fig.1.4) are simple to draw, these are very common.

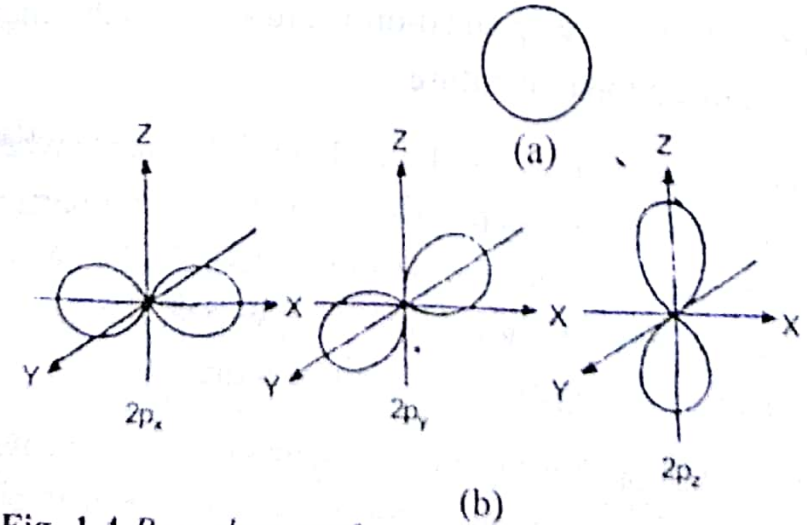

Fig. 1.4 *Boundary surface representation of (a) 1s and (b) 2p orbitals*

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The boundary surfaces of ψ^2 are similar to that of ψ (Fig. 1.2) . However, there is one main difference that ψ may have positive and negative signs whereas ψ^2 is always positive. In other words, ψ^2 can never be negative. Therefore , ***it is*** ψ^2 ***(and not*** ψ***) which can be related with probability because negative probability has no meaning.***

Charge Cloud Picture of Orbitals-Shapes of Orbitals: Another way to interpret the ψ^2 diagrams is to describe the electron as being spread out in the form of a negative charge cloud surrounding the nucleus and the density of the charge cloud is proportional to ψ^2 . In this method, the charge cloud is represented in terms of small dots. To understand this method of representation, we assume the electron as a very small dot. Further, suppose that we take a very large number of photographs of the electron in a hydrogen atom at very short interval of time on the same film. If the film was developed, the picture obtained would be similar to that shown in Fig. 1.5(a).

In terms of charge cloud representation, the value of ψ^2 at any point is taken to represent the density of the charge cloud at that point. The density of the charge cloud is not uniform as ψ^2 is not uniform, i.e., the cloud is more dense in some regions and less dense in other regions. Since the charge cloud is represented as a collection of dots, the density of the dots in any region is proportional to the density of the charge cloud in that region.

For s-orbitals, the density of the charge cloud (at a particular distance) is uniform in all directions, i.e., ***s-orbitals are spherical in shape***. The charge cloud pictures of 1s and 2s orbitals are shown in Fig. 1.5 (a) & (b). In 1s orbital, the density of the charge cloud is greatest at the nucleus and falls off with the distance (Fig.1.3 (a)). The s-orbitals have two important features: *(i) the size of the orbital increases with increasing quantum number 'n', (ii) the s-orbitals of higher energy levels have spherical shells, within them the electron density is zero,* i.e., an electron cannot be found in this region. The sphere about the nucleus where ψ^2 is zero is called ***radial node*** (spherical node). For an ns orbital there will be (n-1) spherical nodes. Thus, in 1s orbital there is no radial node and in 2s orbital (Fig. 1.5)), there is one radial node. *Thus a 2s-orbital differs from 1s-orbital in being larger in size and having one radial node.*

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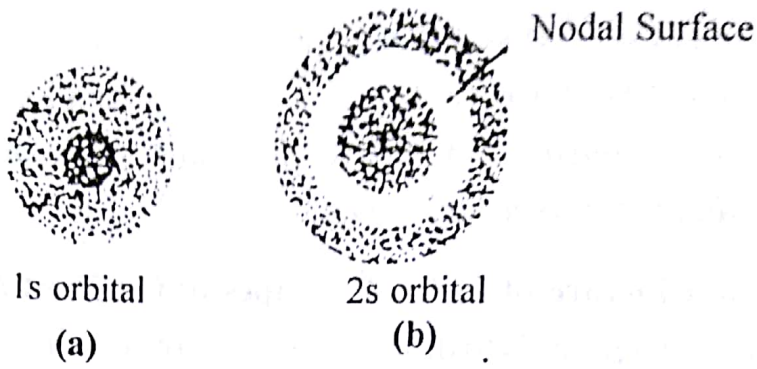

Fig. 1.5 *Charge cloud picture of (a) 1s orbital and (b) 2s orbital.*

For p-orbitals, there are three possible orientations of charge cloud. These three orientations of orbitals are designated as p_x, p_y and p_z. The p_x, p_y and p_z orbitals are oriented along X-axis, Y-axis and Z-axis respectively (Fig. 1.8). Each p-orbital has two lobes which are separated by a plane of zero electron density called ***nodal plane***. Thus, each p-orbital is ***dumb-bell shaped.***

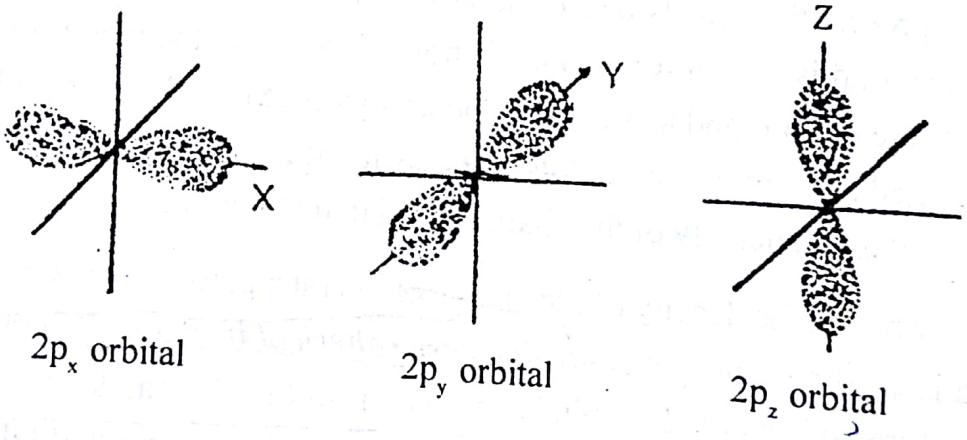

Fig. 1.6 *Shape and orientations of 2p orbitals*

For d orbitals ($l = 2$) there are five possible orientations corresponding to m = -2, -1, 0, +1, +2 values. This means that there are five d orbitals in each d sub shell. These are dxy, dxz, dyz, dx^2-y^2 and dz^2. The five d orbitals are shown in Fig. 1.7. The *dxy, dxz, dyz and* dx^2-y^2 *are double dumb bell shaped.* The dz^2 orbital has a dumb bell shaped curve about the z axis having a circular collar in xy plane.

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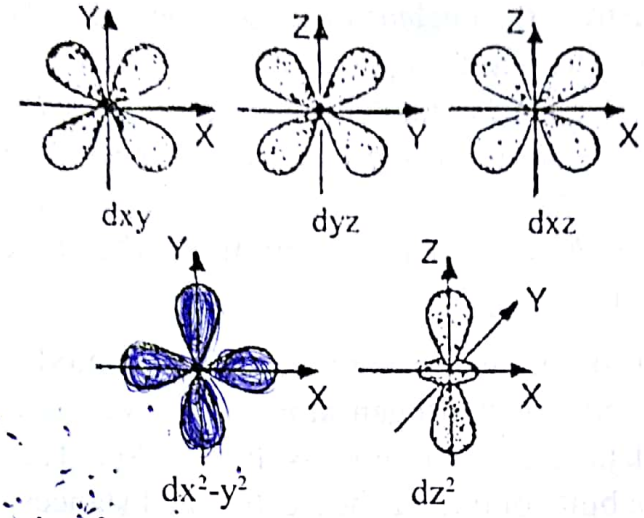

Fig. 1.7 *Shapes and orientations of the five 3d-orbitals.*

Radial Probability Distribution : *The radial probability distribution means the probability of finding an electron within a small radial space around the nucleus of an atom.* Suppose the space around the nucleus is divided into a very large number of thin concentric shells of thickness dr. Now,

radial probability = probability density × volume of the radial shell

The radial probability curves for 1s, 2s and 2p orbitals are shown in Fig. 1.8. It can be seen that radial probability distribution curve of 1s orbital (Fig.1.8) is quite different from its probability density graph (Fig. 1.3(a)).

The probability density (ψ^2) is maximum at the nucleus but the radial probability (or simply probability) is zero at the nucleus. This is because the volume of the shell near the nucleus is very small resulting in zero probability at the nucleus.

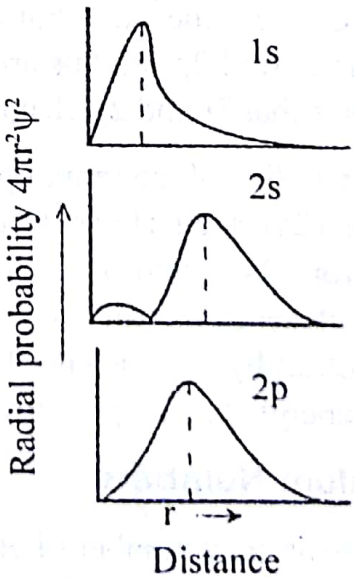

Fig. 1.8 *Radial probability distribution curves for 1s, 2s and 2p orbitals*

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As the distance from the nucleus increases, the volume of the radial shell increases while ψ^2 decreases rapidly (Fig. 1.8). As a result of this, the probability increases, reaches a maximum value and then decreases gradually. *The maximum in the curve gives the distance from the nucleus where the probability of finding the electron is maximum. This distance is called* ***radius of maximum probability***. For hydrogen atom, the radius of maximum probability is 53 pm.

It is important to note that according to Bohr's model, the electron is always at r = 53 pm for hydrogen atom. However, according to wave mechanical model, the electron is most likely to be found at this distance and there is some probability of finding the electron at distances both shorter and larger than 53 pm.

Comparison of Radial Probability Curves of 1s, 2s and 2p orbitals

The radial probability distribution curve for 2s orbital has two maxima, a smaller one near the nucleus and a bigger one at a larger distance (Fig. 1.8). In between these two maxima, there is a minimum where the probability of finding the electron is zero. The point at which the probability of finding the electron is zero is called a ***nodal point.***

The radius of maximum probability for 2s and 2p orbitals are approximately the same but are larger than that for 1s orbital. This means that the 2s and 2p orbitals are at a greater distance than the 1s orbital. This indicates that 2s and 2p electrons have greater energy than the 1s electron.

The radius of maximum probability for a 2p orbital is slightly less than that for a 2s orbital. However, there is a small additional peak in the probability curve for a 2s orbital. This indicates that the electron in 2s orbital penetrates a little closer to the nucleus. Hence the 2s electron is attracted more than the 2p electron by the nucleus. Therefore, a 2s electron is more stable and has lower energy than a 2p electron.

Quantum Numbers

In an atom, a number of electron orbitals are permissible. These orbitals are designated by a set of numbers known as *quantum numbers*. In order to specify energy, size, shape and orientation of the electron orbital, three quantum numbers are required. These are *principal quantum number*, *azimuthal quantum number* and *magnetic quantum number*. These follow from direct solutions of *Schrodinger wave equation*. In order to designate the electron,

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an additional quantum number called *spin quantum number* is needed to specify spin of the electron. Thus, each orbital in an atom is designated by a set of three quantum numbers and each electron is designated by a set of four quantum numbers.

1. The principal quantum number (n). It determines *the energy of an electron as well as the average distance of an electron from the nucleus.* It can have any whole number value such as 1,2,3,4 etc. The energy levels corresponding to these numbers are designated by K,L,M,N etc. As the value of n increases, the electron gets farther away from the nucleus and its energy increases.

The energy of the electron in hydrogen atom is given by the relation

$E_n = -\frac{2\pi^2 me^4 k^2}{n^2 h^2}$ where, m is the mass of the electron e, charge of the electron, h, Planck's constant, n, principal quantum number and $k = \frac{1}{4\pi\varepsilon_0}$, where ε_0 is permitivity of free space (or air) and equals 10^{-12} Fm^{-1} (faraday per metre). Substituting the value of ε_0 and π we get $k = 9 \times 10^9$ Nm^2c^{-2}.

Substituting the values of π, m, e, h and k in the above equation, we get,

$$E_n = -\frac{2.178 \times 10^{-21}}{n^2} \text{ kJ/atom} = -\frac{1.312 \times 10^3}{n^2} \text{ kJ / mole}$$

According to the equation for E_n the largest negative value of energy is obtained when n has the smallest permitted value, i.e., when n is 1. Thus n = 1 gives the lowest energy state (E_1) of the electron in hydrogen atom. This state is called the *ground state of the electron* in the atom. For hydrogen atom , all other states with $n > 1$ are called excited states. The possible energy states of electron in hydrogen atom are :

$E_1 = -1.312 \times 10^3$ kJ mol^{-1} when $n = 1$
$E_2 = -3.28 \times 10^2$ kJ mol^{-1} $n = 2$
$E_3 = -1.46 \times 10^2$ kJ mol^{-1} $n = 3$

Thus, it is clear that the electron in hydrogen atom can have only certain specific values of energy.

2. The angular momentum quantum number or azimuthal quantum number *(l).* This quantum number determines *angular momentum of the*

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electron. The value of *l* gives the sub-level or sub-shell in which the electron is located. It also determines the shape of the orbital in which the electron is located. The number of sub-shells within a principal shell is determined by the value of *n* for that principal energy level. Thus *l* may have all possible whole number values from 0 to n-1 for each principal energy level. The various sub-levels are designated as *s, p, d and f* corresponding to *l* values 0, 1, 2 and 3 respectively. The first energy level has only one sub-shell, i.e., 1s. Second energy level has two sub-shells i.e., 2s and 2p. Third energy level has three sub-shells, 3s, 3p and 3d. Similarly, fourth energy level can have four sub-shells, 4s, 4p, 4d and 4f.

3. The magnetic quantum number (m). This refers to *the different orientations of electron cloud in a particular sub-shell,* around the nucleus, called *orbitals*. The number of orbitals in a particular sub-shell within a principal energy level is given by the number of values allowed to m, which in turn depend on the value of *l*. The possible values of m are all integral values from + *l* to 0 to -*l*, thus making a total of $(2l + 1)$ values. For $l = 0$ (i.e., s-sub-shell), m can have only one value, $m = 0$. It means that s-sub-shell has only one orbital. For $l = 1$ (i.e., p-sub-shell) m can have three values, +1, 0 and -1. This implies that p-sub-shell has three orbitals. For $l = 2$ (i.e., d-sub-shell), m can have five values +2, +1, 0, -1, -2. It means d-sub-shell has five orbitals. For $l = 3$ (i.e., f-sub-shell) m can have seven values +3, +2, +1, 0, -1, -2, -3. Hence, the f-sub-shell has seven orbitals.

4. The spin quantum number,s. This arises from the spectral evidence that *electron in its motion about the nucleus also rotates or spins about its own axis.* The spin can be either clockwise or anticlockwise. The spin quantum number can have only two values which are +½ and -½.

Example 1.1. (a) An atomic orbital has n=4. What are the possible values of *l* ? (b) An atomic orbital has *l*=3. What are the possible values of m? (c) Which shell would be the first to have an f subshell.

Solution: (a) For $n = 4$, *l*=0, 1,2,3, (b) For *l*=3, m = -3, -2, -1, 0, +1, +2, +3 (c) 4th shell will be the first to have an f sub-shell. For f subshell *l*=3. For *l*=3, the minimum possible value of n is 4.

Example 1.2. What are n, *l*, and m values for $2p_x$ and $3p_z$ orbitals?

Solution. For $2p_x$ orbital, n=2, *l*=1, m=-1 or +1

For $3p_z$ orbital n=3, *l*=1, m=0

Example 1.3. Which of the following orbitals are not possible? 1p, 2s, 3f?

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Solution. First shell has only one subshell, 1s. It has only one orbital, 1s orbital. Therefore 1p orbital is not possible. Second shell has two subshells, 2s and 2p. Therefore 2s orbital is possible. Third shell has three subshells, 3s, 3p and 3d. Therefore 3f orbital is not possible.

Example 1.4. Which of the following sets of quantum numbers are not permitted? (i) n=2, *l*=2, m=-1, s=+½ (ii) n=3, *l*=2, m=+1 s=0 (iii) n=3, *l*=2, m=+2, s=-½ (iv) n=2, *l*=1, m=+2, s=+½

Solution. The sets (i) (ii) and (iv) and not permitted because

(i) *l* cannot have a value equal to 2 for n=2 (ii) s cannot be equal to zero. (iv) m cannot be equal to +2 for *l*=1

Self Study Problems

1.1. What are n, *l* and m values of 3s and $3p_x$ electrons?

Ans: For 3s, n=3, *l*=0, m=0. For $3p_x$, n=3; *l*=1 and m=+1 or -1

1.2. What designation is given to an orbital having n=3 and *l*=2?

Ans: 3d

ELECTRONIC CONFIGURATIONS OF ATOMS

The filling of orbitals in an atom is a hypothetical process in which the atom is built up by feeding electrons in orbitals, one at a time and by placing each new electron in the lowest available energy orbital. *The distribution of electrons in different orbitals is known as* **electronic configuration of the atom.**

The filling of orbitals is governed by the following rules:

(1) Aufbau principle. The *aufbau principle* states that *in the ground state of an atom, an electron enters the orbital of lowest energy first and subsequent electrons are fed in the order of increasing energies.* The order of increasing energy of the orbitals is as follows:

1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s.

According to *aufbau principle*, the orbital should be filled in the above sequence.

It is very important to remember that the sequence of energy levels pertains up to *3p* and then *4s* orbital comes first instead of *3d*. In fact, the energy of an orbital is determined by the quantum numbers n and *l* with the help of an important rule known as (n+*l*) rule or **Bohr Bury's rule.** According to this :

(i) Orbitals fill in the order of increasing value of n+l. For example, 3s orbital (n+l=3+0=3) will be filled before 4s (n+l=4+0=4) orbital.

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Similarly, out of 3d and 4s orbitals the 4s (n+l=4+0=4) orbital will be filled before 3d (n+l=3+2=5) orbital.

(ii) If the two orbitals have same value of (n+l), then the orbital with lower value of n will be filled first. For example,

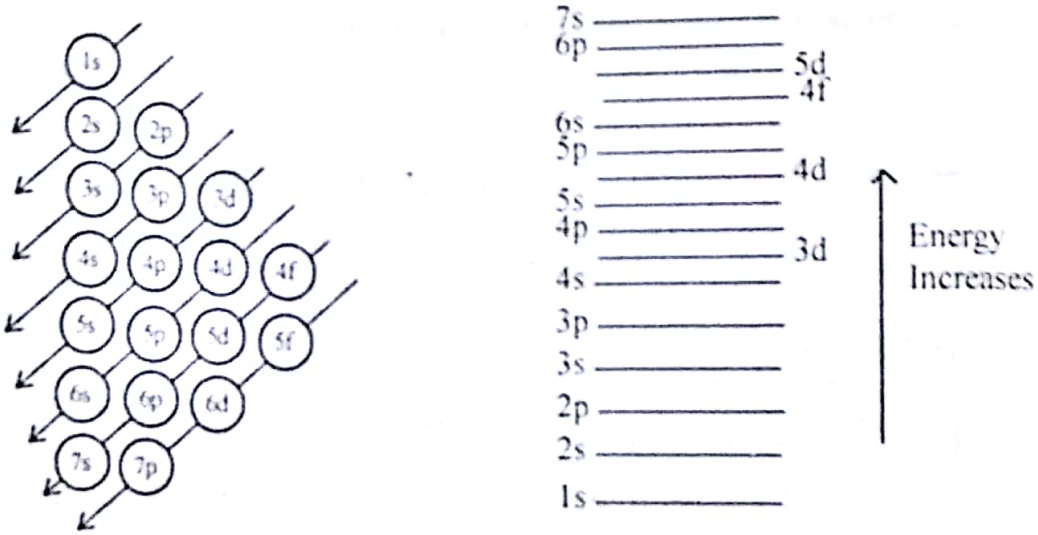

Fig. 1.9. Schematic diagram to remember sequence of filling atomic orbitals

2p orbital (n+*l*=2+1=3) and 3s orbital (n+*l*=3+0=3) have the same (n+*l*) value but 2*p* orbital has lower value of n and therefore, it will be filled first.

This rule also helps to account for the fact that certain orbitals with higher value of n but lower value of *l* have less energy than orbitals with lower value of n and higher value of *l*. For example, let us apply this rule to 4s and 3d orbitals. For 3d orbital, (n+*l*) value=3+2=5 while for 4s orbital (n+*l*) value=4+0=4. Thus, 4s orbital has lesser energy than 3d orbital and, therefore is filled first.

(2) Pauli's Exclusion Principle. Pauli exclusion principle states that:

"No two electrons in an atom can have the same values for all the four quantum numbers."

Thus, in any atom, two electrons may have the same values for three quantum numbers but the fourth must be different. For example, the set of four quantum numbers for the two electrons present in 1s orbital are:

n = 1 l = 0 m = 0 s = +½

n = 1 l = 0 m = 0 s = - ½

Thus, an orbital can have a maximum of two electrons. Moreover if an orbital has two electrons, they must be of opposite spin: for one, s=+½ and

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for other s = -½. Thus, the Pauli exclusion principle may also be stated as:

"no orbital can accomodate more than two electrons and these two must have opposite spin"

According to Pauli exclusion principle, only two sets of quantum numbers are possible for ns level, 6 for np level, 10 for nd level and 14 for nf level.

Therefore, it may be concluded that *the maximum number of orbitals in each energy level is n^2 and the maximum number of electrons is $2n^2$.*

(3) Hund's rule of maximum multiplicity. According to this rule, *electron pairing will not take place in orbitals of same energy (same sub-shell) until each orbital is first singly filled.* This suggests that it is difficult for an electron to enter an orbital which already has an electron than to enter an unoccupied orbital of the same energy. This principle is very important in guiding the filling of *p, d* and *f* orbitals which have more than one kind of orbitals. For example, we know that there are three *p* orbitals (p_x, p_y and p_z) in the *p*-subshell of a principal energy level. According to *Hund's rule*, each of the three *p* orbitals must get one electron of parallel spin before any one of them receives the second electron of opposite spin.

Electronic Configuration of Atoms

Based upon the above rules and the sequence of energy levels, let us write the electronic configurations of some atoms.

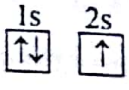
Hydrogen (At. No. = 1). Since hydrogen has only one electron, it must go to 1s orbital which has lowest energy. The electronic configuration for hydrogen is :

H : (Z=1) $1s^1$ or pictorially it may be shown as

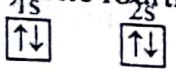
Helium: (At. No. = 2). In helium atom, the second electron can also go into 1s orbital. The two electrons must have opposite spins (*Pauli exclusion principle*).

He: (Z=2) $1s^2$ or

Lithium: (At. No. = 3). Since 1s orbital is filled with two electrons, it cannot have any more. Therefore, the third electron goes to the next lowest energy orbital, namely 2*s*.

Li: (Z=3): $1s^2 2s^1$ or 

Beryllium: (At. No. = 4). The fourth electron in beryllium fills 2*s* orbital

Be (Z=4): $1s^2\ 2s^2$ or 

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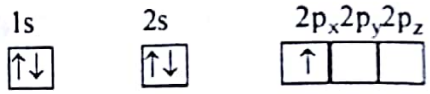
Boron: (At. No. = 5). In this case, the four electrons completely fill 1*s* and 2s orbitals. The fifth electron goes into any one of the 2*p* orbitals (say $2p_x$):

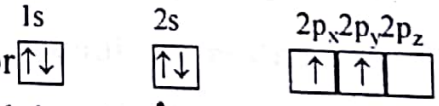
B (Z = 5): $1s^2\ 2s^2\ 2p_x^1$ or 

It may be remembered here that the fifth electron can enter any of the three $2p_x$, $2p_y$ or $2p_z$ orbitals because all are of the same energy. Thus, both the following configurations are also equally correct. It is only a matter of convention that we fill the orbitals in the order of $2p_x$, $2p_y$ and $2p_z$.

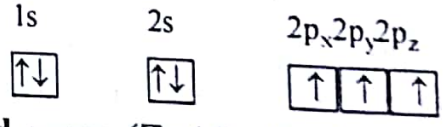
$1s^2\ 2s^2\ 2p_y^1$

and $1s^2\ 2s^2\ 2p_z^1$

Carbon: (At. No. = 6). In carbon atom, the sixth electron is also to go into the 2*p* sub-shell. Here Hund's rule applies, i.e., the electrons enter the orbitals of same energy with parallel spin until all are singly filled. Therefore, the sixth electron cannot enter the $2p_x$ orbital, rather it can go into either $2p_y$ or $2p_z$ in accordance with Hund's rule:

C(Z=6): $1s^2\ 2s^2\ 2p_x^1\ 2p_y^1$ or 

Nitrogen (At. No. = 7). Applying Hund's rule, nitrogen atom has three unpaired electrons in 2*p* orbitals as:

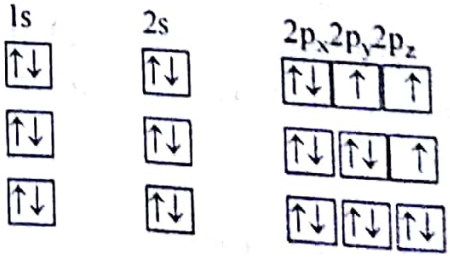
N(Z=7): $1s^2\ 2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^1$ or 

Oxygen (Z=8), fluorine (Z=9) and neon (Z=10). Beginning with oxygen, the 2p orbitals start getting filled by second electron till each of these are completely filled.

O(Z = 8): $1s^2\ 2s^2\ 2p_x^2\ 2p_y^1\ 2p_z^1$

F(Z=9): $1s^2\ 2s^2\ 2p_x^2\ 2p_y^2\ 2p_z^1$

Ne (Z=10): $1s^2\ 2s^2\ 2p_x^2\ 2p_y^2\ 2p_z^2$



Sodium, Z=11 to Argon, Z=18. The electronic configurations of these atoms are written exactly in the same manner as discussed above. Here, 3*s* and 3*p* orbitals are filled as did 2*s* and 2*p* orbitals in Li to Ne.

Furthermore, for simplicity a common convention is used. In this, the detailed electronic configuration of the noble gas core preceding the valence shell is represented by the symbol of the noble gas in square bracket. Then,

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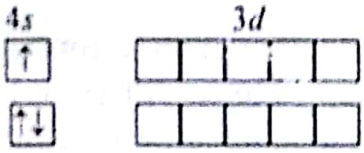
the configuration of the valence shell is written after the symbol. For example, the electronic configuration of sodium may be written as,

$$Na\ (Z=11) : [Ne]\ 3s^1.$$

The electronic configurations of these elements are given below :

Sodium, Na (Z=11)	[Ne] $3s^1$
Magnesium, Mg (Z=12)	[Ne] $3s^2$
Aluminium, Al (Z=13)	[Ne] $3s^2\ 3p_x^1$
Silicon, Si (Z=14)	[Ne] $3s^2\ 3p_x^1\ 3p_y^1$
Phosphorus, P (Z=15)	[Ne] $3s^2\ 3p_x^1\ 3p_y^1\ 3p_z^1$
Sulphur, S (Z=16)	[Ne] $3s^2\ 3p_x^2\ 3p_y^1\ 3p_z^1$
Chlorine, Cl (Z=17)	[Ne] $3s^2\ 3p_x^2\ 3p_y^2\ 3p_z^1$
Argon, Ar (Z=18)	[Ne] $3s^2\ 3p_x^2\ 3p_y^2\ 3p_z^2$

Potassium (Z=19) and calcium (Z=20). In these atoms, electrons enter the 4s orbital in accordance with the fact that 4s orbital is slightly lower in energy than 3d orbital.

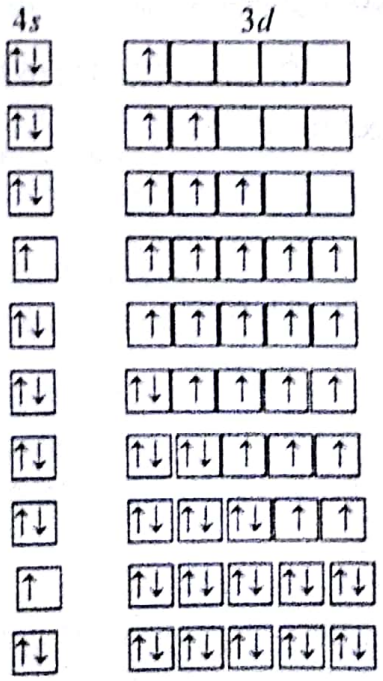


Potassium, K (Z=19) : [Ar] $4s^1$

Calcium, Ca (Z=20) : [Ar] $4s^2$

Scandium (Z=21) to zinc (Z=30). With scandium (Z=21), 3*d* subshell comes into use and from scandium to zinc, this subshell is gradually filled up.

The electronic configurations of the elements from scandium (Z=21) to zinc (Z=30) are given below:



Scandium,	Sc (Z=21) : [Ar] $4s^2\ 3d^1$ or
Titanium,	Ti (Z=22) : [Ar] $4s^2\ 3d^2$ or
Vanadium,	V (Z=23) : [Ar] $4s^2\ 3d^3$ or
Chromium,	Cr (Z=24) : [Ar] $4s^1\ 3d^5$ or
Manganese,	Mn(Z=25) : [Ar] $4s^2\ 3d^5$ or
Iron,	Fe (Z=26) : [Ar] $4s^2\ 3d^6$ or
Cobalt,	Co (Z=27) : [Ar] $4s^2\ 3d^7$ or
Nickel,	Ni (Z=28) : [Ar] $4s^2\ 3d^8$ or
Copper,	Cu (Z=29) : [Ar] $4s^1\ 3d^{10}$ or
Zinc,	Zn (Z=30) : [Ar] $4s^2\ 3d^{10}$ or

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Stability of filled and half filled orbitals

Exceptional configurations of chromium and copper. From the above configurations, it may be noted that there are two irregularities in the general trend. The electronic configurations of Cr and Cu are different from what we expected. The electronic configurations for these atoms are expected to be

Cr : [Ar] $4s^2\ 3d^4$ and Cu : [Ar] $4s^2\ 3d^9$

However, the actual configurations are :
Cr : [Ar] $4s^1\ 3d^5$ and Cu : [Ar] $4s^1\ 3d^{10}$

This is attributed to the fact that half filled (containing one electron per orbital) and completely filled (containing two electrons per orbital) electronic configurations have lower energy and therefore extra stability. Thus, the configurations, d^5, d^{10}, f^7 and f^{14} which are either half filled or compeletely filled, are more stable.

Thus, to acquire increased stability, one of the 4*s* electrons goes into the nearby 3*d* orbitals so that 3*d* orbitals get half filled in Cr and completely filled in Cu.

The stability of these configurations can also be explained due to their larger *exchange energy*. The exchange means the shifting of electrons from one orbital to another of the same subshell. The exchange of electrons also leads to stability. It has been observed that in case of d^5 and d^{10} configurations, the electrons can exchange their positions more than in case of d^4 and d^9 configurations. Thus, the greater exchange of half filled and completely filled configurations gives them more stability.

Gallium (Z=31) to krypton (Z=36). With gallium onwards, 4p orbitals get filled up as:

Gallium	Ga (Z=31)	[Ar] $4s^2\ 3d^{10}\ 4p_x^1$
Germanium	Ge (Z=32)	[Ar] $4s^2\ 3d^{10}\ 4p_x^1\ 4p_y^1$
Arsenic	As (Z=33)	[Ar] $4s^2\ 3d^{10}\ 4p_x^1\ 4p_y^1\ 4p_z^1$
Selenium	Se (Z=34)	[Ar] $4s^2\ 3d^{10}\ 4p_x^2\ 4p_y^1\ 4p_z^1$
Bromine	Br (Z=35)	[Ar] $4s^2\ 3d^{10}\ 4p_x^2\ 4p_y^2\ 4p_z^1$
Krypton	Kr (Z=36)	[Ar] $4s^2\ 3d^{10}\ 4p^6$

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CHEMICAL BONDING

Most of the substances exist in the form of clusters or aggregates of atoms. Any such cluster in which atoms are held together and which is electrically neutral, is called a molecule.

"The force of attraction which binds the atoms together in a molecule is called a ***chemical bond.***"

The process of combination of atoms is called ***chemical bonding.***

Cause of Chemical Bonding

Since long many attempts have been made to understand the cause of chemical combination. The first explanation about the cause of chemical combination between atoms was the tendency of the atoms to attain stable noble gas configurations. It has been observed that atoms of noble gases have little or no tendency to combine with each other or with atoms of other elements. The electronic configurations of valence shell for noble gases can be expressed as ns^2np^6 (except helium, $1s^2$) and this represents the *stable configuration and corresponds to maximum stability.* Due to their stable configuration, the noble gas atoms neither have any tendency to gain nor lose electrons and therefore, they have no combining capacity.

All atoms other than noble gases have less than eight electrons in their outermost shells. In other words, they do not have stable configurations. Therefore, they combine with each other and also with other atoms to acquire stable noble gas configurations (ns^2np^6 or $1s^2$). To a large extent, the elements undergo electronic rearrangements to attain stable noble gas configurations. *Thus, the tendency or urge of atoms of various elements to attain stable configuration of eight electrons in their valence shells, is the* **cause of chemical bonding**. *The principle of attaining maximum of eight electrons in the valence shell of atoms is called* **octet rule.**

However, it may be noted that for hydrogen and lithium atoms, the stable configuration means two electrons which corresponds to the configuration of the nearest noble gas helium.

The cause of chemical bonding may also be understood in terms of decrease in energy during bond formation as discussed below:

Energetics of Bond Formation

Let us consider energy changes in the formation of bonds between two hydrogen atoms to form a hydrogen molecule.

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Hydrogen exists in nature as H_2 molecules. Since a more stable state has lower energy, it means that the bonded state of atoms (molecule) has lower potential energy than isolated atoms. Thus, energy is released when a molecule is formed. For example, when one gram mole of hydrogen is formed from 2 gram atoms of hydrogen, 433 kJ of energy is evolved.

$$H + H \longrightarrow H_2 + 433 \text{ kJ}$$

On the other hand, energy is needed to break a molecule. For example, hydrogen molecule dissociates into atoms when heated to a sufficiently high temperature by absorbing 433 kJ mol^{-1} energy. This may be written as

$$H_2(g) + 433 \text{ kJ} \longrightarrow 2\,H(g)$$

Thus, we can say that hydrogen molecule is more stable than the hydrogen atoms. The greater stability of hydrogen molecule than hydrogen atoms can be explained as follows:

We know that an atom consists of a positively charged nucleus and electrons revolving around the nucleus. As the two atoms approach each other, they start to interact and new types of forces begin to operate. For example, let us consider two hydrogen atoms A and B having nuclei H_A and H_B and the corresponding electrons e_A and e_B respectively approaching each other. When the two atoms are at large distances from each other, no interaction between the two takes place. At this stage, the total energy of the system is the sum of the energies of the individual atoms. When the two atoms come closer, new attractive and repulsive forces begin to operate. Besides the attraction of the nucleus of one atom for its own electrons, i.e., H_A–e_A and H_B–e_B, the following attractive and repulsive forces start operating:

(i) Attractive forces. Attractive forces are between electron of atom A(e_A) and nucleus of B (H_B) and electron of atom B (e_B) and nucleus of A(H_A).

(ii) Repulsive forces. Repulsive forces are between the nuclei H_A and H_B and electrons of two atoms e_A and e_B.

Now we know that attractive forces tend to bring the atoms close while repulsive forces tend to push them apart. *It has been observed experimentally that the magnitude of the new attractive forces is more than the new repulsive forces.* As a result, the two atoms approach each other and the potential energy of the system decreases. As the two atoms come closer and closer, the system becomes more and more stable due to decrease of energy. Ultimately, a stage is reached where the total forces of attraction balance the forces of repulsion

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and the system acquires minimum energy. At this stage, the two hydrogen atoms are said to be bonded together to form a stable molecule and the distance (r_0) between the atoms is known as **bond length**. For hydrogen molecule, the distance between two hydrogen atoms corresponding to minimum energy is 0.074 nm (0.74 Å). The potential energy of the system when two atoms are brought closer and closer is represented in Fig. 1.15. This is called *potential energy diagram*. In this diagram, when the two atoms are separate (stage *a*), there is no attractive or repulsive interactions between them and the potential energy of the system (isolated atoms) is assumed to be zero.

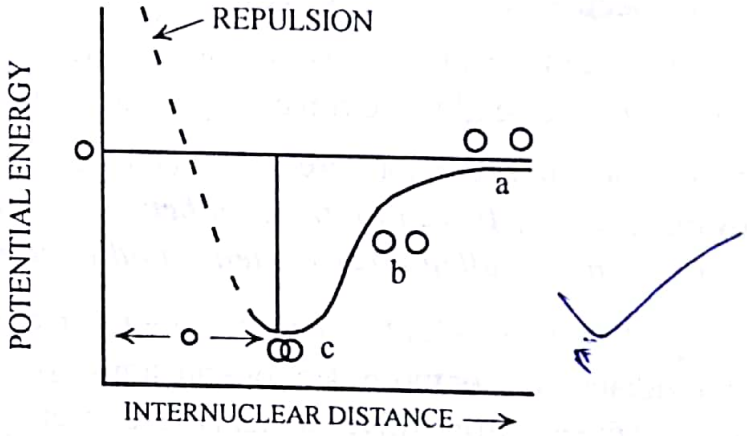

Fig. 1.10. Potential energy diagram for H_2 molecule.

Thus, when the bond is formed, energy is released and therefore, the hydrogen molecule is more stable than the individual hydrogen atoms. The energy corresponding to minimum in the curve is called *bond energy* which is equal to 433 kJ mol^{-1}.

It may be remembered that the two hydrogen atoms cannot be brought closer than 0.074 nm, because the repulsive forces will become large and the potential energy would rise (shown by dotted line in Fig. 1.10).

Types of Chemical Bonds

The chemical combination of two or more atoms involves the redistribution of electrons between themselves so that all of them acquire the stable noble gas configuration.

If the union takes place by the ***transfer of electrons*** from one atom to another, the bonding is said to be ***ionic*** and if it takes place by the ***mutual sharing of electrons*** between the atoms concerned, the bonding is said to be ***covalent***. There is another type of bonding in which both the electrons shared between the atoms are contributed by one atom only. This type of bonding is

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known as *co-ordinate covalent bond*. In addition to the above types, some other types of bonds like *metallic bond* and *hydrogen bond* are also known.

Ionic bond or Electrovalent bond.

In relation to chemical bonding *W. Kossel* pointed out the following facts.

(i) In the periodical table, the highly electropositive alkali metals are separated from the highly electronegative halogens by the noble gases.

(ii) The formation of a positive ion from an atom of alkali metal and a negative ion from an atom of halogen is associated with the loss and gain of an electron by the respective atoms.

(iii) The positive and negative ions so formed attain stable noble gas configurations, i.e., eight electron configuration. (ns^2np^6)

(iv) The positive and negative ions are stabilized by electrostatic attraction. ***This electrostatic force of attraction between positive and negative ions is called electrovalent bond or ionic bond.***

To sum up: When an electronegative element combines with an electropositive element, one or more electrons are transferred from the valency shell of an atom of the electropositive element to the valency shell of the atom of the electronegative element. As a result, the electropositive atom becomes a positive ion and the electronegative atom becomes a negative ion. These oppositely charged ions are held together by an electrostatic force of attraction called *electrovalent bond* or *ionic bond*. Thus, ***ionic bond may be defined as the electrostatic force of attraction holding the oppositely charged ions.***

The number of electrons gained or lost by an atom during the formation of an ionic bond is called electrovalency of the atom. **In short, electrovalency is equal to the charge on the ion.**

To illustrate the formation of ionic bond, let us consider the example of sodium chloride.

Sodium atom ($1s^2\ 2s^2\ 2p^6\ 3s^1$) has only one electron in its valency shell. Chlorine atom ($1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$), on the other hand, has seven electrons in its valency shell. When sodium and chlorine combine, from the valency shell of sodium atom one electron is transferred to the valency shell of chlorine atom. As a result, sodium atom attains the electronic configuration ($1s^2\ 2s^2\ 2p^6$) of the nearest noble gas, neon and it becomes a positive ion. At the same time, Cl atom becomes Cl^- ion with the configuration ($1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$) of the nearest noble gas argon. The Na^+ and Cl^- ions are held together by

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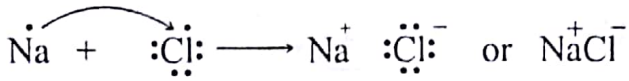
strong electrostatic attraction which constitutes the ionic bond.



Here, sodium atom loses one electron and carries charge +1. ∴ Its electrovalency is +1. On the other hand, Cl atom gains one electron and carries charge -1. So the electrovalency of chlorine is -1.

General Properties of Ionic Compounds

The common properties of ionic compounds are:

1. **Physical state.** Ionic compounds are usually solids with definite shape.
2. **Solubility.** Ionic compounds are soluble in water and other polar solvents but insoluble in nonpolar solvents.
3. **Electrical conductivity.** Do not conduct electricity in the solid state because the ions occupy fixed positions. However, they conduct electricity in the molten state or in solutions because the ions becomes free in that state.
4. **High melting and boiling points.** Ionic solids are hard and have high melting and boiling points.
5. **Ionic reactions.** Chemical reactions of ionic compounds involve the combination of ions furnished in their solutions. Such reactions are called ***ionic reactions*** which are instantaneous and rapid.

Energetics of Ionic Bond Formation

The formation of an ionic bond is accompanied by a decrease of energy. To illustrate the energetics of ionic bond formation let us consider the formation of sodium chloride from gaseous sodium atoms and chlorine atoms. The reaction can be split into three hypothetical stages, each one involving an energy change.

(i) Removal of the valency electron of the sodium atom to form sodium ion. For this, a certain amount of energy is to be supplied. This energy is first ionization energy (I.A) of sodium.

$$Na_{(g)} \longrightarrow Na^{+}_{(g)} + 1e^{-};\ I.E = 495.4kJ\ mol^{-1}$$

(ii) Addition of an electron to chlorine atom to give a chloride ion. During this a certain amount of energy is released. This energy is the *electron affinity* (E.A) of chlorine.

$$Cl_{(g)} + 1e^{-} \longrightarrow Cl^{-}_{(g)};\ E.A = -348.5kJ\ mol^{-1}$$

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(iii) The bringing together of the sodium ion and chloride ion to form solid sodium chloride. During this process energy is released which is known as *lattice energy* (U).

$$Na^+_{(g)} + Cl^-_{(g)} \longrightarrow Na^+Cl^-_{(s)};\ U = -757.3 kJ\ mol^{-1}$$

The overall energy change is, IE + E.A + U = 495.4kJ mol^{-1} -348.5 kJ mol^{-1} -757.3 kJ mol^{-1} = -610.4kJ mol^{-1}.

Thus, it is evident that the formation of NaCl is accompanied by a decrease of energy. This explains the reason for the formation and stability of ionic bond between sodium and chlorine.

From this it is clear that the formation of ionic compound is favoured by

(a) low ionization energy of one atom, (b) high electron affinity of other atom and (c) high lattice energy.

Lattice Energy

Lattice energy of a crystal is defined as the amount of energy released when one gram mole of the crystal is formed from gaseous ions.

The formation of a mole of an ionic solid from constituent gaseous ions may be represented as,

$$A^+_{(g)} + B^-_{(g)} \longrightarrow A^+B^-_{(s)};\ U = -\text{energy (i.e., lattice energy)}$$

When the +ve and -ve ions come together to form the ionic solid, there is a decrease of potential energy. Hence, in the formation of an ionic solid from the constituent gaseous ions, a certain amount of energy is released. Hence, U has a negative sign. The amount of energy released during the formation of one mole of an ionic solid from the constituent gaseous ions, is numerically equal to the energy that will be needed to convert one mole of the same ionic solid into the gaseous ions.

Thus, it is obvious that *the higher the magnitude of lattice energy of an ionic solid, the greater would be its stability.* The lattice energy of ionic solids are quite high. This is because of the strong electrostatic attraction between oppositely charged ions present in the solid. Since the coulombic force of attraction is directly proportional to the product of the charges, the higher the valency of the ions, the greater would be the lattice energy of the ionic solid. As a result, the lattice energy increases from uni-univalent ionic solids to uni-bivalent ionic solids and then to bi-bivalent ionic solids.

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Lattice Energy of Some Ionic Solids

Uni-Uni Valent Solids		Bi-Uni-Valent Solids		Bi-Bivalent Solids	
Ionic Solid	Lattice Energy $kJmol^{-1}$	Ionic Solid	Lattice energy $kJmol^{-1}$	Ionic solid	Lattice energy $kJmol^{-1}$
LiF	-1033	CaF_2	-2581	MgO	-3932
CsF	-748	MgF_2	-2882	BeO	-3125
LiI	-140			MgS	-3254
CsI	-601			ZnO	-4032

Lattice enthalpy

The energy required to completely separate one mole of solid ionic compound into gaseous constituent ions is called ***lattice enthalpy.*** *Thus, lattice enthalpy is equal to lattice energy but with reverse sign.*

Born-Haber Cycle

Since the direct experimental determination of lattice energies is not so easy, these are determined indirectly with the help of a thermochemical cyclic process known as *Born-Haber cycle*. This cycle devised by *Born* and *Haber* in 1919 relates the lattice energy of a crystal to other thermochemical data.

To illustrate this, let us consider the determination of lattice energy of NaCl. The energy terms involved in building a crystal lattice such as sodium chloride be taken in steps. The elements in their standard state are first converted to gaseous atoms, and then to ions, and finally packed into the crystal lattice. Thus, the formation of NaCl(s) involves the following steps.

(i) ***Conversion of metallic sodium to gaseous sodium atoms***

Th amount of energy required to convert one mole of metallic sodium to gaseous sodium atoms is called *sublimation energy.*

$$Na(s) + \Delta Hs \longrightarrow Na(g)$$

(ii) ***Conversion of gaseous chlorine molecule to gaseous chlorine atoms.***

The amount of energy required to dissociate one mole of gaseous chlorine molecules to gaseous atoms is called *dissociation energy*, ΔH_D. Evidently the energy required to dissociate ½ mole of Cl_2 is ½ ΔH_D.

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$$\tfrac{1}{2}Cl_{2(g)} + \tfrac{1}{2}\Delta H_D \longrightarrow Cl_{(g)}$$

(iii) ***Conversion of Gaseous Sodium atoms to Sodium Ions***

The amount of energy required to convert one mole of gaseous sodium atoms to sodium ions is called *ionisation energy*, I.E.

$$Na_{(g)} + I.E. \longrightarrow Na^+_{(g)} + le^-$$

(iv) ***Conversion of gaseous chlorine atoms to chloride ions***

The amount of energy released when one mole of gaseous chlorine atoms are converted into chloride ions in the gaseous state is called *electron affinity*, E.A.

$$Cl_{(g)} + le^- \longrightarrow Cl^-_{(g)} + E.A$$

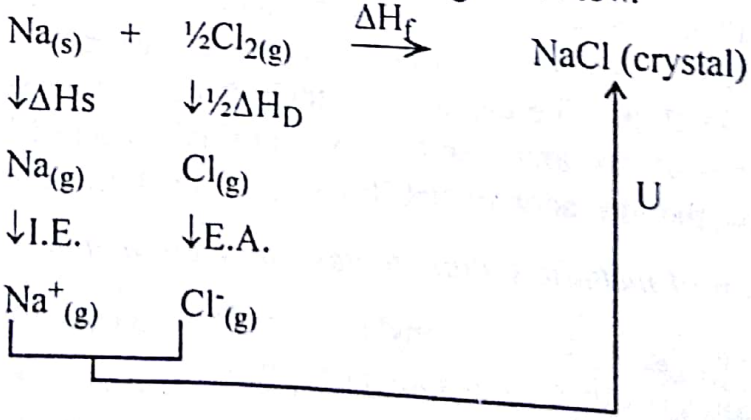
(v) ***Combination of gaseous Na^+ and gaseous Cl^- to form $Na^+Cl^-_{(s)}$***

The amount of energy released when one mole of solid crystalline compound is formed from gaseous Na^+ and Cl^- ions is called *lattice energy*, U

The overall change is given as, $Na_{(s)} + \tfrac{1}{2}Cl_{2(g)} \longrightarrow NaCl_{(s)}$

The enthalpy change for the reaction is called *enthalpy of formation* of NaCl, i.e., ΔH.

The various steps involved in the process may be represented in the form of *Born-Haber cycle* which is given below.



Born-Haber Cycle for the formation of NaCl

According to Hess's law, the overall energy change in a process depends only on the initial and final states and not on the route taken. As can be seen from the *Born Haber cycle*, the enthalpy of formation, ΔH_f is the algebraic sum of the terms going round the cycle, i.e., $\Delta H_f = \Delta Hs + \tfrac{1}{2}\Delta H_D + I.E. + E.A. + U$

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The enthalpies of sublimation and dissociation and the ionization energy are positive since energy is supplied to the system. The electron affinity and lattice energy are negative since energy is evolved in these processes. Thus, for the formation of NaCl(s), we have,

Heat of sublimation of sodium, ΔHs = 108.4 kJ mol^{-1}
Heat of Dissociation of Cl_2, ΔH_D = 241.8 kJ mol^{-1}
Ionization energy of sodium, I.E. = 495.4 kJ mol^{-1}
Electron affinity of chlorine, E.A. = -348.6 kJ mol^{-1}
Enthalpy of formation of NaCl, ΔH_f = -381.2 kJmol^{-1}

Substituting the various values in the above equation, we get -381.2kJ mol^{-1} =108.4 kJ mol^{-1}+½ × 241.8 kJ mol^{-1}+495.4 kJ mol^{-1} -348.6 kJ mol^{-1}+U

∴ Lattice energy of sodium chloride, U = -381.2 kJmol^{+1} -108.4 kJmol^{-1} - ½ x 241.8 kJmol^{-1} -495.4 kJmol^{-1} + 348. 6 kJmol^{-1} = -757.3 kJ mol^{-1}.

Example 1.5. Calculate the lattice energy of MgF_2 from the following data. Heat of sublimation of magnesium, ΔH_S = 146.4 kj mol^{-1}; Dissociation energy of fluorine, ΔH_D = 158.8 kJ mol^{-1}; $IE_1 + IE_2$ of Mg = (737 + 1449) = 2186 kJ mol^{-1}; Electron affinity fluorine = -328 kJ mol^{-1} and ΔH_f of MgF_2 = -1096.5 kJ mol^{-1}.

Solution : $\Delta H_f = \Delta H_S + \Delta H_D + (IE_1 + IE_2) + EA + U$

-1096.5kJ mol^{-1} =146.4 kJ mol^{-1} + 158.8 kJ mol^{-1} + 2186 kJ mol^{-1} - 2 × 328 kJ mol^{-1} + U

∴ U = -1096.5 kJ mol^{-1} - 146.4 kJ mol^{-1} - 158.8 kJ mol^{-1} - 2186 kJ mol^{-1} + 656 kJ mol^{-1} = - 2931.7 kJ mol^{-1}.

Self Study problems

1.3 Calculate the lattice energy and lattice enthalpy of KCl from the given data. Heat of sublimation of potassium = + 89 kJ mol^{-1} ; Dissociation energy of chlorine, ΔH_D = 244 kJ mol^{-1}; Ionization energy of potassium = 425 kJ mol^{-1} ; Electron affinity of chlorine = -355 kJ mol^{-1} and Enthalpy of formation of $KCl_{(s)}$ is - 438 kJ mol^{-1}.

Ans:- Lattice energy = -719 kJ mol^{-1} ; Lattice enthalpy = 719 kJ mol^{-1}.

1.4 Calculate the lattice energy of $MgBr_2$ from the given data. ΔH_f = -524 kJ mol^{-1} ; ΔH_s of Mg = +148 kJ mol^{-1} ; IE_1+IE_2 of Mg = + 2187 kJ mol^{-1} ; ΔH_D of $Br_{2(l)}$=+31 kJ mol^{-1}; E_{DH} of $Br_{2(g)}$ = +193 kJ mol^{-1}; EA of Br = -331 kJ mol^{-1}; ΔH_f $MgBr_{2(s)}$=-524 kJ mol^{-1}.

Ans:- Lattice energy = -2421 kJ mol^{-1}.

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Consequences of Lattice energy. The important consequences of lattice energy are:

(i) The greater the lattice energy, the more stable is the ionic compound.
(ii) Ions having higher charge and smaller radii will have greater lattice energy.
(iii) The value of lattice energy affects the solubilities of ionic compounds.

Solvation Energy and Solubility of Ionic Solids

Ionic solids are only very slightly soluble in most common solvents, except for those that are quite polar. The strong attractive forces between the ions in the crystal must be overcome. This can be accomplished only if the attractive forces between the ions and the solvent molecules are at least comparable to the lattice energy. The energy required for the separation of ions from the crystal comes from the solvation of the ions. The energy released during solvation of ions is called *solvation energy*. Thus, *an ionic crystal will dissolve only when the solvation energy overcomes the lattice energy*. The attractive forces between neutral molecules are usually very weak. Also the forces between an ion and a neutral molecule are not much stronger unless the molecule has a fairly high dipole moment or high polarizability. *The greater the dipole moment, the stronger the attraction by an ion and the greater the solvation energy. Solvation energy also increases with the polarizability of the solvent molecules*. The energy required to separate the ions is diminished by a decrease in the forces between ions, which are dependent on the dielectric constant of the medium. *As the dielectric constant of the medium increases, the attractive force between the ion decreases.* For example, the dielectric constant of water is 78.54 and hence the attractive force between two ions in water is 1/78.54 of the force between the ions separated by the same distance in vacuum. Good solvents for ionic substances usually have high dipole moments and high dielectric constants.

Water is a much better solvent for ionic solids than other solvents with high dielectric constants [e.g., HF.]. This can be explained as follows. A metal ion interacts strongly with a layer of water molecules called primary hydration sphere. The water molecule in the primary hydration sphere form stronger hydrogen bonds to other water molecules than they could otherwise. Thus, a secondary hydration sphere arises. The water molecules in the secondary hydration sphere, in turn, are hydrogen bonded to other water molecules with hydrogen bonds weaker than those formed by water in the primary hydration sphere but stronger than in water itself. [Fig.1.11]. Anions-

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particularly oxoanions also can be strongly hydrogen bonded to water. [Fig. 1.11.]. HF is a poorer solvent for salts, in spite of its high dipole moment and ability to form strong hydrogen bonds because it is a much weaker base than water and hence a poorer electron donor.

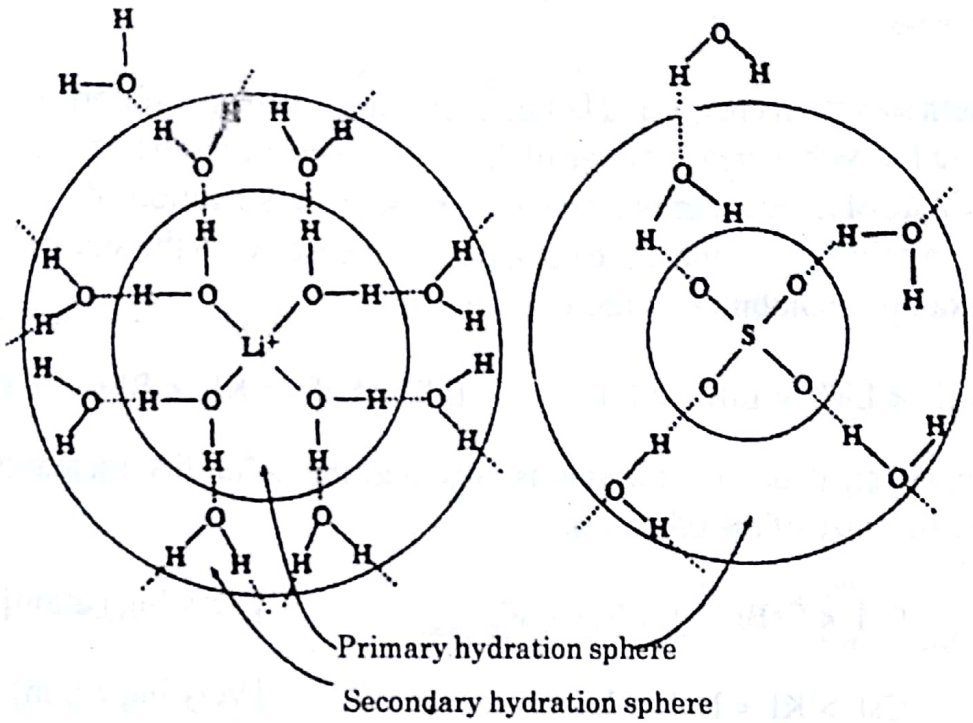

Fig. 1.11
Hydrated Li^+ and SO_4^{2-} ions

The energy changes involved in the dissolution of a salt can be expressed by a Born-Haber type cycle.

$$M + X^-_{(crystal)} \xrightarrow{-U_0} \Delta M^+_{(g)} + X^-_{(g)}$$

$$\Delta H_{soln} \searrow \qquad \downarrow \Delta H_{solv}$$

$$M^+_{(solv)} + X^-_{(solv)}$$

where U_0 is the lattice energy of the crystal, MX, ΔH_{solv} is solvation energy and ΔH_{soln} is the enthalpy of solution. Since the total energy change in going from M^+X^- crystal to the solvated ions is independent of the path, the heat of solution is given by,

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$$\Delta H_{soln} = \Delta H_{solv} - U_0$$

The enthalpy of solution may be positive or negative, depending on the relative magnitudes of the lattice energy and solvation energy. The enthalpy of solution is often, but not always , negative (exothermic) for very soluble substances.

Both solvation energy and lattice energy are inversely proportional to the ionic radii. With increase of size of the ions, the greater is the decrease in the magnitude of lattice energy than the decrease in the solvation energy. Due to this, the solubility of the ionic compound increases with the size of the ion. For example, solubility increases in the order:

$LiF < LiCl < LiBr < LiI$; $LiF < NaF < KF < RbF < CsF$

However, if one of the ions is very big, the solubility increases with decreasing size of the other ion.

$CsI < CsBr < CsCl < CsF$ [Very big cation]

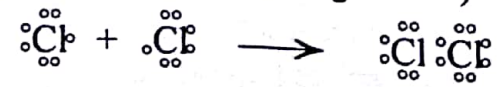
$CsI > KI < NaI < LiI$ [Very big anion]

$CsNO_3 < KNO_3 < NaNO_3 < LiNO_3$ [Very big anion]

COVALENT BONDS

Bonds formed by the mutual sharing of electrons between two atoms are called *covalent bonds*. The atoms involved in covalent bond formation contribute equal number of electrons for sharing. The shared electrons become common to both the atoms. This shared pair of electrons is called *bond pair*. The valence electron pairs, which are not involved in bond formation are called *lone pairs*.

To illustrate the formation of a covalent bond, consider the formation of chlorine molecule from two chlorine atoms. Each Cl atom has seven electrons in the valency shell. The two Cl atoms share one electron each to form Cl_2 molecule. Thus, each Cl atom acquires the electronic configuration of the nearest noble gas, argon (i.e., $ns^2\ np^6$ configuration).



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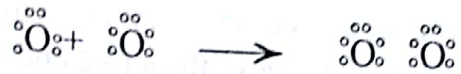
In the formation of oxygen molecule each oxygen atom has six electrons in the valency shell and, therefore, they contribute two electrons each for sharing. Thus, two electron pairs are shared and there is a *double bond* between the two oxygen atoms.



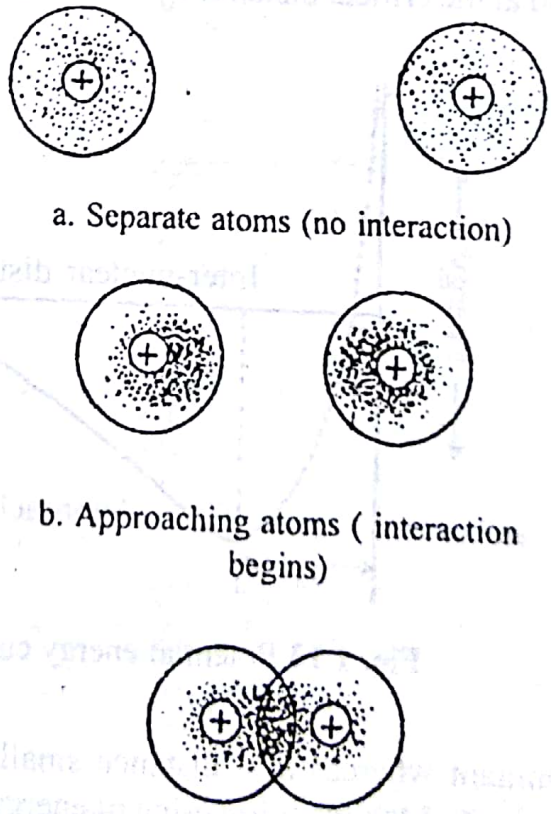
The number of electrons which an atom contributes for sharing in a covalent bond is called its ***covalency.*** Thus, covalency of chlorine is 1 and that of oxygen is 2.

Modern concept of Covalent Bond

Valence Bond Theory

Eventhough the VSEPR model is a useful method for predicting the geometry of simple molecules, it lacks a theoretical basis and has only limited applications. The modern theories of the formation of the covalent bond are essentially based on quantum mechanical principles. Based on quantum mechanics, two theories have been put forward to explain the formation of covalent bonds. These are: *(i) Valence bond theory and (ii) Molecular orbital theory.* Let us discuss the Valence Bond theory (VB theory). For simplicity, we will discuss only the qualitative treatment of VB theory.

This theory is based on the fact that all systems in the universe tend to lower their potential energy. The lowering of energy will increase the stability of the system. Therefore, formation of bonds between the atoms occur only if it is accompanied by decrease of energy. To illustrate this let us consider the formation of hydrogen molecule.



a. Separate atoms (no interaction)

b. Approaching atoms (interaction begins)

c. Bonded atoms (overlap of orbitals)

Fig. 1.12. Formation of hydrogen molecule when two hydrogen atoms approach each other

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Formation of H_2 molecule. Consider two hydrogen atoms A and B having nuclei H_A and H_B and the corresponding electrons e_A and e_B respectively. When the two atoms are at large distance from each other, there is no interaction between them. ***The potential energy of the system at this stage is taken as zero.*** As the two atoms come closer and closer, they will begin to interact with each other. As a result, new attractive and repulsive forces begin to operate. The new forces are:

1. Attractive forces between the nucleus of one atom and the electron of the other atom, i.e., (i) the attraction between H_A and e_B (ii) the attraction between H_B and e_A.

2. Repulsive forces between their electrons and their nuclei, i.e., (i) repulsion between e_A and e_B (ii) repulsion between H_A and H_B.

However, the attractive forces out weigh the repulsive forces and there is a net attraction. Hence the energy of the system decreases as shown in potential energy diagram (Fig. 1.13). Ultimately a stage is reached when the attractive forces balance the repulsive forces. This stage is reached when the atoms are held at the critical distance r_0. At a distance greater than r_0 attractive force is

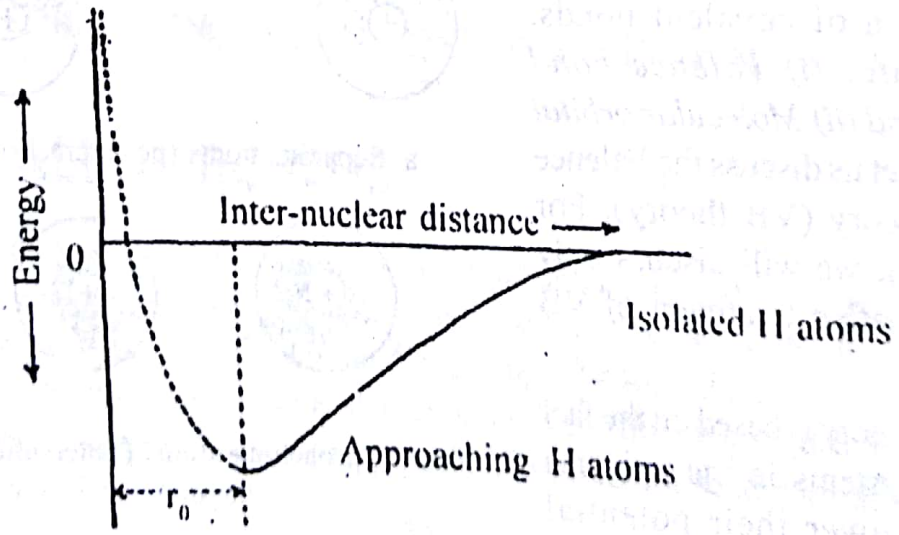

Fig. 1.13 Potential energy curve for hydrogen molecule

dominant whereas at a distance smaller than r_0, the repulsive forces are dominant. Maximum lowering of energy occurs at the critical distance, r_0. At this stage the two H atoms are said to be bonded. The critical distance corresponding to the minimum energy is called ***bond length***. For hydrogen, the value of r_0 is 74 pm (0.74×10^{-10} m). The energy corresponding to the minimum energy is called *bond energy*.

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For hydrogen molecule, the bond energy is 433 kJ per mole. Thus, during the formation of the bond 433 kJ of energy is released. Therefore, the H_2 molecule is more stable than the individual atoms. That is,

$$H + H \rightarrow H_2 + 433 \text{ kJ mol}^{-1}$$

Orbital Overlap Concept for Covalent Bond

When two atoms approach each other, their bonding atomic orbitals undergo partial interpenetration. ***This partial interpenetration of atomic orbitals is known as overlapping of atomic orbitals.*** Overlap of atomic orbitals may be positive, negative or zero depending on the properties of the overlapping atomic orbitals. The various arrangements of s and p orbitals overlap resulting positive, negative and zero overlaps as shown in Fig. 1.14.

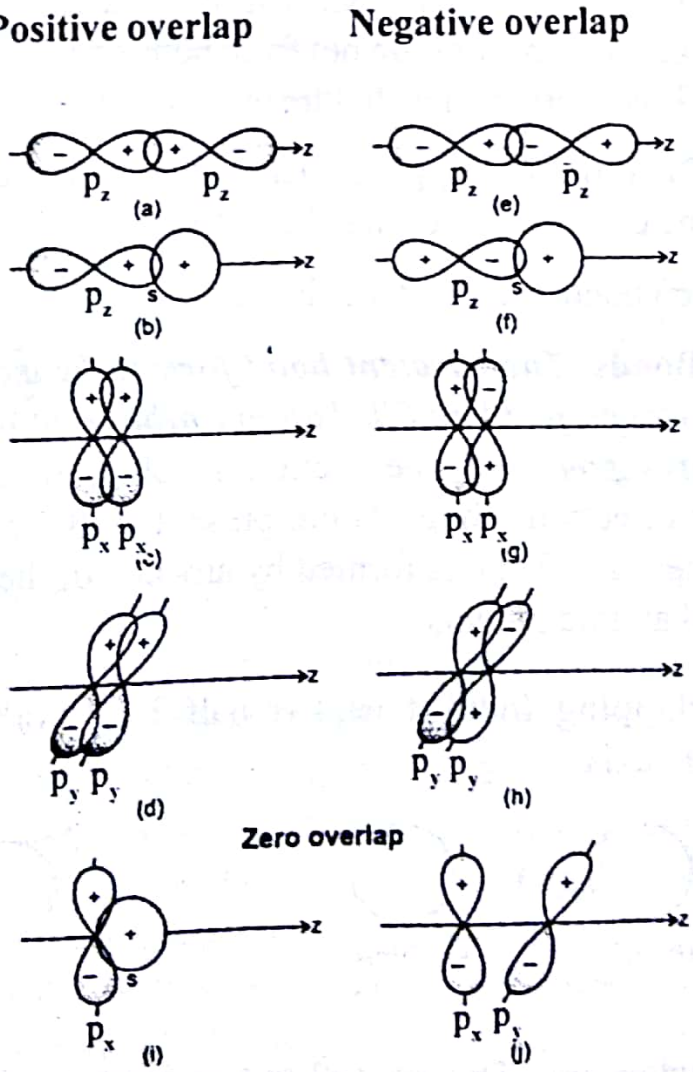

Fig. 1.14 Positive, negative and zero overlaps of s and p orbitals.

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The main ideas of orbital overlap concept of forming covalent bonds are:

(i) *Covalent bonds are formed by the overlapping of half filled atomic orbitals present in their valency shells.*

(ii) *The orbital undergoing overlapping must have electrons with opposite spins. The overlapping results in pairing of the electrons.*

(iii) *Overlapping of atomic orbitals results in the decrease of energy and formation of covalent bond.*

(iv) *The strength of a covalent bond depends on the extent of overlapping. The greater the overlapping, the stronger is the bond formed between two atoms.*

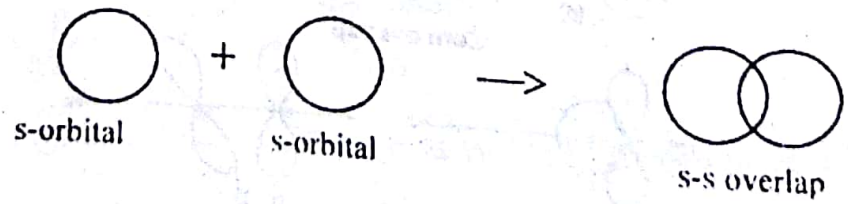
Why is He_2 molecule not formed? Helium atom has two electrons in its 1s orbital. The 1s orbital of one helium atom cannot overlap with the 1s orbital of another helium atom, because it violates the Pauli's exclusion principle. Hence, helium atoms do not form bonds between them. Therefore, He_2 molecule is not formed, i.e., helium exists as a monoatomic gas.

Types of Orbital Overlapping. Depending upon the extent and type of overlapping, the covalent bonds may be of two types.

(i) Sigma (σ) bond and (ii) Pi (π) bond.

1. Sigma Bonds. ***The covalent bond formed by axial overlapping or end to end overlapping of half filled atomic orbitals along the internuclear axis is known as sigma bond.*** The electrons involved in sigma bond formation are called sigma electrons. Single bonds present in all molecule are always of sigma type. The sigma bond is formed by any one of the following types of combination of atomic orbitals.

(i) s-s overlapping. In this type, two half filled s orbitals overlap along the inter nuclear axis.



(ii) s-p overlapping. This type of overlapping occurs between the hal filled s orbital of one atom and half filled p orbital of the other atom.

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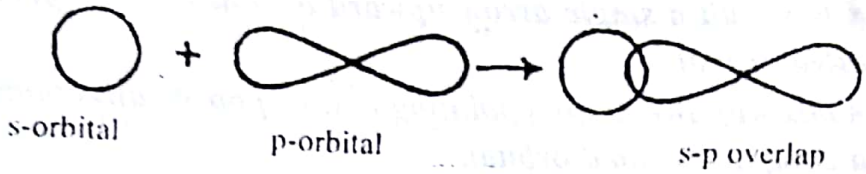
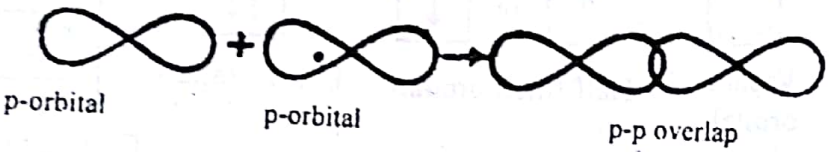
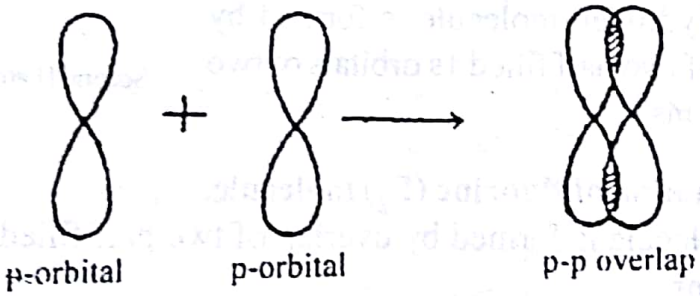


(iii) p-p overlaping. This type of overlapping occurs between the half filled p orbital of one atom and half filled p orbital of the other atom.



π Bonds. ***The covalent bond formed by the side wise or lateral overlap of the half filled atomic orbitals is known as pi bond.*** The electrons involved in pi bond formation are called pi electrons. The atomic orbitals overlap in such a way that their axis remain parallel to each other and perpendicular to the internuclear axis.



The π bond is always present (in addition to the sigma bond) in molecules having multiple bonds, i.e., double or triple bonds. For example, the double bond in O_2 molecule(O=O) consists of one sigma and one π bond.

Strength of sigma and pi bonds. The strength of a covalent bond depends on the extent of overlapping of atomic orbitals forming the bond. During the formation of a sigma bond, the overlapping of orbitals takes place to a large extent. On the other hand, during the formation of a pi bond, the overlapping occurs to a smaller extent. Therefore, *a sigma bond is stronger than a pi bond.*

Representation of covalent bonding by orbital overlap

(i) The orbitals are represented by square boxes.

(ii) An empty box means a vacant orbital.

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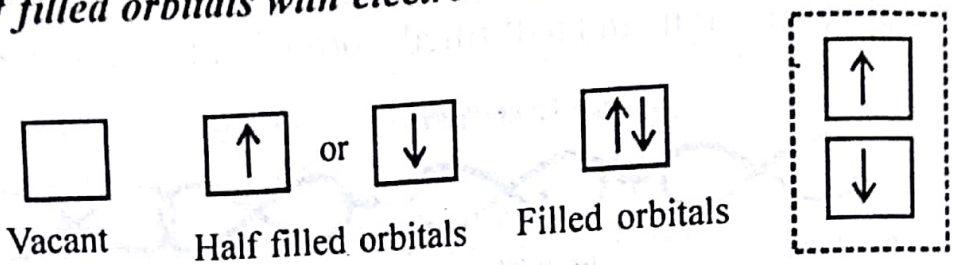
(iii) A box with a single arrow upward or downward represents a half filled orbital.

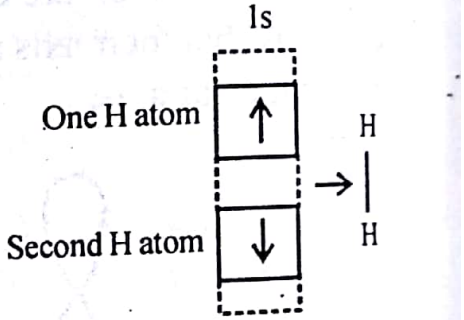
(iv) A box with two arrows pointing in the opposite directions represents a completely filled orbital.

(v) The overlapping of orbitals is shown by dashed lines enclosing the half filled orbitals with electrons having opposite spins.



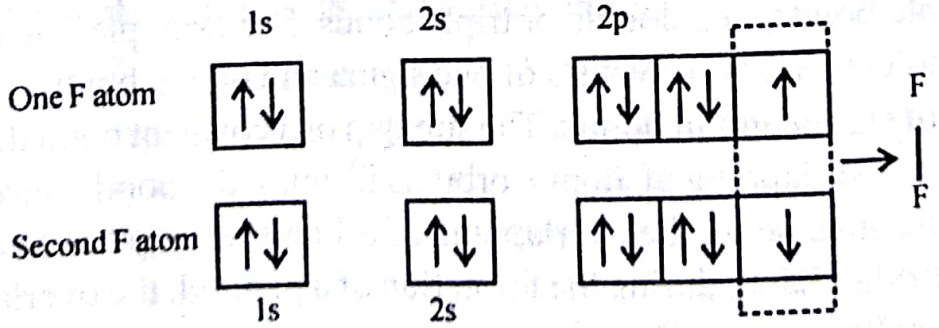
Representation of bonding in some simple molecules

Let us discuss the bonding in some covalent molecules in terms of orbital representation.



(i) Formation of Hydrogen (H_2) molecule. Hydrogen molecule is formed by the overlap of two half filled 1s orbitals of two hydrogen atoms.

(ii) Formation of fluorine (F_2) molecule.
Fluorine molecule is formed by overlap of two half filled p orbitals of two fluorine atoms.



(iii) Formation of Hydrogen (HF) Fluoride molecule. Hydrogen fluoride molecule is formed by the overlap of 1s orbital of hydrogen atom with one of the 2p orbitals of fluorine atom which is half filled.

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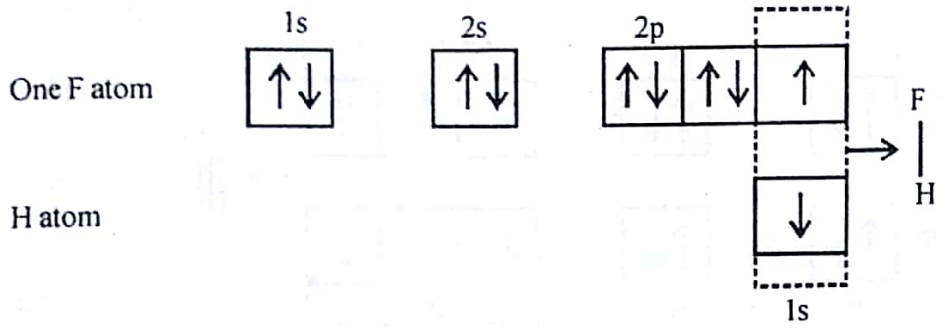
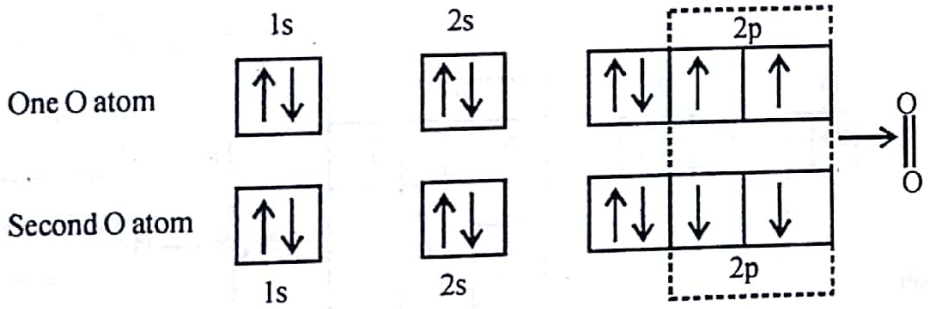
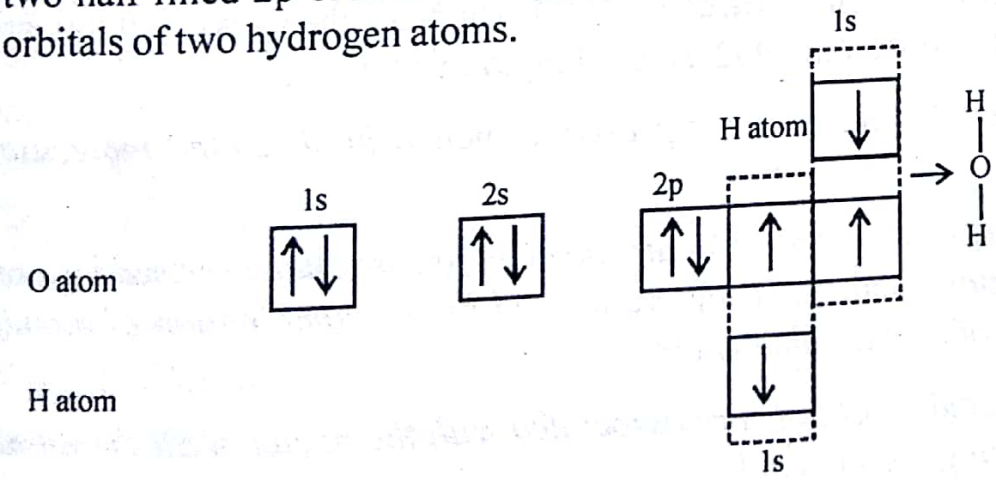


(iv) Formation of oxygen (O_2) molecule. Oxygen molecule is formed when two half filled 2p orbitals of one oxygen atom overlap with two half filled 2p orbitals of the other oxygen atom. This gives rise to a double bond between two oxygen atoms.



(v) Formation of water (H_2O) molecule. Water molecule is formed when two half filled 2p orbitals of oxygen atom overlap with two half filled 1s orbitals of two hydrogen atoms.



(vi) Formation of nitrogen (N_2) molecule. Nitrogen molecule is formed when the three half filled 2p orbitals of one nitrogen atom overlap with three similar orbitals of the other nitrogen atom. As a result, three is a triple bond between nitrogen atoms in N_2 molecule.

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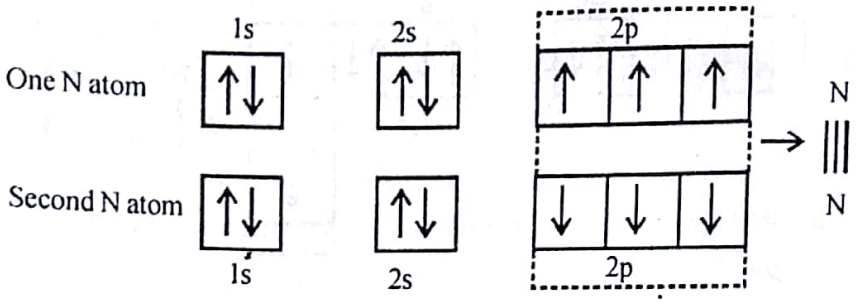
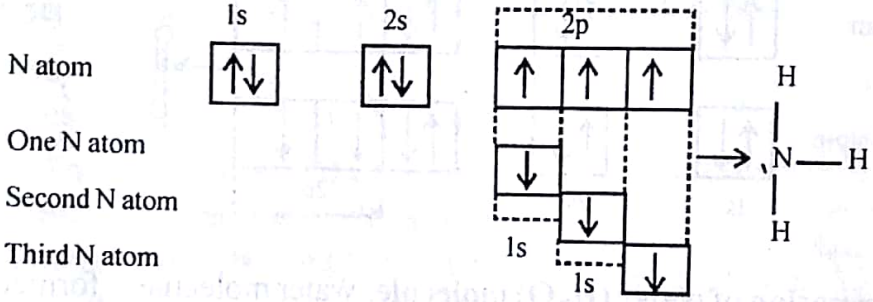


(vii) Formation of Ammonia (NH_3) molecule. Ammonia molecule is formed when three half filled p orbitals of nitrogen atom overlap with three half filled 1s orbitals of three hydrogen atoms.



Molecular Orbital Theory (MOT): The M.O. theory was put forward by *Hund* and *Mulliken* in 1932. According to this theory:

1. *In molecules, electrons are present in new orbitals called* ***molecular orbitals.***
2. *The molecular orbitals, like atomic orbitals, are characterised by a set of quantum numbers. They are formed by the combination of atomic orbitals of nearly same energy.*
3. *The molecular orbitals are associated with the nuclei of all the atoms constituting the molecule.*
4. *The number of molecular orbitals formed is equal to the number of atomic orbitals combined.*
5. *The molecular orbitals are filled in the increasing order of their energies (aufbau principle)*
6. *A molecular orbital, like atomic orbitals, can accomodate only two*

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electrons and these electrons must have opposite spins (Pauli's principle).

7. *Pairing of electrons in molecular orbitals of equal energy does not take place until all such orbitals contain one electron each (Hund's rule).*

The MOT can be applied to homonuclear diatomic, heteronuclear diatomic and polyatomic molecules.

Differences between Atomic Orbitals and Molecular Orbitals

An atomic orbital is *monocentric*, i.e., an electron in an atomic orbital is only influenced by one nucleus. A molecular orbital is *polycentric*, i.e., an electron in a molecualr orbital is influenced by two or more nuclei.

Formation of Molecular Orbitals.

Molecular orbitals are formed by the combination of atomic orbitals. The atomic orbitals can combine by (i) addition and (ii) subtraction of their wave functions. This is called *linear combination of atomic orbitals (LCAO) method.*

If ψ_A and ψ_B are the orbital wave functions of two atoms A and B they can combine by addition to form a molecular orbital, ψ_{MO}. The molecular orbital formed by the addition of atomic orbitals has a lower energy than the combining orbitals. It is called ***bonding molecular orbital*** *since it favours bond formation.* It is also represented as ψ_b. On the other hand, the two atomic orbitals can combine by subtraction to form a molecular orbital ψ^*_{MO}. The molecular orbital formed by subtraction of atomic orbitals has a higher energy than the combining orbitals. It is called ***antibonding molecular orbital*** *as it opposes bond formation.* It is also represented as ψ_a. Thus, the formation of bonding molecular orbital and antibonding molecular orbital can be represented as,

$$\psi_{MO} \text{ or } \psi_b = \psi_A + \psi_B ;$$

$$\psi^*_{MO} \text{ or } \psi_a = \psi_A - \psi_B$$

To illustrate the combination of atomic orbitals, let us consider the combination of two 1s orbitals. They can combine by addition to form the bonding molecular orbital, σ1s and combine by subtraction to form antibonding

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molecular orbital, σ*1s. It is clear from Fig. 1.15 that there is an increase of electron density in between the two nuclei during combination by addition. On the other hand, there is a decrease of electron density in between the two nuclei during combination by subtraction.

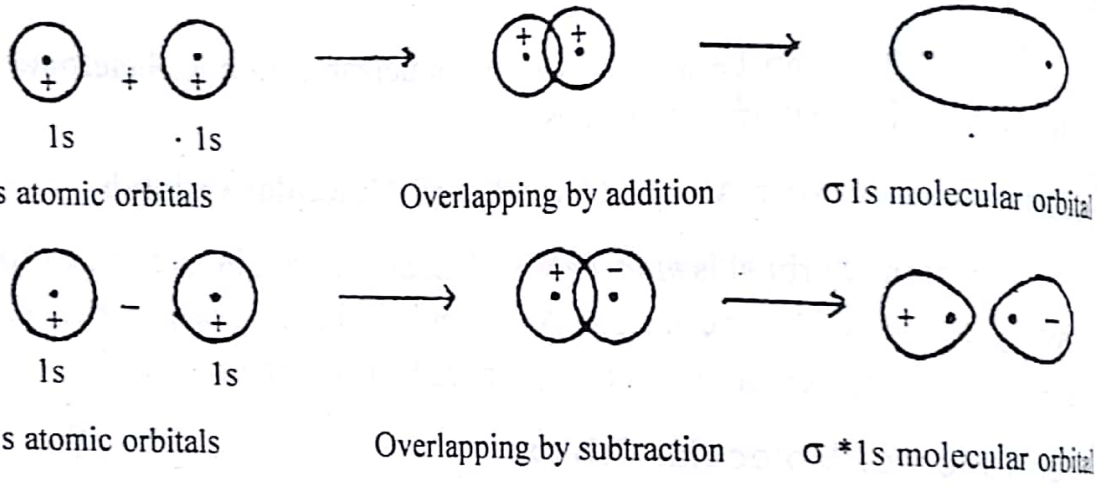

Fig 1.15 Molecular orbitals formed by addition and subtraction of two 1s orbitals

Difference between bonding and antibonding molecular orbitals.

The main points of differences of bonding and antibonding molecular orbitals are summed up in the following Table.

Table 1.4

Bonding M.O	Antibonding M.O
1. Bondig molecular orbital is formed by the addition of atomic orbitals.	1. Antibonding molecular orbital is formed by the subtraction of atomic orbitals.
2. It may or may not have a node.	2. It always has a node in between the nuclei of bonded atoms.
3. Electrons in this orbital lead to the attraction between atoms.	3. Electrons in this orbital lead to repulsion between atoms.
4. It possesses lower energy than the energy of the atomic orbitals.	4. It possesses higher energy than the energy of the atomic orbitals.
5. It is formed when the lobes of the combining atomic orbitals have the same sign.	5. It is formed when the lobes of the combining atomic orbitals have opposite sign

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Conditions for the Combination of Atomic Orbitals: The conditions for the formation of molecular orbitals from atomic orbitals are:

1. The combining atomic orbitals must have nearly equal energies.

2. The atomic orbitals must overlap to a considerable extent.

3. The combining atomic orbitals must have the same symmetry about the molecular axis.

For example, $2p_x$ orbital of one atom can combine with $2p_x$ orbital of the other atom but not with $2p_z$ orbital.

Combination of 2s Atomic Orbitals

2s orbitals combine by addition and subtraction overlap to form bonding molecular orbital, σ2s and antibonding molecular orbital, σ*2s (Fig. 1.16) respectively.

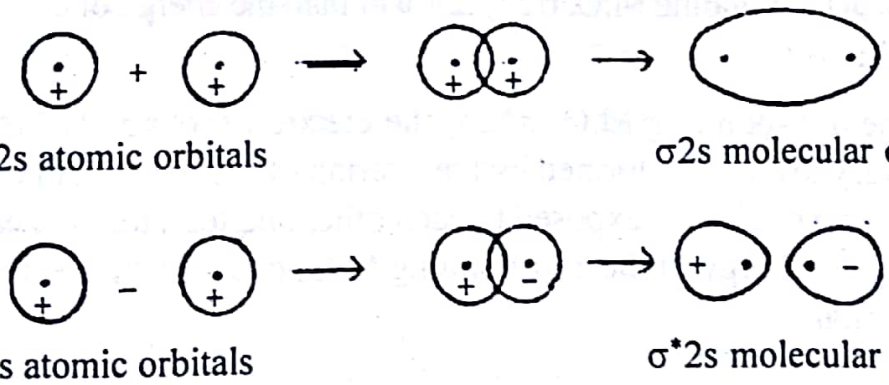

Fig 1.16 Molecular orbitals formed by the addition and subtraction of two 2s orbitals.

The '+' and '–' signs in the figure indicate the algebraical signs of the wave function and they do not have any relation with the charge of the electron cloud which is always negative.

Molecular orbitals from p-orbitals: Since the three p-orbitals of an atom, viz., p_x, p_y and p_z are perpendicular to one another, only one of them (i.e., p_z orbital, since Z axis is taken as inter molecular axis) can overlap with a p orbital of another atom along the axis of the molecule. Thus, the combination of two $2p_z$ orbital can give a sigma bonding molecular orbital, designated as $\sigma 2p_z$ and a sigma anti-bonding molecular orbital designated $\sigma^* 2p_z$ as in Fig. 1.17

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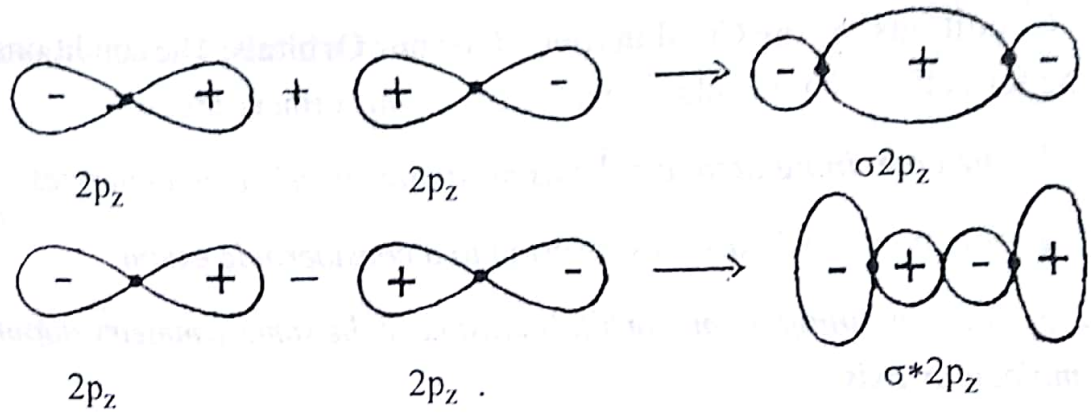

Fig.1.17 Molecular orbitals formed by addition and subtraction overlap of $\sigma 2p_z$ orbitals

The thick dots represent the nuclei. The bonding M.O, $\sigma 2p_z$ is formed when lobes with similar signs overlap and antibonding M.O, $\sigma^* 2p_z$ is formed when lobes with opposite signs overlap.In the bonding M.O,the electron probability between the two nuclei is increased. Since the repulsion between the nuclei in such a case is minimum (as they are well shielded by the electrons), the energy of the bonding M.O, $\sigma 2p_z$ is lower than the energy of the combining atomic orbitals ($2p_z$).

In the anti-bonding M.O, $\sigma^* 2p_z$ the electron probability between the nuclei is very low as it is formed by the overlap of lobes with opposite signs. Hence the two nuclei are exposed to each other and tend to repel each other. Therefore, the energy of the anti-bonding M.O, $\sigma^* 2p_z$ is higher than that of the $2p_z$-orbital.

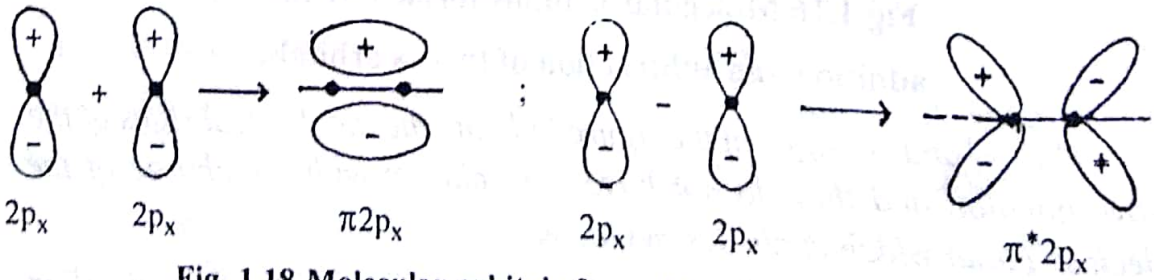

Fig. 1.18 Molecular orbitals formed by the addition and subtraction overlap of $2p_x$ orbitals.

The remaining pairs of p-orbitals (i.e.,$2p_x$ and $2p_y$ orbitals of the two atoms) are perpendicular to the axis of the molecule. So they can overlap only sidewise. *The molecular orbitals which are formed by sidewise overlap of p-orbitals are called pi (π) molecular orbitals.* For example, the addition and subtraction overlap of $2p_x$ orbitals result in the formation of $\pi 2p_x$ and $\pi^* 2p_x$ molecular orbitals respectively (Fig 1.18).

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Similarly, addition and subtraction of $2p_y$ orbitals result in the formation of $\pi 2p_y$ and $\pi^* 2p_y$ molecular orbitals respectively. (Fig. 1.19)

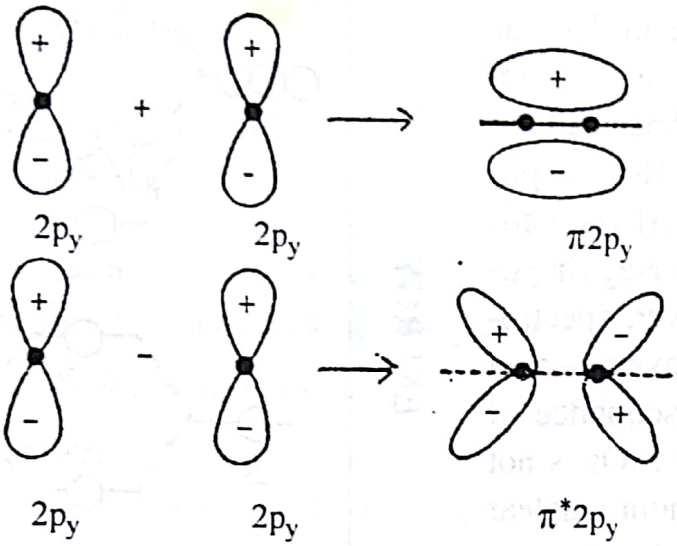

Fig. 1.19. M.O's formed by the addition and subtraction overlap of $2p_y$ orbitals.

Here also, the overlap of lobes with similar signs results in the formation of bonding MO ($\pi 2p_x$ or $\pi 2p_y$) and overlap of lobes with opposite signs result in the formation of anti-bonding MO ($\pi^* 2p_x$ or $\pi^* 2p_y$).

Comparison of Sigma and Pi Molecular Orbitals

σMO	πMO
1. It is formed by the axial overlap of atomic orbitals. 2. Region of overlap is maximum. 3. It leads to the formation of a strong bond. 4. The orbital is symmetrical to rotation about inter nuclear axis	1. It is formed by the lateral overlap of atomic orbitals. 2. Region of overlap is minimum. 3. It leads to the formation of a weak bond. 4. Orbital is not symmetrical to rotation about inter nuclear axis.

Energy Level Diagram for Molecular Orbitals:

The increasing order of energies of the ten molecular orbitals arising from two 1s, two 2s and six 2p atomic orbitals is given below:

$\sigma 1s,\ \sigma^* 1s,\ \sigma 2s,\ \sigma^* 2s,\ \sigma\ 2p_z,\ \pi 2p'_x = \pi 2p_y,\ \pi^* 2p_x = \pi^* 2p_y.\ \sigma^* 2p_z$

The energies of the various molecular orbitals have been determined experimentally from spectroscopic data. The energy levels of different

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molecular orbitals are shown diagramatically in Fig. 1.20.

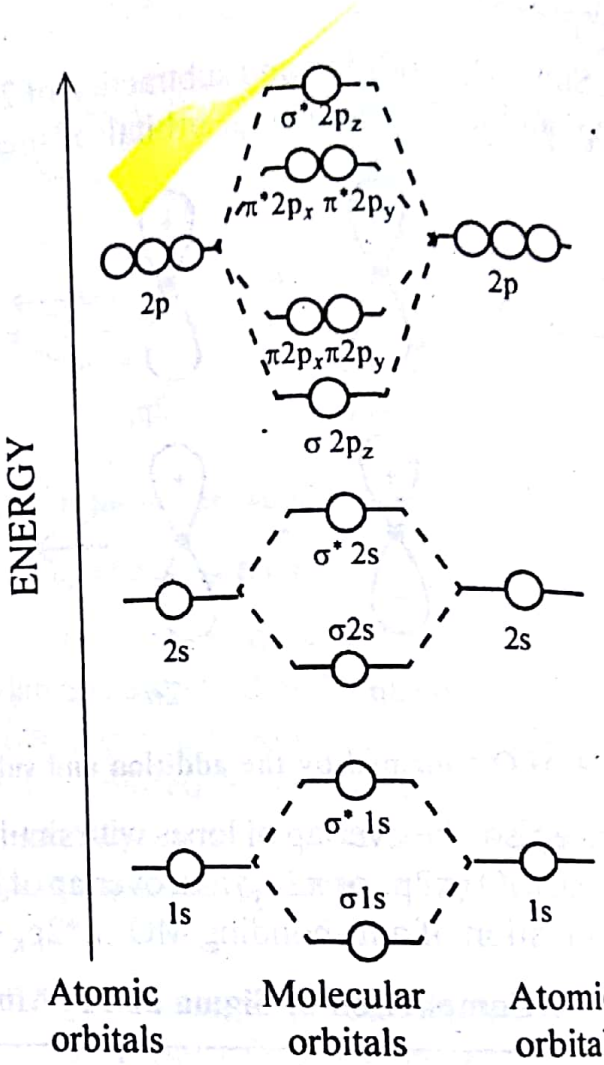

Fig. 1.20. MO energy level diagram for diatomic homonuclear mol

The above molecular orbital energy level diagram is followed for species like, H_2, H_2^+, He_2^+, He_2 (hypothetical), Li_2,Be_2 (hypothetical), O_2, F_2 and Ne_2 (hypothetical). However, spectroscopic studies have revealed that the above sequence of energy levels of MOs is not correct for all homonuclear diatomic molecules. For example, it has been observed experimentally that in some diatomic molecules such as B_2,C_2 and N_2, the energy of $\sigma 2p_z$ MO is higher than that of $\pi 2p_x$ and $\pi 2p_y$ MOs. In these elements, the energy difference between 2s and 2p atomic orbitals is very small. Therefore, the interaction between them is very large. This interaction raises the energy of $\sigma 2p_z$ MO and it occupies a higher position than $\pi 2p_x$ and $\pi 2p_y$ in the energy level diagram. However, the intermixing interaction does not occur for molecules such as O_2 and F_2 as there is large difference in energy between their 2s and $2p_z$ orbitals.

Rules For Adding Electrons to Molecular Orbitals. The molecules are built up by adding electrons to molecular orbitals in the same way as the atoms are built up by adding electrons to atomic orbitals. The principles followed are the same in both the cases

(1) The molecular orbital with the lowest energy is filled first.

(2) The maximum number of electrons in a molecular orbital cannot exceed two.

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(3) If there are two molecular orbitals of the same energy level, electrons will occupy separate orbitals, thus leading to unpaired electron spins.

Electron configuration of molecules and molecular behaviour

The electron configuration of a molecule,i.e., the distribution of electrons in various MOs, can give information about the number of electrons in bonding MOs (N_b) and number of electrons in antibonding MOs (N_a). From this, the behaviour of a molecule can be predicted.

(i) Stability of a molecule: The relative number of bonding and antibonding electrons determine the stability.

1) If Nb is greater than Na, the molecule is stable.

2) If Nb is less than Na, the molecule is unstable.

3) If Nb is equal to Na, the molecule is unstable.

This rule specifies that the effect of anti-bonding electrons is slightly more than that of the bonding electrons. This leads to the non-existence of the molecule.

(ii) Bond Order: The relative stability of molecules is also estimated from a quantity called *bond order.*

Bond order is defined as half of the difference between number of electrons in bonding MOs and anti-bonding MOs.

i.e., Bond order = ½[No. of electrons in bonding MOs -No. of electrons in antibonding MOs] =½ [Nb-Na]

The bond order gives the following informations:

- 1. If the value of *bond order is positive, the molecule is stable and if the value of bond order is negative or zero, the molecule is unstable,* i.e., it is not formed.

2. The dissociation energy of a diatomic molecule is directly proportional to the bond order of molecule, i.e., *greater the bond order, greater is the dissociation energy.*

3. The bond length of a molecule is inversely proportional to the bond order of the molecule, i.e., *greater the bond order, shorter is the bond length.*

4. The bond order of a diatomic molecule is equal to the number of

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covalent bonds in the molecule, i.e., bond order one means a single bond; bond order two means a double bond and so on. *If the bond order is zero no bond is formed.*

iii) Magnetic Behaviour: *A molecule will be paramagnetic if it has unpaired electrons and diamagnetic if it has no unpaired electrons in its molecular orbitals.*

Now let us consider some homonuclear diatomic species.

Hydrogen Molecule, H_2 : This is formed by the combination of two hydrogen atoms. Each hydrogen atom has one electron in 1s orbital. Therefore, the two electrons in hydrogen molecule are present in σ1s molecular orbital. Thus, the electron configuration of H_2 molecule is $\sigma(1s)^2$. The MO diagram of H_2 molecule is given in Fig. 1.21.

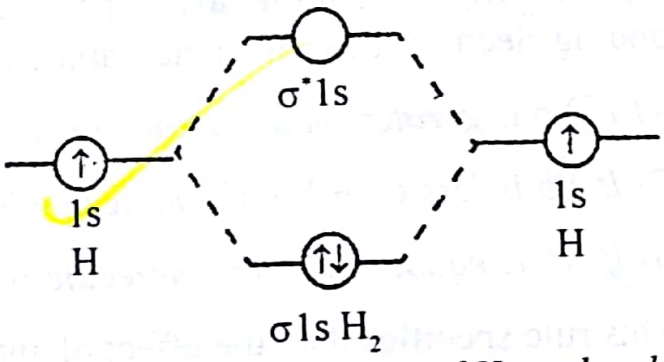

Fig. 1.21 *MO diagram of H_2 molecule*

Bond order of hydrogen molecule = ½ [2-0] = 1.

This means that the two hydrogen atoms are held together by a *single covalent bond*. The H_2 molecule is *diamagnetic as it contains no unpaired electron.*

Nitrogen Molecule, N_2 : The electron configuration of nitrogen atom (Z=7) is $1s^2\ 2s^2\ 2p^3$. Thus, N_2 molecule has 10 valence electrons. These electrons are accommodated in molecular orbitals in the increasing order of energy (Fig.1.22).

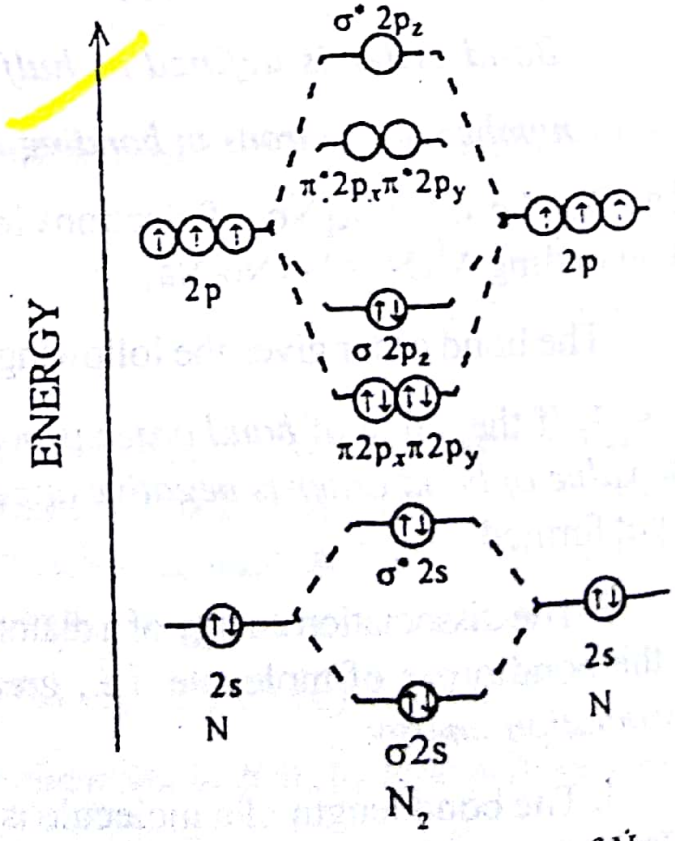

Fig. 1.22 *MO diagram of N_2 molecule*

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Thus, the MO configuration of N_2 molecule is:

KK $\sigma(2s)^2$, $\sigma^*(2s)^2$, $\pi(2p_x)^2=\pi(2p_y)^2$ $\sigma(2p_z)^2$.

Bond order = ½ [8-2] = 3

A bond order of 3 for N_2 molecule suggests the formation of three bonds in it, i.e., N_2 molecule contains a triple bond. *Since all the electrons are paired, the N_2 molecule is diamagnetic.*

The oxygen molecule : Each oxygen atom has 2+6=8 electrons. Thus, the O_2 molecule contains a total of 16 electrons. These are arranged in MOs.

$\sigma 1s^2$ $\sigma^*1s^2\sigma 2s^2\sigma^*2s^2\sigma 2p_z^2\pi 2p_x^2=\pi 2p_y^2\pi^*2p_x^1=\pi^*2p_y^1$

The inner shell of filled $\sigma 1s$ and σ^*1s MOs do not contribute to the bonding. They are essentially the same as the atomic orbitals from which they are formed. So the molecular orbital configuration is sometimes written as:

KK $\sigma 2s^2\sigma^*2s^2\sigma 2p_z^2\pi 2p_x^2=\pi 2p_y^2\pi^*2p_x^1=\pi^*2p_y^1$

The letters KK denote the closed K shell structure. It is assumed that these closed shells do not enter into bonding.

Bond order = ½[8-4]=2

Thus, the two oxygen atoms in O_2 are bonded together by a double bond.

The antibonding π^*2p_x and π^*2p_y orbitals are singly occupied in accordance with *Hund's rule*. Unpaired electrons give rise to *paramagnetism*. Since there are two unpaired electrons with parallel spins, this explains why dioxygen is ***paramagnetic.***

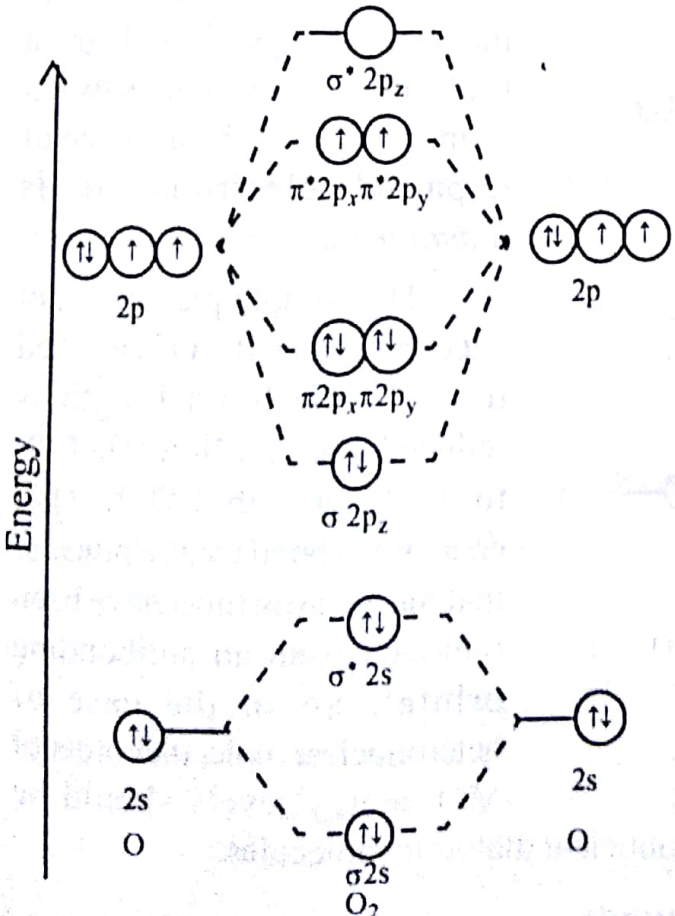

Fig. 1.23. Molecular orbital energy level diagram for O_2 molecule

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Electronic Configuration of Heteronuclear Diatomic Molecules

The principles applied for homonuclear diatomic molecules are applicable here also. However, the molecular orbital diagram will not be symmetrical as in the case of homonuclear diatomic molecules. Because of the difference in the electronegativities of the atoms, the electrons in bonding molecular orbital spend more time near the more electronegative atom. On the other hand, electrons in antibonding molecular orbital are closer to the less electronegative atom. Now, let us consider some examples.

(i) Carbon monoxide molecule, CO : The carbon atom has 6 electrons, and the O atom has 8 electrons, so the CO molecule contains 14 electrons. Thus, it is isoelectronic with N_2. In this case, we are rather less certain about the order of energies of the MOs, since they are different for C and O. Assume the order is the same as for light atoms like C. Thus, the configuration is,

$$\sigma 1s^2\ \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 = \pi 2p_y^2 \sigma 2p_z^2$$

$$\text{or KK } \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 = \pi 2p_y^2 \sigma 2p_z^2$$

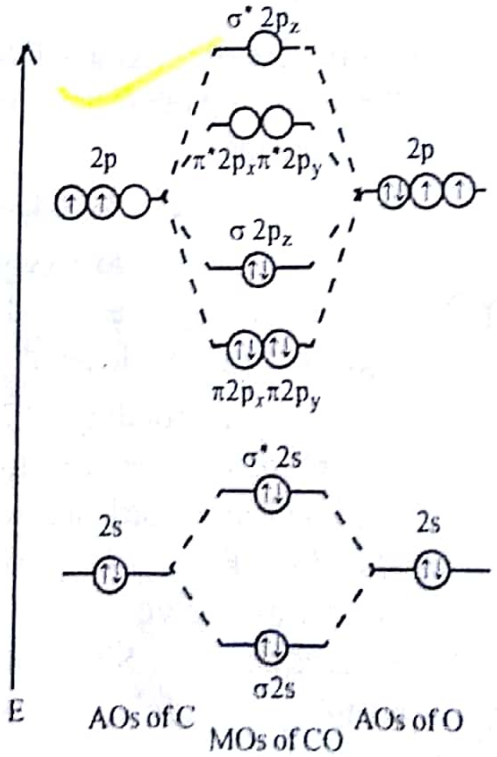

Fig. 1.24 Energy level diagram for CO

The molecular orbital energy level diagram is as shown in Fig. 1.24. The bond order in this case is, ½[8-2]=3. Thus, a triple bond is formed in between C and O. Due to the absence of unpaired electrons, it is *diamagnetic.*

This simple picture is not adequate, since if CO is ionized to CO^+, the bond length is reduced from 1.128 A^0 (in CO) to 1.115 A^0 (in CO^+). This decrease in bond length indicates that the electron must have been removed from an antibonding orbital. So in the case of heteronuclear molecules order of MO energy levels should be different from that of simple homonuclear diatomic molecules.

Properties of Covalent Compounds.

i. ***Physical state.*** Covalent compounds are either gases, liquids or solids.

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ii. ***Solubility.*** They are insoluble in water but soluble in nonpolar solvents.

iii. ***Electrical conductivity.*** Due to nonexistence of ions, covalent compounds do not conduct electricity in any of the states.

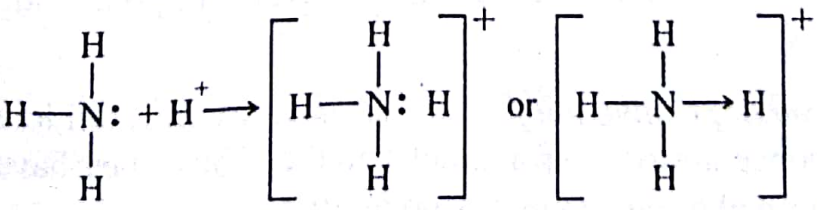
iv. ***Low boiling and melting points.*** The covalent compounds are soft and have low melting and boiling points.

v. ***Molecular reactions.*** The reactions involve the breaking of covalent bonds in reactant molecules and forming new covalent bonds to give molecules of the products. So, these reactions are comparatively slow.

Coordinate or Dative Bond

The covalent bond in which both the electrons in the shared pair is contributed by only one of the two atoms is called coordinate covalent bond. The atom which contribute the electron pair is called **the *donor*** while the other which only shares the electron pair is known as ***acceptor.*** This bond is also known as ***donor acceptor bond.*** This bond is usually represented by an arrow ($\rightarrow$) pointing from donor to the acceptor atom. During the formation of coordinate bond, slight polarity develops in the molecule. The donor acquires a slight positive charge while the acceptor, a slight negative charge. Therefore this bond is also known as the ***semipolar*** bond. The coordinate bond once formed is indistinguishable from a covalent bond because all the electrons are alike irrespective of their source.

To illustrate the formation of coordinate bond, consider the formation of ammonium ion (NH_4^+) from ammonia (NH_3) and positive hydrogen ion (H^+). The nitrogen atom in ammonia contains an unshared pair of electrons known as ***lone pair of electrons.*** Nitrogen atom contributes this lone pair of electrons to the hydrogen ion. This can be represented as shown below.



Combination of NH_3 and H^+

Similarly, NH_3 and BF_3 molecules combines as follows.

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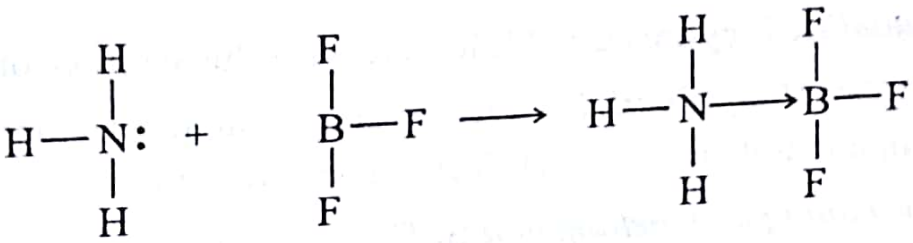
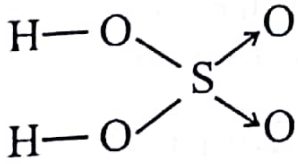


Combination of NH_3 and BF_3

H_2SO_4 has two coordinate bonds as shown below.



The essential *condition for the formation of coordinate bond is that the donor must contains at least a lone pair of electrons* and that the *acceptor must have an incomplete valency shell.*

Partial Covalent Character of the Ionic Bond

When two ions A^+ and B^- are brought together to their equilibrium distance, the nature of bond formed between them depends on the effect of one ion on the other. The cation attracts the electrons on the negative ion and at the same time repels the nucleus. Thus, distorting or polarising the negative ion. Similarly, the negative ion will also polarize the positive ion. But since anions are usually large, and cations small, the effect of a large ion on a small one will be much less pronounced. If the degree of polarisation is quite small, then the bond remains largely ionic. If the degree of polarisation is large, electrons are drawn from the negative ion towards the positive ion, resulting in high concentration of electrons between the two nuclei, and a large degree of covalent character results.

Fajan's Rules : The rules governing the transition from ionic to covalent bonding are called *Fajan's rules*. They are based on the deformation of the interacting ions in the bond A^+B^- (polarization of ions). *Fajans* put forward four rules which summarize the factors favouring polarization and hence covalency.

(i) A small positive ion favours covalency : In small ions the positive charge is concentrated over a small volume. Thus, they have high charge density. This makes the ion highly polarizing.

(ii) A large negative ion favours covalency : Large ions are highly polarizable, i.e., easily distorted by the positive ion, because the outer most

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electrons are away from the nucleus.

(iii) Large charges on either ion, or on both ions, favour covalency : A highly charged cation attracts the electrons of the anion more strongly. Similarly, highly charged anions are easily polarisable. Both these effects increases the amount of polarization.

(iv) Polarization and hence covalency, is favoured if the positive ion does not have a noble gas configuration : Ions which do not have a noble gas configuration include main group elements like Tl^+, Pb^{2+} and Bi^{3+}, transition metal ions like Ti^{3+}, V^{3+}, Cr^{3+}, Mn^{2+} and Cu^+. A noble gas configuration is the most effective in shielding the nuclear charge, so ions without the noble gas configuration will have high charges at their surfaces, and thus be highly polarizing.

Fig. 1.25. Polarization effects (a) Ideal ion pair without any polarization (b) Mutually polarised ion pair (c) Extreme polarization leading to covalent bond formation. Dashed circles represent 'ideal ion'.

Polarity of Covalent Bond

A bond between atoms of two different elements or a heteropolar bond cannot be purely covalent or purely ionic. The bonds in homopolar diatomic molecules like H_2, Cl_2, etc., are purely covalent as the bonding electron pair is equally shared by the bonded atoms. In other words, the electron distribution in covalent bonds joining similar atoms is symmetrical. On the other hand, the electron distribution in covalent bonds joining dissimilar atoms is usually unsymmetrical. That is, the charge cloud of the bonding electron pair is shifted preferentially towards the more electronegative atom. This results in a bond with a partial ionic character. The more electronegative atom acquires a negative charge and the other acquires a positive charge. Bonds like Cl-Cl, H-H, C-C, etc., are purely covalent. These are called *non-polar covalent bonds*. On the other hand,bonds like O-H, C-H, C-Cl, etc., are covalent bonds with partial ionic character. These are called *polar covalent bonds*.

Electronegativity

In 1931, *Pauling* defined electronegativity of an atom as,

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the tendency of the atom to attract electrons to itself when combined in a compound.

This implies that when a covalent bond is formed, the bonding electrons need not be shared equally by both atoms. If the bonding electrons spend more time round one atom, that atom will acquire a δ^- charge and consequently the other atom will have a δ^+ charge. In the extreme case where the bonding electrons are around one atom all of the time, the bond is ionic.

Generally small atoms attract electrons more strongly than large ones, and hence small atoms are more electronegative. Atoms with nearly filled shells of electrons tend to have higher electronegativities than those with scarcely occupied ones.

Factors influencing polarity of covalent bonds

The partial ionic character of a polar covalent bond depends upon two factors. (i) The electronegativity difference of the bonded atoms and (ii) dipole moment of the compound.

Electronegativity difference and partial ionic character : A pure covalent bond (non-polar covalent bond) is formed between atoms A and B when the electronegativities of the two atoms are the same. If A and B have different electronegativities, the resulting bond would be a polar covalent bond since the bonding electron pair is shifted preferentially towards the more electronegative atom making it, slightly negative. As a result, a slight positive charge is developed on the less electronegative atom. Thus, the bond will become a polar covalent bond. If B is more electronegative than A, a polar covalent bond $\overset{\delta^+}{A}-\overset{\delta^-}{B}$ would be formed between A and B. The *greater the difference in the electronegativities of bonded atoms, the greater the polarity or ionic character of the bond.* Thus we should be able to determine the ionic character of the bond from an empirical relationship between ionic character and electronegativity difference.

Pauling Equation: From a plot of electronegativity difference against percentage ionic character, *Pauling* proposed an empirical relation between amount of ionic character and electronegativity difference. The relation is:

$$\text{Amount of ionic character} = 1 - e^{-\frac{1}{4}(\chi_A - \chi_B)^2}$$

Rough and easy to remember values of electronegativity differences and percentage ionic character are:

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χ_A-χ_B	1.0	1.5	2	2.5
Percentage Ionic character	20%	40%	60%	80%

(ii) Dipole moment and partial ionic character

Dipole moment, μ, is the product of charge at one pole, q, and the distance between the poles, d. i.e., $\mu = q \times d$.

Pauling has used, the dipole moment as a measure of the polarity and hence of the ionic character of a bond. The greater the ionic character of a bond, the greater the charge separation and the larger the dipole moment. The percentage ionic character of a bond is obtained from the ratio of the observed dipole moment to the dipole moment for complete electron transfer.

$$\text{i.e., Percentage ionic character} = \frac{\text{Actual dipole moment} \times 100}{\text{Dipole moment of a pure ionic bond}}$$

For example, the observed *dipole moment* for HF is 1.98 Debye units and the dipole moment for the ionic structure H^+F^- (when an electron is completely tranferred from H to F) is 4.4 Debye units.

$$\therefore \text{ Percentage ionic character of HF} = \frac{1.98\,D \times 100}{4.4\,D} = 45\%$$

Similarly, for HCl, internuclear distance is 127 pm, actual dipole moment $= 3.44 \times 10^{-30}$ coulomb metre. Dipole moment when an electron is completely transferred from H to Cl $= 1.6 \times 10^{-19}C \times 127 \times 10^{-12}$ m $= 2.03 \times 10^{-29}$ Cm

$$\therefore \text{ Percent ionic character} = \frac{3.44 \times 10^{-30}\,cm \times 100}{2.03 \times 10^{-29}\,Cm} = 16.94 \approx 17$$

Dipole moment and Geometry of Molecules

In a diatomic molecule, the dipole moment of the bond gives the dipole moment of the molecule. Thus, a diatomic molecule is polar if the bond between the atoms is polar. For example, the dipole moment of HF molecule is the same as that of the H-F bond and is equal to 1.98 D. However, in the case of molecules containing more than two atoms, the dipole moment depends not only on the individual dipole moments of the bonds but also on the spatial arrangement of various bonds in the molecule.

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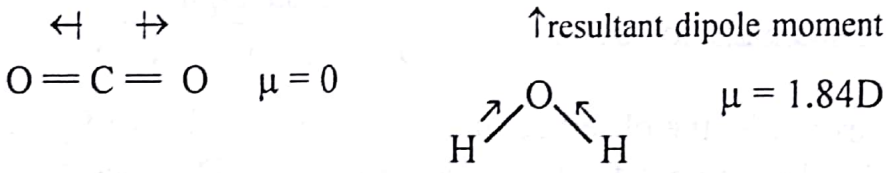
In such cases, the dipole moment of the molecule is the vector sum of the dipole moments of various bonds. Obviously the polar or non-polar nature of a molecule will depend upon the shape of the molecule. In some cases, the bonds may be polar yet the molecule may be non-polar. Thus, dipole moment is used to predict the shape of molecules.

For example, both CO_2 and H_2O are triatomic molecules but the dipole moment of CO_2 is zero while that of water is 1.84 D. This can be explained on the basis of their shapes. In CO_2 both the carbon oxygen bonds are polar. But since the molecule is linear, the dipole moment of C=O bond (2.3D) on one side of the molecule is cancelled by that on the other side. As a result, the net dipole moment of CO_2 is zero. On the other hand, H_2O molecule is not linear but has a bent structure (or V shape). Therefore, the



dipole moment of one O-H bond is not cancelled by that of the other. So, the net dipole moment of water is not zero but is 1.84 D which is the resultant of the dipole moment of two O-H bonds. In the case of CS_2 both the C=S bonds are polar but the molecule as a whole is non-polar and dipole moment is zero because of the linear geometry. Similarly, the BH_3 molecule has a symmetrical, triangular planar geometry and the dipole moments of three B-H bonds cancel one another giving a net dipole moment equal to zero. On the other hand, NH_3 molecule has a pyramidal structure and the dipole moments of the three N-H bonds do not cancel one another but give rise to a net dipole moment of 1.49D. Molecules like CH_4, CCl_4, etc., have symmetrical tetrahedral structures and hence have zero dipole moment. CH_3Cl molecule has a tetrahedral structure, but the dipole moment of C-Cl bond is not equal to that of C-H bond. So its dipole moment is not equal to zero.

Concept of Hybridisation

The ground state electronic configuration of carbon is $1s^2\ 2s^2\ 2px^1\ 2py^1\ 2pz^0$. This may be represented as,

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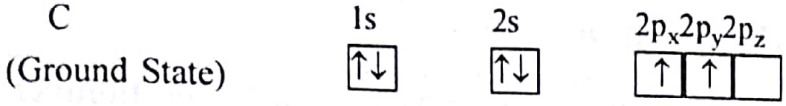
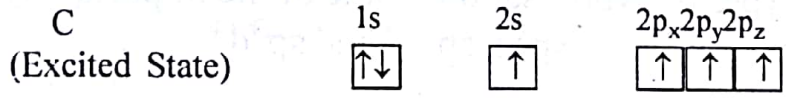


Ground state electronic configuration of carbon.

There are only two half filled orbitals in the valence shell of carbon. Therefore, its bonding capacity should be two. However, in actual practice, carbon exhibits the bonding capacity of four. For example, it is known to form molecules such as CH_4, CCl_4, etc.

This can be explained on the basis of promotion concept. In order to explain the bonding capacity of carbon, it was proposed that one of the electrons from filled 2*s* orbital is promoted to the empty 2*p* orbital ($2p_z$), which is in a higher energy state. Thus, four half filled orbitals get formed in valence shell which account for the bonding capacity of four for the carbon. This state is known as **excited state** and the configuration of carbon in the excited state is:



Excited state electronic configuration of carbon.

The above configuration reveals that all the four bonds formed by carbon will not be identical. For example, in the formation of CH_4 molecule, one C-H bond will be formed by the overlap of 2*s* orbital of C and 1*s* orbital of H whereas the other three C-H bonds will be formed by the overlap of 2*p* orbitals of C and 1*s* orbital of H.

But actually, in most of the organic compounds, such as methane (CH_4), carbon tetrachloride (CCl_4), etc., all the four bonds of carbon atom are equivalent. The equivalent character of the bonds can be explained with the help of a new concept known as **hybridisation**. This may be defined as:

the phenomenon of inter-mixing of the orbitals of slightly different energies so as to redistribute their energies and to give new set of orbitals of equivalent energies and shape.

The new orbitals formed as a result of hybridisation are called **hybrid** or **hybridised orbitals**. Thus, to form four equivalent bonds, one 2*s* and three 2*p* orbitals of carbon hybridise and form four new orbitals called sp^3 hybrid orbitals.

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Characteristics of Hybridisation

(i) The number of hybrid orbitals formed is equal to the number of the orbitals that get hybridised.

(ii) The hybrid orbitals are always equivalent in energy and shape.

(iii) The hybrid orbitals are more effective in forming stable bonds than the pure atomic orbitals.

(iv) The hybrid orbitals are directed in space in some preferred directions to have a stable arrangement. Therefore, the type of hybridisation gives the geometry of the molecule (discussed later on).

(v) Hybridisation never takes place in isolated atoms, but it occurs only at the time of bond formation.

Types of Hybridisation

There are many types of hybridisation. Some of the important types of hybridisation are sp, sp^2, sp^3, dsp^2, sp^3d, sp^3d^2 and sp^3d^3.

(i) sp hybridisation

The intermixing of one s-orbital and one p-orbital of the valency shell of an atom to give rise to two identical orbitals is called ***sp hybridisation.*** *The resulting orbitals are called sp hybrid orbitals.* The two *sp* hybrid orbitals are collinear and thus make an angle of 180° with each other. This hybridisation is often known as **diagonal hybridisation.**

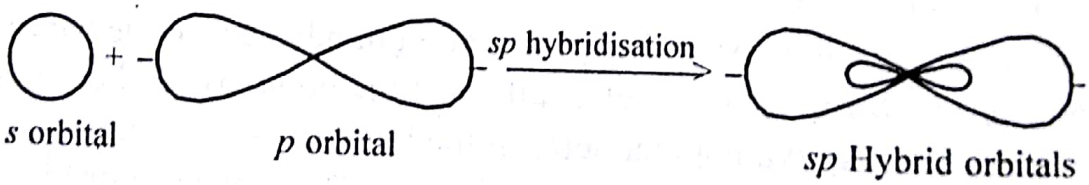

Fig. 1.26. Formation of sp hybrid orbitals

Formation of Beryllium difluoride (BeF_2). Electronic configuration of Be in the ground state is $1s^2\ 2s^2$. Since there are no unpaired electrons in Be atom, it is expected to show zero valency. However, it shows divalency in its compounds such as BeH_2, BeF_2, $BeCl_2$, etc. To account for the divalency of beryllium, one of the electrons from 2*s* orbital is promoted to the next available vacant 2*p* orbital. If these two unpaired electrons are used for bond formation in BeF_2, the bonds formed will not be equivalent because one bond will be formed by the overlap of 2*s* orbital of Be and 2*p* orbital of F while the other will be formed by the overlap of 2*p* orbital of Be and 2*p* orbital of F. But

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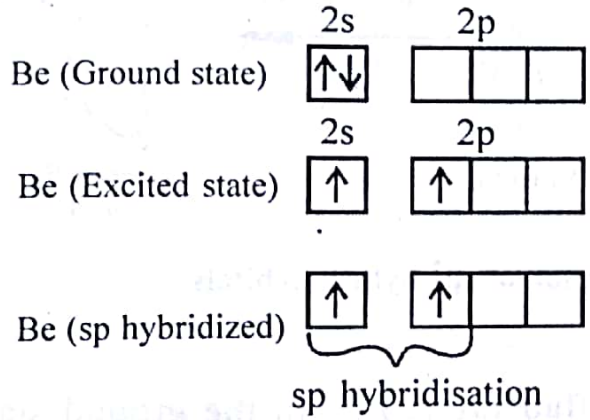
experimentally it is known that both the Be-F bonds in BeF_2 are equivalent. To account for this, it is assumed that one 2s and one 2*p* orbital of Be hybridise to form two *sp* hybrid orbitals.



The two sp hybrid orbitals are directed in space in a linear arrangement making an angle of 180° between the hybrid orbitals. Now, the half filled *p*-orbitals of each fluorine atom overlap with one of the half filled *sp* hybrid orbitals of the Be atom forming two Be-F bonds. This is shown in Fig. 1.27. The bond angle in BeF_2 molecule is 180° and its shape is linear.

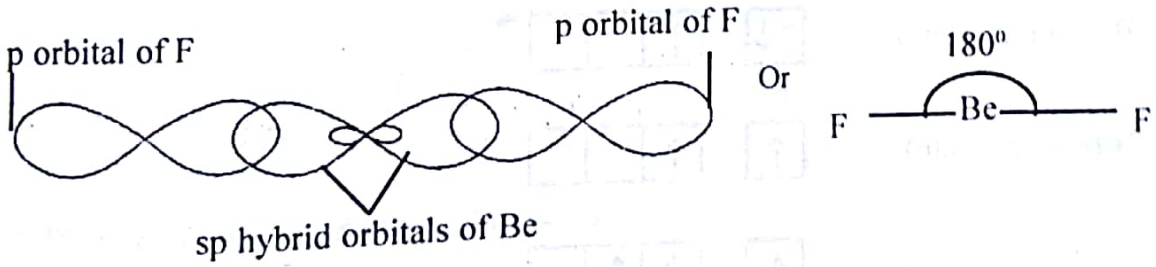


Fig. 1.27 Geometry of BeF_2 molecule.

2. sp^2 hybridisation

The intermixing of one s and two p orbitals of the valency shell of an atom to give rise to three identical orbitals is called sp^2 hybridisation. The resulting orbitals are called sp^2 hybrid orbitals.

The three sp^2 hybrid orbitals are directed towards the corners of an equilateral triangle making an angle of 120° with each other. The hybridisation is, therefore, known as ***trigonal hybridisation***.

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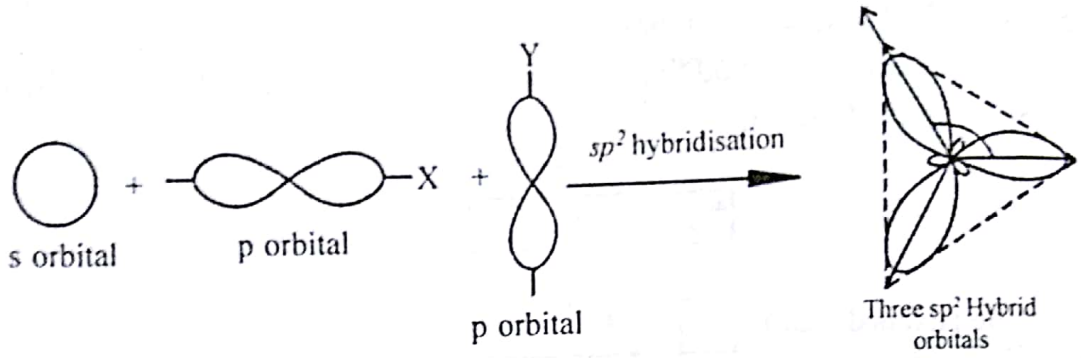
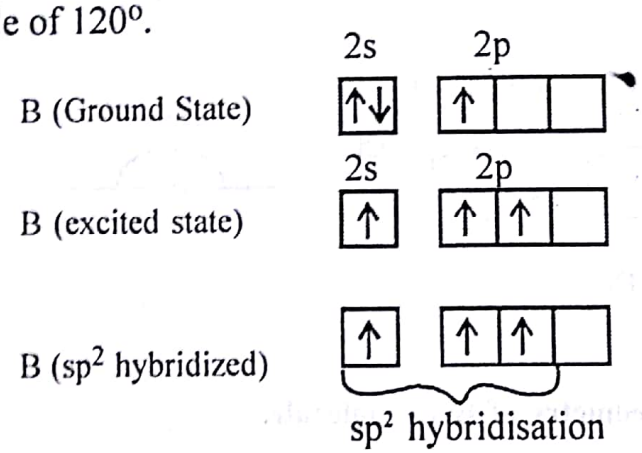


Fig. 1.28 Formation of sp^2 hybrid orbitals.

Formation of Boron trifluoride (BF_3). In the ground state, the electronic configuration of boron is $1s^2\ 2s^2\ 2p_x^1$. Since boron is trivalent in its compounds (such as BF_3), it is assumed that one of the $2s$ electrons is promoted to the next vacant $2p$ orbital, *i.e.*, $2p_y$ orbital. Therefore, the excited state of B has three unpaired electrons. Now, to form equivalent bonds, the three orbitals; $2s$, $2p_x$ and $2p_y$ hybridise to form three sp^2 hybrid orbitals. These three sp^2 hybrid orbitals adopt trigonal planar arrangement making an angle of 120°.



In the formation of BF_3, the three sp^2 hybrid orbitals of boron overlap with half filled $2p$ orbitals of fluorine to form three B-F bonds. Thus, the molecule of BF_3 is planar with each bond angle of 120°.

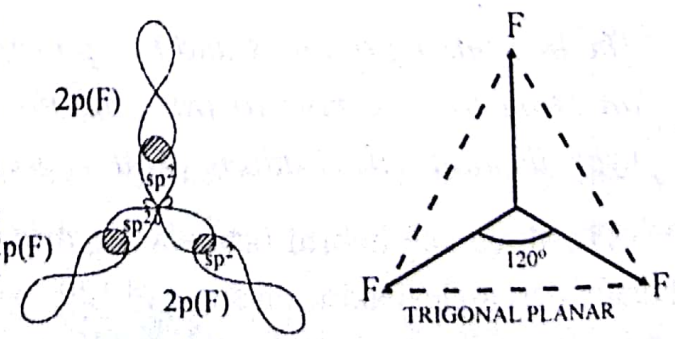


Fig. 1.29. Formation of BF_3

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3. sp^3 hybridisation

The intermixing of one s and three p-orbitals of the valence shell of an atom to form four identical orbitals is called sp^3 hybridisation. The resulting orbitals are called sp^3 hybrid orbitals.

The four sp^3 hybrid orbitals are directed towards the four corners of a regular tetrahedron. The angle between the sp^3 hybrid orbitals is 109°28′. The hybridisation is, therefore, also known as ***tetrahedral hybridisation***.

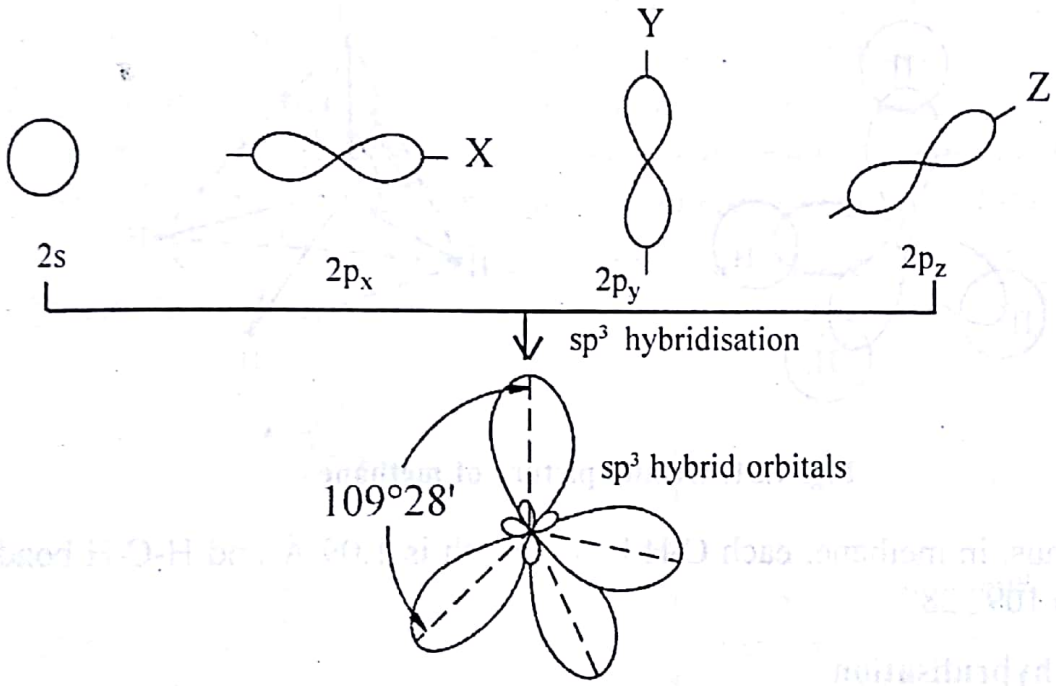

Fig. 1.30. Formation of sp^3 hybrid orbitals

Formation of methane: The electronic configuration of carbon in the ground state is $1s^2, 2s^2, 2p_x^1, 2p_y^1$. To account for its tetravalency, it has four unpaired electrons in the excited state as $1s^2\ 2s^1\ 2p_x^1\ 2p_y^1\ 2p_z^1$. In methane, the carbon atom undergoes sp^3 hybridisation and forms four sp^3 hybrid orbitals. Each of the sp^3 hybrid orbitals of carbon overlaps axially with half-filled 1*s* orbital of hydrogen atoms forming four sigma (σ) C-H bonds.

The four sp^3 hybrid orbitals are directed towards the corners of a regular tetrahedron making an angle of 109° 28′. The overlapping with 1*s* orbitals of hydrogen atom is also shown in Fig. 1.31.

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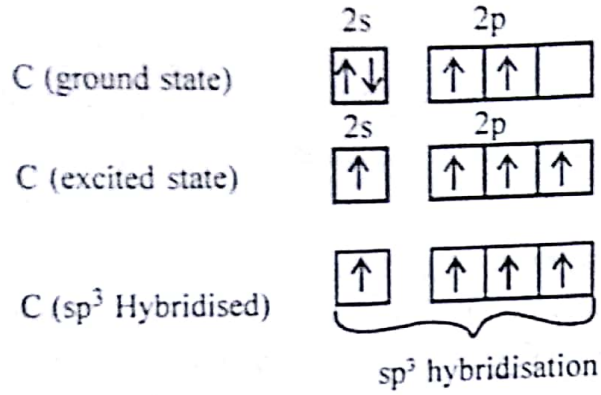


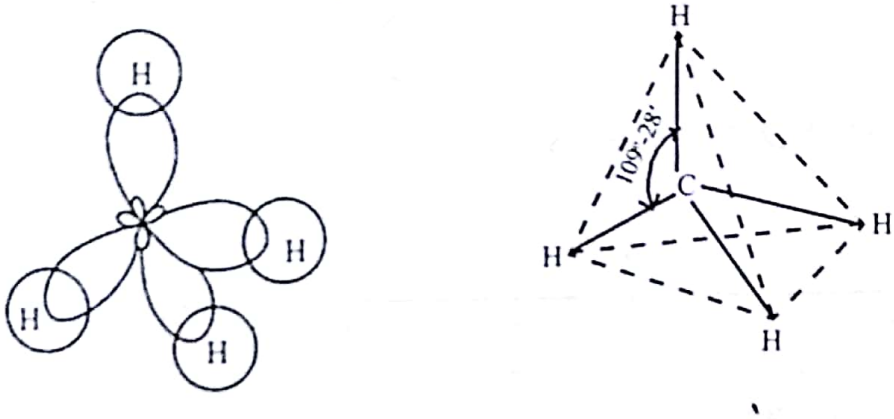


Fig. 1.31. Orbital picture of methane

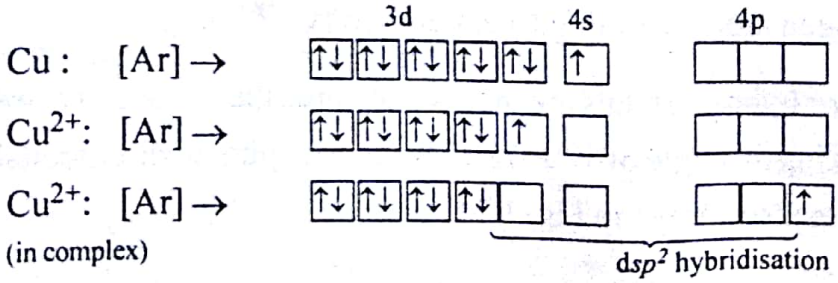
Thus, in methane, each C-H bond length is 1.09 Å and H-C-H bond angle is 109° 28′.

4. dsp^2 hybridisation

The intermixing of the dx^2y^2 orbital, an s orbital and two p orbitals to form four equivalent orbitals is called dsp^2 hybridisation. The resulting orbitals are called dsp^2 hybrid orbitals.

The four dsp^2 orbitals are oriented to the four corners of a square and the angle between the hybrid orbitals is 90°. So the hybridisation is also called ***square planar hybridisation***. The formation of cuprammonium complex involves dsp^2 hybridisation.

Formation of cuprammonium complex, $[Cu(NH_3)_4]^{2+}$



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The electronic configuration of Cu is [Ar] $3d^{10}4s^1$. It loses two electrons to form Cu^{2+} which has the configuration [Ar] $3d^9$. In the formation of complex, the unpaired electron is promoted to a vacant $4p$ orbital. Now, the vacant d orbital, an s orbital, and two p orbitals undergo dsp^2 hybridisation to form four dsp^2 hybrid orbitals which are oriented to the corners of a square. With these vacant dsp^2 hybrid orbitals four NH_3 molecules form co-ordinate bonds giving the complex, $[Cu(NH_3)_4]^{2+}$. The complex is *square planar* in shape. It is *paramagnetic* due to the presence of an unpaired electron on the $4p$ sub level. The complex is deep blue coloured. This can be explained in terms of the excitation followed by deexcitation of the unpaired $4p$ electron.

5. sp^3d hybridisation. *The intermixing of one s-orbital, three p-orbitals and one d-orbital of the valency shell of an atom to form five identical orbitals is called sp^3d hybridisation. The resulting orbitals are called sp^3d hybrid orbitals.*

The five sp^3d hybrid orbitals point towards the corners of a trigonal bipyramidal geometry.

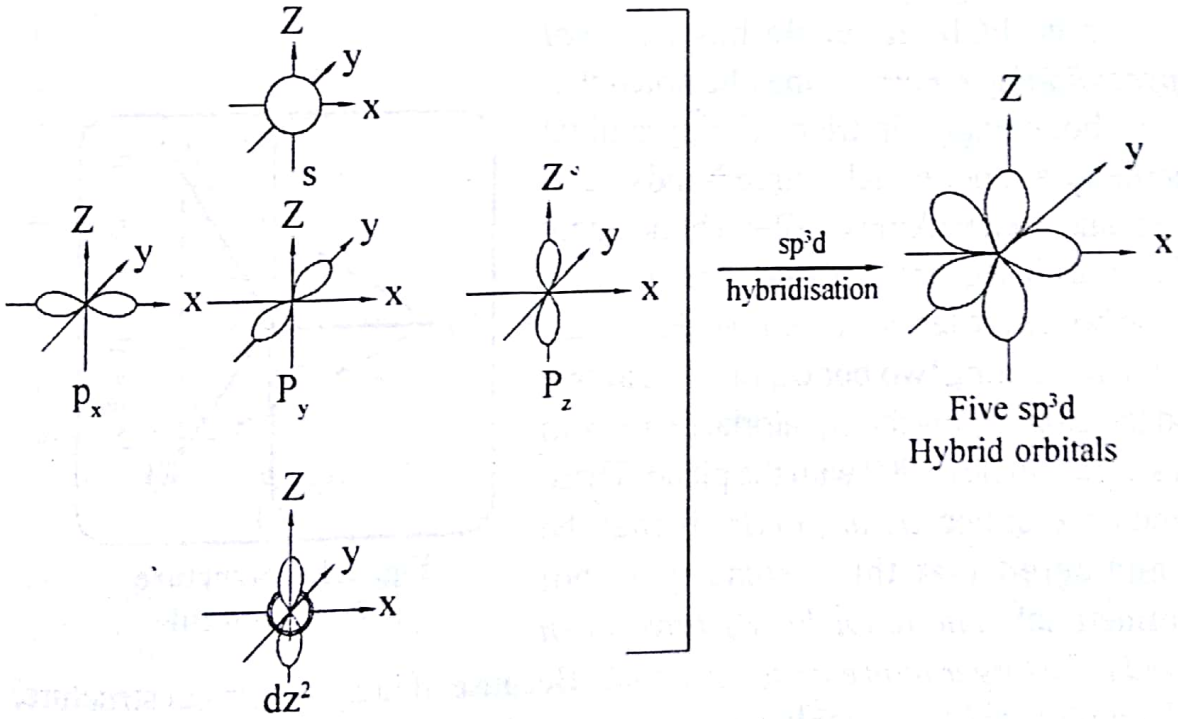

Fig. 1.32. Formation of sp^3 d hybrid orbitals.

In this case, the three orbitals forming a plane are directed towards the corners of an equilateral triangle while the other two are perpendicular to the plane of the triangle lying above and below it. For example, PCl_5 involves sp^3d hybridisation.

Formation of PCl_5: The ground state electronic configuration of phosphorus atom is $3s^2\ 3p_x^1\ 3p_y^1\ 3p_z^1$. It has three unpaired electrons in the

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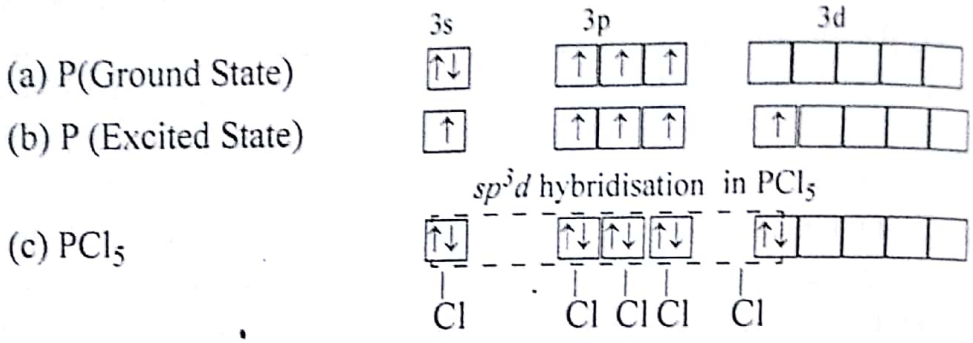
ground state.



Formation of PCl_5 molecule involving sp^3d hybridisation.

To explain the pentavalency of phosphorus in PCl_5, one of the electrons in 3*s* orbital is promoted to the higher 3*d* orbital so that it has five unpaired electrons. This is called the excited state of phosphorus. These five orbitals hybridise to form five sp^3d hybrid orbitals which adopt trigonal bipyramidal arrangement. Each of the sp^3d hybrid orbitals overlaps with 2*p* orbital of fluorine forming five P-Cl bonds.

Thus, PCl_5 molecule has *trigonal bipyramidal geometry*. It may be noted that all the bond angles in trigonal bipyramidal geometry are not equal. Three bonds lie in one plane each making Cl–P–Cl bond angle of 120° as in a trigonal planar arrangement. These bonds are termed as *equatorial bonds*. Of the remaining two bonds, one lies above and the other below the equatorial plane both making an angle of 90° with the plane. These bonds are called *axial bonds*. It may be remembered that this geometry is not symmetrical. *The axial bonds have been found to be larger than equatorial bonds.* Because of unsymmetrical structure, PCl_5 molecule is less stable and highly reactive. Molecules such as PF_5 and $SbCl_5$ have similar geometries.

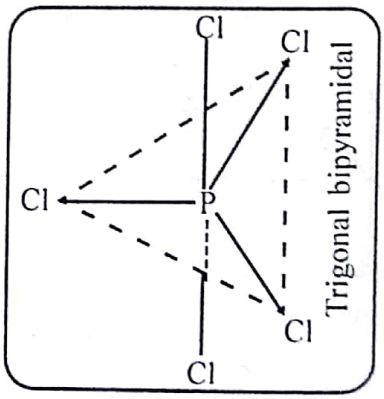


Fig. 1.33. Structure of PCl_5 molecule

The larger bond length of axial bond in comparison to equatorial bond is because of greater repulsion from other bonds in axial position.

6. sp^3d^2 hybridisation

The intermixing of one s-orbital, three p-orbitals and two d-orbitals of the valency shell of an atom, to form six sp^3d^2 hybrid orbitals of identical shapes and equivalent energies is known as sp^3d^2 hybridisation. The sp^3d^2 hybrid orbitals are oriented to the corners of a regular octahedron.

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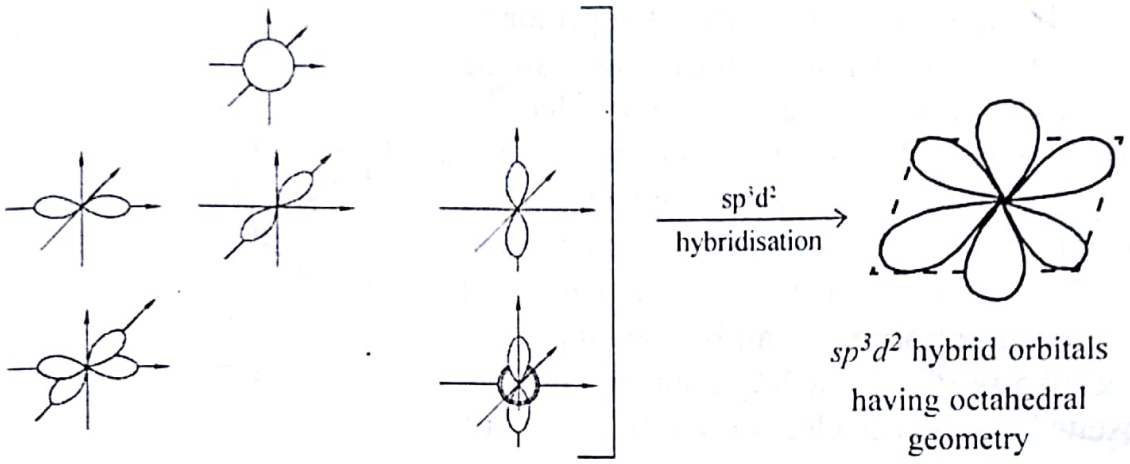


Formation *of* sp^3d^2 hybrid orbitals.

Geometry of SF_6 molecule. In SF_6 molecule, the central sulphur atom has the ground state configuration, $3s^2 3p^4$. To account for the hexacovalency in SF_6, one electron each from $3s$ and $3p_x$ orbitals is promoted to 3d orbitals. These six orbitals get hybridised to form six sp^3d^2 hybrid orbitals. Each of these sp^3d^2 hybrid orbitals overlaps with 2p orbital of fluorine to form S-F bond. Thus, SF_6 molecule has octahedral structure as shown in the Fig. 1.34 In this geometry, all F-S-F bond angles are equal and are of 90° each.

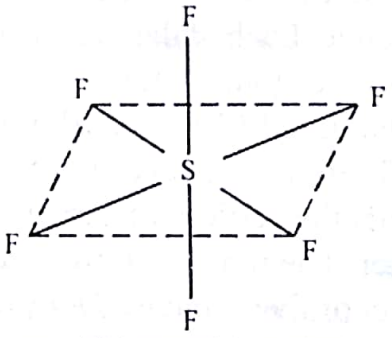
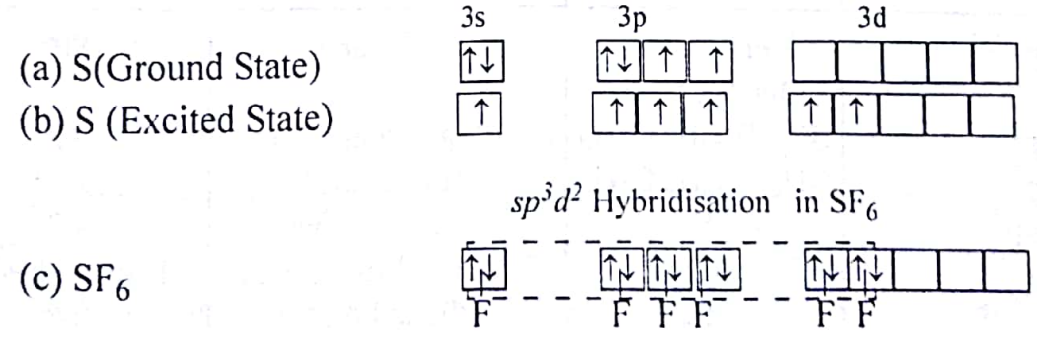


Fig. 1.34. Shape of SF_6 molecule



Formation of SF_6 molecule involving sp^3d^2 hybridisation.

7. sp^3d^3 hybridisation

The intermixing of one s-orbital, three p-orbitals and three d-orbitals of the valency shell of an atom to form seven identical orbitals is known as ***sp^3d^3 hybridisation****. The resulting orbitals are called* ***sp^3d^3 hybrid orbitals.***

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Using sp^3d^3 hybridisation the geometry of iodine heptafluoride molecule can be explained. The ground state outer electron configuration of the central atom iodine is $5s^2\ 5p_x^2\ 5p_y^2\ 5p_z^1$. To explain the heptavalency of iodine it is assumed that during bond formation the $5s$, $5p_x$ and $5p_y$ electrons get unpaired and one electron from each pair is promoted to the vacant $5d_{xy}$, $5d_{xz}$, and $5d_{yz}$ orbitals. Thus the excited state outer electron configuration of iodine is $5s^1\ 5p_x^1\ 5p_y^1\ 5p_z^1\ 5d^1_{xy}\ 5d^1_{xz}\ 5d^1_{yz}$. Now these orbitals hybridise to give seven sp^3d^3 hybrid orbitals. These are directed towards the seven corners of a pentagonal bipyramid. Each of the hybrid orbitals overlaps with the half-filled $2p$ orbitals of fluorine atoms to form IF_7 molecule (Fig. 1.35). Therefore ***IF_7 molecule is pentagonal bipyramidal in shape.***

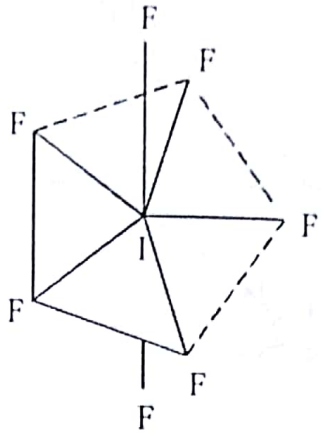

Fig. 1.29. Shape of IF_7 molecule

In IF_7 all the seven bonds are not equivalent. Five of these are directed towards the vertices of a regular pentagon making an angle of 72° with one another. The remaining two are directed at right angles to the plane of the first set of five orbitals. ***Thus, in IF_7, five bond angles are of 72° each and the other two bond angles are of 90° each.***

Type of Hybridisation and Geometry of Covalent Molecules

Type of hybridisation	Name of the compound	Geometry of the molecule	Bond angle
sp	BeH_2, BeF_2, $BeCl_2$, C_2H_2	Linear	180°
sp^2	BF_3, $B Cl_3$, C_2H_4	Plane triangular	120°
sp^3	CH_4, CCl_4, C_2H_6	Tetrahedral	109°28'
dsp^2	$Cu(NH_3)_4^{2+}$	Square planar	90°
sp^3d	PCl_5	Trigonal bipyramidal	120° and 90°
sp^3d^2	SF_6	Octahedral	90°
sp^3d^3	IF_7	Pentagonal bipyramidal	72° and 90°

Valence Shell Electron Pair Repulsion Theory (VSEPR Theory)

In order to explain the shapes of molecules, *Sidgwick* and *Powell* suggested a theory known as ***valence shell electron pair repulsion (VSEPR)***

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theory. This is based upon the repulsive interactions of electron pairs in the valence shell of atoms.

The main postulates of VSEPR theory are:

i) The geometry of a molecule depends on the number of valence shell electron pairs around the central atom in the molecule.

ii) The electron pairs around the central atom repel each other because of the negative charge of their electron clouds. Therefore, the electron pairs stay as far apart as possible to attain maximum stability or minimum energy.

iii) The repulsive interactions between two lone pairs of electrons is greater than those betwen a lone pair and a bond pair, which in turn, is greater than those between two bond pairs of electrons, i.e.,

lone pair - lone pair repulsion > lone pair - bond pair repulsion > bond pair - bond pair repulsion.

The molecules in which the central atom is surrounded only by orbitals containing bond pairs of electrons are said to have ***regular geometry***. On the other hand, the molecules in which the central atom is surrounded also by one or more orbitals containing lone pairs of electrons are said to have ***irregular or distorted geometry***. The distortion in the geometry of the molecule is due to unequal repulsive interactions between the electron pairs around the central atom.

Geometry of molecules containing bond pairs of electrons only

Consider the molecule BeF_2, which has two bond pairs of electrons around the central Be atom. These two electron pairs would be staying as far apart as possible only if they lie on the opposite sides of Be. Thus, BeF_2 has a linear geometry making a bond angle of 180°. It may be noted that any other geometry (making bond angle less than 180° on either side, known as angular) will have more repulsive interactions. Consequently, it will be unstable in comparison to linear geometry.

Similarly, the most stable geometries for 3,4,5 and 6 electron pairs around the central atom are trigonal planar, tetrahedral, trigonal bipyramidal and octahedral respectively as given in the following Table.

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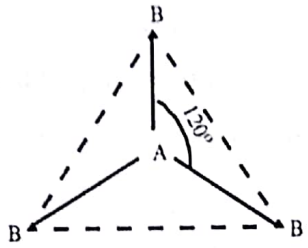
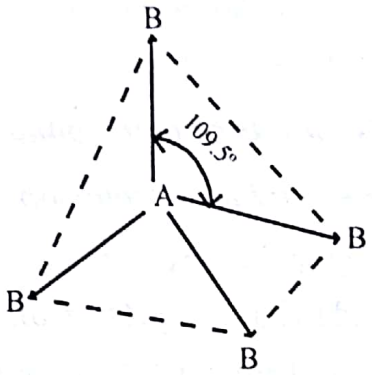
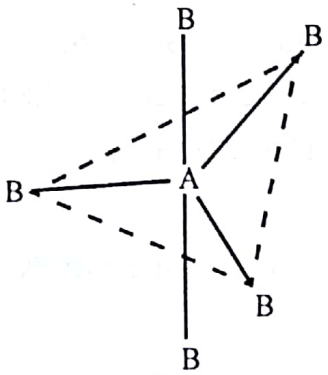
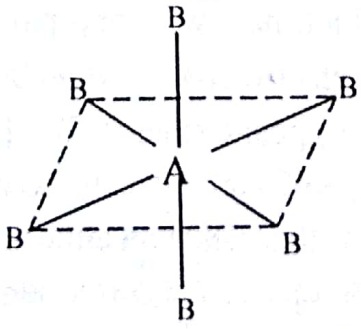
Geometries of molecules on the basis of VSEPR theory

Electron pairs around central atom	Geometrical arrangement		Bond angles	Examples
2		Linear	180°	BeH_2, $BeCl_2$, BeF_2, $ZnCl_2$
3		Trigonal Planar	120°	BF_3, BCl_3, $AlCl_3$
4		Tetrahedral	109.5°	CH_4, SiH_4, SiF_4
5		Trigonal bipyramidal	three 120° two 90°	PF_5, PCl_5, $SbCl_5$
6		Octahedral	90°	SF_6, TeF_6

Geometry of Molecules containing Bond Pairs and Lone Pairs of Electrons

i) Geometry of $SnCl_2$ molecule. The outer electron configuration of the central atom, tin, is $5s^2\ 5p_x^1\ 5p_y^1$. If the two unpaired electrons in the $5p$ orbitals of tin overlap with the half-filled p-orbitals of two chlorine atoms,

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the $SnCl_2$ molecule should be angular with Cl-Sn-Cl bond angle equal to 90°. But, experimentally, the Cl-Sn-Cl bond angle is quite close to 120°. Therefore, it is postulated that one s and two p-orbitals of tin atom hybridise to give three sp^2 hybrid orbitals lying in a plane at 120°. In the formation of $SnCl_2$ two of the hybrid orbitals containing unpaired electrons overlap with the half filled $3p$ orbitals of two chlorine atoms to form two Sn-Cl bonds. The third hybrid orbital is occupied by a lone pair of electrons. Thus the central atom, Sn in $SnCl_2$ is surrounded by two bond pairs and one lone pair of electrons. Since lone pair-bond pair repulsion is greater than bond pair-bond pair repulsion, there is some distortion in the molecule. Therefore, the bond angle is slightly less than 120° and $SnCl_2$ molecule is V-shaped (Fig. 1.36).

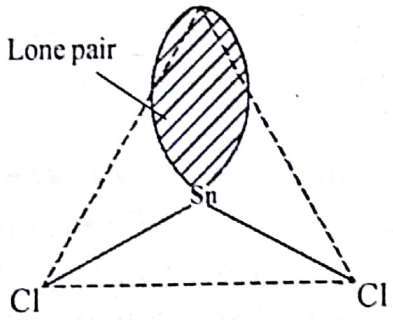

Fig. 1.36. Shape of $SnCl_2$ molecule

ii) Geometry of NH_3 molecule. In ammonia, the central atom is nitrogen. Its electronic configuration in the ground state is $1s^2\ 2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^1$. If the three half-filled p-orbitals in nitrogen overlap with the $1s$ orbitals of three hydrogen atoms to form ammonia, the molecule would have three mutually perpendicular N-H bonds. But actually, NH_3 molecule is pyramidal in shape with H-N-H bond angle of 107°48′. This is explained as follows. The 2s orbital containing a lone pair of electrons and the three 2p orbitals (each containing an unpaired electron) in nitrogen hybridise to form four sp^3 hybrid orbitals. In the formation of ammonia three of the hybrid orbitals (containing one unpaired electron each) overlap with the 1s orbitals of three hydrogen atoms to form three N-H bonds. The fourth sp^3 hybrid orbital is occupied by

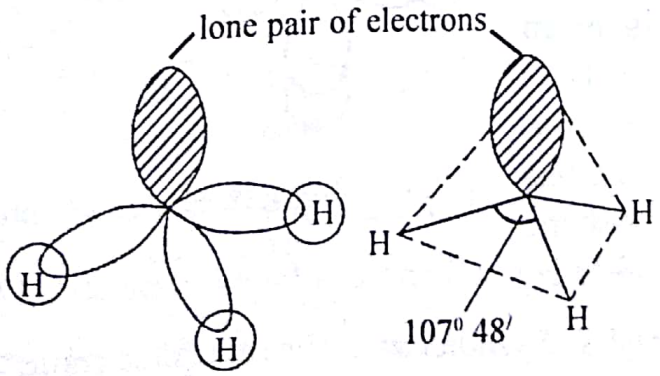

Fig 1.37. Pyramidal structure of ammonia molecule

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a lone pair of electrons. Thus in ammonia the nitrogen atom has three bond pairs and one lone pair of electrons in its valence shell. Since the force of repulsion betwen a lone pair and a bond pair is greater than the force of repulsion between two bond pairs of electrons (VSEPR theory) the molecule becomes a little distorted and the H-N-H bond angle decreases from 109°28′ to 107°48′. Thus the shape of ammonia is pyramidal (Fig. 1.37), in which the nitrogen atom lies at the centre, the three hydrogen atoms form the base and the lone pair of electrons forms the apex of the pyramid, as shown in the Fig. 1.37.

iii) Geometry of H_2O molecule. In this case oxygen is the central atom. The electronic configuration of oxygen in the ground state is $1s^2\ 2s^2\ 2p_x^2\ 2p_y^1\ 2p_z^1$. If the two half filled p-orbitals overlap with 1s orbitals of two hydrogen atoms to form water, the H-O-H bond angle would be 90°. But actually, water molecule has a bent structure with H-O-H bond angle of 104°27′. This is explained by postulating that the 2s, $2p_x$ (both contain lone pairs), $2p_y$ and $2p_z$ (each containing an unpaired electron) hybridise to form four sp^3 hybrid orbitals, arranged tetrahedrally. In the formation of water, two of the hybrid orbitals (containing one unpaired electron each) overlap with the 1s orbtials of two hydrogen atoms to form two O-H bonds. The remaining two hybrid orbitals are occupied by lone pairs of electrons. Thus in water, the oxygen atom has two bond pairs and two lone pairs of electrons. Since the force of repulsion between the two lone pairs of electrons among themselves is more than the force of repulsion between a lone pair and a bond pair which in turn, is more than the force of repulsion between two bond pairs of electrons, there is greater distortion in water molecule than in ammonia. As a result, the H-O-H bond angle in water is even smaller than the H-N-H bond angle in ammonia. Hence water molecule has a bent or angular or V-shape (Fig. 1.38) and the H-O-H bond angle is found to be 104°27′.

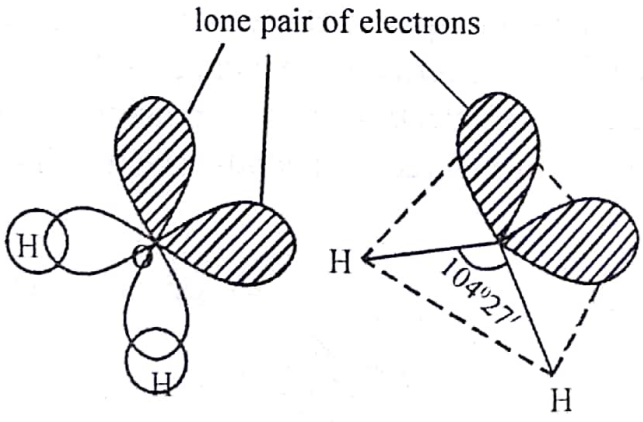

Fig.1.38. Shape of water molecule

iv) Geometry of XeF_2 molecule: The outer electronic configuration of the central atom, xenon, in the ground state is $5s^2 5p_x^2 5p_y^2 5p_z^2$. In the formation of XeF_2 molecule, one of the 5p electrons gets promoted to one of the vacant

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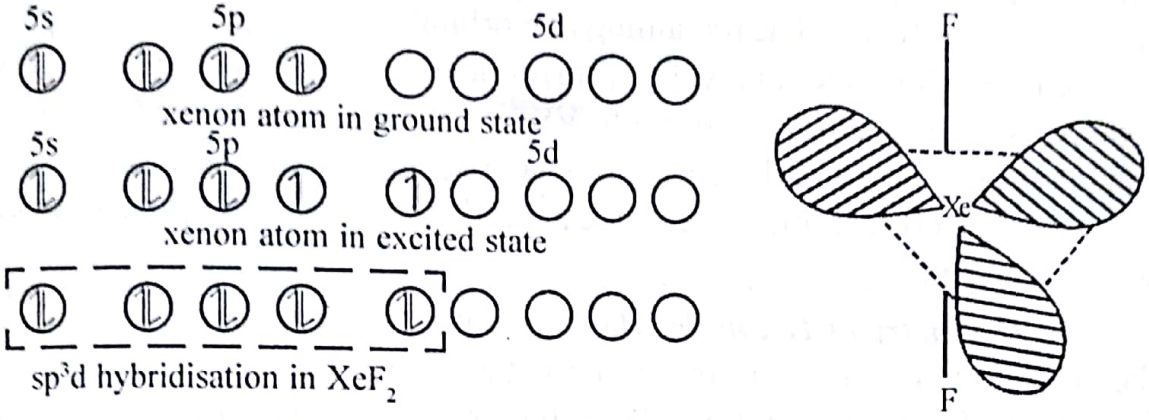

Fig.1.39 Formation of XeF_2 molecule by sp^3d hybridisation

5d-orbitals. So the excited state outer electronic configuration of xenon is $5s^2$ $5px^2$ $5py^2$ $5pz^1$ $5d^1$. The 5s, three 5p and one 5d orbital hybridise to form five sp^3d hybrid orbitals, arranged in a trigonal bipyramidal manner. Two of these hybrid orbitals (containing one unpaired electron each) overlap with the half filled p-orbitals of two fluorine atoms, forming XeF_2. The remaining three hybrid orbitals are occupied by lone pairs of electrons. Thus, in XeF_2 molecule, the Xe atom is surrounded by three lone pairs and two bond pairs of electrons. In order to minimise the electron pair repulsion, the three lone pairs of electrons must be present on equatorial positions. Therefore XeF_2 molecule is linear (Fig. 1.39) in shape and the F-Xe-F bond angle is of 180°.

v) Geometry of XeF_4 molecule: The outer electronic configuration of the central atom, xenon, in the ground state is $5s^2$ $5p_x^2$ $5p_y^2$ $5p_z^2$. In the formation of XeF_4 molecule, two of the 5p electrons are promoted to the vacant 5d orbitals. Thus the excited state outer electronic configuration of Xe is $5s^2$ $5p_x^2$ $5p_y^1$ $5p_z^1$ $5d^1_{x^2-y^2}$ $5d^1_{z^2}$. Now the 5s, three 5p and two 5d orbitals hybridise to form six sp^3d^2 hybrid orbitals. Four of these, containing unpaired

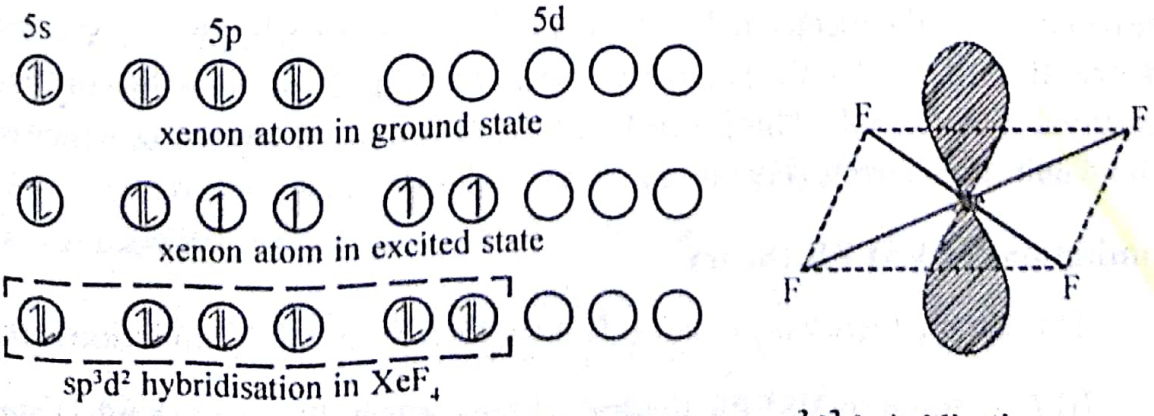

Fig. 1.40. Formation of XeF_4 molecule by sp^3d^2 hybridisation

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electrons, overlap with the half-filled p-orbitals of four fluorine atoms to form four Xe-F bonds. The remaining two orbitals are occupied by lone pairs of electrons. Thus xenon in XeF_4 is surrounded by four bond pairs and two lone pairs of electrons. According to VSEPR theory, a molecule having six orbitals around the central atom should have octahedral geometry. But, XeF_4 is square planar (Fig. 1.40) in shape due to the distortion caused by the presence of two lone pairs.

vi) Geometry of IF_5 molecule: The outer electronic configuration of the central atom, iodine, in the ground state is $5s^2\ 5p_x^2\ 5p_y^2\ 5p_z^1$. In the formation of iodine pentafluoride molecule, two of the 5p electrons are promoted to the vacant 5d orbitals. Thus the excited state outer electronic configuration of iodine is $5s^2\ 5p_x^1\ 5p_y^1\ 5p_z^1\ 5d^1_{x^2-y^2}\ 5d^1_{z^2}$. The 5s, three 5p and two 5d orbitals hybridise to form six sp^3d^2 hybrid orbitals. Five of these (containing unpaired electrons) overlap with the half-filled p-orbitals of five fluorine atoms to form five I-F bonds. The remaining one orbital is occupied by a lone pair of electrons. Thus, iodine in IF_5 is surrounded by five bond

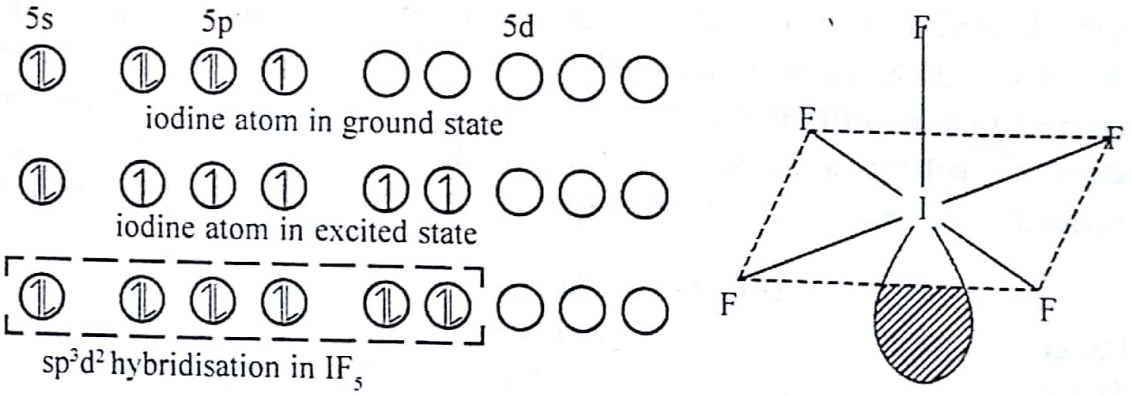

Fig.1.41. Formation of IF_5 by sp^3d^2 hybridisation

pairs and one lone pair of electrons. A molecule containing six orbitals around the central atom is expected to be octahedral in geometry, according to VSEPR theory. But IF_5 molecule is *square pyramidal* in shape, considering the positions of atoms only. This is due to the distortion caused by the presence of a lone pair of electrons (Fig. 1.41).

Limitations of VSEPR theory

This simple theory is not entirely adequate. Some of its limitations are:

(i) According to VSEPR theory, alkaline earth dihalides should have

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linear structure (due to sp hybridisation of the metal atom). However, the structures of alkaline earth dihalides deviate from linearity (e.g., SrF_2 and BaF_2 are bent). This cannot be explained by the VSEPR theory.

(ii) According to VSEPR theory, Li_2O molecule (a molecule of AB_2E_2 type, where E stands for a lone pair) should have a bent structure. Acutally, Li_2O is linear. This linear structure cannot be explained by VSEPR theory.

(iii) The molecular species like $SbCl_6^{3-}$, $BiCl_6^{3-}$, etc., have regular octahedral structures. This cannot be explained by VSEPR theory as there are seven pairs of electrons around the central atom (six bond pairs and one lone pair).

(iv)The theoretical basis of VSEPR theory has not received much support from chemists working in quantum mechanics. There has been some criticism that VSEPR model is no more accurate than a listing of empirical rules.

Metallic Bond

Metals have some very characteristic properties. They are good conductors of heat and electricity. They are lustrous, malleable and ductile. They have high melting and boiling points. It has been observed that metals, generally, have very low ionisation energies because the valency electrons can be taken out relatively easily. This implies that valency electrons in metals are loosely bound to the core or kernel (core or kernel means the nucleus and electrons other than those in the valency shell). Various theories have been proposed to account for the nature of bonding in metals.

(i) The Classical Free-Electron Theory

This theory was proposed by *Drude* and *Lorentz*. It was based on the idea that, high electrical and thermal conductivities of metals are due to the presence of free electrons. According to this model a metal behaves as if it is *an assemblage of positive ions* (*core or kernel*) *immersed in a sea of mobile electrons*. Thus, each electron belongs to a number of positive ions and each positive ion belongs to a number of electrons.

The force of attraction which binds a metal ion (kernel) to a number of electrons within its sphere of influence is known as the metallic bond.

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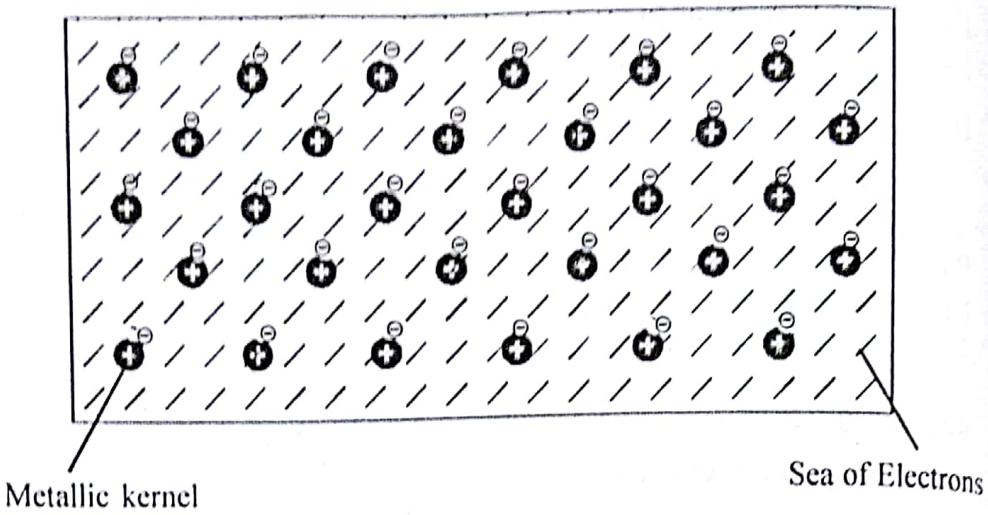

Fig. 1.42
Free electron model of metals

Though the *Drude-Lorentz theory* gives a greatly simplified picture, it affords a useful explanation for many of the properties of metals.

(i) *Electrical conductivity.* Since the free electrons are free to move in an electric field, they give the metal its characteristically high electrical conductivity.

(ii) *Thermal conductivity.* The high thermal conductivity is also due to the presence of free electrons. When one part of the metal is heated, the electrons in that part acquire a large amount of kinetic energy. These electrons move rapidly through the crystal and transfer heat to other parts of the metal.

(iii) *Metallic lustre.* This is also due to the presence of mobile electrons. When a beam of light, falls on the metal surface, the electric field associated with the waves sets the electrons on the metal surface into to and fro oscillations. Now, a moving charge always emits electromagnetic radiation. Hence, the oscillating electrons emit electromagnetic energy in the form of light. Thus, when light falls on a metal surface, it appears as if light is being reflected. Thus, the surface has metallic lustre.

(iv) *Malleability and ductility.* The metallic bond is nondirectional. This means that the force of attraction between M^+ ions and the valency electrons is uniform in all directions. There are no localised bonds. As a result, the M^+ ion can easily move from one lattice site to another. Due to the presence of delocalised bonds, environment of each metal ion remains the same. Thus, the nearest neighbours can be changed easily and new bonds can be formed readily. The ease with which the metal ions can move from one lattice site to

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another is responsible for the properties like malleability and ductility.

(v) *High tensile strength :* Metals can resist stretching without breaking. It is due to strong electrostatic force of attraction between the positive ions (*kernels*) and the *'sea' of electrons.*

(vi) *Elasticity.* Metals possess elasticity. Elasticity is a property by virtue of which a substance can recover its original form soon after the removal of the deforming force. The elasticity is also due to the ease with which metal ions can move from one lattice site to another.

Limitations of free electron theory

This theory cannot explain the vast variations in properties of certain metals. For example, metals are generally good conductors of electricity but Cu is 50 times better conductor than bismuth. Similarly while metals like Na or K are so soft that they can be easily cut with a knife, osmium is so hard that it can scratch even glass. While the melting point of mercury is very low, i.e., only -39^0C, the melting point of tungsten is very high, i.e., 3300^0C.

The Valence Bond Theory

Based on *valence bond theory*, a theory of metallic bonding was developed principally by *Pauling*. According to this model, the structure of metals may be described in terms of covalent bond which resonate among the alternate interatomic positions in the metal. *Pauling* postulates that resonance permits the existence of normal covalencies which resonate between all the possible equivalent pairs of atoms, even if the number of nearest neighbours of an atom in a metal exceeds its valence electrons. For example, potassium has only one valence electron but it crystallises into body centred structure. Thus, each atom has eight equivalent neighbours, together with six next to the nearest neighbours. Since potassium atom has only one valence electron, an electron pair bond can be formed with only one neighbour at one particular time. But the concept of resonance permits the bonding electron pair to move to several other equivalent positions simultaneously. For a crystal of potassium metal, the resonance of K-K bonds can be represented as,

$$\begin{array}{ccc} K-K & K-K & \\ & & \\ K-K & K-K & \\ & & \\ K-K & K-K & \end{array} \quad \leftrightarrow \quad \begin{array}{ccc} & K & \\ & | & \\ K-K & & K-K \\ & & \\ K-K & & K-K \end{array}$$

More stabilisation results from resonance involving ions, as shown below.

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$$\begin{array}{llll} K-K & K-K & K^+ & K \quad K^+ \quad K \\ & & & | \qquad\quad | \\ K-K & K-K \leftrightarrow & K & K \quad K \quad K^- \\ & & | & \qquad | \quad\ | \\ K-K & K-K & K^- - K & K \quad K \end{array}$$

This implies that metallic bond has characteristics of both ionic and covalent bonding. To account for the properties of a metal, the valence bond must be resonating at random among various possible positions.

This model can explain some of the characteristics of metals. For example,

(i) *Metallic conductivity:* The essential requirement for metallic conduction is the presence of vacant orbitals, having energy very close to the occupied orbitals, to accept conductivity electrons. These orbitals are called *metallic orbitals.* Thus, the requirement for a conducting metallic structure is the sufficiency of orbitals to permit the formation of resonating covalent bonds, and in addition, one or more vacant orbitals to receive conductivity electrons. These conditions are satisfied in the example of K given above. Potassium has three vacant 4p orbitals having energy very close to the 4s orbitals occupied by the valency electron. While the 4s and one of the 4p orbitals (in K^- ion) are available for the formation of two covalent bonds, the remaining two 4p orbitals are vacant and hence can act as metallic orbitals for conduction.

Cohesion energy : Metals have high cohesion energy. This can be explained by V.B. theory. Since in any metallic crystal there is an extremely large number of resonance structures, the resonance energy and hence the cohesion energy would be large. As the number of valence electrons increases the number of bonds per atom in any resonance structure increases and hence the cohesion energy increases. Thus, the alkaline earths having two valence electrons per atom have a larger cohesion energy than the alkali metals having one valence electron per atom.

The Band Theory-Molecular Orbital Theory of Metals

The electronic configuration of lithium atom is $1s^2\ 2s^1$. When two lithium atoms approach each other, the 2s orbital of one atom can overlap with 2s orbital of another atom to form two molecular orbitals. One of the molecular orbitals will have lower energy and the other will have higher energy than that of the combining orbitals. When a third lithium atom is added three molecular orbitals are obtained. For n atoms, n molecular orbitals will be

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formed. Thus, the final number of molecular orbitals will be equal to the number of atoms concerned.

As the number of molecular orbitals increases, the spacing between the energy levels of various orbitals decreases. Thus, when there are a large number of atoms, the energy levels of the orbitals are so close together that they almost forms an *'energy band' or 'energy continuum'*.

The number of molecular orbitals must be equal to the number of constituent atomic orbitals. Since there is only one electron per atom in lithium and a MO can hold two electrons, it follows that only half of the MOs in the 2s valence band are filled, i.e., the bonding MOs. It requires only a very small amount of energy to excite an electron to an unoccupied MO.

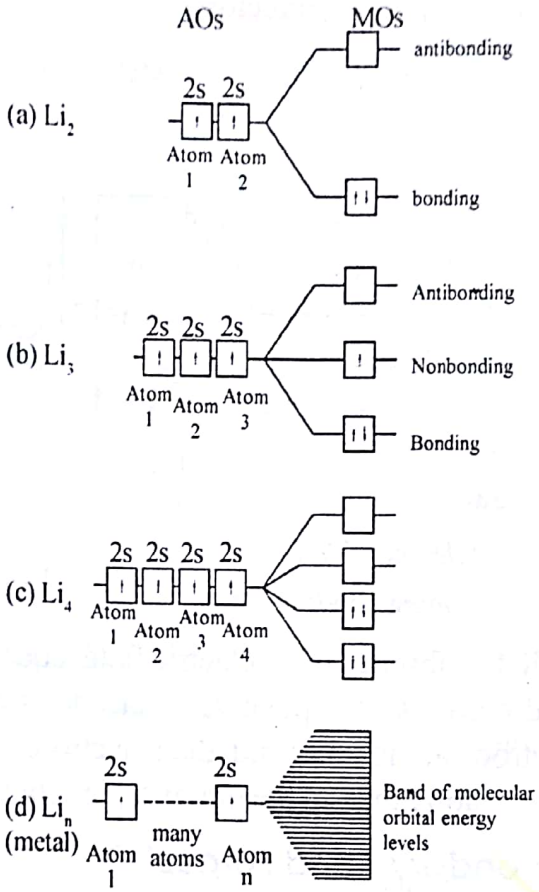

Fig. 1.43

Development of Molecular orbitals into bands in metals.

The molecular orbitals extend in three dimensions over all the atoms in the crystals. So electrons have a high degree of mobility. The mobile electrons account for the high thermal and electrical conduction of metals.

If one end of a piece of metal is heated, electrons at that end gain energy and move to an unoccupied MO where they can travel rapidly to any other part of the metal, which in turn becomes hot. In a similar way, electrical conduction takes place.

In the case of Be the valence orbital is fully filled with two electrons ($1s^2$ $2s^2$). So, no vacant energy level of the 2s energy band is available for the electrons to jump to. In the same way as the 2s AOs form a band of MOs, the 2p AOs form a 2p band. However, at the equilibrium internuclear distance, the 2s energy band and 2p energy band overlap so that some of the vacant

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energy levels of 2p band and the filled energy levels of 2s band fall in the same range of energy. As a result, s electrons can be easily excited to the 2p energy levels. The resulting mobility of electrons makes the alkaline earth metals also good conductors.

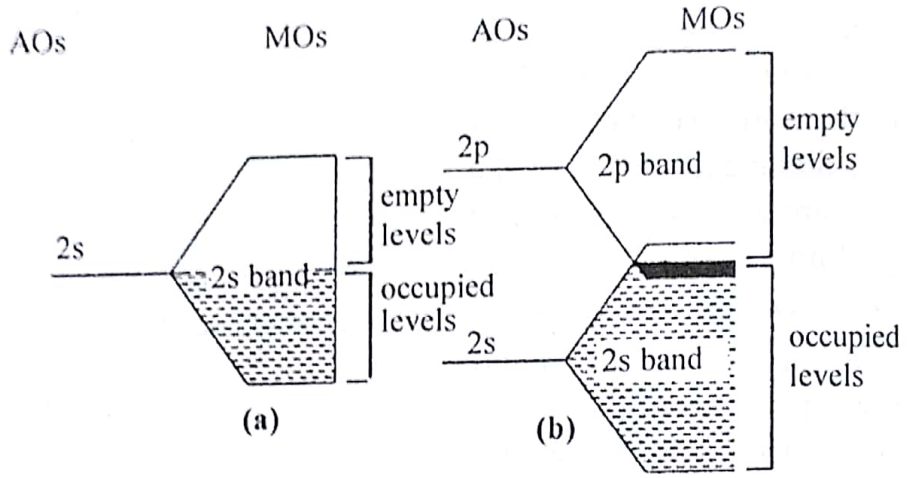

Fig. 1.44

(a) Half filled band in Li

(b) Overlapping bands in Be

In the absence of an electric field, equal number of electrons will move in all directions. If a positive electrode is placed at one end and a negative electrode at the other end, then electrons will move towards the anode much more readily than in the opposite direction, hence an electric current flows.

Secondary Bond Forces

Atoms in a molecule are held together by primary bonds. At the same, the molecules are held together by some other forces. These intermolecular forces are called *secondary bond forces*. These forces are responsible for the aggregation of molecules into solid and liquid phases. The first quantitative statement about the intermolecular forces was made by *van der Waals'*. Hence, the intermolecular forces are known as *van der Waals' forces*. These forces originate from: (i) ion-dipole (ii) dipole-dipole (iii) ion-induced dipole (iv) dipole-induced dipole and (v) induced dipole-induced dipole interactions.

(i) Ion-Dipole Interactions : Ion dipole interactions are responsible for the solvation of ions by polar solvents and the dissolution of ionic substances in such solvents. It has also an important role in the formation of several co-ordination compounds.

(ii) Dipole-Dipole Interactions : Polar molecules have permanent dipoles. Due to the opposite charges at the ends, dipoles will attract each other. The attractive forces due to the electrical interaction between the dipoles is known

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as *dipole-dipole interaction*. For example, gases like ammonia, sulphur dioxide, hydrogen fluoride, hydrogen chloride, etc., have permanent dipoles. As a result of this, there is appreciable dipole-dipole interaction between the molecules of these gases. The magnitude of this type of interaction depends upon dipole moment of the molecule. *The greater the dipole moment the greater is the dipole-dipole interaction.* Due to this attractive interaction, the gases can be easily liquified. The hydrogen bond is a familiar example of this type of attraction.

(iii) Ion-Induced Dipole Interaction : Dipoles may arise due to induction by ions. Thus, when a strongly positive or strongly negative ion is brought in the vicinity of a molecule in which the valence electrons are far away from the positive nucleus, the electronic structure of the molecule is distorted. As a result of this, an induced dipole is formed which is then attracted to the ion. Certain complex ions are formed in this way. For example, the formation to triiodide ion I_3^- is due to the polarisation of I_2 molecule by I^- ion.

$$I_2 + I^- \rightarrow I_3^-$$

(iv) Dipole-Induced Dipole Interaction : A polar molecule may polarise a non-polar molecule having easily deformable electronic structure and thus induce dipoles in that molecule. The induced dipole then interacts with the dipole moment of the polar molecule and as a result the two molecules are attracted together. The magnitude of this interaction depends on the dipole moment of the polar molecule and the polarizability of non-polar molecule.

The attractive forces in dipole-induced dipole interaction would be weaker than those associated with ion-induced dipoles. The formation of hydrates of inert gas elements is due to this type of interaction.

(v) Induced dipole-Induced dipole Interaction [London dispersive forces] : van der Waals forces exist even in non-polar molecules like O_2 and N_2 and also in non polar monoatomic molecules like He, Ne, Ar, etc. In 1930, *F.London* provided an explanation for the existence of this attractive force. According to him, electrons of a neutral molecule keep on oscillating with respect to the nuclei of the atoms. As a result of this, at a given time, positive charge may be concentrated in one region and negative charge in another region of the molecule. Thus, a non-polar molecule may become momentarily self-polarised. This polarised molecule may then induce a dipole moment in a neighbouring molecule. This is shown in Fig.1.45. The forces of electrostatic attraction between induced dipole-induced dipole are known as *London*

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dispersive forces. For a pair of adjacent molecules, London dispersive forces is inversely proportional to the sixth power of the distance (r) between them,

i.e., $F \propto \frac{1}{r^6}$.

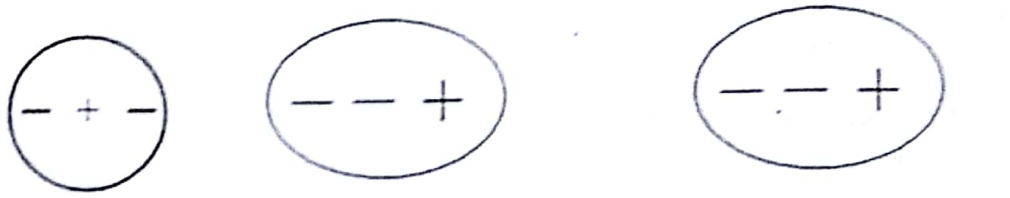

Fig. 1.45

Origin of attractive forces in non-polar molecules

The Hydrogen Bond

The attractive force which binds hydrogen atom of one molecule with electronegative atom (such as fluorine) of the same molecule or another molecule of the same substance or another substance is known as hydrogen bond.

In strength it is intermediate between the weak van der Waals' forces and the strong and ordinary ionic and covalent bonds. This bonding occurs in the simple hydrides of strongly electronegative elements such as O, F and N. In general, it is represented as follows.

.....H – X.....H–X.....H–X.....H–X.....

where, X is the electronegative atom. The short solid line (–) represents the covalent bond, a chemical bond while the dotted line (.....) represents the hydrogen bond which is not a chemical one but only a cohesive force. This bonding arises in the following way.

The electronic configuration of H is $1s^1$. From this, it is expected to form one covalent bond. When this forms a covalent bond with a more electronegative atom such as F, O or N, the electron pair shared between the atoms lies far away from hydrogen atom making the latter highly positive. Thus, the bonds H–F, H–O and H–N are largely polarised and consequently possess a large dipole moment. The net result is that intermolecular attraction caused by these dipoles, i.e., the force of attraction between the hydrogen

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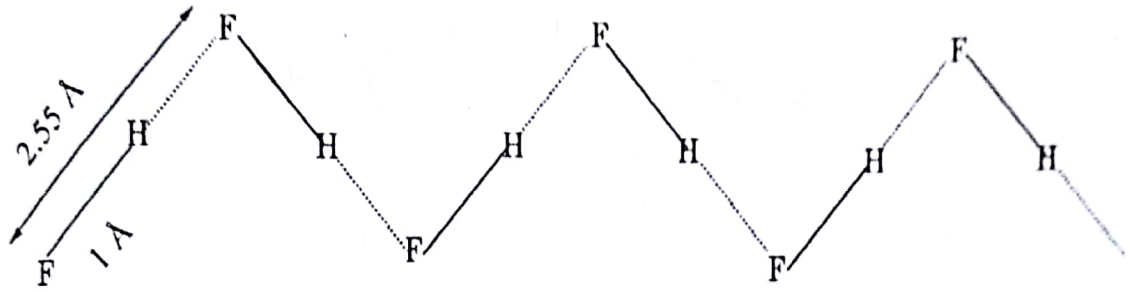
atom of one molecule and the strongly electronegative atom of another molecule is exceptionally strong. This force of attraction is called the *hydrogen bonding*.

Fluorine is the most electronegative element and, therefore, one of the strongest hydrogen bonding is observed in HF. Solid HF consist of a long zig-zag chain of molecules which are associated by hydrogen bonding as shown below:



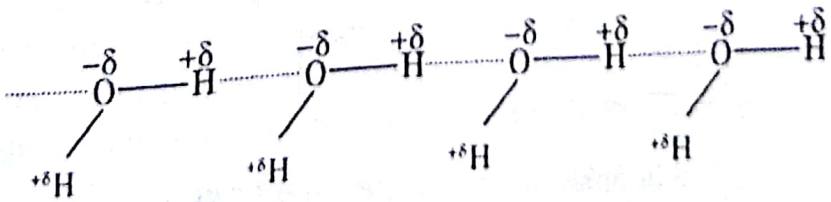
In aqueous solution, HF dissociate and yields the bifluoride ion, HF_2^-, instead of simple fluoride ion, F^-. The HF_2^- ion is a resonance hybrid of the following canonical structures.

$$F^- \cdots H - F \leftrightarrow F - H \cdots F^-$$

As a result of this resonance is that the hydrogen atom is equidistant from the fluorine nuclei.

It should be noted here that the chlorine which is similar to nitrogen in electronegativity has a little tendency for hydrogen bonding. This is perhaps due to the larger size of the Cl atom which gives a less strong electrostatic field.

Oxygen atom of H_2O molecule, since sufficiently electronegative, also forms two hydrogen bonds in addition to its two covalent bonds giving rise to associated molecules, $(H_2O)_n$ as shown below:



Alcohols and carboxylic acids also form associated molecules by hydrogen bonding.

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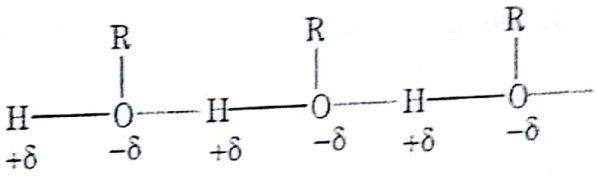
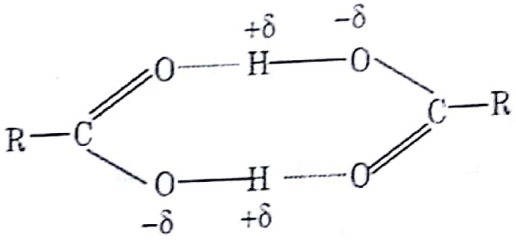


(alcohols)



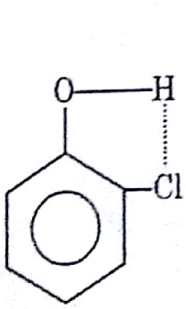
(carboxylic acids)

In water and alcohols, the H-bonding may extend to several molecules as indicated above but in carboxylic acids, the H-bonding is limited to the association of two molecules only.

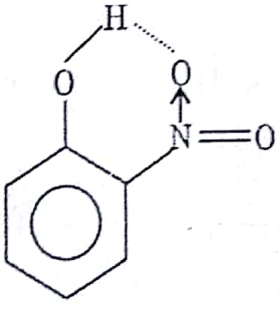
Nitrogen atom is also strongly electronegative. Ammonia, therefore, is also a polar molecule, the nitrogen carrying the negative charge and each hydrogen carrying a positive charge. H-bonding takes place in NH_3 as well yielding associated molecules as, $(NH_3)_n$.

Intramoleculear Hydrogen Bonding

The hydrogen bonding resulting from the electrostatic forces of attraction between hydrogen and an electronegative atom of the same molecule is called intramolecular hydrogen bonding. This type of hydrogen bonding is commonly known as chelation.



O-chlorophenol



O-nitrophenol

Intermolecular Hydrogen Bonding

Hydrogen bonding resulting from electrostatic forces of attraction between the positive hydrogen of one molecule and an electronegative element of

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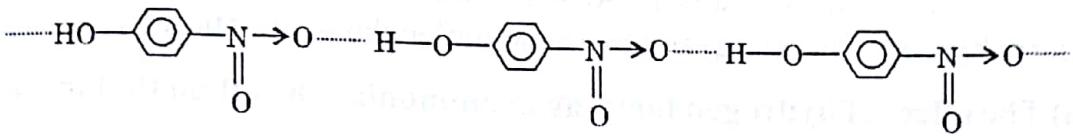
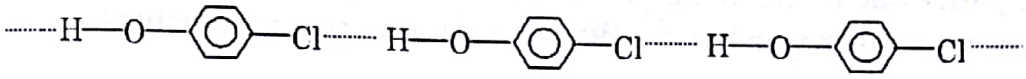
another molecule of the same substance (or different substance) is known as intermolecular hydrogen bonding.

e.g., (i) p-nitrophenol



(ii) p-chlorophenol



As a result of intermoleculear hydrogen bonding, the para derivatives undergo association, resulting in an increase in molecular mass and hence an increase in boiling point. In ortho derivatives, because of intramolecular hydrogen bonds, no molecular association is possible. As a consequence of this, the ortho derivatives are more volatile than the para derivatives. Thus, while ortho nitrophenol is readily volatile in steam, the para nitrophenol is completely non-volatile. The two derivatives can thus be easily separated from each other by steam distillation.

Consequences of hydrogen bonding

The boiling points of H_2O, NH_3 and HF are abnormally high. Let us consider the boiling points of hydrogen compounds of the elements of groups XV, XVI and XVII.

Group XV			Group XVI			Group XVII		
Compound	Molar Mass	B.P.(°C)	Compound	Molar Mass	B.P.(°C)	Compound	Molar Mass	B.P.(°C)
SbH_3	125	-17.0	H_2Te	130	-1.8	HI	128	-35.4
AsH_3	78	-55.1	H_2Se	81	-42.0	HBr	81	-66.7
PH_3	34	-84.6	H_2S	33	-59.6	HCl	36.5	-85
NH_3	17	-33.0	H_2O	18	+100°C	HF	20	+19.9

From the table it is evident that although the boiling point in each group decreases with decrease in molar mass, there is a sudden increase in the boiling points of NH_3, H_2O and HF in groups XV, XVI and XVII respectively as due to strong intermolecular forces caused by intermolecular hydrogen bonding.

(i) The boiling point of water is higher than that of HF

Even if a stronger hydrogen bond exists in HF than in H_2O, the boiling point of water is higher. This is due to the statistical consideration that one molecule of water can form two such bonds, as compared to one for HF.

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(ii) H_2O is a liquid while H_2S is a gas

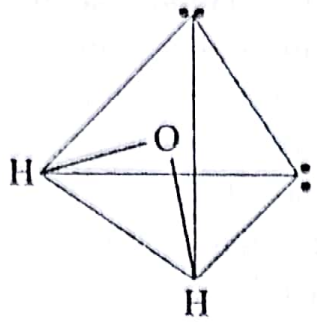
In H_2O, intermolecular hydrogen bonding results in association. So it is a liquid. There is no intermolecular hydrogen bonding in H_2S (since the electronegativity of sulphur is not so high to make the hydrogen positive enough to form hydrogen bonding) and hence no molecular association. So it is a gas.

(iii) The effect of hydrogen bonding in ammonia is less than that in water

It is partly due to smaller electronegativity of nitrogen compared to oxygen and partly due to the fact that nitrogen with only one unshared electron pair forms three hydrogen bonds with NH groups of other molecules.

(iv) Density of ice is less than that of water

It is evident from the shape of water molecule that hydrogen bonds between the oxygen atom of one molecule and two hydrogen atoms of another molecule can be formed only at certain definite angles. This is particularly so in the solid state (ice) since a crystal has a definite space lattice. In ice, each oxygen atom is surrounded tetrahedrally by four H atoms; two of these are bonded covalently and the other two through hydrogen bonds.



Shape of water molecule

In ice, the H_2O molecules themselves are held tetrahedrally with respect to one another.

This gives rise to an *open cage like structure of ice*. This cage like structure, however, breaks down partially as ice melts. The molecules, therefore, come close to one another. In other words, the molecules in water in the liquid state lie closer to one another than in the solid state. Thus, *ice is lighter than water*.

(v) Water contracts when heated between 0^0C and 4^0C

As water is warmed from 0^0C onwards, more and more hydrogen bonds are broken. Consequently the molecules come closer and closer together. This leads to contraction. However, there is the normal effect of expansion also as in other liquids. It appears that upto 4^0C, the former effect predominates and hence there is an overall contraction in volume. Above 4^0C, the second effect predominates and hence there is an increase in volume as the temperature rises beyond 4^0C.

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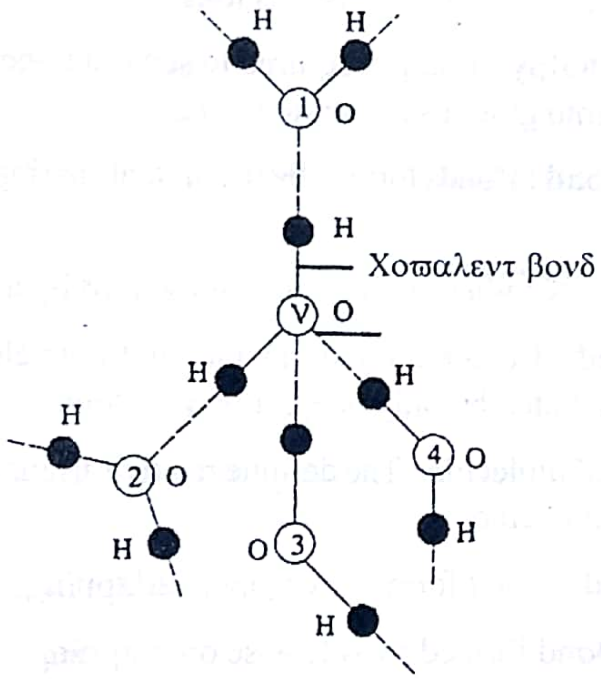

Fig. 3.31
The tetrahedral open cage like crystal structure of ice. The central oxygen atom V is surrounded tetrahedrally by the oxygen atoms marked 1, 2, 3 and 4.

IMPORTANT POINTS FOR QUICK REVISION

Chemical Bond : The attractive force which binds the constituent particles together in a molecule.

Octet Rule : Atoms combine to form molecules because of their tendency to attain an octet of electrons in their valency shell.

Ionic Bond : Electrostatic force of attraction holding the oppositely charged ions.

Electrovalency : Number of electrons gained or lost by an atom during the formation of an ionic bond.

Factors Favouring Ionic Bond Formation : Low ionisation energy of electropositive element, high electron affinity of electronegative element and high lattice energy of the compound to be formed.

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Lattice energy: Amount of energy released when one gram mole of an ionic solid is formed from gaseous ions.

Lattice enthalpy : Energy required to separate one mole of a solid ionic compound into gaseous constituent ions.

Covalent bond : Bonds formed by the mutual sharing of electrons between two atoms.

Covalency : Number of electron pairs shared by an atom.

Dative bond : Covalent bond in which both the electrons in the shared pair is contributed by only one of the two atoms.

Geometry of molecule : The definite relative arrangements of the bonded atoms in a molecule.

Sigma bond : Bond formed by axial overlapping.

Pi bond : Bond formed by side wise overlapping.

Bond angle : Internal angle between the orbitals containing electron pairs in the valency shell of the central atom in a molecule.

Bond length : The average equilibrium distance between the centres of the nuclei of two bonded atoms in a molecule.

Bond energy : Average amount of energy required to break one mole of bonds of same kind so as to separate the bonded atoms in a molecule in the gaseous state.

Bond Order : Number of bonds between the two atoms in a molecule.

With increase of bond order, bond length decreases and bond energy increases.

Hybridisation : Intermixing of atomic orbitals of slightly different energies of same atom to get an equal number of new orbitals of equivalent energies and identical shape.

Non polar molecules : Molecules with zero dipole moment.

Polar molecules : Molecules with dipole moment greater than zero.

Dipole Moment : Product of charge at one centre and the distance between the charged centres.

Resonance : When a molecule cannot be represented by a single structure

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but its characteristic properties can be described by two or more structures, then the aactual structure is said to be a resonance hybrid of these structures.

Hydrogen bond : The attractive force which binds hydrogen atom of one molecule with an electronegative atom of the same or another molecule of the same or different substance.

MODEL QUESTIONS

Section A (Each question carries 2 marks)

1. Mention two uses of dipole moment measurements.
2. NH_3 and CH_4 have sp^3 hybridisation. Shape of these molecules are different. Justify.
3. State VSEPR theory and predict the geometry of XeF_4 molecule.
4. Define dipole moment
5. Predict the shape of IF_7 molecule and draw its shape.
6. State two types of hydrogen bonding with examples.
7. Indicate the hybridisation of carbon atoms in propyne. How many σ and π bonds are present in it?
8. Explain sp hybridisation and the formation of bonds between sp hybridised carbon atoms.
9. Explain polar and non-polar covalent bonds with examples.
10. Mention the type of hybridisation in the following molecules:
    a) BF_3 b) SF_6 c) IF_7
11. The molecule SO_2 has a dipole moment. Is the molecule linear or bent? Justify your answer.
12. On the basis of VSEPR theory explain the shapes of BF_3 and NH_3 molecules.
13. H_2O is a liquid while H_2S is a gas. Justify.
14. The bond angle in H_2O molecule is less than the tetrahedral angle. Why?
15. Predict the shape of XeF_4 molecule on the basis of VSEPR theory.
16. PCl_5 molecule is highly reactive. Why?
17. Which of the two should be more ionic, AlF_3 or $AlCl_3$? Explain.

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18. Which is more polar, H_2S or H_2O? Why?
19. What do you understand by polar and nonpolar molecules?
20. Distinguish between intermolecular and intramolecular hydrogen bonding.
21. What is sp^2 hybridisation? What is the shape of molecules formed by sp^2 hybridisation
22. Why is it necessary to invoke sp^2 hybridisation to explain the bonding in C_2H_4
23. What is the shape of ammonia molecule?
24. What is the type of hybridisation taking place in Xenon atom of Xenon tetrafluoride molecule? What is the shape of the molecule?
25. Which is more polar - HF or HCl ?Justify.
26. State and explain Fajan's rule
27. The bond energy of NO^+ is larger than that of NO. Why?
28. What are bonding and antibonding orbitals?
29. Nitrogen does not form a pentachloride but phosphorus forms a pentachloride. Why?
30. What is bond order? Calculate it for CO molecule.
31. What are the factors favouring the formation of covalent bond?
32. What are the rules for the linear combination of atomic orbitals?
33. Based on VSEPR theory, predict the shape of NH_3 molecule.
34. Explain why sulphur forms SF_6 while oxygen does not form OF_6.
35. State VSEPR theory and predict the geometry of XeF_4 molecule.
36. The bond angle in H_2O molecule is less than the tetrahedral angle. Why?
37. Predict the shape of SF_4 molecule on the basis of VSEPR theory.
38. What is the shape of H_3O^+?

Section B (Each question carries 4 marks)

1. Define hybridisation. Mention the types of hybridisation involved in the following molecules: (a) SF_6 (b) PCl_5 (c) BF_3. Draw the geometry.
2. Write brief notes on polar and nonpolar covalent bonds.
3. State and explain VSEPR theory. What are its limitations?

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4. Explain i) $BeCl_2$ is linear ii) CO_2 is linear iii) H_2O is bent
5. Account for the zero dipole moment of BF_3 and CO_2.
6. CO_2 and H_2O are both triatomic molecules but there is a large difference in their dipole moments. Why?
7. Explain the shape of $SnCl_2$ molecule.
8. Indicate the shapes of the molecules obtained when the following hybridisations occur (i) sp^3 d and ii) sp^3d^2. Give one example for each.
9. Explain why CO_2 and CCl_4 molecules are non polar but $CHCl_3$ molecule is polar.
10. Discuss the influence of hydrogen bonding on physical properties of substances.
11. Explain the shape of SF_6 molecule.
12. Eventhough sp^3 hybridisation takes place in NH_3, the H-N-H bond angle is found to be only $107^o48'$. Why?
13. Explain the following a) NH_3 is polar while BF_3 is non-polar b) H_2O is a liquid while H_2S is a gas c) Sulphur forms hexafluoride whereas oxygen does not.
14. All the bond lengths and bond angles in CH_4 and SF_6 are equal; whereas in PCl_5 and IF_7 there are two sets of bond lengths and bond angles. Explain.
15. Water exists as liquid at room temperature while hydrogen sulphide is a gas at the same temperature. Account for the reason.
16. What are hydrogen bonds? Explain with example. Why water boils at 100^oC whereas hydrogen sulphide boils at 62^oC?
17. Describe hybridisation in iodine atom of IF_7 molecule. What is its shape? What are the bond angles in IF_7 ?
18. Discuss the shape of PCl_5 molecule.
19. Define dipole moment. What is the dipole moment of CO_2?
20. Using MO theory, predict the bond order and the number of unpaired electrons in O_2^{2-}, O_2^-, O_2, O_2^+ and NO.
21. (a) Give a brief account of molecular orbital theory of bonding taking C_2 and N_2 as examples.

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(b) Does He_2 exist? If not why?

22. (a) What are bonding, antibonding and nonbonding orbitals? Give the energy sequence of molecular orbitals derived from s and p orbitals.

(b) Give the MO description of the electronic configuration of CO and O_2 and discuss the bond order and magnetic property of each.

23. (a) What is VSEPR theory? How is it used to predict the geometries of H_2O and NH_3 molecules.

(b) Give an account of VB theory and compare it with MO theory.

24. (a) Using MO theory, predict the bond order and the number of unpaired electrons in O_2, N_2 and CO.

(b) Explain the paramagnetism of O_2 molecule using MO theory.

25. (a) What is LCAO? Discuss this in the formation of HF molecule.

(b) What are the salient features of MO theory of Hund and Mulliken?

26. (a) Construct a plot of potential energy against the inter nuclear distance between two hydrogen atoms and explain what does the minimum in the curve represent.

(b) Give a brief account of metallic bond, giving special emphasis on band theory.

27. (a) What is hydrogen bonding? In what respects is it different from ionic and covalent bonding.

(b) Discuss the band theory of metals.

28. (a) Discuss the nature of metallic bond. How does it account for the physical properties of metals?

29. (a) State and explain VSEPR theory. What are its limitations.

(b) Explain the shape of SF_4 molecule.

30. (a) Eventhough sp^3 hybridisation takes place in NH_3, the H–N–H bond angle is found to be only $107^0 48'$. Why?

(b) Water exists as liquid at room temperature while hydrogen sulphide is a gas at the same temperature. Account for the reason.

31. (a) What are the hydrogen bonds? Explain with example. Why water boils at 100^0C whereas hydrogen sulphide boils at 62^0C?

(b) Describe hybridisation in iodine atom of IF_7 molecule. What is its

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shape? What are the bond angles in IF_7?

32. (a) Discuss the shape of XeX_2 molecule.

(b) Using MO theory, predict the bond order and the number of unpaired electrons in O_2^{2-}, O_2^-, O_2, O_2^+ and NO.

Section C (Each question carries 8 marks)

1. a) What is VSEPR theory ?

b) How can it be applied to explain the shapes of the following molecules (i) PCl_5 (ii) SF_6 (iii) XeF_4.

2. a) What are sp, sp^2 and sp^3 hybridisations? (b) Write the geometry of $BeCl_2$, PCl_5, SF_6 and IF_7

3. What is meant by dipole moment? How is it helpful in explaining the structure of molecules?

4. a) Discuss intermolecular and intramolecular hydrogen bonds with one example for each. (b) Discuss the consequences of hydrogen bonding.

5. a) Draw the MO diagram for O_2 molucule. Why is O_2 paramagnetic? b) Write notes on Born-Haber cycle

6. (a) (i) Draw the MO diagram for CO. Comment on its bond order and magnetic property

(ii) Define (i) bond order and (ii) bond energy. How are they related?

(b) (i) Draw the MO diagram for N_2 molecule and O_2 molecule. Why is O_2 paramagnetic?

7. (a) Give an account of MO theory of bonding with suitable examples. On the basis of the above theory explain the magnetic behaviour of O_2 and CO molecules. Calculate the bond order of above molecules.

(b) (i) Use energy level diagrams to show that nitrogen molecule would be expected to have a triple bond, fluorine molecule a single bond and neon molecule no bond.

8. (a) Compare and contrast valence bond theory and MO theory of bonding.

(b) What is VSEPR theory? How can it be applied to explain the shapes of the following molecules (i) PCl_5 (ii) SF_6 (iii) XeF_4.

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9. (a) Discuss intermolecular and intramolecular hydrogen bonds with one example for each. Discuss the consequences of hydrogen bonding.

   (b) Draw the MO diagram for O_2 molecule. Why is O_2 paramagnetic?

10. (a) (i) How do band theory explain the conducting property of metals?

   (ii) What is hydrogen bonding? In what respect is it different from ionic and covalent bonding.

   (b) (i) What are hybrid orbitals? Discuss the advantages of hybrid orbitals over atomic orbitals in forming bonds (ii) Discuss the band theory of metals.

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Module 2

A. THERMODYNAMIC PARAMETERS

STOICHIOMETRY

Stoichiometry a section of chemistry which deals with the relationships between reactants and / or products in a chemical reaction to determine the desired quantitative data. In other words, stoichiometry is the calculation of relative quantities of reactants and products in chemical reactions. In Greek, *Stoikhein* means element and *metron* means measure, so stoichiometry literally means the *measure of elements.*

Stoichiometry rests upon very basic laws, i.e., law of conservation of mass, the law of definite proportions, the law of multiple proportions and the law of reciprocal proportions. In general, in chemical reactions chemicals combine in definite ratios. Since chemical reactions can neither create nor destroy matter, nor transmute one element into another, the number of atoms of a given element on the reactant side must be equal to the number of atoms of the same element on the product side. As the reacting molecules consists of a definite set of atoms in an integer ratio, the ratio between reactants in a complete reaction is also in integer ratio.

Different elements have a different atomic mass, and as collection of atoms, molecules have a definite molar mass, measured with the unit mole. Thus, to calculate the stoichiometry by mass, the number of molecules required for each reactant is expressed in moles and multiplied by the molar mass of each to give the mass of each reactant per mole of the reaction. The mass ratios may be calculated by dividing each by the total in the whole reaction.

Describing the quantitative relations among substances as they participate in chemical reaction is called *reaction stoichiometry.*

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Atomic Mass

Atoms are extremely small and light. For example, the actual mass of a hydrogen atom is 1.67×10^{-24}g. For practical purposes the relative masses of atoms are used instead of their actual masses. In 1961, the International Union for Chemists selected C-12 isotope as the standard. Based on this,

atomic mass of an element is defined as a number which expresses how many times the mass of one atom of the element is greater than 1/12th the mass of a carbon -12 atom.

$$\text{Atomic mass} = \frac{\text{mass of 1 atom of the element}}{\frac{1}{12} \times \text{mass of 1 atom of carbon} - 12}$$

By the above definition, atomic mass turns out to be a mere number. Then it expresses the relative mass of the atom.

Atomic mass unit scale. However, by another approach, atomic mass can also be expressed in a unit called *atomic mass unit (amu)*. Today, 'amu' has been replaced by 'u' which is called ***unified mass.*** Atomic mass unit is a special unit of mass used for expressing the mass of atoms and subatomic particles such as electrons, protons, neutrons, etc.

Atomic mass unit is defined as exactly 1/12th the mass of a carbon 12 atom.

$$1 \text{ amu} = \frac{\text{mass of 1 carbon} - 12 \text{ atom}}{12}$$

Since the mass of 6.023×10^{23} (Avogadro number) atoms of ^{12}C is 12 g,

$$\text{mass of one } ^{12}C \text{ atom} = \frac{12\,g}{6.023 \times 10^{23}}$$

$$\therefore \quad 1 \text{ amu} = \frac{1}{12} \times \frac{12\,g}{6.023 \times 10^{23}} = 1.66 \times 10^{-24} g$$

On the amu scale, the mass of a ^{12}C atom = 12 amu = 12 u

$$= 12 \times 1.66 \times 10^{-24} g = 1.992 \times 10^{-23} g$$

Therefore, ***the atomic mass of an element can also be defined as the mass of one atom of the element expressed in atomic mass units.***

It was once believed that all atoms of a given element have exactly the same mass. But later studies showed this to be false. It is found that atoms of the same element may have different masses. These are known as ***isotopes.*** In such cases, the atomic mass of an element is taken as the average of the atomic masses of the various isotopes. Taking chlorine as example, it is found that there are some atoms with mass 35 amu or 35 u and others with 37 amu

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or 37 u. These isotopes are present in the ratio 3:1. Therefore, the average atomic mass would be $\frac{35 \times 3 + 37 \times 1}{4} = 35.5$. *This is the reason that atomic mass of many elements have fractional values.* Thus, atomic mass can better be defined as,

> ***the average mass of the atoms of an element expressed in atomic mass units, carbon-12 being taken as the standard, with an atomic mass of 12 amu.***

Example 1. Neon have three naturally occurring isotopes having isotopic mass and relative abundance 20(90.51%), 21(0.28%), and 22(9.21%). Calculate the average atomic mass of the element.

Solution: Average atomic mass of Ne $= \frac{90.51 \times 20 + 0.28 \times 21 + 9.21 \times 22}{(90.51 + 0.28 + 9.21)} =$ 20.187

Gram Atomic Mass or Gram Atom. Atomic mass expressed in grams is called ***gram atomic mass or simply gram atom.***

Thus, atomic mass of oxygen = 16 amu or 16 u

1gram atom of oxygen = gram atomic mass of oxygen = 16 g.

2 gram atoms of oxygen = 2 × gram atomic mass of oxygen

= 2 × 16 g = 32g.

or Number of gram atoms $= \frac{\text{Mass in grams}}{\text{Gram atomic mass}}$

Example 2. Find the number of gram atoms in 24g carbon.

Solution. No of gram atoms in 24g carbon $= \frac{\text{Mass in grams}}{\text{Gram atomic mass}}$

$= \frac{24g}{12g} = 12$

Example 3. Calculate the mass of 1.5 gram atoms of calcium (atomic mass of calcium =40).

Solution : 1gram atom of calcium = 40g.

1.5 gram atoms of calcium = 40g × 1.5 = 60g

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Self Study problems

1. Calculate the mass of 1.8 gram atoms of Mg (atomic mass of Mg = 24)
Ans. 43.2g

2. Calculate the number of gram atoms in 12.8g oxygen (atomic mass = 16) **Ans.** 0.8 gram atom.

Molecular Mass

Relative masses of molecules may be obtained in terms of the same standard used for defining atomic mass. ***The molecular mass of an element or a compound is a number which expresses how many times the mass of one molecule of that element or compound is greater than 1/12th the mass of a carbon-12 atom.***

$$\text{Molecular mass} = \frac{\text{Mass of one molecule of the substance}}{\frac{1}{12} \times \text{mass of one atom of carbon} - 12}$$

Molecular mass can be obtained by adding up the atomic masses of all the atoms present in the molecule. For example, one molecule of water consists of two atoms of hydrogen and one atom of oxygen. Therefore molecular mass of water = (2×1.008) + 16 = 18.016. This value is the relative mass of water molecule. The actual mass of one molecule of water can be expressed as 18.016.amu of 18.016 u.

Example 4. Calculate the molecular mass of glucose ($C_6H_{12}O_6$) molecule.

Solution: Molecular mass is obtained by multiplying atomic mass of each element by the number of its atoms and adding them together. Thus,

$$\begin{aligned}\text{molecular mass of glucose} &= 6 \times (12.011\text{ u}) + 12 \times (1.008\text{ u}) + 6 \times (16.00\text{ u}) \\ &= (72.066\text{ u}) + (12.096\text{ u}) + (96.00\text{ u}) \\ &= 180.162\text{ u}.\end{aligned}$$

Formula Mass : In the case of ionic compounds, it is better to use the term *formula mass* instead of the term molecular mass. Formula mass of an ionic compound is the sum of the atomic masses of all atoms in a formula unit of the compound. For example,

$$\begin{aligned}\text{the formula mass of NaCl} &= \text{atomic mass of Na} + \text{atomic mass of Cl} \\ &= 23.0\text{u} + 35.5\text{ u} = 58.5\text{ u}\end{aligned}$$

Gram Molecular Mass or Gram Mole. *Molecular mass expressed in grams is called gram molecular mass or gram mole.* Thus, 18.016g of water

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is 1gram mole of water, or 1gram mole of water is 18.016g. 2gram moles of water = 2 × 18.016g. Thus,

$$\text{Number of gram moles} = \frac{\text{Mass in grams}}{\text{Gram molecular mass}}$$

Example 5. Calculate the mass of 1.2 gram moles of H_2SO_4

Solution : Gram molecular mass of $H_2SO_4 = (1\times2) + (32\times1) + (16\times4)$

$= 98$

1gram mole of $H_2SO_4 = 98$ g

1.2 gram moles of $H_2SO_4 = 98\text{ g} \times 1.2 = 117.6$ g

Example 6. Calculate the number of gram moles in 27 g H_2O.

Solution

$$\text{No. of gram moles of water in 27 g} = \frac{\text{Mass in grams}}{\text{Gram molecular mass}} = \frac{27\text{g}}{18\text{g}} = 1.5$$

Self Study Problems

3. Calculate the mass of 1.5gram moles of glucose ($C_6H_{12}O_6$)

Ans: 270g

4. **Calculate the number of gram moles in 68.4g cane sugar ($C_{12}H_{22}O_{11}$)** **Ans:** 0.2

Mole concept and Avogadro's number

The number of atoms in one gram atom of an element as well as the number of molecules in one gram mole of any compound is a constant. This is known as *Avogadro's number*, usually denoted by N. The value of this constant is found to be 6.022×10^{23}. Since atomic mass and molecular mass are defined with reference to carbon-12, Avogadro's number also is defined relative to carbon -12. On this basis,

Avogadro's number is defined as the number of atoms present in exact 12 grams of carbon-12.

Mole. The amount of any substance containing Avogadro's number of particles (atoms, ions, molecules, etc.) is known as a **mole**. A mole is also defined in terms of carbon-12.

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A mole is that amount of substance which contains as many particles as there are atoms in 12 grams (or 0.012 kg) of carbon-12

Thus, Avogadro's number or Avogadro constant

$$= \frac{12\,g}{\text{mass of a C-12 atom}}$$

$$= \frac{12\,g}{1.992648\times10^{-23}\,g}$$

$$= 6.0221367 \times 10^{23} = 6.022 \times 10^{23}$$

It is the unit amount of substance. Just as 'a dozen' or 'a gross' represents a certain number of objects, a mole represents a specific number of particles like atoms, ions, molecules, electrons, etc.

Molar mass. ***The mass of one mole of a substance is the molar mass.*** It is usually expressed as:

(i) gram atomic mass (gram atom) when atoms are considered.
(ii) gram molecular mass (gram mole) when molecules are considered and
(iii) gram ionic mass (gram ion) when ions are considered.

The molar mass can be calculated by adding up the atomic masses indicated by the corresponding symbol or formula and by expressing the value in grams. The table given below will illustrate the meaning of mole and molar mass.

Mole and Molar Masses

Name	Symbol or Formula	Mass of one mole	Number and kind of particles in 1 mole
Oxygen	O	16g	6.022×10^{23} atoms
Oxygen	O_2	32g	6.022×10^{23} molecules
Carbon dioxide	CO_2	44g	6.022×10^{23} molecules
Sodium ion	Na^+	23g	6.022×10^{23} ions

Molar Volume. By Avogadro's hypothesis equal number of molecules occupy the same volume at a given temperature and pressure. For comparing the densities or volume of gases, the temperature and pressure fixed are 273 K and 1 atmosphere respectively. Under these standard conditions, the gas is said to be at standard temperature and pressure (S.T.P). One gram mole of

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oxygen (32g), that is 6.022×10^{23} molecules is found to occupy a volume of 22.414 litres at S.T.P. Similarly 28g of nitrogen, 44g of carbon dioxide or one gram mole of any gaseous element or compound must occupy a volume of 22.414 litres at S.T.P. ***The volume occupied by one mole of a gas is the gram molecular volume or molar volume.*** At STP it is 22.414 litres. The molar volume of any gas contains 1 mole of the substance.

The mole concept is useful in calculating the actual mass and number of atoms and molecules present in a definite amount of a substance.

Example 7. Calculate the actual mass of a C-12 atom.

Solution: Gram atomic mass of C-12 = 12 g

Number of atoms in 1 gram atom of C-12 = Avogadro's number
$= 6.022 \times 10^{23}$

$$\text{Mass of one C-12 atom} = \frac{\text{Gram atomic mass}}{\text{Avogadro's number}} = \frac{12\text{g}}{6.022 \times 10^{23}}$$

$$= 1.992 \times 10^{-23}\text{g}$$

Example 8. Calculate the mass of 1 amu in grams

Solution: 1 amu = $\frac{1}{12}$th the mass of a C-12 atom

Mass of one C-12 atom = 1.992×10^{-23}g

$$1 \text{ amu} = \frac{1}{12} \times 1.992 \times 10^{-23} = 1.66 \times 10^{-24}\text{g}$$

Example 9. Calculate the mass of one molecule of water

Solution : Gram molecular mass of H_2O = 18g

Number of molecules in 1 gram molecular mass of $H_2O = 6.022 \times 10^{23}$

$$\text{Mass of 1 molecule of } H_2O = \frac{\text{Gram atomic mass}}{\text{Avogadro's number}} = \frac{18\text{g}}{6.022 \times 10^{23}}$$

$$= 2.99 \times 10^{-23}\text{g}$$

Example 10. How many molecules are present in 1 gram water

Solution : Gram molecular mass of H_2O = 18g

Number of molecules in 18g = Avogadro's number = 6.022×10^{23}

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$$\text{Number of molecules in 1 g} = \frac{\text{Avogadro's number}}{\text{Gram molecular mass}} = \frac{6.022 \times 10^{23}}{18}$$

$$= 3.344 \times 10^{22}$$

Example 11. Calculate the mass of 0.5 mole of sodium chloride

Solution: Gram molecular mass of NaCl = 58.45g

i.e., Mass of one mole NaCl = 58.45g

$\therefore$ Mass of 0.5 mole NaCl = Number of moles × Gram molecular mass

$= 0.5 \times 58.45\text{g} = 29.225\text{g}$

Example 12. Calculate the number of moles in (a) 50g $CaCO_3$ (b) 8g O_2

Solution : (a) Gram molecular mass of $CaCO_3$ = 100g

$$\text{Number of moles} = \frac{\text{Mass of grams}}{\text{Gram molecular mass}} = \frac{50\text{g}}{100\text{g}} = 0.5$$

(b) Gram molecular mass of O_2 = 32g

Mass of O_2 = 8g

$$\text{Number of moles} = \frac{\text{Mass in grams}}{\text{Gram molecular mass}} = \frac{8\text{g}}{32\text{g}} = 0.25$$

Example 13. How many moles of a gas are there in a) 44.828 litres at S.T.P. b) 112.07 ml at S.T.P.

Solution: a) Volume of gas at S.T.P. = 44.818 litres

Molar volume at STP = 22.414 litres

$$\text{Number of moles} = \frac{\text{Volume at STP}}{\text{Molar volume at STP}} = \frac{44.828\,\text{lit.}}{22.414\,\text{lit.}} = 2$$

b) $$\text{Number of moles} = \frac{\text{Volume at STP}}{\text{Molar volume at STP}} = \frac{112.07\,\text{ml}}{22.414\,\text{ml}} = 0.005$$

Example 14. Calculate the volume at S.T.P. of a) 0.2 mole NO b) 8.5 g NH_3

Solution: a) Volume at STP= Number of moles × Molar volume at STP

$= 0.2 \times 22.414 = 4.4828$ litres

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b) Mass of NH_3 = 8.5 g ; Gram molecular mass of NH_3 = 17 g.

No. of Moles of NH_3 = $\frac{8.5g}{17g}$ = 0.5 Volume at STP

= Number of moles × 22.414 litres

= 0.5 x 22.414 = 11.212 litres

Self Study Problems

5. Calculate the mass of a molecule of CO_2 **Ans:-** 7.31×10^{-23}g
6. Calculate the number of atoms in each of the following (i) 52 moles He (ii) 52 amu of He (iii) 52 g He

   Ans:- (i) 3.13×10^{25} atoms (ii) 13 atoms (iii) 7.83×10^{24} atoms
7. Calculate the mass of (i) 0.01 mole water (ii) 0.02 mole CO_2 (iii) 0.06 mole Cl_2 **Ans:-** (i) 0.18g (ii) 0.88 g (iii) 4.26 g
8. Calculate the number of atoms of carbon present in 1 mol of ethanol. **Ans:-** $2 \times 6.023 \times 10^{23}$
9. Calculate the number of chloride ions in 0.2 g of $BaCl_2\ 2H_2O$. **Ans:-** 9.87×10^{20}
10. What mass of magnesium contains the same number of atoms as in 24g carbon? **Ans:-** 48g
11. Calculate the number of molecules in 56 ml CO_2 at STP. **Ans:-** 1.45×10^{23}
12. 1050 ml of a gas at STP weighs 3g. Calculate the mass of one molecule of the gas. Ans:- **1.063×10^{-24}g**

Chemical Equations

A chemical equation is a representation of a chemical change by using symbols and fomulae. For constructing an equation, the symbols or formulae of reactants are written on the left hand side of an arrow and those of products on the right hand side with plus signs wherever necessary. We can state a reaction as follows. Hydrogen reacts with chlorine to form hydrogen chloride. This statement can be much abbreviated to,

$H_2 + Cl_2 \rightarrow HCl$ .....(i) where the arrow means "yeilds" or "forms".

In every process, the law of conservation of mass must be obeyed. Therefore

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the total number of atoms of each kind must be the same on both sides of the arrow. This can be achived by multiplying the formula HCl by 2.

Thus, $H_2+Cl_2 \rightarrow 2HCl$.....(ii)

This is now a correct equation or a true equation since the two sides have been equated. The expression (i) is called *unbalanced equation or skelton equation. The equation (ii) is called a balanced equation.*

Balancing of Chemical Equations

The process of converting a skeleton equation to a balanced equation is called balancing an equation. For balancing chemical equations, the following guidelines can be used.

(i) If any elementary gas appears, change it to its atomic state.

(ii) The formula containing the maximum number of atoms is selected first and the atoms present in it are balanced.

(iii) If the above step fails, the atoms of that element which occurs at minimum number of places are balanced first.

(iv) Atoms of elementary gases are balanced last of all.

(v) Once all the atoms are balanced, change the equation into the molecular form.

For example, consider the equation, $Fe+H_2O \rightarrow Fe_3O_4+H_2$

Changing the elementary substance hydrogen to atomic form

$Fe+H_2O \rightarrow Fe_3O_4+H$

Fe_3O_4 has largest number of atoms. To balance Fe atoms, multiply Fe by 3 and to balance oxygen atoms multiply H_2O by 4. In four molecules of H_2O, there are 8 atoms of H which are balanced by multiplying H on the R.H.S by 8. Thus,

$3Fe+4H_2O \rightarrow Fe_3O_4+8H$

Coverting to molecular form, $3Fe+4H_2O \rightarrow Fe_3O_4+4H_2$

This is the balanced equation.

Example 15. Balance the equation, $C_2H_5OH+O_2 \rightarrow CO_2+H_2O$

Solution : Converting oxygen to elementary state.

$C_2H_5OH+O \rightarrow CO_2+H_2O$

Balancing C atoms, $C_2H_5OH+O \rightarrow 2CO_2+H_2O$

Balancing H atoms, $C_2H_5OH+O \rightarrow 2CO_2+3H_2O$

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Balancing O atoms, $C_2H_5OH+6O \rightarrow 2CO_2+3H_2O$

Changing to molecular form, $C_2H_5OH+3O_2 \rightarrow 2CO_2+3H_2O$

Self Study Problems

13. Balance, $As_2O_3+SnCl_2+HCl \rightarrow SnCl_4+As+H_2O$

Ans:- $As_2O_3+3SnCl_2+6HCl \rightarrow 3SnCl_4+2As+3H_2O$

14. Balance, $C_4H_{10}+O_2 \rightarrow CO_2+H_2O$

15. Balance, $C_6H_6+O_2 \rightarrow CO_2+H_2O$

Significance of a Chemical Equation

A chemical equation has both qualitative and quantitative meanings. It tells what substances undergo chemical change to form what products. It also represents the relative masses of each substance involved. Consider the following equation, $C+O_2 \rightarrow CO_2$

Qualitative meaning. Carbon reacts with oxygen to form CO_2.

Quantitative meanings (i) 1 atom of carbon combines with 1 molecule of oxygen to form 1 molecule of CO_2. (ii) 1 mole of carbon atoms combine with 1 mole of oxygen molecules to form 1 mole of CO_2 molecules. (iii) 12 amu of carbon combines with 32 amu of oxygen to form 44 amu CO_2 (iv) 12 g carbon (1 gram atom) combines with 32 g oxygen (1 gram mole) to form 44g CO_2 (1 gram mole).

Reactions involving gases have also quantitative meanings in terms of volumes. For example, the equation,

$H_2+Cl_2 \rightarrow 2HCl$ means that

(i) 1 volume H_2 combines with 1 volume Cl_2 to form 2 volumes of HCl.

(ii) 1 gram molecular volume (22.4 litres) of H_2 reacts with 1 gram molecular volume (22.4 litres) of Cl_2 at STP to form 2 gram molecular volumes (2×22.4 litres) of HCl at STP.

Calculations based on Chemical Equations

Quantities of subtances corresponding to the chemical equations are called stoichiometric quantities. From balanced equations, we can calculate the masses of reactants and products involved in a reaction.

Example 16. Calculate the mass of calcium oxide obtained by heating 25g $CaCO_3$

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Solution :

$$\underset{40+12+(3\times16)}{CaCO_3} \rightarrow \underset{40+16}{CaO} + CO_2$$

The equation states that 1 gram mole (100g) of $CaCO_3$ produces 1 gram mole (56g) of CaO.

$\therefore$ 25g $CaCO_3$ produces $\frac{56g \times 25g}{100g}$ of CaO = 14g CaO

Example 17. What mass of $KClO_3$ must be heated to produce 9.6g oxygen according to the equation, $2KClO_3 \rightarrow 2KCl + 3O_2$

Solution:

$$\underset{2\times[39.1+35.5+(3\times16)]\ g}{2KClO_3} \rightarrow 2KCl + \underset{3\times32g}{3O_2}$$

According to the equation, 2 gram moles ($2\times122.6g$) of $KClO_3$ is required to produce 3 gram moles ($3\times32g$) of oxygen.

$\therefore$ Amount of $KClO_3$ required to produce 9.6g O_2

$$= \frac{2\times122.6g\times9.6g}{3\times32g} = 24.52g$$

Example 18. Calculate the mass of O_2 required for the complete burning of 4g Mg.

Solution :

$$\underset{2\times24g}{2Mg} + \underset{32g}{O_2} \rightarrow 2MgO$$

According to the equation, 2gram atom of (48g) Mg requires 1gram mole (32g) of O_2

$\therefore$ 4g of Mg requires $\frac{32g \times 4g}{2 \times 24g}$ $O_2 = 2.66g\ O_2$

Example 19. How many grams of Cl_2 are required to completely react with 0.4g of H_2 to yield HCl? Also calculate the amount of HCl formed.

Solution

$$\underset{2\times1}{H_2} + \underset{2\times35.5}{Cl_2} \rightarrow \underset{2(1+35.5)}{2HCl}$$

According to the equation, 1gram mole (2g) of H_2 requires one gram mole (71g) of Cl_2 to form 2gram moles (73g) HCl.

$\therefore$ 0.4g H_2 requires $\frac{71 \times 0.4}{2}$ g Cl_2 = 14.2g Cl_2

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0.4g H_2 forms $\frac{73 \times 0.4}{2}$ g HCl = 14.6g HCl

Self Study Problems

16. 1.2g of a sample of chalk ($CaCO_3$) on treating with excess of dilute HCl liberate 230 ml CO_2 at STP. Calculate the percentage purity of the sample.
Ans:- 85.58

17. 0.62g of an impure sample of NaCl gave 0.81 g AgCl on treating with $AgNO_3$. Calculate the percentage impurity in the sample of NaCl. **Ans:-** 46.77

18. Calculate the mass of HgO required to produce 22g oxygen by thermal decomposition.
Ans:- 297g

19. Calculate the mass of zinc required to produce enough H_2 to reduce 8.5g CuO to Cu.
Ans:- 7.002g

20. Hydrogen reacts with nitrogen to form ammonia according to the equation: $N_{2(g)}+3H_{2(g)} \rightarrow 2NH_{3(g)}$

Determine how much ammonia would be produced if 200.0 g hydrogen react.
Ans:- 11322g

Limiting Reactant

Very often the reacting substances are not present in exactly the same proportions as required by the balanced equation. Consider the reaction, $2H_2+O_2 \rightarrow 2H_2O$. Suppose the reaction mixture consists of 2 moles of H_2 and 2 moles of O_2. From the equation, it is evident that 2 moles of H_2 requires only one mole of O_2 for complete reaction. Therefore, 1 mole of O_2 will be left over.In such a case, H_2 is said to be the limiting reactant because its concentration limits the amount of the product.

Thus, *the reactant which is completely consumed in the reaction is called the limiting reactant.*

Example 20. 3g H_2 reacts with 29g $O_2$to yield H_2O.

(i) Which is the limiting reactant?

(ii) Calculate the maximum amount of H_2O that can be formed.

(iii) Calculate the amount of one of the reactants which remains unreacted

Solution.

$2H_2$	+	O_2	$\rightarrow$	$2H_2O$
2 moles		1 mole		2 moles

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$$\text{Number of moles of } H_2 \text{ present} = \frac{\text{Mass}}{\text{Molar mass}} = \frac{3}{2} = 1.5 \text{ mol}$$

$$\text{Number of moles of } O_2 \text{ present} = \frac{\text{Mass}}{\text{Molar Mass}} = \frac{29}{32} = 0.906 \text{ mol}$$

(i) Calculation of limiting reactant

According to the equation, 2 mols of H_2 requires 1 mol of O_2

$\therefore$ 1.5 mols of H_2 requires $\frac{1}{2}$ x 1.5 mol of O_2 = 0.75 mol O_2

Number of moles of O_2 actually present = 0.906 mols

O_2 is in excess and H_2 is the limiting reactant.

(ii) Calculation of maximum amount of H_2O formed.

2 mols of H_2 form H_2O = 2 mols

$\therefore$ 1.5 mols of H_2 form H_2O = 1.5 mols

Calculation of amount of one of the reactants (i.e., O_2) which remains unreacted. Number of moles of O_2 added = 0.906 mols.

Number of moles of O_2 used up = 0.75 mols

$\therefore$ Number of moles of O_2 unreacted = 0.906 - 0.75 = 0.156 mol

Self Study Problems

21. If 20g of $CaCO_3$ is treated with 20.0 g of HCl, how many grams of CO_2 can be generated according to the following equation:

$$CaCO_{3(g)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + H_2O_{(l)} + CO_2$$

Ans: 8.8g CO_2 ; $CaCO_3$ is the limiting reactant

Calculations involving Solutions.

Molarity. There are several ways of expressing the concentration of a solution.A commonly used way of expressing the concentration is in terms of molarity. *The molarity of a solution is defined as the number of moles of the solute per litre of the solution.* It is denoted by M.

$$\text{Molarity, M} = \frac{\text{Number of moles of solute}}{\text{Volume of solution in litres}}$$

$$= \frac{\text{Mass in gram of the solute per litre}}{\text{Molar mass of the solute}}$$

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The unit of molarity is mol L^{-1} or mol dm^{-3}

Example 21. A solution is prepared by dissolving 8g of NaOH in distilled water to give one litre of solution. Calculate the molarity of the solution.

Solution. Mass per litre = 8g

Molar mass = 40

$$\text{Molarity M} = \frac{\text{Mass of solute per litre}}{\text{Molar mass of solute}} = \frac{8}{40} = 0.2\,\text{M}$$

Example 22. A solution is prepared by dissolving 4g of NaOH in distilled water to give 200 cm^3 solution. Calculate the molarity of the solution.

Solution. Mass per litre $= \frac{4\text{g} \times 1000\,\text{ml}}{200\,\text{ml}} = 20\,\text{g}$; Molar mass = 40

$$\text{Molarity} = \frac{\text{Mass per litre}}{\text{Molar mass}} = \frac{20}{40} = 0.5\text{M}$$

Example 23. How many moles and how many grams of NaCl are present in 250 cm^3 of 0.5 M NaCl solution.

Solution. Number of moles of NaCl in 1 litre = 0.5

$$\therefore \text{Number of moles of NaCl in 250 cm}^3 = \frac{0.5 \times 250}{1000} = 0.125$$

$\therefore$ Molar mass of NaCl = 58.44

Mass of 0.125 mol NaCl, i.e., mass in 250 cm^3 solution

$$= 0.125 \times 58.44\text{g} = 7.3\text{g}$$

Example 24. How many grams of $BaCl_2$ are needed to prepare 100 cm^3, 0.1 M solution of $BaCl_2$

Solution. Molar mass of $BaCl_2$=208.2g

$\therefore$ Mass of $BaCl_2$ required to prepare 1000cm^3 1 M solution = 208.2g

Mass of $BaCl_2$ required to prepare 100 cm^3 1 M solution

$$= \frac{208.2\text{g}}{10} = 20.82\text{g}$$

Mass of $BaCl_2$ required to prepare 100 cm^3 0.1 M solution

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$$= \frac{20.82g}{10} = 2.082g$$

Example 25. Concentrated sulphuric acid is 98% by mass and has a density of 1.84g cm^{-3}. What volume of concentrated acid is required to make 5 litres of 0.5 M H_2SO_4 solution.

Solution. Number of moles of H_2SO_4 in one litre 0.5 M solution = 0.5

Number of moles of H_2SO_4 in 5 litre 0.5 M solution = $0.5 \times 5 = 2.5$.

Molar mass of H_2SO_4 = 98g

$\therefore$ Mass of H_2SO_4 in 2.5 mole,

i.e., in 5 litre of 0.5M solution = $2.5 \times 98 = 245g$

Concentration of H_2SO_4 = 98% by mass

The mass of H_2SO_4 containing 245g acid by mass = $\frac{100 \times 245}{98} = 250g$

Density of the solution = 1.84 gcm^{-3}

Volume of 250 g concentrated H_2SO_4 solution = $\frac{250\,g}{1.84\,g\,cm^{-3}} = 136\,cm^3$

Thus, volume of concentrated H_2SO_4 required to prepare 5 litres of 0.5 M solution = 136 cm^3

Self Study Problems

22. 500 cm^3 of 0.25 M Na_2SO_4 is added to an aqueous solution of 15g of $BaCl_2$ resulting in the formation of a white precipitate of insoluble $BaSO_4$. How many moles and how many grams of $BaSO_4$ are formed?
Ans:- 16.8g $BaSO_4$

23. Commercially available concentrated HCl contains 38% HCl (i) What is the molarity of the solution? (ii) What volume of con. HCl is required to make 1 L of 0.1 M HCl? **Ans:-** 12.38M (ii) 8.08 cm^3

24. How many grams of Na_2CO_3 are required to prepare 100 cm^3 of 0.25 M solution. **Ans:-** 2.65 g

25. The reaction, $2\,Br^-(aq) + Cl_2(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$ is used for the preparation of bromine. If we have 50 ml of 0.06 M solution of NaBr, what

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volume of 0.05 M solution of Cl_2 is needed to react completely with the Br^-?
Ans:- **3.00×10^{-2} L Cl_2 solution.**

Thermodynamics

Physical and Chemical reactions are normally accompanied by energy changes. Sometimes, the energy changes associated with the chemical reactions are more significant than the products of the reaction, e.g., burning of fuel, coal, etc. Therefore, the study of energy changes in chemical reactions is of great importance in our everyday life. *The branch of chemistry which deals with energy changes associated with chemical reactions is called chemical thermodynamics.*

Some Basic Concepts.

System and surroundings: A selected portion of the universe which is under investigation is called the *system.* The rest of the universe is called the *surroundings.* The system and the surroundings are separated by real or imaginary boundaries. For example, if we are studying the reaction of two substances A and B placed in a beaker, the substances A and B will constitute the system. The walls of the beaker is the boundary and everything else is the surroundings.

Types of systems: Systems are of three types:

(i) **Isolated system :** *A system which can exchange neither energy nor matter with the surroundings is called an isolated system.* Hot tea placed in a thermosflask is an example of an isolated system.

(ii) **Closed system:** *A system which can exchange energy but not matter with the surroundings is called a closed system.* Hot tea placed in a closed steel vessel is an example of a closed system.

(iii) **Open system :** *A system which can exchange energy as well as matter with the surroundings is called an open system.* Hot tea placed in an open cup is an example of an open system. All living systems are open systems because they are continuously exchanging matter and energy with the surroundings.

Macroscopic properties : *The properties which depend on the bulk of the system are called macroscopic properties.* These are two types.

(i) **Intensive properties :** *The properties which are independent of the amount of the substance are called intensive properties.* eg., temperature,

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pressure, density, viscosity, refractive index, etc.

(ii) **Extensive properties :** *The properties which depend on the amount of substance are called extensive properties.* eg., mass, volume, internal energy, enthalpy, entropy, free energy, etc.

State of a system : *It is the condition of existence of the system which is described in terms of macroscopic properties such as pressure, temperature,* etc.

State variables : *The macroscopic properties required to describe the state of the system are called state variables.* e.g., pressure, temperature, volume and composition.

State Functions: *A state function is a property of the system whose value depends only on the state of the system and independent of the path by which the state is reached.* e.g., pressure, volume, internal energy, enthalpy, entropy, free energy, etc.

Path functions : *A path function is a property whose value depends on the path followed to reach the system.* e.g., work, heat, etc.

Process: *The path or operation by which a system changes from one state to another is called the process.*

Types of processes.

(i) **Isothermal process :** *It is a process which occurs at constant temperature.* i.e., $\Delta T=0$. In this, heat exchange occurs. i.e., $dq \neq 0$.

(ii) **Adiabatic process:** *It is a process in which no heat exchange occurs.* i.e., $dq=0$. In this, temperature changes, i.e., $dT \neq 0$

(iii) **Isobaric process :** *It is a process which occurs at constant pressure.* i.e., $dP=0$

(iv) **Isochoric process :** *It is a process which occurs at constant volume.* i.e., $dV=0$.

(v) **Cyclic Process :** *It is a process in which the system returns to the original state.*

(vi) **Reversible process :** *It is a process in which the direction may be reversed by a very small change in any one of the variables.*

(vii) **Irreversible process :** *It is a process which cannot be reversed*

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by small changes in variables.

First law of Thermodynamics

We are familiar with different forms of energies such as kinetic energy, potential energy, thermal energy, light energy, electrical energy, chemical energy, etc. During a physical or chemical change, the energy may be converted from one form into the other. For example, water stored in a tank behind the dam at a high level has a tremendous amount of potential energy by virtue of its height (position). When water is allowed to fall down, the height decreases and consequently its potential energy decreases. However, the falling water acquires larger kinetic energy due to its motion. Therefore, it can rotate the turbines and produce electricity. In electric bulb, electrical energy is converted to light energy. Similarly, an electric heater converts electrical energy into heat energy and an electric fan converts it to mechanical energy. Electrical energy can also be converted into chemical energy during electrolysis. This means that the different forms of energy are interconvertible. But, it may be noted that the total energy remains constant though one form of energy may be converted into the other. Thus, whenever energy in one form disappears, an equal amount of energy in some other form appears. This is known as **conservation of energy** and has been put in the form of a law known as **first law of thermodynamics**. It states that:

energy can neither be created nor destroyed although it can be converted from one form into another.

This law holds good in chemical reactions also. Energy may be absorbed or evolved in a chemical reaction but the total energy of the reacting system and the surroundings remains constant. This law has been stated in several other forms but the basic idea remains the same, i.e., conservation of energy. For example, the law can also be stated as

the total energy of the universe remains constant.

Internal energy or Intrinsic energy

Every substance possesses a definite quantity of energy which depends upon factors such as chemical nature of the substance, temperature and pressure. This is known as **intrinsic energy or internal energy.** It is represented by the symbol E. It is the sum of different forms of energies associated with atoms or molecules. Some of these forms are:

(i) kinetic energy (i.e., translational energy) possessed by the molecules

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by virtue of their motion (E_k).

(ii) vibrational energy associated with the vibrations within the molecules (E_v).

(iii) rotational energy due to rotation of the molecules about their own axis (E_r).

(iv) electronic energy associated with the electronic states of the molecules (E_e).

(v) chemical bond energy due to the presence of forces which bind atoms together within a molecule (E_b).

(vi) potential energy due to electrostatic interactions between different molecules (E_p).

(vii) nuclear energy due to nuclei of atoms (E_n) in the molecule, etc.

Thus, internal energy, E is

$$E = E_k + E_v + E_r + E_e + E_b + E_p + E_n, \text{ etc.}$$

The internal energy of a system depends upon the state of the system and not upon how the system attains the state. Therefore, *internal energy is a state function*. The value of internal energy also depends upon the amount of matter. Therefore, it is an *extensive property*. The actual value of internal energy of a system cannot be determined. But the change in internal energy for a process can be determined.

Internal energy change, ΔE

The absolute value of internal energy possessed by a substance cannot be calculated because it is not possible to find the exact values for the constituent energies. The change in internal energy for a process,

$$A \longrightarrow B$$

may be represented as the difference between the internal energies of the two states, i.e.,

$$\Delta E = E_B - E_A$$

where E_A is the internal energy in state A, E_B is the internal energy in state B and ΔE gives the change in internal energy. It can be shown that

The change in internal energy (ΔE) of a reaction is equal to heat evolved or absorbed during the reaction at constant temperature and constant volume, i.e., $\Delta E = qv$. Thus,

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ΔE = Heat absorbed or evolved in a reaction at constant temperature and volume

The change in internal energy in a chemical reaction is determined with the help of an apparatus known as ***bomb calorimeter.***

Enthalpy

Enthalpy may be defined as *the sum of the internal energy and the product of its pressure and volume.*

It is denoted by the symbol H and is given by

$$H = E + PV$$

where E is the internal energy, P is the pressure and V is the volume of the system.

Every substance has a definite value of enthalpy in a particular state. Thus, like internal energy, enthalpy is also a state function. Its value depends upon the amount of matter present in the system. So it is an extensive property.

Enthalpy Change

We have, $H = E + PV$ ..... (i)

When the system undergoes a change, its enthalpy changes.

i.e., $\Delta H = \Delta E + \Delta(PV)$ ... (ii)

i.e., $\Delta H = \Delta E + P\Delta V + V\Delta P$ ... (iii)

when the process is carried out at constant pressure, $\Delta P = 0$ and $\therefore V\Delta P = 0$. Then,

$$\Delta H = \Delta E + P\Delta V \text{ .... (iv)}$$

But from first law,

$$\Delta E = q + w \text{ ..... (v)}$$

If the work done during the process is only expansion work,

$$w = -P\Delta V \text{ .... (vi)}$$

$$\therefore \Delta E = q - P\Delta V \text{ .... (vii)}$$

Substituting (vii) in (iv), we get

$$\Delta H = q - P\Delta V + P\Delta V = q \text{ ..... (viii)}$$

i.e., $\Delta H = q_p$, heat change at constant pressure. Thus,

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the enthalpy change of a reaction is equal to the heat absorbed or evolved during a reaction at constant temperature and pressure, i.e.,

ΔH = **Heat evolved or absorbed in a reaction at constant temperature and constant pressure.**

Spontaneous Processes : The term spontaneity means the ***feasibility of a process***. *A process which can take place by itself under a given set of conditions is said to be a* ***spontaneous process***. The spontaneous processes are also called ***feasible*** or ***probable processes***. It may be noted that the term spontaneous should not be taken to mean that the process occurs *instantaneously*. It simply denotes that the process has an urge to proceed or it is practically feasible.

It may be noted that some processes may take place under ordinary conditions while there are some processes, which require some initiation before they can proceed. But once initiated, they proceed by themselves. These processes are also regarded as spontaneous processes. A few examples of spontaneous processes are :

(a) Process where no initiation is needed

Water keeps on evaporating from ponds, rivers, sea and open vessels.

$$H_2O\ (l) \longrightarrow H_2O(g)$$

(b) Process where some initiation is required

In domestic oven, coal (carbon) keeps on burning once ignited

$$C(s) + O_2(g) \longrightarrow CO_2(g)$$

From the above reactions, it can be concluded that the spontaneity (or feasibility) means *the inherent tendency of a process to occur of its own in a particular direction under a given set of conditions.* On the other hand, *the processes which are forbidden and are made to take place only by supplying energy continuously from outside the system are called* **non-spontaneous process**. For example, water can be made to flow up a hill (reverse of natural tendency) with the help of a machine. This can continue as long as the external energy is supplied to it. When the supply of energy is cut off, the process stops.

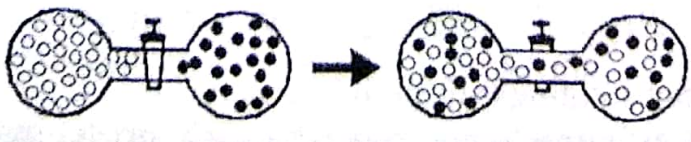

Fig. 2.1 Increase in disorder on mixing of gases

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Direction of spontaneous processes. Most of the spontaneous chemical reactions are found to be exothermic. For example,

$C(s) + O_2(g) \longrightarrow CO_2(g)$; $\Delta H = -393.5$ kJ

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$; $\Delta H = -286$ kJ

From these examples it may be concluded that reactions proceed in the direction of decrease in energy. However, all spontaneous processes need not be exothermic. Some endothermic changes also proceed spontaneously. For example, evaporation of water, melting of ice, dissolution of ammonium chloride in water, etc., are endothermic but occur spontaneously. Again the mixing of two non-reacting gases placed separately in two containers connected by a narrow tube occurs spontaneously.

Due to mixing, the freedom of movement or randomness of the gas molecules increases. But there is no energy change. In all the above examples, the final state is more disordered than the initial state. Thus it may be concluded that *the direction of a spontaneous process is determined not only by the direction in which energy is lowered but also by the direction in which disorder is increased.* Thus, an endothermic reaction can occur spontaneously if the extent of disorder outweighs the increase of energy.

Entropy

The extent of disorder in a system is expressed by a property called entropy. *Entropy is defined as a property which measures the disorder of the system.* It is denoted by the symbol S. It is a *state function*. Further it is an *extensive property*. The change in entropy is denoted as ΔS. Thus,

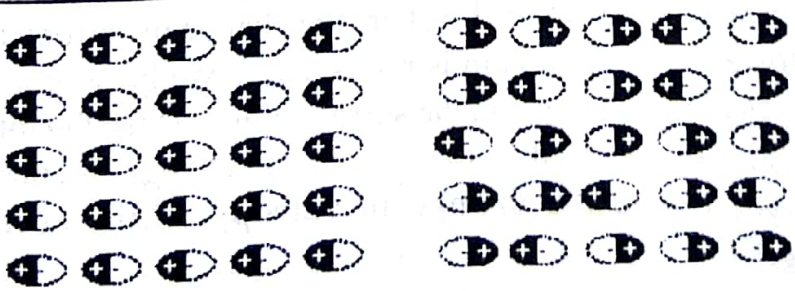

Fig.2.2.

(a) Perfect crystal of CO at 0K (All dipoles oriented in same direction)

(b) Crystal of CO above OK. (Some of the dipoles are disoriented)

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$\Delta S = S_{final} - S_{initial\ state}$. On heating a system its entropy will change. For example, in a perfect crystal of CO, at 0K, all the C-O dipoles are aligned in the same direction [Fig. 2.2 (a)]. Thus there is perfect order or no disorder. Hence the entropy will be minimum. On heating, some dipoles may orient in opposite directions [Fig. 2.2 (b)].

The entropy change in a process is the ratio of the reversible heat change (q_{rev}) to the temperature (T) at which the heat change occurs.

$$\text{i.e., } \Delta S = \frac{q_{rev}}{T}$$

Entropy has the units of energy per temperature. i.e., $JK^{-1}mol^{-1}$.

Physical Significance of Entropy

The physical significance of entropy is that *entropy is a measure of degree of disorder (or randomness) of a system.* The greater the disorder in a system, the higher is the entropy. Obviously, for a given substance the solid state is the state of lowest entropy (i.e., the most ordered state), the gaseous state is the state of highest entropy and the liquid state is intermediate between the two. In the case of mixing of two gases, the gases mix to achieve more randomness or disorder. In this case, there is no exchange of matter or energy between the system and the surroundings. The change occurs from ordered state (less entropy) to disordered state (higher entropy). Thus, the change in entropy is positive.

Second Law of Thermodynamics and Entropy

The second law of thermodynamics gives the relationship between entropy and spontaneity. In an isolated system such as mixing of gases, there is no exchange of energy or matter between the system and surroundings. But due to increase in disorder, there is an increase in entropy. Thus, for a spontaneous process in an isolated system, the change in entropy is positive, i.e., $\Delta S > 0$. However, if the system is not isolated, we have to consider the entropy changes of the system and the surroundings. Thus, the total entropy change (ΔS_{total}) will be equal to the sum of the change in entropy of the system (ΔS_{system}) and the change in entropy of the surroundings ($\Delta S_{surroundings}$).

$$\text{i.e., } \Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings}$$

For a spontaneous process, ΔS total must be positive, i.e.,

$$\Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings} > 0$$

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i.e., ΔS universe > 0

Thus,

> *"the entropy of the universe always increases in the course of every spontaneous (natural) change."*

This is another statement of 2nd law. The essentials of the first and second laws of thermodynamics were thus summed up by Clausius as:

The energy of the universe remains constant - 1st law

The entropy of the universe tends towards a maximum - 2nd law

Entropy and Equilibrium state

During a spontaneous process, the entropy change of the system increases. When the system reaches the equilibrium state, the entropy of the system becomes maximum and therefore, no more increase in the entropy of the system is possible. Thus, the condition for any system to achieve equilibrium stage is that its entropy becomes maximum. The mathematical condition for entropy (S) to be maximum is that the change in entropy (ΔS) is zero, i.e.,

$\Delta S = 0$ (at equilibrium)

Let us apply the entropy criterion to decide the feasibility of the process of conversion of water to ice, at 1 atm pressure.

$$H_2O_{(l)} \longrightarrow H_2O(s)$$

The entropy changes for three different temperatures 272, 273 and 274K are given below.

Temperature (K)	ΔS_{system} $JK^{-1}\,mol^{-1}$	$\Delta S_{surroundings}$ $JK^{-1}mol^{-1}$	$\Delta S_{universe}$ $JK^{-1}mol^{-1}$
272	-21.85	+21.93	+0.08
273	-21.99	+21.99	0
274	-22.13	+22.05	-0.08

(i) At 272 K, $\Delta S_{universe}$ is positive. Therefore, freezing of liquid water to ice is *spontaneous*.

$H_2O(l) \longrightarrow H_2O_{(s)}$; Spontaneous at 272K

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(ii) At 274K, $\Delta S_{universe}$ is negative. Therefore, the freezing of water is not spontaneous at 274K

$H_2O_{(l)} \longrightarrow H_2O_{(s)}$; Non-spontaneous at 274K

(iii) At 273K, $\Delta S_{universe}$ is zero. Therefore, the process is at equilibrium, i.e., neither freezing nor melting is spontaneous. At this temperature, water and ice are in equilibrium and no net change is observed.

$H_2O_{(l)} \rightleftharpoons H_2O_{(s)}$; Equilibrium at 273K. Thus, $\Delta S_{universe}$ is a criterion for spontaneity of a change.

Work Function and Free energy Function

We have seen that the total of the entropy change of the system and the surroundings (i.e., $\Delta S_{universe}$) serves as a criterion of spontaneity of a process. However, in order to decide about the feasibility of a process, we shall have to know the entropy change of the system as well as that of the surroundings. This is not always convenient. We may, therefore, consider entropy change in terms of other state functions which can be determined more conveniently. Two such functions are the *work function* (A) and the free energy function (G).

Work Function, A

The work function A is defined as,

$A = E - TS$ ..... (i) where E is the internal energy, T is the temperature in kelvin, and S is the entropy. Since E, T and S are state functions, A is a *state function.*

Let us consider a system in state 1 having work function A_1 at a constant temperature, T.

$$\text{i.e., } A_1 = E_1 - TS_1 \qquad \text{..... (ii)}$$

When it undergoes a change to state 2 at the same temperature, the work function in state 2 is given as,

$$A_2 = E_2 - TS_2 \qquad \text{..... (iii)}$$

Subtracting (ii) from (iii), we get

$$A_2 - A_1 = (E_2 - E_1) - T(S_2 - S_1) \qquad \text{..... (iv)}$$

$$\text{i.e., } \Delta A = \Delta E - T\Delta S \qquad \text{..... (v)}$$

If the change is carried out reversibly at the constant temperature T and

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that the heat absorbed is equal to q_{rev}, then $\Delta S = \frac{q_{rev}}{T}$..... (vi)

Substituting (vi) in (v), we get

$$\Delta A = \Delta E - T \times \frac{q_{rev}}{T}$$

$$\text{i.e., } \Delta A = \Delta E - q_{rev} \quad \text{..... (vii)}$$

From first law of thermodynamics,

$\Delta E = q + W$, it follows that

$$W_{rev} = \Delta E - q_{rev} \quad \text{..... (viii)}$$

If during the change, work is done by the system, it would carry a negative sign so that

$$-W_{rev} = \Delta E - q_{rev} \quad \text{..... (ix)}$$

From equations (vii) and (ix), we have

$$\Delta A = -W_{rev} \text{ or } -\Delta A = W_{rev} \quad \text{..... (x)}$$

Since the process is carried out reversibly, W represents the maximum work. Thus, the significance of the work function is that its decrease during an isothermal reversible process gives the *maximum work*. This enables us to define the work function. Thus,

"the work function is defined as a thermodynamic property whose decrease during an isothermal reversible process gives the maximum work that can be done by the system during the given change."

The work function A is also referred to as *Helmholtz free energy* or the *Helmholtz function*.

Free energy change and feasibility of a reaction.

The free energy change, ΔG provides the overall criterion for the feasibility of chemical process. It gives the balance of the two tendencies, ΔH and $T\Delta S$ as,

$$\Delta G = \Delta H - T\Delta S$$

This equation is called *Gibbs-Helmholtz equation* which is useful in predicting spontaneity of a process.

Even if the reaction is not favoured energetically (endothermic reaction), the driving force may be provided by a favourable change in entropy. It can be shown that only those processes are spontaneous or feasible in which free

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energy undergoes a decrease i.e., ΔG is negative. By calculating ΔH and $T\Delta S$ and hence ΔG, we can easily predict whether the reaction is spontaneous or not.

For example, when energy and entropy factors are favourable, i.e., ΔH is negative and $T\Delta S$ is positive, then ΔG *must be negative i.e.,* $\Delta G = (-)-(+) = -$ ve *for the spontaneous process.*

On the other hand, if both the tendencies oppose, i.e., ΔH is positive and $T\Delta S$ is negative, then ΔG is positive.

$$\text{i.e., } \Delta G = (+) - (-) = + \text{ ve}$$

Thus, ΔG *is positive for a non-spontaneous process*

If both the tendencies are equal and opposite, for example, if ΔH is positive (unfavourable) and $T\Delta S$ is also positive (favourable) or vice-versa and both are of equal magnitude, then ΔG becomes zero. At this *stage, the process does not proceed in either direction and the reaction, is said to be in* **equilibrium state.**

In short, we can say

(i) **If ΔG is negative,** *the process is* **spontaneous.**

(ii) **If ΔG is zero,** *the process is in* **equilibrium state.** There is no net reaction in either direction.

(iii) **If ΔG is positive,** *the process* **does not occur** *in the forward direction. It may, however, go in the reverse direction.*

Conditions for Spontaneous Processes i.e., ΔG negative

If a change is to be spontaneous in the forward direction, ΔG must be negative. This is possible only under the following conditions:

(i) Both the energy and the entropy factors are favourable and may have any magnitude, i.e., ΔH is negative and $T\Delta S$ is positive.

(ii) When energy factor favours the change (ΔH = -ve) but entropy factor opposes the change ($T\Delta S$ = -ve), then the magnitude of ΔH should be more than $T\Delta S$.

$$\Delta H \text{ (-ve)} > T\Delta S \text{ (-ve)}$$

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(iii) When energy factor is not favouring (ΔH = +ve) but entropy factor favours the change ($T\Delta S$ = +ve), then the magnitude of $T\Delta S$ should be more than the energy factor.

$$\text{i.e., } T\Delta S \text{ (+ve)} > \Delta H \text{ (+ve)}$$

These conditions are summed up in Table 2.1

Table 2.1. Tendencies for spontaneous processes, i.e., ΔG negative

ΔH	$T\Delta S$	Magnitude of the tendencies for ΔG -ve
– (Favourable)	+ (Favourable)	Any magnitude
– (Favourable)	– (Unfavourable)	$\Delta H > T\Delta S$
+ (Unfavourable)	+ (Favourable)	$T\Delta S > \Delta H$

Example 26. For the change $H_2O_{(s)} \longrightarrow H_2O(l)$, $\Delta H_f = 6.97 kJ\ mol^{-1}$ and $\Delta S_f = 25.4\ JK^{-1}\ mol^{-1}$ at 25°C. Calculate the free energy change and predict whether melting of ice is spontaneous or not at this temperature,

Solution: $\Delta G = \Delta H - T\Delta S = 6.97 - 298 \times 25.4 \times 10^{-3} = 6.97 - 7.57 = -0.6 kJ\ mol^{-1}$

Since ΔG is negative, the melting of ice is spontaneous at 25°C.

Example 27. For the reaction, $M_2O_{(s)} \longrightarrow 2M_{(s)} + \frac{1}{2}O_{2(g)}$. $\Delta H = 30\ kJmol^{-1}$ and $\Delta S = 0.07\ kJK^{-1}\ mol^{-1}$. Calculate at what temperature the reaction would be spontaneous?

Solution: In this ΔH and $T\Delta S$ are positive. Therefore, ΔG will be negative only at temperatures higher than the temperature corresponding to $\Delta H = T\Delta S$. The temperature corresponding to $\Delta H = T\Delta S$ is given as,

$$T = \frac{\Delta H}{\Delta S} = \frac{30}{0.07} = 428.6\ K$$

$\therefore$ The reaction would be spontaneous at temperatures above 428.6K

Self Study Problems

26. For the reaction, $2H_2O_{2(g)} \longrightarrow 2H_2O_{(g)} + O_2(g)$, ΔH = -211.29 $kJmol^{-1}$ and ΔS = 131.98 $Jmol^{-1}$. Predict the feasibility of the reaction at 298K.

Ans: ΔG = -250.62 k $Jmol^{-1}$. As ΔG is -ve, the process is spontaneous at 298 K.

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27. For a reaction $\Delta H = 40.63$ kJmol^{-1} and $\Delta S = 108.8$ JK^{-1} mol^{-1} Predict the feasibility of the reaction at 27°C.

Ans: $\Delta G = 7.99$ kJmol^{-1}. As ΔG is +ve, the reaction is not feasible.

Thermochemistry

Chemical reactions are normally accompanied by energy changes. These energy changes appear, ordinarily, in the form of evolution or absorption of heat. A chemical reaction which proceed with the absorption of heat energy is called *endothermic reaction*. For example, when one mole of nitrogen reacts with one mole of oxygen to form two moles of nitric oxide, 180.5 kJ of heat is absorbed at constant temperature and constant pressure.

On the other hand, a chemical reaction which proceed with the evolution of heat energy is called *endothermic reaction*. For example, when one mole of soild carbon combines with one mole of oxygen to form one mole of carbon dioxide, 393.5 kJ of heat is evolved at constant temperature and pressure.

The branch of chemistry which deals with the study of heat changes in chemical reactions is called ***thermochemistry.***

Conventions used in Thermochemistry. Before discussing various aspects of thermochemistry, it is necessary to define various conventions used. Some of the important ones are:

(1) Since physical states of the reactants or products influence the magnitude of the heat evolved or absorbed in the reaction, these are indicated by symbols (*g*), (*l*) and (*s*) placed after the chemical formulae. These stand for gaseous, liquid and solid states respectively.

(2) Differrent allotropic forms of the same element are associated with different energies. Therefore, in case of elements which exhibit allotropy, the name of the allotropic modification must be mentioned. For example, in the case of carbon we must write $C_{(diamond)}$ or $C_{(graphite)}$ as the case may be. Similarly, in the case of sulphur, we should write as $S_{(rhombic)}$ or $S_{(monoclinic)}$ to indicate the allotropic modification.

(3) In the case of solutions we use the symbol *aq* (aqua, water) which signifies larger excess of water to ensure that there is no heat change on adding more water.

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4) Two systems of sign convention are being used for enthalpy changes. In the older system, the sign was used with reference to the surroudings. In the reaction,

$$A + B \longrightarrow C + D + x \text{ kJ}$$

$+x$ kJ indicated x kJ gained by the surroundings and lost by the reacting system. Similarly in the reaction,

$$A+B \longrightarrow C+D - x \text{ kJ}.$$

$-x$ kJ indicated x kJ absorbed by the reacting system from the surroundings.

The opposite of this convention is used in thermodynamics and this is the one used at present in thermochemistry. In this system, the enthalpy change in a reaction is indicated by ΔH which carries + or - sign with reference to the reacting system. Thus '$\Delta H = -x$ kJ' means that x kJ of heat have been lost by the reacting system. Similarly, '$\Delta H = + x$ kJ' means that x kJ of heat have been gained by the reacting system.

Thus, burning of carbon in oxygen to produce carbon dioxide was previously represented as:

$$C_{(\text{graphite})} + O_{2(g)} \longrightarrow CO_{2\,(g)} + 393.5 \text{ kJ}$$

At present it is represented as:

$$C_{(\text{graphite})} + O_{2(g)} \longrightarrow CO_{2(g)}; \Delta H = -393.5 \text{ kJ}$$

Each one of these states that 12 g of graphite carbon reacts with 32 g of gaseous oxygen to produce 44 g of gaseous carbon dioxide when 393.5 kJ of heat are *lost* by the reacting system (and absorbed by the surroundings).

In this chapter, we shall use only the modern convention.

Heat Capacity and Specific Heat Capacity

Heat capacity (C) of a system is defined as the quantity of heat required to raise the temperature of the system through 1^0. If q is the quantity of heat required to raise the temperature of the system from T_1 to T_2, the heat capacity C is given as,

$$C = \frac{q}{T_2 - T_1} = \frac{q}{\Delta T}$$

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Since heat capacity varies with temperature, the value of C is considered over a small range of temperature. If dq is the quantity heat required to raise the temperature by dT, then $C = \frac{dq}{dT}$.

We know that heat is a path function. Likewise heat capacity is also a path function. So, conditions such as constant pressure or constant volume have to be specified to define the heat capacity of a system, particularly if it is a gas. Heat capacity at constant volume is denoted as Cv and heat capacity at constant pressure is denoted as Cp.

For infinitesimally small changes, the expression for first law may be written as,

$$dE = dq + dw = dq - PdV \quad \therefore \quad dq = dE + PdV$$

At constant volume, $dV = 0$ and $\therefore PdV = 0 \quad \therefore dq_V = dE$

$$\therefore CV = \frac{dq_V}{dT} = \frac{dE}{dT} \text{ or } \left(\frac{\partial E}{\partial T}\right)_V$$

Thus, *heat capacity at constant volume is the rate of change of internal energy with temperature.*

But, at constant pressure, $dV \neq 0$

$$\therefore dq_V = dE + PdV \quad \text{.....(i)}$$

we have, $H = E + PV \quad \therefore dH = dE + PdV$ (at constant pressure)

or $dE = dH - PdV$ ..... (ii). Substituting (ii) in (i), we get

$$\therefore dq_p = dH - PdV + PdV = dH \quad \therefore Cp = \frac{dq_p}{dT} = \frac{dH}{dT} \text{ or } \left(\frac{\partial H}{\partial T}\right)_p$$

Thus, ***heat capacity at constant pressure is the rate of change of enthalpy with temperature.***

In chemical calculations, we often use the terms ***specific heat capacity*** and ***molar heat capacity.***

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Specific heat capacity

Specific heat capacity, c, is defined as the quantity of heat required to raise the temperature of one unit mass of a substance by one degree.

Specific heat capacity of water, for example, is 4.18 $J\,g^{-1}K^{-1}$

If q is the quantity of heat required to raise the temperature of a system by ΔT, then specific heat capacity, $c = \frac{q}{m \times \Delta T}$

Example 28 The quantity of heat required to raise the temperature of 25.0g of ethylene glycol by 1K is 60.8J. Calculate the specific heat capacity of ethlene glycol.

Solution: Specific heat capacity, $c = \frac{q}{m \times \Delta T}$.

Here, q=60.8J, m=25g & ΔT=1K

$$\therefore c = \frac{60.8\ J}{25g \times 1K} = 2.43\ Jg^{-1}K^{-1}$$

Molar heat capacity

Molar heat capacity of a substance is the quantity of heat required to raise the temperature of one mole of a substance by one degree. The heat capacity of one mole of a gaseous system at constant volume is represented as Cv and that at constant pressure as Cp.

Relation between Cp and Cv

We have, $Cp = \frac{dH}{dT}$ and $Cv = \frac{dE}{dT}$

$$\therefore Cp - Cv = \left(\frac{dH}{dT}\right) - \left(\frac{dE}{dT}\right) \quad \ldots\ldots\ldots\ldots (i)$$

But, $H = E + PV$ and $PV = RT$

$\therefore H = E + RT$

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$$\therefore \frac{dH}{dT} = \frac{dE}{dT} + R. \quad \frac{dT}{dT} = \frac{dE}{dT} + R \quad \ldots\ldots\ldots\ldots \text{(ii)}$$

Substituting (ii) in (i), we get

$$Cp - Cv = \frac{dE}{dT} + R - \frac{dE}{dT} = R \qquad \text{Thus, } Cp - Cv = R$$

Ratio of Cp and Cv. The ratio of molar heat capacity at constant pressure to that at constant volume is denoted as γ.

$$\text{i.e., } \gamma = \frac{Cp}{Cv}$$

For monatomic gases γ is 1.67, for diatomic gases it is 1.40 and for triatomic gases the value is 1.33.

ENTHALPY CHANGES IN DIFFERENT CHEMICAL REACTIONS

The amount of heat evolved or absorbed during a chemical reaction when the number of moles of the reactants indicated by the balanced chemical equation, have been completely reacted is called enthaply of reaction.

For example, the enthalpy change for the reaction of one mole of methane with two moles of oxygen to form one mole of CO_2 and two moles of water is -890.3 kJ, i.e., 890.3 kJ of heat is evolved during this reaciton. Thus,

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l) \; ; \; \Delta H = -890.3 \text{ kJ}$$

It may be noted that the enthaply of a reaction varies with the temperature. Therefore, it is convenient to fix up a standard or a reference state. According to thermodynamic convention, *a substance is said to be in standard state when it is present in its most stable state at 298 K under a pressure of one atmosphere.* The change in enthalpy that occurs when a process is carried out at 298 K and one atmospheric pressure is called **standard enthalpy change**. It is represented as ΔH^o where the superscript 'o' indicates, the standard state.

The enthalpy changes taking place during the chemical reactions are generally expressed in different ways depending upon the nature of the reaction. The various types of enthalpies of reactions are discussed below:

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Enthalpy of formation

The enthalpy change accompanying the formation of one mole of a compound from its elements is called enthalpy of formation.

It is denoted by ΔH_f. For example, enthalpies of formation of carbon dioxide, methane (CH_4) and water may be expressed as

$C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)}$; $\Delta H = \Delta H_f = -393.5$ kJ

$C_{(s)} + 2H_{2(g)} \longrightarrow CH_{4(g)}$; $\Delta H = \Delta H_f = -74.8$ kJ

$H_{2(g)} + \frac{1}{2}O_{2(g)} \longrightarrow H_2O_{(l)}$; $\Delta H_f = -286.0$ kJ

It may be noted that the enthalpies of formation of compounds may be negative or positive. A compound which has a negative enthalpy of formation is called an *exothermic compound* while a compound which has a positive enthalpy of formation is called an *endothermic compound.*

If all the species of the chemical reaction are in their standard states (i.e., at 298K and one atmospheric pressure) the enthalpy of formation is expressed as ΔH^o_f. Thus,

the enthalpy change accompanying the formation of one mole of a compound from its elements, all the substances being in their standard states (1 atm pressure and 298 K) is called standard enthalpy of formation.

The standard enthalpy of formation of a compound is also called its *standard enthalpy*. By convention, the heat of formation of an element in its standard state is taken as zero. Thus, the standard enthalpy of formation,

ΔH_f^o = H (compound in standard state) -H (elements in standard states)

i.e., ΔH_f^o = H (compound in standard state)

Thus, the standard enthalpy of formation of a compound is the enthalpy of the compound.

The knowledge of standard enthalpies of formation of various substances, can be used to calculate the enthalpies of reactions. In general, the standard enthalpy of any reaction (ΔH^o) is equal to the difference between the ΔH_f^o values of all the products and ΔH_f^o values of all the reactants, i.e.,

ΔH^o for a reaction = standard enthalpies of formation of all products-

standard enthalpies of formation of all reactants.

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$\Delta H^o = \Delta H_f^o$ (products) - ΔH_f^o(reactants)

Example 29. *Calculate the enthalpy change in the reaction:*

$4NH_3(g) + 3O_2(g) \longrightarrow 2N_2(g) + 6H_2O(l)$

at 298 K. Given that the enthalpies of formation at 298 K for $NH_3(g)$ and $H_2O(l)$ are - 46.0 and -286.0 kJ mol^{-1} respectively.

Ans: ΔH^o for the reaction,

$4NH_3(g) + 3O_2(g) \longrightarrow 6H_2O(l) + 2N_2(g)$ is given as,

$\Delta H^o = \Delta H_f^o$ (products) - ΔH^o (reactants)

$= \{6\Delta H_f^o[H_2O(l)] + 2\ \Delta H_f^o[N_2(g)]\} - \{4\Delta H_f^o\ [NH_3(g)] + 3\Delta H_f^o\ [O_2(g)]\}$

$\Delta H_f^o[H_2O(l)] = -286.0$ kJ mol^{-1}

$\Delta H_f^o[NH_3(g)] = -46.0$ kJ mol^{-1}

$\Delta H_f^o[O_2(g)] = 0$ and $\Delta H_f^o[N_2(g)] = 0$ (by convention)

$\therefore\ \Delta H^o = \{6(-286) + 2(0)\} - \{4(-46.0) + 3\ (0)\}$

$= -1716$ kJ + 184 kJ = -1532 kJ

Self Study Problems

28. *Calculate the enthalpy change for the reaction;*

$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$

Given that the standard enthalpies of formation of $CH_4(g)$, $CO_2(g)$ and $H_2O(l)$ are -74.8 kJ mol^{-1}, -393.5 kJ mol^{-1} and -285.8 kJ mol^{-1} respectively.
Ans: -890.3 kJ

Enthalpy of combustion

The enthalpy change accompanying the complete combustion of one mole of a substance in excess of oxygen or air is called ***enthalpy of combustion.***

It is generally denoted by ΔH_c. Some examples of combustion reactions are given below:

(i) Combustion of carbon

$C(s) + O_2(g) \longrightarrow CO_2(g)$; $\Delta H = \Delta H_c = -393.5$ kJ

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(ii) Combustion of methane

$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O_{(l)}$; $\Delta H = \Delta H_c = -890.3$ kJ

(iii) Combustion of benzene (C_6H_6)

$C_6H_6(l) + 15/2\ O_2(g) \rightarrow 6CO_2(g) + 3H_2O(l)$; $\Delta H = \Delta H_c = -3268$ kJ

(iv) Combustion of ethyl alcohol

$C_2H_5OH(l) + 3½O(g) \longrightarrow 2CO_2(g) \longrightarrow + 3H_2O(l)$

$\Delta H = \Delta H_c = -1367.4$ kJ

It may be noted that the enthalpy of combustion (ΔH) is always negative since heat energy is always involved in combustion.

Enthalpy of Neutralisation (Heat of Neutralisation)

Enthalpy of neutralisation is the enthalpy change when one gram equivalent of an acid is neutralised by one gram equivalent of a base in dilute solution. For example,

$HCl(aq) + NaOH\ (aq) \longrightarrow NaCl(aq) + H_2O(l)$; $\Delta H = -57$ kJ

$HNO_3(aq) + KOH_{(aq)} \longrightarrow KNO_3(aq) + H_2O(l)$; $\Delta H = -57$ kJ

It is interesting to note that the enthalpy of neutralisation of all strong acids by strong bases is almost a constant and is equal to -57 kJ

This can be explained by *Arrhenius theory of ionisation*. According to this theory, strong acids and strong bases are completely ionised in dilute solutions. Thus, the above reactions can be written as,

$H^+ + Cl^- + Na^+ + OH^- \longrightarrow Na^+ + Cl^- + H_2O$; $\Delta H = -57$ kJ

i.e., $H^+ + OH^- \longrightarrow H_2O$; $\Delta H = -57$ kJ

$H^+ + NO_3^- + K^+ + OH^- \longrightarrow K^+ + NO_3^- + H_2O$; $\Delta H = -57$ kJ

i.e., $H^+ + OH^- \longrightarrow H_2O$; $\Delta H = -57$ kJ

So neutralisation reactions simply involve the formation of one mole of water from H^+ and OH^- ions. Therefore, the enthalpy change also should be a constant.

However, if the acid or base is weak, the value is different. The difference in value will be equal to the amount of heat change involved in the ionisation of the weak electrolyte and is known as the *heat of ionisation* of the weak electrolyte. For example,

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$CH_3COOH(aq) + NaOH(aq) \longrightarrow CH_3COONa(aq) + H_2O(l)$; $\Delta H = -55.5kJ$

In general, heat of neutralisation of a weak electrolyte = -57 kJ + heat of ionisation of weak electrolyte.

∴ Heat of ionisation of the weak electrolyte, acetic acid = Heat of neutralisation of weak electrolyte, acetic acid + 57 kJ = -55.5 + 57 kJ = 1.5 kJ

Heat of reaction at constant volume or constant pressure.

When a chemical reaction takes place at constant volume, the heat change is called heat of reaction at constant volume. It is denoted as Qv. At constant volume, there is no pressure volume work. The heat change is only due to the change of internal energy, i.e., $Q_v = \Delta E$. However, in most cases chemical reactions are carried out at constant pressure, e.g., at atmospheric pressure. The heat change is then termed as heat of reaction at constant pressure. Here, the heat change is due to change of internal energy as well as pressure volume work, i.e., due to enthalpy change. Thus, $Q_p = \Delta H$. In the case of reactions involving only solids and liquids, the difference between the two quantities is negligible. But, in reactions which involve gases, this is not so.

Relation between heats of reaction at constant volume and at constant pressure.

When gases interact, the reaction may be carried out at constant pressure or at constant volume in a closed vessel. If the reaction takes place at constant pressure, the heat changes can be expressed as $Q_p(\Delta H)$ and similarly heat changes at constant volume as Q_v. We know that heat change at constant pressure is the change in the heat contents (enthalpy) of the system

$$Q_p = \Delta H = \Delta E + P\Delta V \ldots\ldots (1)$$

ΔV being the change in the volume of the system in the chemical process, the suffix p denotes that pressure is being maintained constant.

On the other hand, if the process takes place at constant volume, there is no mechanical work done. Hence

$$Q_v = \Delta E \ldots\ldots(2)$$

Substracting (2) from (1)

$$Q_p - Q_v = P\Delta V \qquad \ldots\ldots(3)$$

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But $PV = n_g RT$ or $V = \frac{n_g RT}{P}$

At constant temperature and pressure

$$\Delta V = \Delta n_g \frac{RT}{P}$$

where Δn_g is the change in the number of moles of gaseous species = No. of moles of gaseous products - No. of moles of gaseous reactants.

Substituting this value of ΔV in Eqn. (3), we get.

$$Q_P = Q_v + \Delta n_g RT \quad .....(4)$$

Another way of writing Eqn. (4) is

$$\Delta H = \Delta E + \Delta n_g RT \quad .....(5)$$

Eqn. (4 or 5) can be used for calculating Q_v from Q_p and vice versa

Example 30. The enthalpy of formation of methane at constant pressure at 300K is -75.83 kJ. What will be the heat of formation at constant volume? ($R = 8.3\ JK^{-1}\ mol^{-1}$)

Solution. Here $\Delta H = Q_p = -75.83$ kJ ; $R = 8.3\ JK^{-1}\ mol^{-1}$; $T = 300$ K

For the formation of methane according to the reaction,

$$C_{(graphite)} + 2H_{2(g)} \longrightarrow CH_4(g)$$

$$\Delta n_g = 1 - 2 = -1$$

Substituting these values in the equation,

$$Q_p = Q_v + \Delta ng\ RT, \text{ we get}$$

$$-75.83 = Q_v + (-1)\ (8.3 \times 300) / 1000 \text{ kJ}$$

$$\therefore\ Q_v = -75.83 + 2.49 = \mathbf{-73.34\ kJ}$$

Self Study Problems

29. One mole of naphthalene was burnt in oxygen gas at constant volume to give CO_2 gas and liquid water at 25°C. The heat evolved was found to be 5138.8 kJ. Calculate the enthalpy of reaction at constant pressure.

Ans: $Q_p = -5143.8$ kJ.

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Variation of heat of reaction with temperature - Kirchhoff's equation.

The heat change of any process varies with temperature. The exact influence of temperature can be worked out as follows. Let us consider the reaction,

$$A \longrightarrow B$$

The energy change at constant volume is given by

$$\Delta E = E_B - E_A \qquad \text{.....(1)}$$

The rate of change of internal energy with temperature is

$$\left(\frac{\delta(\Delta E)}{\delta T}\right)_v = \left(\frac{\delta E_B}{\delta T}\right)_v - \left(\frac{\delta E_A}{\delta T}\right)_v \qquad \text{.....(2)}$$

i.e., $\left(\frac{\delta(\Delta E)}{\delta T}\right)_v = (C_v)_B - (C_v)_A$ ..... (3) [By definition of C_v]

$$\text{or } \left(\frac{\delta(\Delta E)}{\delta T}\right)_v = \Delta C_v \qquad \text{.....(4)}$$

The rate of change of internal energy w.r.t. temperature is thus equal to difference in heat capacities at constant volume of the products and the reactants.

Equation (4) can also be written as,

$$\delta(\Delta E) = \Delta Cv.dT \qquad \text{.....(5)}$$

Integrating within the limits T_1 to T_2,

$$\int_{T_1}^{T_2} \delta(\Delta E) = \int_{T_1}^{T_2} \Delta Cv.dT \qquad \text{.....(6)}$$

i.e., $\Delta E_2 - \Delta E_1 = \Delta Cv\ (T_2 - T_1)$ ..... (7)

Similarly, for the reaction carried out at constant pressure

$$\Delta H = H_B - H_A \qquad \text{..... (8)}$$

$$\text{or } \left(\frac{\delta(\Delta H)}{\delta T}\right)_p = \left(\frac{\delta H_B}{\delta T}\right)_p - \left(\frac{\delta H_A}{\delta T}\right)_p = (C_p)_B - (C_p)_A \qquad \text{..... (9)}$$

$$\text{or } \left(\frac{\delta(\Delta H)}{\delta T}\right)_p = \Delta C_p \qquad \text{..... (10)}$$

Eqns. (4) and (10) are known as the Kirchhoff's equations and are useful

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for calculation of the effect of temperature on ΔH and ΔE. This can be seen by integrating Eqn. (10).

$$\text{or } d(\Delta H) = \Delta C_p dT \qquad \ldots\ldots (11)$$

$$\therefore \int_{H_1}^{H_2} d(\Delta H) = \int_{T_1}^{T_2} \Delta Cp.dT \qquad \ldots\ldots (12)$$

Assuming ΔCp to be independent of temperature, we get

$$\Delta H_2 - \Delta H_1 = \Delta C_p(T_2 - T_1) \qquad \ldots\ldots (13)$$

where ΔH_2 and ΔH_1 are the heats of reaction at constant pressure at temperatures T_2 and T_1 respectively. It is evident from eqn. (13) that the extent of variation of heat of reaction with temperature depends upon the magnitude and the sign of ΔC_p.

Example 3. *The enthalpy of reaction* $N_2 + 3H_2 \longrightarrow 2NH_3$ *at 300 K was found to be -91.97 kJ. What will be the enthalpy of reaction at 323K? The molar heat capacities at constant pressure and 300K for nitrogen, hydrogen and ammonia are 28.46, 28.33 and 37.08 J* K^{-1} mol^{-1} *respectively.*

Solution. According to Kirchhoff's equation

$$d(\Delta H) = \Delta C_p dT$$

which on integration gives

$$\Delta H_2 - \Delta H_1 = \Delta C_p (T_2 - T_1)$$

Here, $\Delta H_1 = -91.97$ kJ at $T_1 = 300$ K

and we want to find ΔH_2 at $T_2 = 323$K

ΔC_p for the reaction : $N_2 + 3H_2 \rightleftharpoons 2NH_3$ is given by

$$\Delta C_p = 2(C_p \text{ for } NH_3) - [C_p \text{ for } N_2 + 3 \times C_p \text{ for } H_2]\ JK^{-1}$$

$$= (2 \times 37.08) - (28.46 + 3 \times 28.33)\ J\ K^{-1}$$

$$= 74.16 - (28.46 + 84.99) = (74.16 - 113.45) = -39.29\ JK^{-1}$$

Substituting the values in Kirchhoff's equation,

$$\Delta H_2 = \Delta H_1 + \Delta C_p (T_2 - T_1), \text{ we get}$$

$$\Delta H_2 \text{ at } 323K = -91.97 - \frac{39.29 \times (323-300)}{1000} \text{ kJ}$$

$$= -91.97 - 0.904 = \mathbf{-92.874\ kJ}$$

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Self Study Problems

30. The enthalpy of reaction (ΔH) for the formation of ammonia according to the reaction. $N_2 + 3H_2 \rightleftharpoons 2NH_3$ at 27°C was found to be -91.94 kJ. What will be the enthalpy of reaction (ΔH) at 50°C? The molar heat capacities at constant pressure and at 27°C for nitrogen, hydrogen and ammonia are 28.45, 28.32 and 37.07 joules respectively. **Ans:** -92.85 kJ

Hess's Law

The energy changes accompanying chemical processes are governed by a general principle known as Hess's law. This law states that.

"the enthalpy change of a chemical reaction is the same, whether the reaction takes place in one step or in several steps."

This means that the amount of heat evolved or absorbed in a chemical reaction is always the same whether the reaction takes place in one step or in several steps. Suppose in a reaction, the reactant A changes to product B in one step and the enthalpy change during this process is ΔH. Now, suppose the same process is carried out in two steps.

(i) the reactant A changes to C (intermediate stage) and the enthalpy change during this step is ΔH_1.

(ii) C changes to the product B and the enthalpy change during this step is ΔH_2.

Thus, according to Hess's law, the enthalpy change for the reaction is the same whether the reaction follows path I or path II, i.e.,

$$\Delta H = \Delta H_1 + \Delta H_2$$

Theoretical Justification of the Hess's Law. The theoretical justification can be obtained from the following considerations:

The substance A can be converted into B by two paths. In path I, A is converted directly into B and let heat evolved in this step be q. On the other hand, when path II is followed, let the heats evolved during the two steps A to C and C to B be q_1 and q_2 respectively. The total heat evolved in path II is $q' = q_1 + q_2$. According to Hess's law q' must be equal to q.

Let us suppose that q' is greater than q. This means that by converting, A to B by path II and then converting B back to A by path I, the heat energy

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equal to q′-q can be created. This directly comes in conflict with the law of conservation of energy (first law of thermodynamics) according to which the energy can neither be created nor destroyed. Hence, q must be equal to q′. In other words, heat changes during the process must be independent of the path. This is Hess's law.

Illustration of Hess's Law

Let us illustrate the law by considering the formation of carbon dioxide from carbon and oxygen. Carbon can be converted into CO_2 by two ways:

Path 1. $C(s) + O_2(g) \longrightarrow CO_2(g)$; $\Delta H = -393.5$ kJ

Path 2. $C(s) + \frac{1}{2}O_2(g) \longrightarrow CO(g)$; $\Delta H_1 = -110.5$ kJ

$CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$; $\Delta H_2 = -283.0$ kJ

$C(s) + O_2(g) \longrightarrow CO_2(g)$; $\Delta H = -393.5$ kJ

$\therefore \Delta H = \Delta H_1 + \Delta H_2$

Thus, the net enthalpy change for the formation of CO_2 is the same in both the methods, hence this proves the Hess's law.

Thus, it may be concluded that according to Hess's law, the thermochemical equations can be added, subtracted or multiplied like algebraic equations to obtain the desired equation.

Applications of Hess's Law

Hess's law has been found to be very useful in determining the enthalpy changes for the chemical reactions which are not suitable for direct measurements with calorimeter. Therefore, it has many applications in the study of chemical energetics. Some of the important applications of Hess's law are:

1. Determination of enthalpy of formation of substances which cannot be measured experimentally. There are a large number of compounds such as methane (CH_4), carbon monoxide (CO), benzene (C_6H_6), etc., which cannot be synthesised directly from their elements. Therefore, their enthalpies of formation cannot be determined experimentally by calorimetric method. The enthalpies of formation of such substances can be determined by an indirect method based on Hess's law.

For example,enthalpy of formation of carbon monoxide can be calculated from the enthalpy of combustion data for carbon and carbon monoxide.

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(i) $C(s) + O_2(g) \longrightarrow CO_2(g)$ $\Delta H = -393.5$ kJ

(ii) $CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$ $\Delta H = -283.0$ kJ

By subtracting eq. (ii) from (i), we get

$$C(s) + \frac{1}{2}O_2(g) \longrightarrow CO(g) \qquad \Delta H = -110.5 \text{ kJ}$$

Thus, the enthalpy of formation of carbon monoxide is -110.5 kJ which has been obtained from the experimentally determined enthalpy of combustion values of C(s) and CO(g).

Example 32. Calculate the heat of formation of ethane from the following data: (i) $C(s) + O_2(g) \longrightarrow CO_2(g)$; $\Delta H = -393.5$ kJ

(ii) $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$: $\Delta H = -285.8$ kJ

(iii) $C_2H_6(g) + 7/2\ O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$; $\Delta H = -1560.0$ kJ

Solution. The required equation for the formation of ethane is

$$2C(s) + 3H_2(g) \longrightarrow C_2H_6(g) \qquad \Delta H = ?$$

Multiply eq. (i) by 2 and eq. (ii) by 3 and add

(i) x 2, $2C(s) + 2O_2(g) \longrightarrow 2CO_2(g)$; $\Delta H = -787$ kJ

(ii) x 3, $3H_2(g) + 3/2O_2(g) \longrightarrow 3H_2O(l)$; $\Delta H = -857.4$ kJ

(iv) $2C(s) + 3H_2(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$; $\Delta H = -1664.4$ kJ

Subtract eq. (iii) from the above eq. (iv)

(iii) $C_2H_6(g) + \frac{7}{2}\ O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$; $\Delta H = -1560.0$ kJ

Subtract, $2C(s) + 3H_2(g) \longrightarrow C_2H_6(g)$; $\Delta H = -84.4$ kJ

Thus, heat of formation of ethane = -84.4 kJ

Example 33. *Calculate the enthalpy of formation of benzene represented by the following reaction;*

$$6C(s) + 3H_2(g) \longrightarrow C_6H_6(l)$$

The standard enthalpy of combustion of benzene is -3266.0 kJ and the standard enthalpies of formation of $CO_2(g)$ and $H_2O(l)$ are -393.1 and -286.0 kJ respectively.

Solution. The desired equation is

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(i) $6C(s) + 3H_2(g) \longrightarrow C_6H_6(l)$; ΔH=?

Other equations representing various reactions can be written as

(ii) $C_6H_6(l) + 7½\ O_2(g) \longrightarrow 6CO_2(g) + 3H_2O(l)$; ΔH = -3266.0 kJ

(iii) $C(s) + O_2(g) \longrightarrow CO_2(g)$; ΔH = -393.1 kJ

(iv) $H_2(g) + ½O_2(g) \longrightarrow H_2O(l)$; ΔH = -286.0 kJ

Multiply eq. (iii) by 6 and eq. (iv) by 3 and add

(iii) x6, $6C(s) + 6O_2(g) \longrightarrow 6CO_2(g)$; ΔH = -2358.6 kJ

(iv) x3, $3H_2(g) + 3/2\ O_2(g) \longrightarrow 3H_2O(l)$; ΔH = -858.0 kJ

(v) $6C(s) + 3H_2(g) + 15/2\ O_2(g) \longrightarrow 6CO_2(g) + 3H_2O(l)$; ΔH = -3216.6kJ

Subtract eq. (ii) from eq. (v)

(ii) $C_6H_6(l) + 15/2\ O_2(g) \longrightarrow 6CO_2(g) + 3H_2O(l)$; ΔH = -3266.0

$6C(s) + 3H_2(g) \longrightarrow C_6H_6(l)$

Enthalpy of formation of benzene = +49.6 kJ

Self Study Problems

31. Calculate the enthalpy of formation of acetic acid (CH_3COOH) if its enthalpy of combustion is 867 kJ mol^{-1}. The enthalpy of formation of $CO_2(g)$ and $H_2O(l)$ are -393.5 kJ mol^{-1} and -285.9 kJ mol^{-1} respectively.

Ans: -491.8 kJ

32. Calculate the standard enthalpy of formation of propane (C_3H_8) if its enthalpy of combustion is -2220.2kJ mol^{-1}. The enthalpy of formation of $CO_2(g)$ and $H_2O(l)$ are -393.5 kJ mol^{-1} and -285.8 kJmol^{-1} respectively.

Ans: 103.5 kJ

33. Calculate the heat of formation of $H_2O(g)$ from the following data:

(i) $H_2(g) + ½O_2(g) \longrightarrow H_2O(l)$; ΔH = -284.5 kJ

(ii) $H_2O(l) \longrightarrow H_2O(g)$; ΔH = +41.8 kJ **Ans :** -242.7 kJ

34. Calculate the enthalpy of formation of benzene from the following data.

1. $C_6H_6(l) + 7½\ O_2(g) \longrightarrow 6CO_2(g) + 3H_2O(l)$; ΔH = -3264.6 kJ
2. $C(\text{graphite}) + O_2(g) \longrightarrow CO_2(g)$; ΔH = -393.5 kJ

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3. $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$; $\Delta H = -285.9$ kJ

Ans: Enthalpy of formation of benzene is + 45.9 kJ

3. Predicting enthalpies of various reactions. The practical utility of Hess's law lies in the fact that thermochemical equations may be added, subtracted, multiplied or divided by numerical factors like algebraic equations. As a result, it is possible to calculate enthalpies of many reactions whose direct measurement would be difficult or even quite impossible.

Example 34. Calculate the enthalpy of formation of anhydrous aluminium chloride, Al_2Cl_6 from the following data:

(i) $2Al(s) + 6HCl\ (aq) \longrightarrow Al_2Cl_6(aq) + 3H_2(g)$; ΔH=-1004.0 kJ

(ii) $H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$; ΔH=-183.9 kJ

(iii) $HCl(g) + aq \longrightarrow HCl(aq)$; $\Delta H = -73.2$ kJ

(iv) $Al_2Cl_6(s) + aq \longrightarrow Al_2Cl_6(aq)$; $\Delta H = -643$ kJ

Solution. The formation of Al_2Cl_6 from its elements may be represented as :

$2Al(s) + 3Cl_2(g) \longrightarrow Al_2Cl_6(s)$; ΔH=?

Multiply eq. (ii) by 3 and eq (iii) by 6 and add

(ii) x 3, $3H_2(g) + 3Cl_2(g) \longrightarrow 6HCl(g)$; $\Delta H = -551.7$ kJ

(iii) x 6, $6HCl(g) + aq \longrightarrow 6HCl(aq)$; $\Delta H = -439.2$ kJ

(v) $3H_2(g) + 3Cl_2(g) + aq \longrightarrow 6HCl(aq)$; $\Delta H = -990.9$ kJ

Add eq. (i) to eq. (v)

(i) $2Al(s) + 6HCl(aq) \longrightarrow Al_2Cl_6(aq) + 3H_2(g)$; $\Delta H = -1004.0$ kJ

Add, (vi) $2Al(s) + 3Cl_2(g) + aq \longrightarrow Al_2Cl_6(aq)$ $\Delta H = -1994.9$ kJ

Subtract eq. (iv) from eq. (vi)

(iv) $Al_2Cl_6(s) + aq \longrightarrow Al_2Cl_6(aq)$ $\Delta H = -643$ kJ

$2Al(s) + 3Cl_2(g) \longrightarrow Al_2Cl_6(s)$ $\Delta H = -1351.9$ kJ

Thus, enthalpy of formation of anhydrous $Al_2Cl_6 = -1351.9$ kJ

Example 35. Calculate the heat of combustion of glucose from the following data:

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(1) $C_{(graphite)} + O_2(g) \longrightarrow CO_2(g)$; $\Delta H = -395.0kJ$

(2) $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$; $\Delta H = -269.4$ kJ

(3) $6C_{(graphite)} + 6H_2(g) + 3O_2(g) \rightarrow C_6H_{12}O_6(s)$; $\Delta H = -1169.8$ kJ

Solution. The required equation is,

$C_6H_{12}O_6 + 6O_2(g) \longrightarrow 6CO_2(g) + 6H_2O(l)$

Multiplying equation (1) by 6 and equation (2) by 6.

(4) $6C_{(graphite)} + 6O_2(g) \longrightarrow 6CO_2(g)$; $\Delta H = -2370.0$ kJ

(5) $6H_2(g) + 3O_2(g) \longrightarrow 6H_2O(l)$; $\Delta H = -1616.4$ kJ

Reversing equation (3),

(6) $C_6H_{12}O_6(s) \longrightarrow 6C_{(graphite)} + 6H_2(g) + 3O_2(g)$; $\Delta H = +1169.8$kJ

Adding equations (4), (5) and (6)

$C_6H_{12}O_6(s) + 6O_2(g) \longrightarrow 6CO_2(g) + 6H_2O(l)$; $\Delta H = -2816.6$ kJ

Self Study Problems

35. The heats of solution of anhydrous $CuSO_4$ and hydrated $CuSO_4$, $5H_2O$ are -66.5 and 11.7 kJ mol respectively. Calculate the heat of hydration of $CuSO_4$ to $CuSO_4$ $5H_2O$ **Ans:** -78.2 kJ

Bond dissociation energies and bond energies

We know that when a bond is formed between the atoms energy is released. Obviously, the same amount of energy will be needed to break the bond. This is called the bond dissociation energy and is measure of the bond strength.

The amount of energy required to break one mole of a bond between the atoms in the gaseous state is called **bond dissociation energy.**

It is generally expressed in terms of kJ/mole. For example, 433 kJ of energy is required to break or dissociate the H-H bonds in one mole of H_2 molecules. Therefore, the bond dissociation energy of H-H bond is 433 kJ mol^{-1}. Similarly, the bond dissociation energies of Cl-Cl bond in Cl_2 is 242.5 kJ mol^{-1}.

The bond dissociation energy depends upon the nature of the bond and also the molecule in which the bond is present.

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In the case of a diatomic molecule, the bond dissociation energy is the same as bond energy. Thus, the bond energy of H-H bond is 433 kJ mol^{-1}. Similarly, the bond energy of Cl-Cl bond is 242.5 kJ mol^{-1}. In the case of a polyatomic molecule, the bond dissociation energy is not the same for successive bonds although they may be of the same type. In such a case, the bond energy is not equal to the bond dissociation energy. For example, in water, H_2O, the bond dissociation energies of two O-H bonds differ from one another as described below.

$$HO\text{-}H(g) \longrightarrow H(g) + OH(g) \; ; \; \Delta H = 498 \text{ kJ}$$

$$O\text{-}H(g) \longrightarrow H(g) + O(g) \; ; \; \Delta H = 430 \text{ kJ}$$

Here, the bond energy is taken as the average of the bond dissociation energies of the various similar bonds. For example, in the above case, the bond energy of O-H bond is expressed as,

$$\Delta H_{O-H} = \frac{498 + 430}{2} = 464 \text{ kJ mol}^{-1}$$

Thus,

bond energy or bond enthalpy for any particular type of bond in a compound may be defined as the average amount of energy required to break one mole (i.e., Avogadro's number) of bonds of that type present in the compound.

Bond energy is also called the **enthalpy of formation of the bond.**

Calculation of bond energy of heteronuclear diatomic molecules. Hess's law may be used for calculating the bond energy (bond dissociation energy) for diatomic molecules containing two different atoms as shown in the following example.

Example 36. Calculate the bond energy for H-Cl bond. Given that the enthalpies of formation of HCl(g), H(g) and Cl(g) are -92.2 kJ mol^{-1}, 217.7 kJ mol^{-1} and 121.4 kJ mol^{-1}.

Solution. Required equation is,

$$HCl(g) \longrightarrow H(g) + Cl(g) \; ; \; \Delta H^o = ?$$

But $\Delta H^o = \Delta H_f^o$ of products - ΔH^o_f of reactants

= 217.7 kJ + 121.4 kJ - (-92.2 kJ) = 431.3 kJ mol^{-1}. Thus, $\Delta H^o_{H\text{-}Cl}$ = 431.3 kJ mol^{-1}

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That is, bond energy of H-Cl bond is 431.3 kJ mol^{-1}.

Calculation of bond energy of bonds in polyatomic molecules

In case of polyatomic molecules (containing more than one bond of the same type) bond energy is obtained by taking the average of the bond dissociation energies.

Example 37. Calculate the bond energy of C-H bond in CH_4 from the following data.

$CH_3\text{-}H(g) \longrightarrow CH_3(g) + H(g)$; $\Delta H = 425$ kJ mol^{-1}

$CH_2\text{-}H(g) \longrightarrow CH_2(g) + H(g)$; $\Delta H = 470$ kJ mol^{-1}

$CH_2(g) \longrightarrow CH(g) + H(g)$; $\Delta H = 416$ kJ mol^{-1}

$CH(g) \longrightarrow C(g) + H(g)$; $\Delta H = 335$ kJ mol^{-1}

Solution : Bond energy of C-H bond $= \frac{425 + 470 + 416 + 335}{4} = 411.5$ kJ

Example 38. *Calculate the bond energy of HBr bond. Given that enthalpy of formation of HBr is -36.2 kJ mol^{-1}. The bond energies of H-H and Br-Br bond are 431 kJ mol^{-1} and 188 kJ mol^{-1} respectively.*

Solution. (1) $H + Br \longrightarrow HBr$; $\Delta H_f = -36.2$ kJ mol^{-1}

(2) $H_2(g) \longrightarrow 2H$; $\Delta H = 431$ kJ mol^{-1}

(3) $Br_2(g) \longrightarrow 2Br$; $\Delta H = 188$ kJ mol^{-1}

In order to calculate the bond energy of HBr bond, let us consider the reaction.

$$HBr(g) \longrightarrow H(g) + Br(g)$$

$$\Delta H^o = \Sigma \Delta H^o_f(\text{products}) - \Sigma \Delta H^o_f(\text{reactants})$$

$$= \left(\frac{431}{2} + \frac{188}{2}\right) - (-36.2) = 345.7$$

Thus, $\Delta H^o_{H-Br} = 345.7$ kJ mol^{-1}

That is, the bond energy of H-Br bond is 345.7 kJmol^{-1}

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Use of bond energy to calculate enthalpy change of reactions: A chemical reaction involves the breaking of bonds in the reactants and formation of new bonds in products. The enthalpy change of the reaction may be given as;

Enthalpy of reaction = sum of bond energies of reactants - sum of bond energies of products.

Example. 39. Calculate ΔH for the reaction, $H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$. Given that bond energies of H-H, Cl-Cl and H-Cl bonds are 435 kJ, 243 kJ and 431 kJ respectively.

Solution: ΔH = Bond energy of reactants - Bond energy of products

= B.E. (H-H) + B.E. (Cl-Cl) - 2BE (H-Cl)

= 435 kJ + 243 kJ - (2 x 431 kJ) = -184 kJ

Example 40. Calculate ΔH for the reaction, $CH_2Cl_2(g) \longrightarrow C(g) + 2H(g) + 2Cl(g)$. Given that the bond energies of C-H bond and C-Cl bond are 413 kJ mol^{-1} and 326 $kJmol^{-1}$ respectively.

Solution: This involves the breaking of two C-H bonds and two C-Cl bonds and formation of no bond.

$\therefore$ ΔH = B.E.(Reactants) - B.E. (Products)

= 2B.E. (C-H) + 2 B.E. (C-Cl) - 0 = 2 x 413 + 2 x 326 - 0 = 1482 kJ

Self Study Problems

36. Calculate ΔH for the reaction, $CH_2=CH_2(g) + 3O_2(g) \longrightarrow 2CO_2(g) + 2H_2O(g)$. Given that bond energies of C-H, O=O, C=O, O-H and C=C bonds are 414 kJ, 499 kJ, 724 kJ, 460 kJ and 619 kJ respectively.

Ans: $\Delta H^o = 964\ kJmol^{-1}$

37. Calculate ΔH for $C_2H_4(g) + H_2(g) \longrightarrow C_2H_6(g)$. Given that bond energies of C=C, C-H, H-H and C-C bonds are 619 kJ, 413 kJ, 435 kJ and 347 kJ respectively. **Ans:** -119kJ

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IMPORTANT POINTS FOR QUICK REVISION

Law of Conservation of Mass : In a chemical reaction, total mass of the reactants is equal to the total mass of the products.

Law of Definite Proportions : A chemical compound always contains the same elements combined in the same ratio by mass.

Law of Multiple proportions : If two elements combine to form more than one compound, the different masses of one of the elements that combine with a fixed mass of the other bear a simple ratio.

Law of Reciprocal Proportions : The ratio of the masses of two elements which separately combine with a given mass of a third element is the ratio in which the two elements react with each other or a simple multiple of this ratio.

Law of Gaseous Volumes : When gases combine to form gaseous products a simple ratio exists between the volumes of the reactants and the products at constant temperature and pressure.

Atomic Mass : Atomic mass of an element is defined as a number which expresses how many times the mass of one atom of the element is greater than 1/12 the mass of a carbon-12 atom.

Gram atomic Mass of Gram atom : Atomic mass expressed in grams.

Molecular Mass : Molecular mass of an element or a compound is a number which expresses how many times the mass of one molecule of that element or compound is greater than 1/12 the mass of a carbon-12 atom.

Gram Molecular Mass or Gram Mole : Molecular mass expressed in grams is called gram molecular mass or gram mole.

Mole : Amount of any substance containing Avogadro's number of particles, i.e., atoms, ions, molecules, etc.

Molar Mass : Mass of one mole.

Avogadro's Number, N : Number of atoms present in exact 12 grams of carbon-12. $N = 6.023 \times 10^{23}$.

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Molar Volume : Volume occupied by one mole of a gas. Its value at NTP = 22.414 litres.

Limiting Reactant : The reactant which is completely consumed in a reaction is called the limiting reactant.

Molarity : Number of moles of solute per litre of the solution

Molarity = Mass per litre / Molar Mass

First Law : Energy can neither be created nor destroyed. $\Delta E = q + W$

Enthalpy : $H = E + PV$.

Relation between enthalpy change and internal energy change:

$\Delta H = \Delta E + \Delta nRT$.

Hess's Law : Enthalpy change in a chemical reaction in the same whether the reaction takes place in one step or in several steps.

Spontaneous process : Process which occur without the help of an external agency. It is irreversible.

Second law : Entropy of the universe always increase in the course of every spontaneous change.

Enetropy change : $\Delta S = \Delta q_{rev} / T$

Entropy of fusion : $\Delta S_f = \Delta H_f / T_f$

Entropy vapourisation : $\Delta S_V = \Delta H_V / T_f$

Entropy sublimation : $\Delta S_{sub} = \Delta H_{sub} / T$

Free energy : Maximum energy available from a system that may be converted into work during a process.

Gibbs-Helmholtz equation : $\Delta G = \Delta H - T\Delta S$

Prediction of spontaneity : $\Delta G = -ve$ (Spontaneous); $\Delta G = 0$, process will be at equillibrium, $\Delta G = +ve$ (Nonspontaneous).

Third Law : At absolute zero, entropy of a perfectly crystalline substance is zero.

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If ΔH is –ve and ΔS is +ve, ΔG will always be –ve and the process will be spontaneous.

If both ΔH and ΔS are –ve, ΔG can be –ve only when ΔH>TΔS. Hence such reactions will be spontaneous at sufficiently low temperatures.

If both ΔH and ΔS are +ve, ΔG will be –ve only when ΔH<TΔS and such reactions will be spontaneous at sufficiently high temperatures.

If ΔH is +ve and ΔS is –ve, ΔG will always be +ve and the process will be nonspontaneous

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Model Questions

Section A [Each question carries 3 marks]

1. One mole of an ideal gas at 25°C is allowed to expand isothermally and reversibly form a volume of 10 litres to 20 litres. Calculate the work done by the gas
2. Define the terms (a) Cp and (b) Cv of an ideal gas. How are they related?
3. State (i) first law of thermodynamics and (ii) second law of thermodynamics.
4. Why does Cp exceed Cv for an ideal gas. Give a molecular explanation.
5. Based on Gibbs-Helmholtz equation give the conditions for feasibility of a process.
6. Predict the state of a reaction when (i) $\Delta G < O$ and (ii) $\Delta G = O$
7. Define (a) entropy (b) free energy
8. For spontaneous changes, at low temperature. indicate the signs for ΔH and ΔS
9. What is entropy? The standard enthalpy of melting of ice at 273 K is 6.00 kJ mol^{-1}. Calculate the standard entropy of melting of ice
10. Write the relation between ΔG, ΔH and ΔS. What are the conditions for spontaneity of a process?
11. Based on Gibbs- Helmholtz equation give the conditions for feasibility of a process
12. State the first law of thermodynamics. Give its mathematical formulation.

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13. Define (i) work function (ii) Gibbs free energy function
14. State the first law of thermodynamics. What are its limitations?
15. What do you understand by the terms (i) internal energy and (ii) enthalpy of a system
16. Define enthalpy of formation. State Hess's law
17. The standard heat of formation of SO_3 and SO_2 are -395.6 and -296.6 kJ respectively. Calculate ΔH for the reaction, $SO_2(g) + \frac{1}{2}O_2(g) \longrightarrow SO_3(g)$
18. State and illustrate Hess's law
19. Heat of neutralisation of acetic acid with NaOH is less than that of the neutralisation of HCl with NaOH. Justify.
20. Define (i) Heat of combustion (ii) Heat of hydration.
21. Define (i) bond dissociation energy and (ii) bond energy?
22. Calculate the enthalpy of hydrogenation, $C_2H_4(g) + H_2(g) \longrightarrow C_2H_6(g)$
    Given that bond energy of H-H = 433 kJ, C=C = 615 kJ and C-C = 347 kJ and C-H = 413 kJ.
23. State and explain Hess's law
24. Define ΔH and ΔE. How are they related to each other?
25. Write an expression showing variation of heat of reaction with temperature.
26. Define Hess's law of constant heat summation. Derive it from the first law of thermodynamics.
27. The heat of reaction $\frac{1}{2}H_2 + \frac{1}{2}Cl_2 \longrightarrow HCl$ at 27°C is -22.1 k.cal. Calculate the heat of reaction at 77°C. The molar heat capacity at constant pressure at 27°C for H, Cl and HCl are 6.82, 7.70 and 6.80 cal. mol^{-1} respectively
28. Calculate ΔH for the following reaction at 298K.
    $SO_2(g) + \frac{1}{2} O_2(g) \longrightarrow SO_3(g)$; ΔE = -97.027 kJ.
29. Calculate the standard enthalpy change for the reaction, $CH_4(g) + 2Cl_2(g) \longrightarrow CCl_4(l) + 4HCl(g)$. Given that ΔH°_f of $CH_4(g)$, $CCl_4(l)$ and $HCl(g)$ are -74.9 kJ -139 kJ and -92.3 kJ respectively
30. Define (a) Enthalpy of formation and (b) Enthalpy of solution.

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Section B (Each question carries 5 marks)

1. What do you understand by heat capacity of a system? Show from thermodynamic considerations that Cp- Cv = R.
2. Derive Gibbs- Helmholtz equation
3. Derive an expression for work done in the reversible isothermal expansion of an ideal gas.
4. Derive Clausius- Clapeyron equation.
5. What is spontaneity? How will you predict spontaneity in terms of entropy and free energy.
6. Define enthalpy of formation. Calculate the enthalpy of formation of diborane from the following data.

   $B_2H_6 + 3O_2 \longrightarrow B_2O_3 + 3H_2O$; $\Delta H = -2020$ kJ

   $2B + 3/2\ O_2 \longrightarrow B_2O_3$; $\Delta H = -1264$ kJ

   $H_2 + \frac{1}{2}O_2 \longrightarrow H_2O$; $\Delta H = -285.8$ kJ
7. Derive the relation between heats of reaction at constant volume and at constant pressure.
8. Derive Kirchhoff's equation showing the variation of heat of reaction with temperature.
9. Define bond energy and bond dissociation energy. Calculate the enthalpy of formation of water, given that bond energies of H-H, O=O and O-H bonds are 433 kJ mol^{-1}, 492 kJ mol^{-1} and 464 kJ mol^{-1} respectively.
10. Calculate the heat of formation of ethane from the following data.

    $C_2H_6(g) + 3\frac{1}{2}O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$; $\Delta H = -1560$ kJ

    $C + O_2 \longrightarrow CO_2$; $\Delta H = -393.3$ kJ

    $H_2 + \frac{1}{2}O_2 \longrightarrow H_2O$; $\Delta H = -284.5$ kJ

Section C (Each Question carries 8 marks)

1. a) Why does heat of capacity of a gas at constant volume differ from that at constant pressure? Derive thermodynamically the relation between Cp and Cv.

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b) In a certain process 675 J of heat is absorbed by a system while 290 J of work is done on the system. What is the change in internal energy for the process?

2. a) What is meant by reversible process?

Derive an expression for work done in the reversible isothermal expansion of an ideal gas.

b) Calculate the work done in expanding one mole of an ideal gas from a volume of 2 dm^3 to 20 dm^3 at 27°C.

3. a) State Kirchhoff's equation. Indicate how it can be used to evaluate ΔH of a reaction from heat capacity data of reactants and products.

b) Calculate the heat of formation of CO_2. Given that $CO(g) + H_2O(l) \rightleftharpoons CO_2(g) + H_2(g)$; $\Delta H = 0.7$ k. cal. Heat of formation of $H_2O(l)$ and CO(g) are -68.3 and -26.4 k.cal mol^{-1} respectively.

4. a) Define ΔH and ΔE. How are they related? Under what condition ΔH becomes equal to ΔE?

b) State and explain Hess's law.

5. A system at 400 K exchanges 2J of heat to the surroundings at 300 K.

(a) Calculate the entropy change of the system and the surroundings.

(b) State whether the process is spontaneous or not. Justify.

(c) Mention the law behind this.

Ans. (a) $\Delta S_{system} = \dfrac{-2J}{400\ K}$

$$\Delta S_{surroundings} = \frac{+2J}{300\ K}$$

(b) Spontaneous

As $\Delta S_{surroundings} > \Delta S_{system}$,

$$\Delta S_{system} + \Delta S_{surroundings} = \frac{-2J}{400\ K} + \frac{+2J}{300\ K} > 0, \text{ i.e., +ve}$$

That is, the net entropy increases. So the process is spontaneous.

(c) Second law of thermodynamics.

The law states that all spontaneous processes tend to increase the entropy of the universe.

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6. (a) For the change, $H_2O_{(s)} \rightleftharpoons H_2O_{(l)}$, $\Delta H = 6.97$ kJ mol^{-1}.

If ΔS is positive, predict the condition of temperature under which the change is spontaneous.(1½)

(b) For the reaction, $MgO_{(s)} + C_{(s)} \rightarrow Mg_{(s)} + CO_{(g)}$
$\Delta H^0 = +491.18$ kJ mol^{-1}, $\Delta S^0 = 197.67$ JK^{-1} mol^{-1}. Find out whether it is possible to reduce MgO using carbon at 298 K. If not, at what temperature it becomes spontaneous.

Ans. (a) For a process to be spontaneous, ΔG should be negative. By Gibbs-Helmholtz equation, $\Delta G = \Delta H - T\Delta S$. Here both ΔH and ΔS are positive.
i.e., ΔG = +ve – (+ve) = +ve – ve.
Thus, ΔG will be negative only at high temperature conditions.

(b) $\Delta G = \Delta H - T\Delta S = 491.18$ kJ mol^{-1} – (298 K × 0.198 kJ K^{-1} mol^{-1})
= 491.18 kJ mol^{-1} – 59 kJ mol^{-1} = 432.18 kJ mol^{-1}
As ΔG is positive, it is not possible to reduce MgO at 298 K.
At $\Delta G = 0$, $\Delta H = T\Delta S$

$$\therefore T = \frac{\Delta H}{\Delta S} = \frac{491.18 \text{ kJ mol}^{-1}}{0.198 \text{ kJ mol}^{+1}\text{K}^{-1}} = \mathbf{2483.8\ K}$$

Thus, ΔG will become negative at temperature above 2483.8 K. That is, the reduction will be spontaneous above 2483.8 K.

7. (a) "Entropy of the universe remains constant whereas the total energy increase". Do you agree with this statement. If not, give the correct statement.
(1)

(b) For the reaction, PbO + C → Pb + CO,
$\Delta H = 108.4$ kJ mol^{-1} and $\Delta S = 190$ JK^{-1} mol^{-1}. State whether the reaction will be spontaneous at 300 K. Justify. At what temperature the value of ΔG will be zero?
(2+1)

Ans. (a) Do not agree. The correct statements is,
'Total energy of the universe remains constant whereas the entropy increases.'

(b) The process will be spontaneous only if ΔG is negative.
At 300 K, $\Delta G = \Delta H - T\Delta S$ = 108.4 kJ mol^{-1} – 3000 K × 0.19kJ K^{-1} mol^{-1}
= + 51 kJ mol^{-1}
As ΔG is +ve, the reaction will not be spontaneous at 300 K.

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$\Delta G = \Delta H - T\Delta S$,
when $\Delta G = 0$, we get $0 = \Delta H - T\Delta S$ or $\Delta H = T\Delta S$

$$\therefore T = \frac{\Delta H}{\Delta S} = \frac{108.4 \text{ kJ mol}^{-1}}{0.190 \text{ kJmol}^{-1}} = 570.5 \text{ K}$$

That is, ΔG will be zero at 570.5 K.

8. At the melting point of ice, ΔG decreases. Do you agree with this. Justify your answer

Ans: Not agrees.
At the melting point, $\Delta G = 0$ because at equilibrium $H_2O_{(s)}$ is in equilibrium with $H_2O_{(l)}$.

9. The entropy of a perfectly crystalline substance is zero at absolute zero temperature. Is it correct or not? Why?

Ans: Correct. At absolute zero temperature, there is no disorder or randomness in a perfectly crystalline susbstance. That is, the entropy (measure of disorder or randomness) is zero.

10. For a reaction, both ΔH and $T\Delta S$ are positive. Under what conditions will it be spontaneous?

Ans: For a reaction to be spontaneous, ΔG should be negative. But, $\Delta G = \Delta H - T\Delta S$. So, ΔG can be negative only if the magnitude of $T\Delta S$ is greater than that of ΔH. This may occur at high temperature.

11. Predict the spontaneity of the process when:

a. Both ΔH and ΔS are +ve b. Both ΔH and ΔS are –ve.

c. ΔH –ve and ΔS +ve. d. ΔH +ve and ΔS –ve

Ans: The process will be spontaneous only when ΔG is negative, ie., $\Delta G = \Delta H - T\Delta S = -ve$.

a. When both ΔH and ΔS are positive, $\Delta G = \Delta H - T\Delta S$ will be negative and the process will be spontaneous only if $T\Delta S > \Delta H$. This is possible at high temperature.
b. When both ΔH and ΔS are –ve, $\Delta G = \Delta H - T\Delta S$ will be negative and the process will be spontaneous only if $\Delta H > T\Delta S$. This is possible at low temperature
c. When ΔH is –ve and ΔS is +ve, $\Delta G = \Delta H - T\Delta S$ will be negative and the process will be spontaneous under all conditions, i.e, at any temperature.
d. When ΔH +ve and ΔS is –ve , $\Delta G = \Delta H - T\Delta S$ will be positive and the process will not be feasible under any condition.

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B. KINETICS

Chemical kinetics is the branch of chemistry which deals with the study of rates of reactions, the factors which influence them and the mechanism by which the chemical reactions proceed.

Rate of a Chemical Reaction

The rate of a reaction may be defined as the change in concentration of a reactant or a product per unit time. Let us consider a simple reaction,

$$A \longrightarrow B$$

The rate of this reaction may be expressed in either of the following two ways.

(i) The rate of decrease in concentration of the reactant, A. Suppose the concentration of A at time t_1 is $[A]_1$ and at time t_2 is $[A]_2$. Therefore the rate of reaction may be expressed as,

$$\text{Rate of reaction} = \frac{\text{Decrease in concentration of A}}{\text{Time Taken}}$$

$$= \frac{[A]_2 - [A]_1}{t_2 - t_1} = -\frac{\Delta[A]}{\Delta t}$$

Significance of -ve sign : The concentration of reactants decreases during a reaction ; i.e., $\Delta[A] = [A]_2 - [A]_1$ is negative. Therefore, $\frac{\Delta[A]}{\Delta t}$ will be negative. But the rate of reaction is always positive. So a (-ve sign) is put before $\frac{\Delta[A]}{\Delta t}$ such that $-\frac{\Delta[A]}{\Delta t}$ is positive.

(ii) The rate of increase in concentration of the product, B. Suppose the concentration of B at time t_1 is $[B]_1$ and at time t_2 is $[B]_2$. Therefore the rate of reaction may be expressed as,

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$$\text{Rate of reaction} = \frac{\text{Increase in concentration of B}}{\text{Time taken}} = \frac{[B]_2 - [B]_1}{t_2 - t_1}$$

$$= \frac{\Delta[B]}{\Delta t}$$

Average rate and instantaneous rate :

We know that the rate of a chemical reaction depends on the concentration of the reactants. As the reaction progresses, the concentration of the reactant changes and the rate of reaction also changes. Thus, the rate of reaction calculated by dividing the total change in concentration by time needed for the change is really the *average rate*. Thus,

$$\text{Average rate of reaction} = \frac{\text{Total change of concentration}}{\text{Time needed for the change}}$$

The best method to express the rate of reaction as accurately as possible is to make time interval Δt very small so that the rate of reaction remains almost constant during that time interval. Therefore, the rate of reaction may be expressed as the rate of change of concentration of any one of the reactants or products at a particular moment of time. Such a rate of reaction is called instantaneous rate. Thus, if $d[A]$ or $d[B]$ represents the infinitesimally small changes in concentration of the reactant A or product B in infinitesimally small time, dt, the instataneous rate may be expressed as,

$$\text{instantaneous rate} = \frac{-d[A]}{dt} = \frac{d[B]}{dt}$$

Units of Rate

Rate has the units of concentration divided by time. We express the concentration in mol L^{-1} but time may be given in seconds, minutes, hours, days or years. Consequently the units of reaction rates may be, mol L^{-1} h^{-1} or mol L^{-1} min^{-1} or mol $L^{-1}s^{-1}$, and so on.

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2.2 Factors influencing the Rate of a Chemical Reaction

There are a number of factors which influence the rate of a reaction. Some of the important factors are :

(i) Concentration of the reactants : The rate of a reaction increases with increase in concentration of the reactants.

(ii) Temperature : Generally, there is an increase in the rate of reaction with the increase in temperature,

(iii) Nature of reactants and products: Rate of reaction is influenced by the nature of reactants. For example, ionic reactions occur suddenly.

(iv) Catalyst : A catalyst is a substance which changes the rate of a reaction without undergoing any permanent change by itself. The presence of a positive catalyst lowers the activation energy and hence increases the rate of reaction. The presence of a negative catalyst, on the other hand, enhances the activation energy and hence decreases the rate of the reaction.

(v) Presence of radiation : Radiations play an important role in the course of many reactions. The reactions which occur under the influence of light radiations are known as photochemical reactions.

(vi) Surface area of reactants : The rate of surface reactions increases with increase of surface area.

Dependence of Reaction rates on concentration :

When a reaction proceeds, the reactants are converted into products and hence the concentration of the reactants decreases while that of products increases as shown in Fig. 2.3. From the figure, it is evident that in the beginning of the reaction, only reactants are present. As the reaction progresses, the concentration of the reactants decreases while the concentration of the products increases.

It is also clear that the change in concentration of the various species occurs rapidly in the beginning, but very slowly as the reaction approaches the final stage. Thus the rate of reaction is large in the beginning but it decreases as the reaction progresses. This means that *the rate of a reaction is directly proportional to the concentration of the reactants.*

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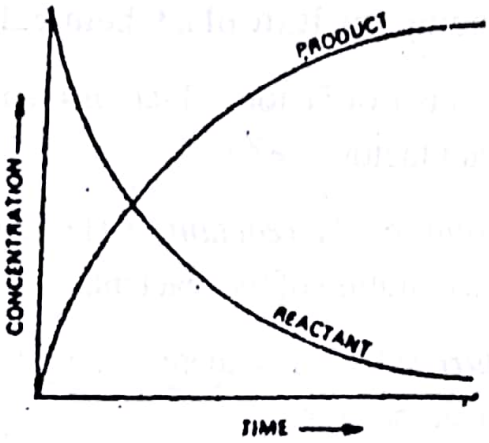

Fig. 2.3 Concentration versus time plot

The quantitative relationship between the rate of a reaction and the molar concentration of the reactants is given by the *law of mass action* which states that *the rate at which a substance reacts is proportional to its active mass and the rate of a chemical reaction is proportional to the product of the active masses of the reactants.* Active mass of a substance means the molar concentration (i.e., number of moles per litre). The active mass of a substance, A is expressed as C_A or [A].

Let us consider a simple reaction A+B → products.

The rate of this reaction can be represented as,

$r \alpha [A][B]$ i.e., $r = k[A][B]$ where, k is the rate constant.

If the molar concentration of each of the reactants involved in the reaction is unity, i.e., [A] = 1 and [B] = 1 then,

$r = k \times 1 \times 1 = k$

Thus, *rate constant of a reaction is defined as the rate of the reaction when the molar concentration of each of the reactants is unity.*

The Rate Law :

The rate law is defined as the mathematical expression which denotes the actual, i.e., the experimentally observed rate of the reaction in terms of the molar concentration of the reacting species which influences the rate of the reaction.

Let us consider the hypothetical reaction,

$$aA + bB \rightarrow cC + dD.$$

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From the kinetic study of the reaction, the dependence of the concentration of reactants on the rate of the reaction have been found to be,

$$\text{rate} = k[A]^m [B]^n \quad \ldots\ldots\ldots\ldots\ldots\ldots \text{(i)}$$

Where 'm' and 'n' are powers of the concentrations of the reactants A and B respectively. It may be noted that the values of m and n are determined experimentally and may or may not be equal to the co-efficients a and b in the balanced equation. The expression (i) is called the *rate law*. Thus, the rate law gives the experimentally observed dependence of reaction rates on concentration.

Now, let us write the rate law of some specified reactions.

(i) Thermal decomposition of nitrogen pentoxide.

$$2N_2O_{5(g)} \rightarrow 4NO_{2(g)} + O_{2(g)}$$

The rate of this reaction is experimentally observed to be proportional to the concentration of N_2O_5. Thus the rate law is,

$$\text{Rate} = k\,[N_2O_5]$$

This means that the rate doubles, when the concentration of N_2O_5 is doubled.

(ii) $2NO_{2(g)} + F_{2(g)} \rightarrow 2NO_2F_{(g)}$

Experimental observations reveal that the rate of the reaction is proportional to the concentrations of NO_2 and F_2. Thus the rate law is,

$$\text{rate} = k[NO_2][F_2]$$

This means that the rate doubles when the concentration of either NO_2 or F_2 is doubled.

Order and Molecularity of a Reaction

Order of a reaction

The order of a reaction is defined as the sum of the powers of the concentration terms in the rate equation. Thus, for the hypothetical reaction.

$$aA + bB \rightarrow cC + dD$$

The rate law is : $\text{rate} = k[A]^m[B]^n$

The order of this reaction is m+n. To be more precise, we can say that the order of the reaction with respect to A is 'm' and the order with respect to

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B is 'n' while the overall order of the reaction is m+n. The overall order of a reaction may range from 1 to 3 and can be zero or fractional. Now let us take some examples:

(i) $N_2O_{5(g)} \rightarrow 2NO_{2(g)} + \frac{1}{2}O_{2(g)}$; Rate = $k[N_2O_5]$: order = 1

(ii) $H_2O_2 \rightarrow H_2O + \frac{1}{2}O_2$: Rate = $k[H_2O_2]$ $\therefore$ order = 1

(iii) $2NO_2 \rightarrow 2NO + O_2$: Rate = $k[NO_2]^2$ $\therefore$ order = 2

(iv) $H_2 + I_2 \rightarrow 2HI$; Rate = $k[H_2][I_2]$ $\therefore$ order = 2

(v) $2NO + Cl_2 \rightarrow 2NOCl$: Rate = $k[NO]^2[Cl_2]$ $\therefore$ order = 3

(vi) $2NH_3 \xrightarrow{Pt} N_2 + 3H_2$; Rate = $k[NH_3]^0$ $\therefore$ order = 0

(vii) $CH_3CHO \rightarrow CH_4 + CO$: Rate = $k[CH_3CHO]^{3/2}$ $\therefore$ order = $\frac{3}{2}$

Molecularity of a reaction :

Molecularity of a reaction is defined as the number of reacting species (atoms, ions or molecules) which undergo simultaneous collision to bring about a chemical reaction. If a reaction involves the decomposition of only a single species, the molecularity is one or it is called a *unimolecular reaction.* For example, the decomposition of hydrogen peroxide involves single species and it is a unimolecular reaction.

$H_2O_2 \rightarrow H_2O + \frac{1}{2}O_2$ (unimolecular)

Similarly, if the reaction involves the collision of two species, it is *bimolecular* and if three species take part in a collision leading to the formation of products, it is called *trimolecular* and so on. For example, the dissociation of hydrogen iodide is a bimolecular reaction.

$2HI_{(g)} \rightarrow H_{2(g)} + I_{2(g)}$ (bimolecular)

The reaction of nitric oxide and oxygen is a trimolecular reaction because

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it involves collision of three reacting molecules.

$$2NO_{(g)} \rightarrow O_{2(g)} + 2NO_{2(g)} \qquad \text{(trimolecular)}$$

In a similar way, we can expect reactions which involve the collisions of four or more molecules. However, it has been observed that reactions involving three or more molecules are uncommon. The reason for this is that the occurrence of such reactions requires the simultaneous collisions of three or more than three molecules. The chances of the occurrence of such collisions are very less.

Difference between order and molecularity

The important differences between order and molecularity are given below :

Order	Molecularity
1. Order is the sum of the powers of the concentration terms in the rate law.	1. Molecularity is the number of reacting species undergoing simultaneous collision.
2. Order is determined experimentally.	2. Molecularity is a theoretical concept.
3. Order can have fractional values also.	3. Molecularity can have integral value only
4. Order can be zero.	4. Molecularity cannot be zero.

Units of Rate Constants

The rate is the change in concentration with time. Therefore, the rate of reaction is expressed by concentration units divided by time units. If the concentrations are expressed in moles per litre and time in seconds, then the units for rate of a reaction are mol $L^{-1}s^{-1}$.

$$\text{i.e., rate} = \frac{\text{Concentration}}{\text{time}} = \frac{\text{mol}\,L^{-1}}{s} = \text{mol}\,L^{-1}\,s^{-1}$$

The units of rate constants of different orders are different. For first order reaction, rate = k[A]

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$$\therefore \quad k = \frac{\text{rate}}{[A]} = \frac{\text{mol}\,L^{-1}s^{-1}}{\text{mol}\,L^{-1}} = s^{-1}$$

Thus, *the unit of rate constant of first order reaction is s^{-1} or $time^{-1}$.*

For second order reaction,

$$\text{rate} = k \times [A]^2 \qquad \therefore \quad k = \frac{\text{rate}}{[A]^2}$$

$$= \frac{\text{mol}\,L^{-1}s^{-1}}{(\text{mol}\,L^{-1})^2} = L\,\text{mol}^{-1}\,s^{-1}$$

Thus, *the unit of rate constant of second order reaction is $L\ mol^{-1}$ s^{-1} or $L\ mol^{-1}\ time^{-1}$.*

For third order reaction,

$$\text{rate} = k \times [A]^3$$

$$\therefore \quad k = \frac{\text{rate}}{[A]^3} = \frac{\text{mol}\,L^{-1}s^{-1}}{(\text{mol}\,L^{-1})^3} = L^2\text{mol}^{-2}\,s^{-1},$$

Thus, *the unit of rate constant of third order reaction is, L^2mol^{-2} s^{-1}or L^2mol^{-2} $time^{-1}$.*

For the n^{th} order reaction, rate = $k[A]^n$

$$\text{rate} = k \times [A]^2 \qquad \therefore \quad k = \frac{\text{rate}}{[A]^n}$$

$$= \frac{\text{mol}\,L^{-1}s^{-1}}{(\text{mol}\,L^{-1})^n} = \text{mol}^{1-n}\,L^{n-1}\,\text{time}^{-1}$$

Various Orders of Reactions - Integrated Rate Law

First Order Reactions

Let us consider a first order reaction,

$$A \rightarrow \text{products.}$$

Let the initial concentration (at t = 0) of A is 'a' moles per litre. If after time t, x moles of A have reacted, the concentration of A is (a-x). Then the

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rate dx/dt of reaction is proportional to the concentration of A.

$$\therefore \frac{dx}{dt} \alpha (a-x) \quad : \quad \text{i.e.,} \quad \frac{dx}{dt} = k_1 (a-x)$$

Where k_1 is the rate constant

$$\therefore \frac{dx}{a-x} = k_1 dt \qquad \text{.....(i)}$$

On integrating this we get,

$$-\ln(a-x) = k_1 t + I \qquad \text{.....(ii)}$$

Where I is the constant of integration

When t = 0, x = 0 and equation (ii) will become,

$$-\ln a = I \qquad \text{..... (iii)}$$

Substituting (iii) in (ii) we get, $-\ln (a-x) = k_1 t - \ln a$

Hence, $\ln a - \ln (a-x) = k_1 t$; $\ln a/a-x = k_1 t$

$$\therefore k_1 = {}^{1}/_{t} \ln a/a-x \qquad \text{.....(iv)}$$

$$k_1 = {}^{2.303}/_{t} \log a/a-x \qquad \text{..... (v)}$$

This is the integrated rate law for first order reaction.

Second order reactions :

Let us consider a second order reaction, 2A → products. Let the initial concentration of A is 'a' moles per litre. If after time 't', 'x' moles of A have reacted, the concentration of A is (a-x). For such a second order reaction rate is proportional to the square of the concentration of the reactant. Thus,

$$\frac{dx}{dt} = k_2 (a-x)^2 \qquad \text{.....(i)}$$

$$\text{i.e.,} \quad \frac{dx}{(a-x)^2} = k_2 dt \qquad \text{.....(ii)}$$

On integrating (ii) we get,

$$\frac{1}{a-x} = k_2 t + I \qquad \text{.....(iii)}$$

Where I is the constant of integration

When t = 0, x = 0. Therefore equation (iii) will become,

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$$\frac{1}{a} = I$$

Substituting the value of I in (iii) we get

$$\frac{1}{a-x} = k_2 t + \frac{1}{a}$$

Hence, $k_2 t = \frac{1}{a-x} - \frac{1}{a}$ $\therefore k_2 = \frac{1}{t}\left[\frac{1}{a-x} - \frac{1}{a}\right]$

i.e., $k_2 = \frac{1}{t} \cdot \frac{x}{a(a-x)}$ .....(iv)

This is the integrated rate law for second order reaction.

Third order reactions

Let us consider the reaction, 3A → Products.

Let the initial concentration of A be 'a' moles per litre and after time t, x moles have reacted. Then the concentration of A becomes (a-x). Therefore,

$$\frac{dx}{dt} = k_3(a-x)^3 \quad \text{..... (i)}$$

i.e., $\frac{dx}{(a-x)^3} = k_3 dt$ .....(ii)

Integrating eq. (ii), we get,

$$\frac{1}{2(a-x)^2} = k_3 t + I \quad \text{.....(iii)}$$

Where I is the integration constant. When t = 0, x = 0

$$\therefore \frac{1}{2a^2} = I$$

Substituting this value of I in (iii) we get,

$$\frac{1}{2(a-x)^2} = k_3 t + \frac{1}{2a^2}$$

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Hence,

$$k_3 t = \frac{1}{2(a-x)^2} - \frac{1}{2a^2} \quad \therefore k_3 t = \frac{1}{2}\left[\frac{1}{(a-x)^2} - \frac{1}{a^2}\right]$$

$$k_3 = \frac{1}{2t}\left[\frac{1}{(a-x)^2} - \frac{1}{a^2}\right] \quad \text{or } k_3 = \frac{1}{2t}\left[\frac{x(2a-x)}{a^2(a-x)^2}\right] \quad \text{.....(iv)}$$

This is the integrated rate law of third order reaction.

Zero order reactions

Consider a reaction, A → products. Let the initial concentration of A be 'a' moles per litre and after time t, x moles have reacted. Then the concentration of A beocmes (a-x)

$$\therefore \quad \frac{dx}{dt} = k_0 (a-x)^0 = k_0 \quad \text{.....(i)}$$

$$dx = k_0 \; dt \quad \text{.....(ii)}$$

Integrating eq. (ii) we get

$$x = k_0 \; t + I \quad \text{.....(iii)}$$

Where I is the integration constant.

When t = 0, x=0

∴ (iii) will become, $0 = k_0 \times 0 + I$ i.e., $I = 0$

$$\therefore x = k_0 \; t$$

This is the integrated rate law for zero order reaction.

Pseudo order Reactions

A reaction in which one of the reactants is present in a large excess shows an order different from the actual order. *The experimental order which is not the actual order is known as the pseudo order.* Since for elimentary reactions, molecularity and order are identical, pseudo order reactions are also called *pseudo-molecular reactions.*

Let us consider a reaction, A+B → Products, in which the reactant B is

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present in large excess. Its rate law is given by, rate = k[A][B].

However, since B is present in large excess, its molar concentration remains practically constant. Thus, the rate law can be written as,

rate = k'[A], where k'=k[B]

Thus the actual order is two but in practice it will be one. Therefore, the reaction is called *pseudo first order*.

Examples of Pseudo order Reaction

(i) Hydrolysis of an ester : For example, hydrolysis of ethyl acetate in aqueous solution in presence of a mineral acid as catalyst.

$$CH_3COOC_2H_5 + H_2O \rightarrow CH_3COOH + C_2H_5OH$$

Its rate law may be written as, rate = $k[CH_3COOC_2H_5]\,[H_2O]$

But H_2O is present in large excess so that $[H_2O]$ = constant

$\therefore$ rate = $k'[CH_3COOC_2H_5]$, where $k' = k[H_2O]$

Thus the reaction is actually second order but in practice it is found to be first order. So it is a pseudo first order reaction. However, it is a bimolecular reaction.

(ii) Hydrolysis of cane sugar : Hydrolysis of cane sugar in aqueous solution in the presence of mineral acid is another typical example of a pseudo first order reaction.

$$C_{12}H_{22}O_{11} + H_2O \rightarrow C_6H_{12}O_6 + C_6H_{12}O_6$$

Its rate law is given as, rate = $k\,[C_{12}H_{22}O_{11}]\,[H_2O]$

But H_2O is present in large excess and hence $k\,[H_2O] = k'$

$\therefore$ rate = $k'\,[C_{12}H_{22}O_{11}]$

Thus, the reaction is experimentally found to be first order. However, its molecularity is two.

Fractional life time of a reaction : Fractional life time is the time required to complete a definite fraction of the reaction. The fraction generally taken is one half of a given reactant.

Half-life Period of a Reaction

Half-life period of a reaction is defined as the time required for the concentration of a reactant to decrease to half its initial value. It is denoted

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by the symbol $t_{\frac{1}{2}}$ or $t_{0.5}$. The half-life period of a reaction may be calculated by using the integrated rate equation for its order.

Half-life of zero order reaction :

For zero order reaction, the integrated rate law is,

$x = k_0 t$ ..... (i)

When $t = t_{\frac{1}{2}}$,

$x = a/2$ ..... (ii)

Substituting (ii) in (i) we get,

$\frac{a}{2} = k_0 t_{\frac{1}{2}}$ ..... (iii)

$\therefore \quad t_{\frac{1}{2}} = \frac{a}{2 k_0}$ ..... (iv)

Half-life of first order reaction

For first order reaction, rate constant is given as,

$$k_1 = \frac{2.303}{t} \log \frac{a}{a-x} \quad \text{.....(i)}$$

When $t = t_{1/2}$, $x = \frac{a}{2}$ .....(ii)

Substituting (ii) in (i), we get,

i.e., $k_1 = \frac{2.303}{t_{\frac{1}{2}}} \log \frac{a}{a - \frac{a}{2}}$; i.e., $k_1 = \frac{2.303}{t_{\frac{1}{2}}} \log \frac{a}{a/2}$

$k_1 = \frac{2.303}{t_{\frac{1}{2}}} \log a \times \frac{2}{a}$; $k_1 = \frac{2.303}{t_{\frac{1}{2}}} \log 2$

$= \frac{2.303 \times 0.3010}{t_{1/2}}$; i.e., $k_1 = \frac{0.693}{t_{1/2}}$ $\therefore t_{1/2} = \frac{0.693}{k_1}$

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Thus, for a first order reaction, $t_{1/2}$ is independent of initial concentration.

Half-life of second order reaction

The integrated rate law of second order reaction is,

$$k_2 = \frac{1}{t} \frac{x}{a(a-x)} \quad \text{.....(i)}$$

When $t = t_{1/2}$, $x = a/2$ .....(ii)

Substituting (ii) in (i), we get, $k_2 = \frac{1}{t_{1/2}} \frac{a/2}{a(a-\frac{a}{2})}$ .....(ii)

$$= \frac{1}{t_{1/2}} \cdot \frac{a/2}{a\frac{a}{2}} = \frac{1}{t_{1/2}\ a} \quad \therefore\ t_{1/2} = \frac{1}{k_2 a} \quad \text{.....(iii)}$$

Thus, $t_{1/2}$ of a second order reaction is inversely proportional to the initial concentration of the reactant.

Half-life for a third order reaction : For a third order reaction, the half life period is found to be related to the initial concentration, 'a' as,

$$t_{1/2} \propto \frac{1}{a^2} \qquad \therefore \quad t_{1/2} \times a^2 = \text{a constant}$$

Half-life for a n^{th} order reaction : For a n^{th} order reaction, the half life period is related to the initial concentration, 'a' as,

$$t_{1/2} \propto \frac{1}{a^{n-1}} \qquad \therefore \quad t_{1/2} \times a^{n-1} = \text{a constant}$$

Example 41 Identify the order of the reaction for which the rate constants are : (i) 5.5×10^{-2} mol $L^{-1}s^{-1}$ (ii) 3.1×10^{-2} mol^{-1} Ls^{-1} (iii) $2.5 \times 10^{-2}s^{-1}$ (iv) $1.5 \times 10^{-2}L^2s^{-1}$ (v) $1.5 \times 10^{-2}atm^{-1}s^{-1}$

Solution (i) zero (ii) 2 (iii) 1 (iv)3 (v) 2

Self Study Problems

38 : Identify the order of the reaction for which the rate constants are (i) $3 \times 10^{-1}s^{-1}$ (ii)9.2×10^{-4} mol L^{-1} s^{-1} **Ans :** (i) 1 (ii) 0

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Example 42 : The decomposition of a substance follows first order kinetics. The half life period of the reaction is 35 minutes, what is the rate constant of the reaction ?

Solution : For a first order reaction, we have $t_{½} = \frac{0.693}{k_1}$

$$\therefore k_1 = \frac{0.693}{t_{½}}; \quad \text{But } t_{½} = 35\,\text{min}; \therefore k_1 = \frac{0.693}{35\,\text{min}} = 0.0198\ \text{min}^{-1}$$

i.e., rate constant, $k_1 = 1.98 \times 10^{-2}\ \text{min}^{-1}$

Example 43: The rate constant of a first order reaction is $10^{-3}\ s^{-1}$. How much time will it take for 10g of the reactant to reduce half the quantity?

Solution : Time required to reduce half the quantity = $t_{½}$

$$\text{But, } t_{½} = \frac{0.693}{k_1} \text{ and } k_1 = 10^{-3}\ s^{-1}$$

$$t_{½} = \frac{0.693}{10^{-3}\ s^{-1}} = 0.693 \times 10^{3}\ s = 693 \text{ seconds}$$

Example 44 : The half life period for thermal decomposition of acetone at 873 K at different initial pressures are :

P_o (mm Hg) :	98	230	296	
$t_{½}$ (s) :	86	85	86	Find the order of reaction?

Solution : The data shows that $t_{½}$, is independent of the initial concentration. ∴ It is a first order reaction.

Self Study Problems

39 : Calculate the half life period of a first order reaction whose rate constant is $200\ s^{-1}$. **Ans :** 3.465×10^{-3}s

40 : A substance decomposes following first order kinetics. If the half life of the reaction is 35 minutes, what is the rate constant of the reaction? **Ans :** $1.98 \times 10^{-2}\ \text{min}^{-1}$

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Example 45 The decomposition of H_2O_2 has been found to be first order with a rate constant $7.3 \times 10^{-4} s^{-1}$. (a) Calculate the rate of the reaction when $[H_2O_2] = 0.25$ molL^{-1} (b) What concentration of H_2O_2 would give a rate of 7.49×10^{-4} mol $L^{-1} s^{-1}$.

Solution (a)Rate=$k[H_2O_2]$; $[H_2O_2] = 0.25$ mol L^{-1} and $k=7.3 \times 10^{-4} s^{-1}$

$\therefore$ Rate = $7.3 \times 10^{-4} s^{-1} \times 0.25$ mol $L^{-1} = 1.825 \times 10^{-4}$ mol $L^{-1}s^{-1}$

(b) Rate = 7.49×10^{-4}mol litre$^{-1}s^{-1}$; $k = 7.3 \times 10^{-4} s^{-1}$

$$\therefore [H_2O_2] = \frac{\text{rate}}{k} = \frac{7.49 \times 10^{-4} \text{ mol } L^{-1} s^{-1}}{7.3 \times 10^{-4} s^{-1}} = 1.026 \times 10^{-4} \text{ mol } L^{-1}$$

Self Study Problems

41. The decomposition of N_2O_5 in a CCl_4 solution has been investigated

$N_2O_5 \rightarrow 2NO_2 + \frac{1}{2}O_{2(g)}$

The reaction has been found to be of the first order in N_2O_5 with a first order rate constant $6.2 \times 10^{-4} s^{-1}$. Calculate the rate of the reaction when (a)$[N_2O_5]$=1.25 mol L^{-1} and (b) $[N_2O_5] = 0.25$ mol L^{-1}? (c) What concentration of N_2O_5 would give a rate of 2.4×10^{-3} mol $L^{-1}s^{-1}$?

Ans : a) 7.75×10^{-4} mol $L^{-1}s^{-1}$ b) 1.55×10^{-4} mol $L^{-1} s^{-1}$
c) 3.8 mol L^{-1}

Example 46 The decomposition of benzene diazonium chloride follows first order kinetics at 232 K. The value of specific reaction rate is 7.09×10^{-2} min^{-1}. Calculate the rate of reaction if the concentration of the reactant is 1.9×10^{-2} mol L^{-1}.

Solution : For the first order reaction, r = k[A]

i.e., r = k[Benzene diazonium chloride]. But $k = 7.09 \times 10^{-2}$ min^{-1}

[Benzene diazonium chloride] = 1.9×10^{-2} mol L^{-1}

$\therefore$ Rate of reaction = 7.09×10^{-2} $min^{-1} \times 1.9 \times 10^{-2}$ mol L^{-1}

$= 1.347 \times 10^{-3}$ mol L^{-1} min^{-1}.

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Self Study Problems

42: The rate constant of a first order reaction has been found to be 0.0435 min^{-1}. The initial concentration of the reactant is 0.12 mol L^{-1}. How much time will it take for the concentration of the reactant to become 0.03 mol^{-1}.

Ans : 31.9 min

43 : The rate constant for the decomposition of N_2O_5 is $6 \times 10^{-4} s^{-1}$. At what time will the initial concentration of 1M be reduced to 0.2 M if the reaction is of first order ?

Ans : 2683 s.

Example 47 : A substance decomposes following first order kinetics. The half life period of the reaction is 35 minutes, what is the rate constant of reaction?

Solution : For a first order reaction ,

$$t_{1/2} = \frac{0.693}{k} \text{ or } \qquad k = \frac{0.693}{t_{1/2}}$$

Now, $t_{1/2} = 35$ min.

$$k = \frac{0.693}{35\,\text{min}} = 1.98 \times 10^{-2}\ \text{min}^{-1}$$

Example 48 : The thermal decomposition of a compound is of first order. If 50% of the compound is decomposed in 120 minutes. How long will it take for 90% of the compound to decompose?

Solution :

Let us first calculate, k as $k = \frac{0.693}{t_{1/2}}$; $t_{1/2}$ 120 min.

$$\therefore \qquad k = \frac{0.693}{120\ \text{min.}} = 5.77 \times 10^{-3}\ \text{min}^{-1}$$

Now, for the first order reaction, $t = \frac{2.303}{k} \log \frac{[A]_0}{[A]}$

If the initial concentration of A, i.e., [A]o = a, then [A] = a-a x 0.90 = 0.10 a, because 90% of compound decomposes and only 10% is left.

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$$\therefore t = \frac{2.303}{5.77 \times 10^{-3}} \log \frac{a}{0.1a} = \frac{2.303}{5.77 \times 10^{-3}} \log 10 = 399 \text{ min.}$$

Example 49 : Calculate two third life of a first order reaction having $k = 5.48 \times 10^{-14}\ s^{-1}$.

Solution : For two third of a reaction, $a\text{-}x = a - \frac{2a}{3} = \frac{a}{3}$

$$t_{2/3} = \frac{2.303}{k} \log \frac{a}{a/3} = \frac{2.303}{k} \log 3$$

$$t_{2/3} = \frac{2.303}{5.48 \times 10^{-14}} \log 3 = 2.01 \times 10^{13} s$$

Self Study Problems

42 : A 50% of a reaction gets completed in 16 minutes. What fraction of reaction would occur in 32 minutes? **Ans :** 75%

43 : If the half life period of a first order reaction in A is 2 min. How long will it take [A] to reach 25% of its initial concentration and 10% of its initial concentration? **Ans :** 4 min, 6.65 min

44 : The half life period for a reaction of first order is 2.31×10^3 min. How long will it take for 1/5 of the reactant to be left behind? **Ans :** 5.36×10^3 min.

45 : Show that in case of a first order reaction, the time taken for completion of 99.9% reaction is ten times the time required for half change of the reaction.

Characteristics of First, Second, Third and Zero order reactions

1. First Order Reactions : The characteristics of first order reactions are : (i) The rate of reaction depends on the first power of a concentration term. (ii) The integrated rate law is, $k_1 = \frac{2.303}{t} \log \frac{a}{a-x}$ (iii) A plot of log (a-x) against t will be a straight line with slope = -2.303/k and

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intercept - log a. (iv) $t_{½}$ is independent of initial concentration. (v) Unit of rate constant is time $^{-1}$.

2. Second Order Reactions : The characteristics of second order reactions are : (i) The rate of reaction depends on two concentration terms. (ii) The integrated rate equation is $k_2 = \frac{1}{t} \frac{x}{a(a-x)}$ (iii) A plot of 1/(a-x) against t will be a straight line with slope k_2. (iv) $t_{½}$ is inversely proportional to the initial concentration. (v) Unit of rate constant is L mol^{-1} time^{-1}.

3.Third Order Reactions : The characteristics of third order reactions are :

(i) Rate of reaction depends on three concentration terms. (ii) The integrated rate law is $k_3 = \frac{1}{2t} \frac{x(2a-x)}{a^2(a-x)^2}$ (iii) $t_{½}$ is inversely proportional to the square of the initial concentration. (iv)The unit of rate constant is L^2 mol^{-2} time^{-1}.

4. Zero Order Reactions : The important characteristics of zero order reactions are : (i) The rate of reaction is independent of the initial concentration of reactants. (ii) The integrated rate law is $x = k_0 t$ (iii) A plot of x agianst t would be a straight line passing through the origin. (iv) Unit of k is mol L^{-1} time^{-1}. (v) $t_{½}$ is proportional to the initial concentration.

Just understand the meaning of each reactn no derivation required.

Kinetics of Complex Reactions

There are some reactions which proceeds in a series of steps instead of a single step. These reactions are called *complex reactions*. The study of chemical kinetics of such reactions are highly complicated. Important among such reactions are:

1. Opposing or Reversible Reactions

Reactions in which the products formed react to give back the reactants are called reversible or opposing reactions. Initially, the rate of forward reaction is very large which decreases with passage of time. On the other hand, the rate of backward reaction is initially zero which increases with passage of time. As a result, a stage is reached at which the two rates become equal; the overall rate is then zero. This situation is called chemical equilibrium. It is dynamic in nature, i.e., all the species are formed at the same rate at which they are reacting.

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Consider the simplest case of an opposing reaction in which both the forward and backward reactions are of first order:

$A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} B$, where k_1 and k_{-1} are the rate constants of the forward and backward reactions respectively. Let the initial concentration of A be 'a' and that of B zero. After time t, the concentration of A will be (a–x) and that of B will be x. Thus, the rate of forward reaction is equal to $k_1(a-x)$ and the rate of backward reaction is $k_{-1}x$. So the net rate of formation of B will be,

$$\frac{dx}{dt} = k_1(a-x) - k_{-1}x \quad \text{.....(1)}$$

If the concentration of B at equilibrium is x_e when the net rate is zero, then

$$k_1[a-x_e] - k_{-1}x_e = 0 \quad \text{.....(2)} \quad \therefore \quad k_1(a-x_e) = k_{-1}x_e$$

$$\therefore \quad k_{-1} = \frac{k_1(a-x_e)}{x_e} \quad \text{.....(3)}$$

Substituting (3) in (1), we get

$$\frac{dx}{dt} = k_1(a-x) - k_1\left(\frac{a-x_e}{x_e}\right)x = k_1\frac{(x_e-x)a}{x_e} \quad \text{.....(4)}$$

i.e, $\frac{dx}{x_e - x} = \frac{a}{x_e}k_1 dt$ .....(5) Integrating (5), we get

$$-ln(x_e - x) = \frac{a}{x_e}(k_1 t + I) \quad \text{.....(6), where I is the constant of integration.}$$

or $$-\frac{x_e}{a}ln(x_e - x) = k_1 t + I \quad \text{.....(7)}$$

At t = 0, x = 0 so that, $I = \frac{x_e}{a}ln\, x_e$

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$\therefore$ (7) becomes, $-\frac{x_e}{a}[ln(x_e - x)] = k_1 t - \frac{x_e}{a} ln\, x_e$

$\therefore\ k_1 t = \frac{x_e}{a} ln\left(\frac{x_e}{x_e - x}\right)$ .....(8) $\quad\therefore\ k_1 = \frac{x_e}{at} ln\left(\frac{x_e}{x_e - x}\right)$ .....(9)

From (3), $k_{-1} = k_1 \frac{a}{x_e} - k_1 \frac{x_e}{x_e}$ i.e., $k_{-1} = k_1 \frac{a}{x_e} - k_1$

i.e., $k_1 + k_{-1} = k_1 \frac{a}{x_e}$ .....(10)

From (9) and (10), $k_1 + k_{-1} = \frac{x_e}{at} ln \frac{x_e}{x_e - x} \frac{a}{x_e} = \frac{1}{t} ln \frac{x_e}{x_e - x}$ .....(11)

By knowing k_1, a and x_e, the rate constant, k_{-1} of the reverse reaction may be calculated.

The above equation, i.e., $k_1 + k_{-1} = \frac{1}{t} ln \frac{x_e}{x_e - x}$ is an expression for the first order reaction where the initial concentration is being replaced by x_e, the equilibrium concentration. Thus, a reversible first order reaction may be treated as if it were a simple first order reaction.

2. Consecutive Reactions

The reactions which proceed from reactants to products through one or more intermediate steps are called *consecutive reactions*. Consider a simple first order consecutive reaction scheme,

$$A \xrightarrow{k_1} B \xrightarrow{k_1'} C$$

Initial concentration	a	0	0
Concentration after time,t	x	y	z

In this reaction, each stage has its own rate constant, i.e., k_1 and k_1'. The overall rate of the reaction will depend upon the relative magnitudes of k_1 and k_1'. If $k_1 >> k_1'$, the formation of C from B will control the overall rate of

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reaction. On the other hand, if $k_1' >> k_1$, the conversion of A to B will control the overall rate of reaction. However, when k_1 and k_1' are of comparable magnitudes, the overall rate of the reaction will depend on both k_1 and k_1' and the problem will become complicated.

The rate of disappearance of A, $-\frac{d[A]}{dt} = -\frac{dx}{dt} = k_1 x$ .....(1)

The rate of formation of B, $\frac{d[B]}{dt} = \frac{dy}{dt} = k_1 x - k_1' y$ .....(2)

The rate of formation of C, $\frac{d[C]}{dt} = \frac{dz}{dt} = k_1' y$ .....(3)

Integrating (1) within the limits x = a, t = 0 and x =x, t = t, we get

$$k_1 t = ln\frac{a}{x} \quad \text{or} \quad x = ae^{-k_1 t} \quad .....(4)$$

As in the case of first order reaction, the concentration of A decreases exponentially with time. Substituting (4) in (2), we get

$$\frac{dy}{dt} = k_1 ae^{-k_1 t} - k_1' y \quad .....(5)$$

This is a linear first order differential equation in y and on solving this, we get

$$y = \frac{k_1 a}{k_1' - k_1}[e^{-k_1 t} - e^{-k_1' t}] \quad .....(6)$$

Knowing the values of x and y, the concentration of C (i.e., z) can be calculated. That is,

$$z = \frac{a}{k_1' - k_1}\left[(k_1' - k_1' e^{-k_1 t}) - (k_1 - k_1 e^{-k_1' t})\right] \quad .....(7)$$

The concentrations of A, B and C, i.e., x, y and z are plotted against time in the following Fig.12. From the Fig. it is evident that the concentration of A decreases exponentially while that of C increases gradually to the value of 'a'

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when A has changed completely to C. The concentration of B increases to a maximum value and then decreases asymptotically to zero. When $k_1 >> k_1'$, $e^{-k_1 t}$ is much smaller than $e^{-k_1' t}$ in equation (7) and so may be neglected. Thus, $z \approx a(1-e^{-k_1' t})$. The formation of C therefore involves only the smaller rate constant k_1' and is hence the rate determining step. On the other hand, when $k_1' >> k_1$, equation (7) becomes, $z \approx a(1-e^{-k_1 t})$. This involves the smaller rate constant k_1, i.e., the rate constant of the slower step.

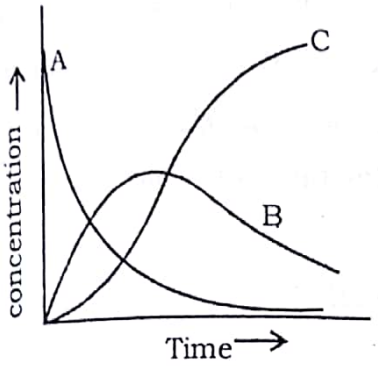


Fig.2.4 *Concentration changes in a consecutive first order reaction.*

Examples of Consecutive Reactions

(i) Decomposition of sodium hypochlorite

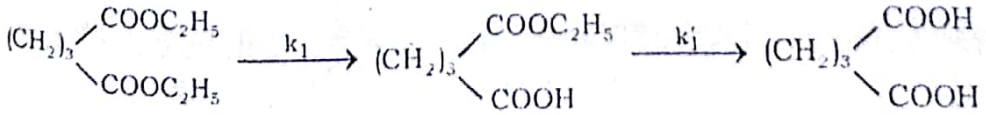
It occurs in two steps :

(a) $2NaClO \xrightarrow[\text{slow}]{k_1} NaCl + NaClO_2$

(b) $NaClO + NaClO_2 \xrightarrow[\text{fast}]{k_1'} NaCl + NaClO_3$

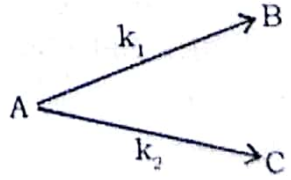
The rate of the overall reaction is controlled by the rate of reaction (i), i.e., the slower step.

(ii) Acid Hydrolysis of Diethyl Adipate



3. Parallel Reactions or Side Reactions

These are reactions in which the reacting molecules react or decompose in different ways giving different products. The reaction which gives the highest yield



Occurrence of side or parrallel reactions

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of the product is called the main reaction and the reactions which gives smaller yields of products are called side reactions. For example, consider a reaction, in which the reactant A gives two products B and C through two different reactions with rate constant k_1 and k_1'. If $k_1 > k_1'$, then the reaction $A \rightarrow B$ would be the *main reaction* and the reaction $A \rightarrow C$ would be the *side reaction*. However, if $k_1 \approx k_1'$, then the two reactions are called *parallel reactions*. To calculate the overall rate constant in such cases, the rate of the side reactions should be taken into account.

Assume that both the reactions are of first order.

Then, $r_1 = \frac{d[B]}{dt} = k_1 C_A$ .....(1) for $A \rightarrow B$ and $r_2 = \frac{d[C]}{dt} = k_1' C_A$ ....(2) for $A \rightarrow C$.

The overall rate of disappearance of A is give by,

$$-\frac{d[A]}{dt} = r_1 + r_2 = k_1 C_A + k_1' C_A = [k_1 + k_1'] C_A = k' C_A \text{ .....(3)}$$

Integrating (3), under conditions that $C_A = C_0$ at $t = 0$ and $C_A = C_A$ at $t = t$, we get

$$C_A = C_0 e^{-k't} = C_0 e^{-(k_1 + k_1')t} \text{ .....(4)}$$

By knowing C_0, the initial concentration of A and C_A, the concentration at time t, $(k_1 + k_1')$ can be calculated. From equations (1) and (2),

$$\frac{k_1}{k_1'} = \frac{r_1}{r_2} = \frac{\text{yield of product B}}{\text{yield of product C}}$$

By measuring the yields of B and C, the ratio $\frac{k_1}{k_1'}$ can be calculated. By knowing $k_1 + k_1'$ and $\frac{k_1}{k_1'}$, the rate constants k_1 and k_1' can be calculated.

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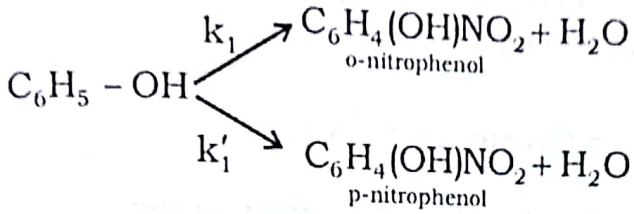
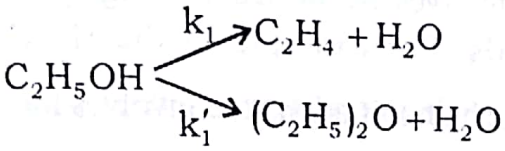
Examples of Parallel Reactions

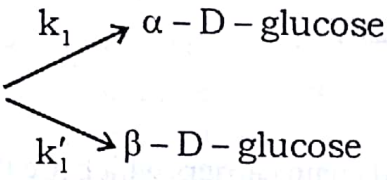
1. Nitration of phenol



2. Dehydration of ethanol



3. D glucose (Aldehyde form)



Qualitative Idea of Chain Reactions

A chain reaction is defined as a reaction in which highly reactive species are formed as intermediates which carry on the reaction at a rapid rate for a long time. These reactions were first proposed by ***Bodenstein*** to account for the high yields of photochemical reaction between H_2 and Cl_2.

Principle of Chain Reactions

In chain reactions there is an initial slow step in which certain highly reactive species are produced. These reactive species combine with other molecules to form product species and at the same time some more of the reactive species. The reactive species thus formed react, in turn, with other molecules forming once again more of the product species and generating again more of the reactive species. This continues as a chain reaction at a rapid rate since highly reactive species are involved in the reaction. The

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reaction goes on until the chain terminates in one of the steps involved.

High temperature decomposition of hydrocarbons (gas phase pyrolysis), of aldehydes, and several carbonyl compounds, reactions of hydrogen with halogens, etc., belong to chain reactions.

Different steps in a Chain Reaction

The different steps involved in a chain reaction are:

(i) Chain initiation

The chain initiation step involves the formation of highly reactive species like atoms or free radicals. For example in the H_2–Cl_2 reaction, $H_2 + Cl_2 \xrightarrow{h\nu} 2HCl$ the chain initiation step involves the formation of Cl atoms (free radicals).

$$Cl_2 + h\nu \rightarrow 2Cl^{\bullet}$$

(ii) Chain Propagation

The initially generated chain carriers which are transient, i.e., short lived take part in the subsequent step or steps to carry on the reaction and give rise to intermediate steps leading to the formation of final products of the reaction. These are called chain propagation steps. For example, in the H_2–Cl_2 reaction the propagation steps are:

(a) $Cl^{\bullet} + H_2 \rightarrow HCl + H^{\bullet}$

(b) $H^{\bullet} + Cl_2 \rightarrow HCl + Cl^{\bullet}$

In (a), hydrogen atoms and in (b) chlorine atoms are formed which continue the propagation of the chain. These atoms are called chain carriers.

(iii) Chain transfer

In some chain propagation steps initial chain carriers are consumed and new chain careers are generated by reaction with some existing molecules. These steps are called *chain transfer steps*. For example, in the H_2–Cl_2 reaction, chain is initiated by chlorine atoms in step (a) and propagated by hydrogen atoms in step (b). Thus, initiation of chain is transferred from the

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chlorine atoms in step (a) to the hydrogen atoms in step (b). The consecutive steps (a) and (b) are thus called chain transfer steps.

(iv) Chain termination

At the end, when there are no more or not enough molecules of the reactants to be converted to the products, the active chain carriers are converted to the nonactive molecular species. This step or steps are called chain termination step or steps. For example, in $H_2 - Cl_2$ reaction the chain termination steps are;

$$H^{\bullet} + H^{\bullet} \rightarrow H_2 ; \quad Cl^{\bullet} + Cl^{\bullet} \rightarrow Cl_2 ; \quad H^{\bullet} + Cl^{\bullet} \rightarrow HCl$$

Influence of temperature on the Rate of Reactions

The Arrhenius Equation

It has been found that an increase of temperature increases the rate of a reaction. In majority of cases an increase of temperature by 10^0C increases the rate of a reaction by a factor 2 or 3. Thus, the ratio of rate constants of a reaction at two different temperatures differing by 10 degree is called *temperature coefficient.*

$$\text{i.e., Temperature coefficient} = \frac{k_{t+0}}{k_t} = \frac{k_{35^0}}{k_{25^0}} \approx 2 \text{ to } 3$$

However, in 1889 *Arrhenius* suggested a simple and more correct relationship between the rate constant, k, for a reaction and the temperature of the system, i.e.,

$$\frac{d\ln k}{dT} = \frac{E_a}{RT^2} \quad \text{.....(1), where } E_a \text{ is called the } \textit{activation energy} \text{ of the}$$

reaction and is characteristic of the reaction. *Activation energy is defined as the additional amount of energy over the average energy of the molecules at a given temperature which a molecule must possess to take part in a chemical reaction.* Integrating (1), we get

$$\ln k = -\frac{E_a}{RT} + I \quad \text{.....(2)}$$

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or $\log k = -\frac{E_a}{2.303\,R} + I'$ .....(3), where I or I' is the integration constant.

or $k = Ae^{-E_a/RT}$ .....(4), where A is the pre-exponential factor. E_a and A are called the Arrhenius parameters.

Equation (3) suggests that a plot of $ln\ k$ versus 1/T should be a straight line as shown in the Fig.2.5. The slope of the plot is equal to $-E/2.303\ R$ and the intercept is equal to I'. From the slope, the activation energy can be calculated. However, if we integrate equation (1) between the limits $k = k_1$ at $T = T_1$ and $k = k_2$ at $T = T_2$, we get

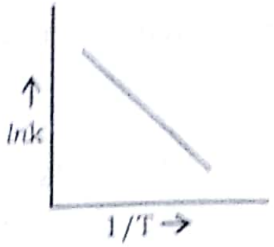

Fig.2.5 *Plot of ln k × 1/T*

$$ln\frac{k_2}{k_1} = \frac{E_a}{R}\left[-\frac{1}{T}\right]_{T_1}^{T_2} = +\frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] = \frac{E_a}{R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

or $$\log\frac{k_2}{k_1} = \frac{E_a}{2.303\,R}\left[\frac{T_2 - T_1}{T_1 T_2}\right] \quad .....(5)$$

From this equation it is evident that the activation energy, E_a can be calculated from the value of the rate constants at two different temperatures.

Knowing E_a, A can be calculated from the Arrhenius equation, $k = Ae^{-E_a/RT}$. Taking log, $ln\ k = ln A - E_a/RT$ or $ln\ A = ln\ k + E_a/RT$. Taking antilog, we get A.

EXAMPLE 50

The rate constants of a reaction at two temperatures 273 K and 303 K are 2.46×10^{-5} and 1.63×10^{-3}. Calculate the activation energy of the reaction.

Ans. $$\log\frac{k_2}{k_1} = \frac{E_a}{2.303\,R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

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$T_1 = 273$ K, $T_2 = 303$ K; $k_2 = 1.63 \times 10^{-3}$; $k_1 = 2.46 \times 10^{-5}$, $R = 8.314$ JK mol^{-1}

$$\therefore \log \frac{1.63 \times 10^{-3}}{2.46 \times 10^{-5}} = \frac{E_a}{2.303 \times 8.313} \times \frac{30}{273 \times 303}$$

$$\therefore E_a = \frac{2.303 \times 8.314 \times 273 \times 303}{30} \log \frac{1.63 \times 10^{-3}}{2.46 \times 10^{-5}}$$

$= 96510$ J mol^{-1} = **96.51 kJ mol^{-1}**

Self Study problems

46. For the reaction,

$$CH_{4(g)} + 2S_{2(g)} \rightarrow CS_{2(g)} + 2H_2S_{(g)}$$

The rate constants are 1.1 *l* mol^{-1} s and 6.4 *l* mol^{-1} s at 823 K and 898 K respectively. Calculate the energy of activation

[**Ans.** $ln \frac{k_2}{k_1} = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$; $E_a = 1.4 \times 10^5$ J mol^{-1}]

47. The rate constant of a reaction at 25^{0}C and 40^{0}C are 5.7×10^{-5} dm^3 mol^{-1} s^{-1} and 1.64×10^{-4} dm^3 mol^{-1}s^{-1}at 40^{0}C. Calculate the activation energy and pre-exponential factor.

[**Hint:** $ln \frac{k_2}{k_1} = \frac{E_a}{R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$ $\therefore$ $E_a = 54.48$ kJ mol^{-1}.

$k = Ae^{-E_a/RT}$; $lnk = ln\,A - E_a/RT$ $\therefore$ $lnA = ln\,k + E_a/RT$

$\therefore$ $A = 2.01 \times 10^5$ dm^3 mol^{-1}s^{-1}]

The Energy of Activation

The molecules must acquire a certain minimum amount of energy (threshold energy) before they undergo chemical reactions. Most of the molecules,

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however, have much less kinetic energy than the threshold energy. The excess energy that the reactant molecules, having energy less than the threshold energy, must acquire in order to react to give products is called *activation energy*. Thus,

Activation Energy = Threshold Energy – Energy actually possessed by the molecules.

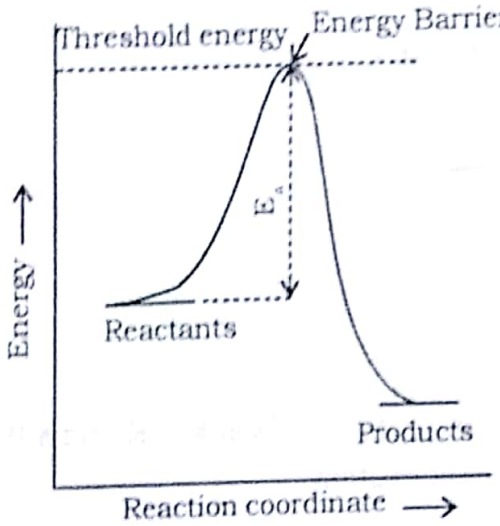

Fig.2.6 *Energy of Activation*

This means that non-active molecules, i.e., molecules having energy less than the threshold energy can be activated by absorption of some extra energy. This extra energy is the *activation energy*. According to the concept of activation, the transformation of the reactants into the products does not take place directly. There is an energy barrier in between the reactants and products. The reactant molecules must cross this energy barrier to yield the products. This barrier determines the magnitude of threshold energy (and also activation energy) [Fig. 2.6]. From the Fig. it is evident that E_a is the activation energy of the reactants. The reactants must absorb this much energy before they react and change into products. Different reactions require different amounts of activation energy. A reaction which has lower activation energy will occur at a faster rate at a given temperature. For example, the reaction

$NO_{(g)} + \frac{1}{2}O_{2(g)} \rightarrow NO_{2(g)}$ is fast at ordinary temperatures as it requires a comparatively lower activation energy. On the other hand, the reaction,

$CO_{(g)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{2(g)}$, is slow as it needs a comparatively higher activation energy. Thus, the differences in activation energies are largely responsible for the differences in the rates of different reactions.

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Collision Theory

For a chemical reaction to occur molecules of the reacting substances must collide with one another. This idea forms the basis of collision theory of reactions. According to collision theory the rate of a reaction is proportional to the collision frequency, Z.

i.e., rate of a reaction $\propto Z$.

The collision frequency is the number of collisions per unit volume per second. Normally the value of Z is very high. Thus, if all collisions are effective in producing products, all chemical reactions must be completed almost instantaneously. However, most of the reactions are found to be slow. This implies that all encounters between the reactant molecules are not effective in producing products. In fact, only a very small fraction 'f' of the total number of collisions are effective. The collisions which produce products and hence result in chemical reactions are known as *effective collisions*.

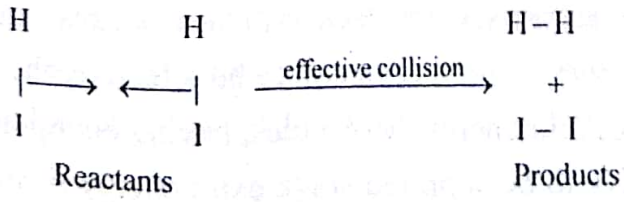
The fraction 'f' is determined by the energies of the molecules as well as their orientations. In order to occur a reaction,

(i) the colliding molecules must possess a certain minimum amount of energy. The minimum amount of energy which must be possessed by the colliding molecules to undergo effective collisions is called *threshold energy*.

(ii) the molecules must collide with the proper orientation.

For example, in the reaction $2HI \rightarrow H_2+I_2$, the reactant molecules may collide in two ways.

(i) Collisions with proper orientation



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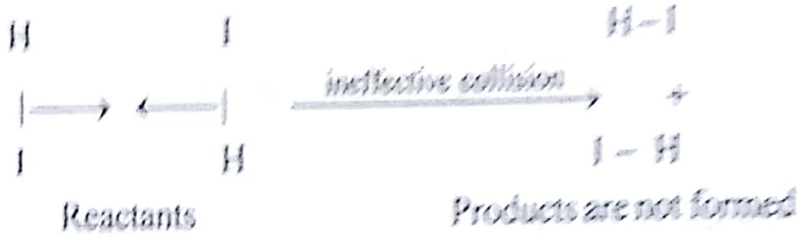
(ii) Collision with improper orientation



Thus, products are formed only in those collisions in which the colliding molecules are properly oriented and possess the threshold energy. If 'f' is the fraction of molecules possessing threshould energy and p is the probability factor (steric factor) i.e., the probability of a particular collision being favourable to reaction, the rate of the reaction is :

rate = pfZ ; But $Z \propto C$, concentration of reactants

$\therefore$ rate = pf X concentration of reactants,

or, rate = k X concentration of reactants, where k = pf.

$\therefore$ rate $\propto$ concentration of reactants

Thus, according to collision theory, for a chemical reaction to occur:

i. the molecules of the reacting substances must collide.

ii. the colliding molecules must possess threshold energy and

iii. the colliding molecules must be proprely oriented.

The effect of temperature on rates of reactions may be explained on the basis of collision theory.

According to the collision theory, the molecules of the reactants must collide for a reaction to occur. However for a collision to be effective, the reactant molecules must have threshold energy. Molecules, having energy less than this minimum energy, have to be supplied some extra energy so that these collisions become effective to form products.

The extra energy that molecules require in order to react is called the

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activation energy.

It is equal to the difference between the threshold energy needed for a collision to be effective and the average kinetic energy of the molecules.

i.e., Activation energy = Threshold energy - Average kinetic energy of the reactant molecules.

As the temperature increases, more and more molecules gain the activation energy and hence the number of molecules having threshold energy increases.

Generally, the fraction of molecules having energies equal to or greater than the threshold energy (E) is nearly doubled, by a ten degree rise in temperature. Consequently, the number of effective collisions and the rate of reaction are also approximately doubled. Thus, *the increase in the rate of reaction with increase in temperature is because of the increase in the number of effective collisions.* Base of ↘ is collision theory.

Theory of Unimolecular Reactions - Lindemann Mechanism

The collision theory can satisfactorily account for the bimolecular collisions. Thus, when two molecules A and B collide, their relative kinetic energy exceeds the *threshold energy* and as a result the collision causes the breaking of existing bonds and the formation of new bonds. When we consider a unimolecular reaction of the type, $A \rightarrow P$, the molecule A requires the necessary activation energy by colliding with another molecule, then the reaction should obey second order kinetics and not the first order kinetics. However, a number of unimolecular reactions have been studied, which were found to be of first order. For example, the decomposition of $CH_3CHO_{(g)}$ is unimolecular and follows first order kinetics.

$$CH_3CHO_{(g)} \rightarrow CH_4(g) + CO_{(g)}$$

A satisfactory theory of these reactions was proposed by *F.A. Lindemann* in 1922. He proposed that the behaviour of unimolecular reactions can be explained on the basis of bimolecular collisions. According to Lindemann mechanism:

(i) The energy required to break the bonds of a gaseous molecule is very high. This energy is acquired by a gas molecule A as a result of collisions

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with its neighbour (M). The energy gained by a molecule is thus at the cost of energy of its neighbour.

$$A + M \xrightarrow{k_1} A^* + M \text{ (Activation)}$$

where A* represents the activated molecule of the reactant and M represents another normal molecule which may be of the reactant, product or inert substance that is also present.

(ii) The activated molecules do not decompose immediately but remain in the activated form for a finite period. In other words, there may be a time lag before the decomposition of the activated molecule. During this time lag the activated molecule may get deactivated by colliding with another molecule.

$$A^* + M \xrightarrow{k_2} A + M \text{ (Deactivation)}$$

(iii) The activated molecule may decompose to give the products,

$$A^* \xrightarrow{k_3} \text{Products (Decomposition)}$$

Thus the observed rate of decomposition of a gas is the rate of appearance of the product and is given according to step (iii) as,

$$\frac{d}{dt}[\text{Product}] = k_3[A^*] \quad \text{.....(1)}$$

The overall rate may be measured experimentally but k_3 cannot be determined without knowing the concentration of activated molecules.

Steps (ii) and (iii) of the above reaction involving the disappearance of A* by deactivation or decomposition are much faster than the formation of A^* in step (i) by collision. As a consequence it is reasonable to assume that the number of activated molecules (A*) present at any time during the course of reaction would be very small and remains constant over a reasonable period of time. This is the *steady state approximation*. By applying this, the rate of formation and rate of decomposition of A* are equal, i.e.,

$$k_1[M][A] = k_2[A^*][M] + k_3[A^*]$$

$$\therefore [A^*] = \frac{k_1[M][A]}{k_2[M] + k_3} \quad \text{.....(2)}$$

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Substituting the value of [A*] from (1) in (2), we get

$$\text{Rate} = \frac{d}{dt}(\text{product}) = \frac{k_3 k_1 [M][A]}{k_2[M] + k_3} \quad \ldots\ldots(3)$$

This equation appears to be complicated but there are two limiting possibilities.

Case 1. When $k_2[M] >> k_1$, i.e., the rate of deactivation of A* is much faster than the rate of decomposition of A* to A, k_3 in the denominator may be neglected. So equation (3) becomes,

$$\text{Rate} = \frac{k_3 k_1 [A]}{k_2} = k'[A] \quad \ldots\ldots(4)$$

Thus, the reaction is first order with respect to A.

Case 2. When $k_3 >> k_2[M]$, i.e., the rate of decomposition of A is much greater than the rate of its deactivation, $k_2[M]$ in the denominator may be neglected. Now equation (3) becomes,

$$\text{Rate} = k_1[M][A] \quad \ldots\ldots(5)$$

Thus, the reaction is of second order.

In a gaseous reactions, the increase of pressure increases the number of collisions so that $k_2[M]$ is large and therefore such reactions are first order reactions.

Enzyme Catalysis

Many organic reactions are taking place in the body of animals and plants to maintain the life processes. These reactions being slow are catalysed by the organic substances called *enzymes*. Enzymes are complex protein molecules having high molar mass of the order of 10,000 or even more and are derived from living organisms. Each enzyme can catalyse a specific process. Processes catalysed by enzymes are termed *enzyme catalysis*. For example, the enzyme urease present in soyabeans catalyses the hydrolysis of urea to CO_2 and ammonia.

$$NH_2CONH_2 + H_2O \xrightarrow{\text{urease}} 2NH_3 + CO_2$$

Catalyst defⁿ in book.

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The enzyme zymase, produced by living yeast cells, in fermentation of sugar converts glucose to alcohol.

$$C_6H_{12}O_6 \xrightarrow{\text{Zymase}} 2C_2H_5OH + 2CO_2$$

The enzyme diastase produced in the germinated barley seeds converts starch into maltose.

$$2(C_6H_{10}O_5)_n + nH_2O \xrightarrow{\text{Diastase}} nC_{12}H_{22}O_{11}$$

The enzyme maltase converts maltose into glucose

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{\text{maltase}} 2C_6H_{12}O_6$$

The enzyme invertase present in yeast converts cane sugar to glucose and fructose.

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{\text{invertase}} C_6H_{12}O_6 + C_6H_{12}O_6$$

Enzyme catalysts are even more specific than inorganic catalysts. For example, urease is capable of converting only urea into ammonia and carbondioxide and cannot catalyse the similar conversion of N-methylurea. The specificity of an enzyme is explained on the assumption that the shape of the given enzyme is such that only molecules of the reactants in a specific reaction it controls, fit at the active sites on the enzyme surface. After fitting there, these reactants come in close contact and proper orientation for the reaction to proceed. (Fig.2.7)

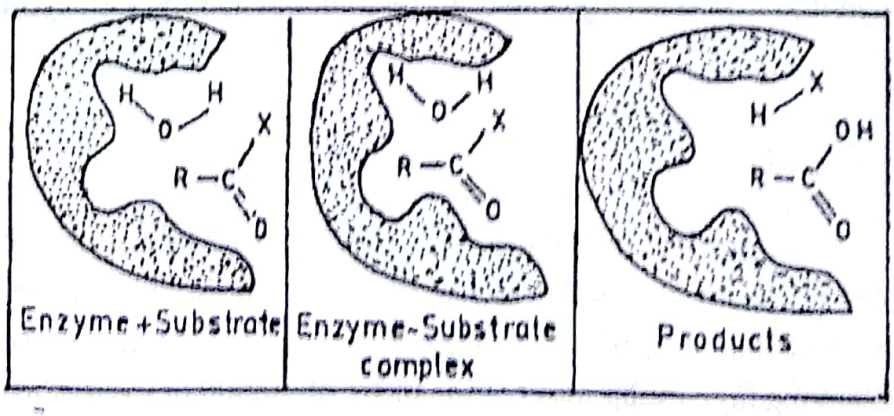

Fig.2.7 *Specificity of an enzyme which brings the reactants in close contact and proper orientation for the reaction to proceed. (a) Enzyme and substrate approach each other. (b) A complex is formed when substrate molecules fit on active sites of the enzyme surface. (c) Products separate from the enzyme after the reaction is complete.*

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The enzyme reactions are sensitive to temperature. The enzymes act most efficiently in the temperature range 298 K to 310 K. The temperature at which the reaction rate is maximum is called the *optimum temperature*. For example, the optimum temperature of enzyme reactions occurring in human body is 37^0C. At much higher temperatures, all physiological reactions will cease due to loss of enzymatic activity. This is one reason why high body temperature (i.e., fever) is very dangerous.

The rate of enzyme catalysed reaction varies with pH of the system. The rate of enzyme catalysed reactions is maximum at a particular pH called *optimum pH*. The enzyme activity is lower at all other values of pH. Thus, many enzymes of the body function best at pH of about 7.4, the pH of the blood and body fluids.

The catalytic activity of an enzyme is inhibited (reduced) or poisoned (completely destroyed) by the addition of other substances. These inhibitors or poisons interact with the active functional groups on the surface of the enzyme. For instance, heavy metal ions like Ag^+ and Hg^{2+} react with the –SH groups of the enzyme and poison it.

$$\text{Enz} - \text{SH} + \text{Ag}^+ \xrightarrow{\text{Poisoning}} \text{Enz} - \text{S} - \text{Ag} + \text{H}^+$$

The physiological activity of many drugs is related to their action as enzyme inhibitors in the body. For example, sulpha drugs, penicillin and streptomycin inhibit the action of several bacteria and are proved to be effective in curing pneumonia, dysentery, cholera and many other infections diseases.

Presence of *activators or* coenzymes greatly enhance the catalytic activity of enzymes. Activators are metal ions like Na^+, Mn^{2+}, Cu^{2+}, etc., which get weakly bonded to enzyme molecules and promote their catalytic action. For example, the addition of Na^+ (NaCl) enhances the catalytic action of *amylase*.

The presence of a *coenzyme,* a small nonprotein (vitamin) along with an enzyme often promotes the catalytic activity of the enzyme.

Mechanism of enzyme-catalysed reactions : Michaelis - Menten Mechanism

For enzyme catalysed reactions, *Michaelis* and *Menten* suggested the following mechanism. Let E be the enzyme and S be the substrate molecule

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undergoing the reaction. In the first step an intermediate complex, i.e., enzyme-substrate complex (ES) is formed.

$$\underset{\text{Enzyme}}{E} + \underset{\text{Substance}}{S} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \underset{\text{intermediate complex}}{ES} \quad \text{(fast)}$$

In the second step, the complex dissociate to give the enzyme and the product, P. That is,

$$ES \xrightarrow{k_2} E + P \quad \text{(slow)}$$

In the overall reaction $S \rightarrow P$, the enzyme is consumed in step 1 and regenerated in step 2. True equilibrium is not achieved in step 1 because the subsequent slow reaction is constantly removing the intermediate complexes. The enzyme concentration, in general, is far less than the substrate concentration. i.e., $[E] << [S]$ so that $[ES] << [S]$. Hence the steady state approximation may be used for ES. According to the slow rate determining step, the rate of the reaction is given by, $r = -\frac{d[S]}{dt} = \frac{d[P]}{dt} = k_2[ES]$ ....(1)

Using steady state approximation for ES,

$$\frac{d[ES]}{dt} = k_1[E][S] - k_1[ES] - k_{-1}[ES] = 0 \quad \text{.....(2)}$$

If $[E]_0$ is the initial concentration of the enzyme, then at any time during the course of the reaction,

$$[E]_0 = [E] + [ES] \quad \text{or} \quad [E] = [E]_0 - [ES] \quad \text{.....(3)}$$

Substituting this value of [E] from equation (3) in equation (2), we get

$$\frac{d[ES]}{dt} = k_1\{[E]_0 - [ES]\}[S] - k_{-1}[ES] - k_2[ES] = 0 \quad \text{.....(4)}$$

Thus, $k_1[E]_0[S] = \{k_{-1} + k_2 + k_1[S]\}[ES]$ .....(5)

$$\therefore [ES] = \frac{k_1[E]_0[S]}{k_{-1} + k_2 + k_1[S]} \quad \text{.....(6)}$$

Substituting this value of [ES] from equation (6) in equation (1), we get

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$$r = \frac{k_1 k_2 [E]_0 [S]}{k_{-1} + k_2 + k_1 [S]} \quad \text{.....(7)}$$

Dividing the numerator and the denominator by k_1,

$$r = \frac{k_2 [E]_0 [S]}{(k_{-1} + k_2)/k_1 + [S]} = \frac{k_2 [E]_0 [S]}{K_m + [S]} \quad \text{.....(8)}$$

Equation (8) is known as the *Michaelis-Menten Equation* or *Michaelis-Menten Law*. The constant K_m is called the *Michaelis constant*. That is, $K_m = \frac{(k_{-1} + k_2)}{k_1}$.

When all the enzyme has reacted with the substrate at high concentration, the rate of reaction will be maximum. No free enzyme will remain and hence $[E]_0 = [ES]$. Thus, from equation (1),

$r_{max} = V_{max} = k_2 [E]_0$ .....(9), where V_{max} is the maximum rate, using the notation of enzymology.

∴ The Michaelis-Menten equation may be written as,

$$r = V_{max} [S]/(K_m + [S]) \quad \text{.....(10)}$$

when $K_m >> [S]$, [S] may be neglected in the denominator such that,

$r = V_{max} [S]/K_m = k'[S]$ .....(11)

Thus, the reaction will be of first order.

When $[S] >> K_m$, K_m can be neglected in the denominator such that,

$$r = V_{max} = \text{constant} \quad \text{.....(12)}$$

Thus, the reaction will be of zero order.

When $K_m = [S]$, $r = \frac{1}{2} V_{max}$

Thus, Michaelis constant is equal to that concentration of S at which the rate of formation of product is half the maximum rate obtained at high concentration of S. The constant k_2 in equation (9) is called the *turnover number* of the enzyme. *Turnover number is the number of molecules converted in unit time by one molecule of enzyme.* The values of k_2, in general,

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are 100 to 1,000 per second though they may be as large as 10^5 to 10^6 per second.

IMPORTANT POINTS FOR QUICK REVISION

Rate of reaction : Change in concentration of any one of the reactants or products per unit time.

Rate law : Mathematical expression showing dependence of rate of reaction on concentration of reacting substance(s).

Rate constant : Rate of reaction when the molar concentration of each of the reactant is unity.

Molecularity : Number of reacting molecules which undergo simultaneous collision to produce a chemical change.

Order of reaction : Sum of the powers of the concentration terms in the rate law.

Rate law of first order reaction : $K_1 = \frac{2.303}{t} \log \frac{[A_0]}{[A]}$

Half life period of first order reaction : Time taken to reduce the concentration of a reactant to half of the initial value.

Rate law of zero order reaction : $[A] - [A]_0 = -k_0 t$ or $[A] = [A]_0 - k_0 t$

or $k_0 = \frac{[A]_0 - [A]}{t}$

Half life period of zero order reaction : $t_{½} = \frac{[A_0]}{2k_0}$

Order from $t_{½}$ value : For 1st order, $t_{½}$ = constant. For 2nd order, $t_{½} \propto \frac{1}{a}$ or $t_{½} \times a$ = constant. For nth order, $t_{½} \propto \frac{1}{a^{n-1}}$ or $t_{½} \times a^{n-1}$ = constant.

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Temperature coefficient : Ratio of rate constants of a reaction at two temperatures differing by 10^0.

Activation Energy : Extra energy required by the reactant molecules undergo the reaction.

Threshold energy : Minimum energy possessed by molecule to undergo effective collision.

Activation energy = Threshold energy – Average energy of reactant molecule.

Arrhenius equation : (Relation between rate constant and temperature) $k = e^{-E_a/RT}$.

Elementary reaction : Reaction taking place in a single step.

Complex reaction : Reaction taking place through a sequence of elementary steps.

Rate determining step : Slowest step of the reaction.

Mechanism of reaction : The sequence of elementary steps leading to the overall reaction.

Arrhenius equation may also be written as,

$$\log\frac{k_2}{k_1} = \frac{E_a}{2.303\,R}\frac{T_2 - T_1}{T_1 T_2}$$

Factors influencing rate of reaction : Concentration, temperature, nature of reactants, surface area, catalyst and radiation.

Difference between order and molecularity

	Order		Molecularity
1.	Experimental quantity	1.	Theoretical concept
2.	Sum of the powers of the concentration terms in rate law	2.	Number of species which simultaneously collide
3.	May have whole number or fractional values	3.	It has only whole number values
4.	Can be zero	4.	Cannot be zero

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MODEL QUESTIONS

Section A (Each question carries 2 marks)

I. 1. Arrhenius equation is .....
2. Half life period of a first order reaction is .....
3. Unit of second order rate constant is .....
4. The change in concentration of a reactant or a product per unit time is called .....

II. 1. Total change of concentration/time needed for the change = ...
2. The sum of the powers of the concentration terms in the rate equation is called .....
3. The number of reacting species which undergo simultaneous collision to bring about a chemical reaction is called .....
4. A reaction involving the collision of two species is called .....

III. 1. Hydrolysis of ester in the presence of a mineral acid is a ..... reaction.
2. For a third order reaction, $t_{\frac{1}{2}} \times$ ..... = a constant.
3. For a second order reaction, $t_{\frac{1}{2}} = \frac{1}{.....}$
4. K_{35}/K_{25} = ...

IV. 1. According to ..... theory, for a chemical reaction to occur, the molecules of the reacting substances must collide.
2. The extra energy that molecules require in order to react is called .....
3. Threshold energy – Average kinetic energy of the reactant molecules = ...
4. According to collision theory, for a chemical reaction to occur the colliding molecules must possess.....

Section B (Short answer type) (Each question carries 2 marks)

1. Define a zero order reaction. Give an example.
2. What are the factors influencing the rate of the reactions?
3. Give one example of second order reaction.
4. Give an example of third order reaction.

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5. Define (i) order of a reaction. (ii) Molecularity of equation.
6. How is the half life of first order reaction related to rate constant?
7. Write the expression for half life period of a zero order reaction.
8. Define molecularity of a reaction. How does it differ from order of the reaction?
9. What is the unit of rate constant of first order reaction?
10. Write down the integrated rate law of first order reaction.
11. Define half life period of a reaction.
12. Brief about activation energy.
13. Why are reactions of order higher than three are uncommon?
14. Give one example each of second order and third order reactions.
15. Distinguish between order and molecularity.
16. Explain pseudo-unimolecular reactions. Give examples.
17. Mention two general characteristics of a first order reaction. Define activation energy.
18. Define order of a reaction. How is the half-life of first order reaction related to rate constant?
19. Name an enzymes. Which is the transformation brought about by the enzyme?
20. Explain the difference between order and molecularity of a reaction.
21. Define order of a reaction. How is the half-life of first order reaction related to rate constant?
22. How does order and molecularity differ?
23. A first order reaction is found to have a rate constant $k=7.39 \times 10^{-5}$. Find the half life of the reaction.
24. Derive an expression for the half life period of a zero order reaction.
25. What is half life period of a reaction? How does it vary with the initial concentration for a first order and second order reaction?
26. A first order reaction is 20% complete in 20 minutes. How much time will it take to complete 80% of the reaction.
27. Why are reactions of order higher than 3 uncommon?
28. Define order of a reaction. For the reaction $\alpha A + \beta B \rightarrow$ Products, the rate laws was find to be rate = $k(A)^m(B)^n$. What is the overall order?

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29. Write down the rate law of zero order reaction? What is the unit of its rate constnt.
30. What is "Energy of activation"? Can it be negative? Can it be zero?

Section C (Short Essay) (Each question carries 4 marks)

1. What is the collision theory of reaction rate?
2. Derive an expression for the first order rate constant.
3. How does molecularity and order differ?
4. Define order of a reaction. How is the half life of a first order reaction related to rate constant?
5. What is half life period of a reaction? How does it vary with the initial concentration for a first order and second order reaction?
6. Why are the reactions of order higher than 3 uncommon?
7. Define order of a reaction. For the reaction $\alpha A + \beta B \rightarrow$ products. The rate law was find to be rate = $k(A)^m(B)^n$. What is the overal order?
8. Write down the rate law of zero order reaction? What is the unit of its rate constant?
9. What is "Energy of activation"? Can it be negative? Can it be zero?
10. A first order reaction is 20% complete in 50 seconds. How long will it take for the reaction to be 90% complete?
11. (a) When is a reaction said to be bimolecular? (b) A first order reaction is 50% complete in 30 seconds at a certain temperature. How long will it take for the reaction to be 99% complete at the same temperature?
12. A first order reaction is found to have a rate constant, $k = 7.39 \times 10^{-5}$. Find the half life of the reaction.
13. A first order reaction is 20% complete in 50 seconds. How long will it take for the reaction to be 90% complete?

Section D (Long Essay) (Each question carries 7 marks)

1. (a) Derive an expression for rate constant of a first order reaction.
   (b) Explain the influence of temperature on reaction rates.
2. (a) Give the important characteristics of a second order reaction.
   (b) Derive an expression for the half life of a zero order reaction.

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Module 3

A. ELECTROCHEMISTRY

Electrochemistry is the branch of Chemistry which deals with the study of the relationship between chemical energy and electrical energy and the interconversion of one form to another.

Electrolytes

Aqueous solutions of certain substances conduct electricity while those of some others do not conduct electricity. On this basis, *Michael Faraday*, in 1824, classified substances into *electrolytes* and *nonelectrolytes*. *An* ***electrolyte*** *is defined as a compound whose aqueous solution or melt conducts electricity.* eg., sodium chloride. On the other hand, *a compound whose aqueous solution or melt does not conduct electricity is called a* ***nonelectrolyte***. eg., glucose.

When an electrolyte is dissolved in water, it splits up into ions. However, all the electrolytes do not ionize to the same extent. Depending on the extent of ionisation, the electrolytes are divided into two classes : *strong electrolytes and weak electrolytes.*

(i) **Strong electrolytes.** The substances which ionise almost completely into ions in aqueous solution are called *strong electrolytes*. For example, HCl, H_2SO_4, NaOH, KOH, KNO_3, etc., are strong electrolytes.

(ii) **Weak electrolytes :** The substances which ionise to a small extent in aqueous solution are called *weak electrolytes*. For example, CH_3COOH, NH_4OH, $(NH_4)_2CO_3$, HCN, etc., are weak electrolytes.

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Ionisation of Weak Electrolytes - Ostwald's Dilution law

We know that the strong electrolytes are almost completely ionised. However, the weak electrolytes are only partially ionised and a dynamic equilibrium is established between the unionised molecules and the ions in solution. This is called ***ionic equilibrium***. Thus,

the ionic equilibrium is the equilibrium which is established between the unionised molecules and the ions in solution of weak electrolytes.

Degree of Ionisation or Dissociation

The fraction of the total number of molecules of an electrolyte which ionise into ions is called ***degree of ionisation or degree of dissociation.*** It is generally denoted by the symbol α (alpha). Thus

Degree of ionisation,

$$\alpha = \frac{\text{number of molecules of the electrolyte which ionise}}{\text{total number of molecules of the electrolyte}}$$

Let us consider the ionisation of a weak electrolyte AB in water. Let the initial concentration of the electrolyte be c and the degree of ionisation be α. Thus, the concentrations of various species at equilibrium will be as shown below.

$$AB_{(aq)} \rightleftharpoons A^+_{(aq)} + B^-_{(aq)}$$

Ionic conc.	c	o	o
Equilibrium conc.	$c(1-\alpha)$	$c\alpha$	$c\alpha$

By the law of chemical equilibrium,

$$K = \frac{[A^+][B^-]}{[AB]} = \frac{c\alpha \cdot c\alpha}{c(1-\alpha)} = \frac{c\alpha^2}{1-\alpha}$$

If one mole of an electrolyte be dissolved in V litre of the solution, then C=1/V, where V is known as the dilution for the solution. Thus, the above expression becomes

$$K = \frac{\alpha^2}{(1-\alpha)V}$$

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This expression which correlates the variation of the degree of dissociation of an electrolyte with dilution is called ***Ostwald's dilution law.***

For weak electrolyte, α is very small as compared to 1. Therefore, 1-α may be taken approximately equal to 1. Then, the Oswald's dilution law expression becomes

$$K = c\alpha^2 \quad \text{or} \quad \alpha^2 = \frac{K}{c} \qquad \therefore \ \alpha = \sqrt{\frac{K}{c}} \quad \text{or} \quad \alpha \propto \sqrt{\frac{1}{c}}$$

$$\text{or } K = \frac{\alpha^2}{V} \quad \text{or} \quad \alpha^2 = KV \quad \text{or} \quad \alpha = \sqrt{KV} = K'\sqrt{V} \quad \text{or} \quad \alpha \propto \sqrt{V}$$

Thus, the ***degree of ionisation is inversely proportional to the square root of concentration or directly proportional to the square root of dilution.*** In other words, ***as the concentration decreases or dilution increases, the degree of ionisation increases.*** That is, as ***c approaches zero or dilution approaches infinity, the degree of ionisation approaches unity.*** This generalisation is called ***Ostwald's dilution law.***

Limitations of Ostwald's Law

Ostwald's dilution law holds good only for weak electrolytes and fails completely when applied to strong electrolytes.

Electrolysis

The phenomenon of decomposition of an electrolyte by the passage of electric current is called ***electrolysis.***

The device used to carry out electrolysis is called ***electrolytic cell.***

An electrolytic cell consists of an electrolytic solution and two metal electrodes connected to a battery. The electrode connected to the positive terminal of the battery is called *anode* while the electrode connected to the negative terminal of the battery is called *cathode*. During electrolysis positive ions migrate to the negative electrode (cathode) and negative ions migrate to the positive electrode (anode). The anions give up electrons to the anode and are discharged. This process is called *anodic oxidation.* The cations accept electrons and are discharged at the cathode. This process is called *cathodic reduction.*

Electrolysis of molten electrolytes. Consider the electrolysis of molten NaCl. Molten NaCl contains free Na^+ and Cl^- ions. When it is electrolysed, sodium is deposited at the cathode and Cl_2 is liberated at the anode. The reactions are:

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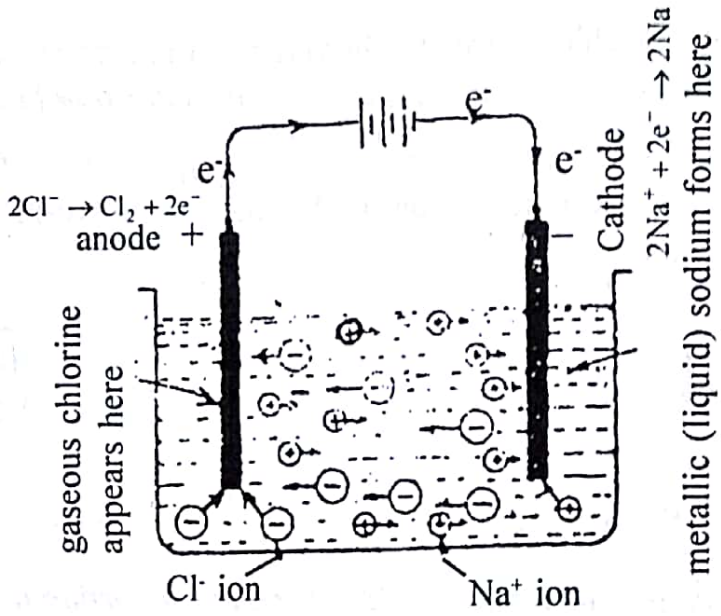

Fig. 3.1 *Electrolysis of molten NaCl*

At anode : $2Cl^- \rightarrow Cl_2 + 2e^-$

At cathode : $2Na^+ + 2e^- \rightarrow 2Na$

The overall reaction is,

$$2Na^+ + 2Cl^- \rightarrow 2Na + Cl_2$$

Electrolysis of aqueous solution. Consider the electrolysis of aqueous NaCl. Aqueous NaCl contains Na^+, Cl^-, H^+ and OH^- ions. During electrolysis of the solution, Cl_2 is liberated at the anode and hydrogen at the cathode and NaOH is formed in the solution. The reactions involved are,

At anode : $2Cl^- \rightarrow Cl_2 + 2e^-$ (oxidation)

At cathode : $2H_2O + 2e^- \rightarrow 2OH^- + H_2$ (reduction)

The overall reaction is, $2Cl^- + 2H_2O \rightarrow 2OH^- + H_2$

Faraday's laws of electrolysis

Michael Faraday discovered a quantitative relationship between the extent of electrolysis and the quantity of electricity passed.

Faraday's First Law. *It states that the mass (m) of the substance deposited or liberated at an electrode is directly proportional to the quantity of*

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electricity passed.

i.e., $m \propto Q$. But, $Q = c \times t$ where c is the current in amperes and t the time in seconds.

$\therefore m \propto ct$ or $m = zct$, where z is a constant called ***electrochemical equivalent.***

When c = 1 ampere and t = 1s or Q = 1 coulomb

$\therefore m = z$

Thus,

> *electrochemical equivalent of a substance is the mass of the substance liberated or deposited by 1 ampere current in 1 second or 1 coulomb of electricity.*

Faraday's Second Law. *It states that when the same quantity of electricity is passed through solutions of different electrolytes, the masses of the substances liberated at the electrode are in the ratio of their equivalent masses.*

$$\text{i.e., } \frac{\text{Mass of A deposited}}{\text{Mass of B deposited}} = \frac{\text{Equivalent mass of A}}{\text{Equivalent mass of B}}$$

Faraday. *The quantity of electricity required to liberate or deposit one gram equivalent of a substance is called a Faraday (F).* It is found to be equal to 96500 C.

Example 3.1 Calculate the quantity of electricity required to deposit 6.345 g of Cu.

Solution : Quantity of electricity required to deposit 63.54 g Cu = 2F = 2×96500 C

$$\text{Quantity of electricity required to deposit 6.354g Cu} = \frac{2 \times 96500 \times 6.354}{63.54}$$

= 19300 C

Example 3.2 How much Ag is deposited by 965 coulombs of electricity?

Solution : 96500 C deposit Ag = 108 g

$$\therefore \text{965 C deposits} = \frac{108 \times 965}{96500} = 1.08 \text{ g}$$

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Self Study Problems

3.1 Calculate the quantity of electricity required to deposit 10.8 g of Ag from silver nitrate solution. Ans. 9650 C

THEORY OF STRONG ELECTROLYTES

To be explained with fig.

The Interionic Attraction Theory : Debye - Huckel Theory

The strong electrolytes which exist as ions even in the solid state must be completely ionised in solution at all concentrations. This means that in a solution containing one gram equivalent of electrolyte, the number of ions would be the same at all concentrations. And yet the equivalent conductance increases with dilution. This is due to the change in the ionic speed which alters with dilution. At ordinary concentration, the ions are in close proximity and hence there are interionic forces which render them unable to move with a speed as would otherwise be possible with dilution, the inter ionic forces decreases and hence the conductance increases.

In 1923, *Peter Debye* and *E. Hückel* treated the interionic effects for a solution carrying an electric current quantitatively. This is known as ***Debye-Huckel theory of interionic attraction***. The treatment was later modified in some respects by *Lars Onsager* (1927).

The basic principle of this theory is that as a result of electrostatic attraction between oppositely charged ions, each ion in solution is surrounded by an *ionic atmosphere* of opposite charge. In the absence of an external applied electric field, the ionic atmosphere is spherically and symmetrically distributed as shown in Fig. 3.2(a). On applying an electric field, the central ion moves in one direction and the ionic atmosphere in the opposite direction. Thus, a central positive ion tends to move towards the cathode on the right while its negative ionic atmosphere tends to move towards the anode on the left. The ionic atmosphere of the oppositely charged ions around the central ion does not have time to adjust itself to remain spherically symmetrical around the central ion and it lags behind. (Fig. 3.2(b)). As a consequence the central ion is retarded in its motion by the atmosphere, for the negative charges behind would be greater than those in front of the central ion. The force causing

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retardation will be in a direction opposite to that of the motion of the central

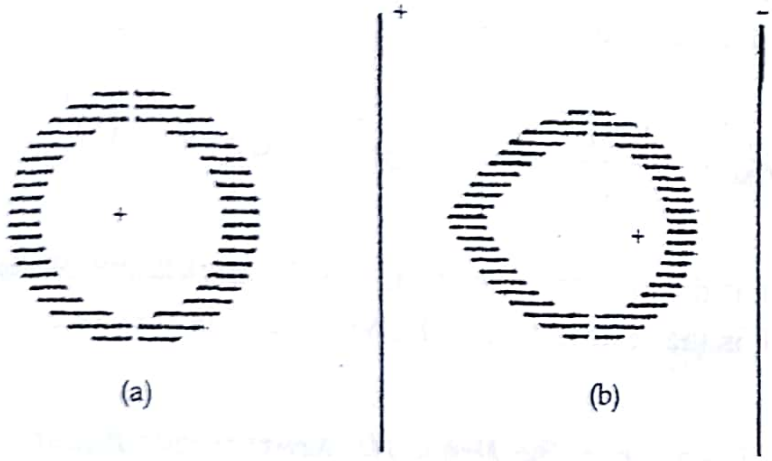

Fig.3.2 *Ionic atmosphere. (a) Field off (b) Field on*

ion. The ionic atmosphere during the motion of the central ion becomes unsymmetrical. This effect on the velocity (i.e., the drag on the moving ion) of an ion is called the ***asymmetry effect*** or ***relaxation effect.***

There is another factor which opposes the speed of an ion and is responsible for lowering the conductivity. This is the tendency of the ionic atmosphere associated with the solvent molecule to move in a direction opposite to that of the ion. Thus, the cations have to move through a medium moving with the negative ions towards the anode. In the same way, anions have to move through the solvent carrying cations in the opposite direction. Thus, the moving ions are forced to swim upstream against the flow of solvent. As a result of this, the speed of the ion decreases. This retarding effect on the velocity of the moving ion is called the ***electrophoretic effect***.

The third factor which affects the mobility of ions is the ***viscous effect*** caused by the viscous drag of the solvent on the movement of ions. The ion tends to move in the direction of the applied electric field. The electric force is opposed by the frictional viscous drag exerted by the solvent. As a consequence of this the velocity of the ion decreases. The greater the viscosity of the solvent, the greater is the viscous drag and hence the smaller is the ionic mobility and the conductance.

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No need to study derivation or equation Conly theory)

Debye, Hückel and *Onsager* showed that these retarding effects on an ion produce a decrease in molar conductance and depend on its concentration. For a uni-univalent electrolyte like KCl, the following equation was derived by *Debye, Hückel* and *Onsager,*

$$\lambda_m = \lambda_M^0 - \left[\frac{82.4}{(\varepsilon T)^{1/2}\eta} + \frac{8.20\times 10^5 \lambda_M^0}{(\varepsilon T)^{3/2}}\right]\sqrt{C} \quad \text{.....(1)}$$

where ε is the dielectric constant, η is the coefficient of viscosity of the medium, T is the temperature in Kelvin and C is the concentration of the solution.

Eq. (1) is known as the ***Debye-Hückel-Onsager Equation.*** From this equation it is evident that λ_M is less than λ_M^0. This is partly due to electrophoretic effect and partly due to asymmetry effect. The first term in bracket gives a measure of electrophoretic effect while the second term gives a measure of asymmetry effect. For a given solvent at a given temperature, we may write the equation as,

$$\lambda_M = \lambda_M^0 - [A + B\lambda_M^0]\sqrt{C} \quad \text{.....(2)}$$

where A and B which replace the constant quantities within the bracket of equation (1) are called *Debye-Hückel constants*. In aqueous solution at 25^0C, A = 60.2 and B = 0.229.

Therefore in aqueous solution at 25^0C the equation will become,

$$\lambda_m = \lambda_M^0 - (60.2 + 0.229\lambda_M^0)\sqrt{C} \quad \text{—(3)}$$

The validity of the equation is tested by plotting the experimental values of λ_M against $\sqrt{C}$. If the equation is correct, a straight line of slope equal to (60.2 +

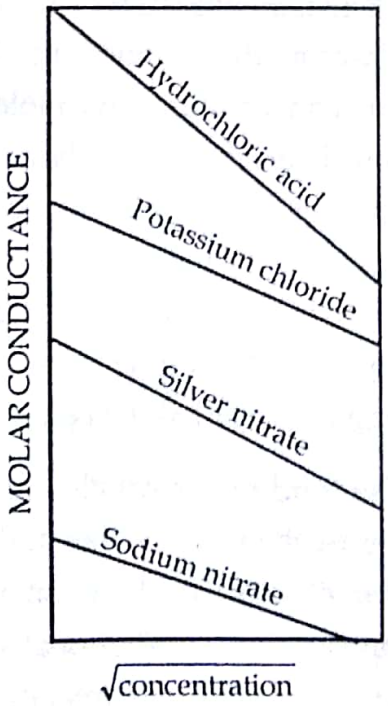

Fig.3.3 *Verification of Debye-Huckel equation*

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$0.229\ \lambda_M^0$) should be obtained (Fig.3.3). This has been found to be true upto concentrations of the order of 0.02 M. However in more concentrated solutions the conductance is higher than that predicted by the Debye-Huckel-Onsager equation. If a solution is at infinite dilution, i.e., if c is almost zero, then the second term on the right hand side of the equation may be neglected. As a result, λ_M approaches λ_M^0 as expected.

ELECTROMOTIVE FORCE

Chemical changes involving production or consumption of electrical energy are called *electrochemical changes.* A cell is an electrochemical device for the conversion of electrical energy into chemical energy or viceversa. The cells in which energy consuming chemical changes take place are known as *electrolytic cells* whereas the cells in which energy producing chemical changes take place are known as *galvanic* or *voltaic cells*.

Galvanic cell

A galvanic cell or voltaic cell is a device which converts chemical energy into electrical energy.

A galvanic cell consists of two electrodes combined in such a way that when they are connected by a metallic wire, an electric current will flow through it. Such an electrode consists of a metallic conductor in contact with an electrolytic conductor. When a metal rod is dipped in a solution containing its own ions, an equilibrium will be established between the metal and the metal ions. Due to this equilibrium, an electric potential is attained by the metal with respect to the solution. Such an arrangement in which a metal is in equilibrium with its own ions is called a metal electrode. When two such metal electrodes are connected through a porous partition, a galvanic cell is formed.

Daniell Cell : It is a typical example of a galvanic cell. It consists of a Zn rod dipped in $ZnSO_4$ solution (1M) and a Cu rod dipped in $CuSO_4$ solution (1M). The Zn and Cu rods are called *electrodes* and these are connected by a metallic wire through a high resistance voltmeter or an ammeter which is used to detect the passage of current. The two solutions are connected by a salt bridge, an inverted U tube containing an inert electrolyte like KCl, KNO_3 or NH_4NO_3. As soon as the two electrodes are connected and the salt bridge

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To be explained by drawing diag. and oxan & redn reactn

is placed in position as in Fig.3.4, Zn starts to dissolve in solution due to oxidation.

$$Zn \rightarrow Zn^{++} + 2e^{-} \text{ (oxidation)}$$

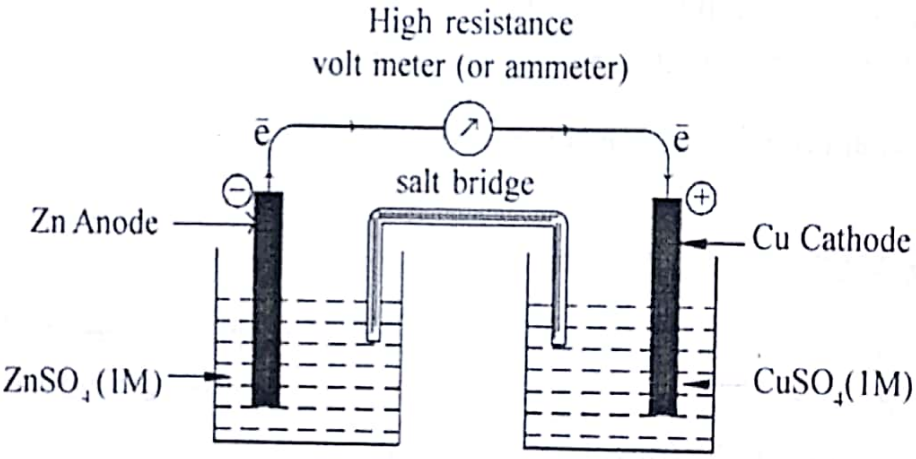

Fig.3.4. *Daniell Cell*

The released electrons accumulate on the Zn rod, hence it becomes negatively charged. These electrons move through the external circuit to the Cu rod. The Cu^{++} ions in contact with Cu electrode receive these electrons and get reduced to Cu atoms which are deposited on Cu electrode.

$$Cu^{++} + 2e^{-} \rightarrow Cu \text{ (reduction)}$$

Due to the consumption of electrons, the Cu rod becomes positively charged.

The electrode at which reduction occurs is called *cathode* and the electrode at which oxidation occurs is called *anode*. Thus, in Daniell cell, Zn acts as the *anode* and Cu acts as the cathode.

An electrochemical cell consists of two half-cells. For example, in Daniell cell, the Zn rod placed in $ZnSO_4$ solution constitutes one half-cell, whereas the other half-cell is Cu rod dipped in $CuSO_4$ solution. The reaction taking place in a half-cell is called *half-cell reaction*. The half-cell reaction in which electrons are released is called *oxidation half reaction* or *anode half reaction*. On the other hand, the half-cell reaction in which electrons are gained is called *reduction half reaction* or *cathode half reaction*. On combining the two half reactions, the cell reaction is obtained. Thus in Daniell cell,

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$Zn \rightarrow Zn^{++} + 2e^-$ (oxidation half reaction)

$Cu^{++} + 2e^- \rightarrow Cu$ (reduction half reaction)

$Zn + Cu^{++} \rightarrow Zn^{++} + Cu$ (cell reaction, i.e., redox reaction)

In a galvanic cell, anode is the negatively charged electrode whereas cathode is the positively charged electrode. To sum up, in a galvanic cell:

1. Chemical energy is converted into electrical energy.

2. Electrons flow from anode to cathode.

3. Anode is the electrode at which oxidation occurs; it is the -ve electrode.

4. Cathode is the electrode at which reduction occurs; it is the +ve electrode.

Salt bridge and its functions

A salt bridge is an inverted U tube containing a concentrated solution of an inert electrolyte like KCl, KNO_3, NH_4NO_3, etc., or solidified solution of such an electrolyte in agar-agar.An inert electrolyte is one whose ions do not take part in the redox reaction and also do not react with the electrolyte used. The mobilities of the cation and the anion of the electrolyte should be nearly the same. The important functions of the salt bridge are: (i) It eliminates the liquid junction potential (i.e., the potential difference at the junction of two solutions.)

(i) It completes the electric circuit with out mixing the two solutions in the half-cells.

(ii) It maintains the electrical neutrality of the solutions in the two half-cells.

Reversible and Irreversible Cells

An electrochemical cell may be *reversible* or *irreversible*. In a ***reversible*** cell, the reactions are always in a state of equilibrium and only infinitesimally small amount of currents are drawn from it. When such a cell is connected to an external source of emf (driving force) exactly equal to that of the cell (opposing force), no chemical reaction should take place within the cell. When the driving force is decreased by an infinitesimally small amount, then the current should flow from the cell to the external source. On the other hand, if

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the driving force is increased by an infinitesimally small amount then, the current should flow from the external source to the cell in the opposite direction and the chemical reaction in the cell should be reversed. If the above conditions are not satisfied, the cell is said to be ***irreversible***.

Daniell cell is a typical example of a reversible cell. When the two electrodes are connected to an external source of emf infinitesimally smaller than the emf of the cell, the current flows from zinc electrode to copper electrode and the cell reaction is,

$$Zn + Cu^{++} \rightleftharpoons Cu + Zn^{++}$$

When the external emf is infinitesimally greater than that of the cell emf the current flows in the opposite direction and the cell reaction is reversed. If the external emf is exactly equal to that of the opposing emf no current flows and no reaction occurs.

A cell consisting of zinc and copper electrode dipped into a solution of sulphuric acid is an example of irreversible cell. When the cell is connected to an external source of emf infinitesimally smaller than that of the cell emf, zinc dissolves in sulphuric acid to form zinc sulphate with the evolution of hydrogen gas as,

$$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$$

When the external emf is infinitesimally greater than that of the cell emf, copper dissolves in sulphuric acid to form copper sulphate with the evolution of hydrogen gas as,

$$Cu + H_2SO_4 \rightarrow CuSO_4 + H_2$$

From these reactions it is evident that the cell reaction is not reversed; the two reactions give different sets of products. Thus, the cell is not reversible. In other words, it is an irreversible cell.

Cell diagram or Representation of a cell

In 1953, IUPAC recommended the following conventions for writing cell diagrams.

i) To represent the anode, write the metal or solid phase first and then the metal ion or electrolyte.

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ii) To represent the cathode, write the ion or electrolyte first and then the metal or solid phase.

iii) The interface across which a potential difference exists is represented by a single vertical line (|) or semicolon (;). Thus, the two half-cells in a Daniell cell are indicated as,

$Zn \mid Zn^{++}$ or $Zn; Zn^{++}$ (anode half-cell)

$Cu^{++} \mid Cu$ or $Cu^{++}; Cu$ (cathode half-cell)

iv) Sometimes two materials taken together constitute the electrode. It is represented by a comma (,). For example, in the half-cell in which Ag and AgCl constitute an electrode, we can write $Ag,AgCl \mid Cl^{-}$.

v) The symbol for an inert electrode is often enclosed in a bracket. For example, in $H^{+} \mid H_2(Pt)$,Pt is the inert electrode.

vi) The salt bridge which separates the electrolytic solutions of half-cells is indicated by two vertical lines (||). For example, the salt bridge separating the two solutions containing Zn^{++} and Cu^{++} ions in Daniell cell is represented as: $Zn^{++} \parallel Cu^{++}$.

vii) Anode half-cell is written on the left and cathode on the right.

By using the above convention, the Daniell cell may be represented as either,

$Zn \mid Zn^{++} \parallel Cu^{++} \mid Cu$ or $Zn; Zn^{++} \parallel Cu^{++}; Cu$

Single Electrode potential

When a metal is in contact with a solution containing its ions, a potential difference is created at the metal-solution interface. Metal has a tendency to lose electrons and go into solution. At the same time ions in the solution have a tendency to gain electrons and get converted into the metal atoms. For any system one of these tendencies will pre-dominate. If the metal has greater tendency to get oxidized (i.e., to lose electrons), its atom will loss electrons readily and form positive ions which go into solution. The electrons lost on the electrode would be accumulated on the metal electrode and the electrode acquires a slight negative charge with respect to the solution. This results in separation of charges. Similarly, if the ions have a greater tendency to get

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reduced, they will take electrons from the electrode. As a result, a net positive charge will be developed on the electrode with respect to the solution. This will also result in the separation of charges.

Due to separation of charges between the electrode and the solution, an electric potential is set up between the metal electrode and its solution.

> *The potential difference between a metal electrode and the solution containing its ions surrounding the electrode is called the* ***electrode potential.***

Thus, the electrode potential is a measure of the tendency of an electrode to gain or lose electrons. The potential of one half-cell is called the ***single electrode potential***. The electrode potential may be of two types:

1) **Oxidation potential.** *The tendency of an electrode to lose electrons or to get oxidised is called its* ***oxidation potential.***

i.e., $M(s) \rightarrow M^{n+}(aq) + ne^-$

ii) Reduction potential. *The tendency of an electrode to gain electrons or to get reduced is called its* ***reduction potential.***

i.e., $M^{n+}_{(aq)} + ne^- \rightarrow M_{(s)}$

It is evident that the oxidation potential is the reverse of the reduction potential. For example, if the reduction potential of Zn is-0.76 volt, its oxidation potential is +0.76 volt.

> **To sum up:** *The tendency of an electrode to gain or lose electrons when it is in contact with its own ions is called* ***electrode potential.***

It is not possible to determine experimentally the potential of a single electrode (half cell). It is only the difference of potentials between two electrodes that we can measure by combining them to a complete cell. By arbitrarily fixing the potential of one electrode as zero, numerical values may be assigned to potentials of other electrodes. Accordingly, the potential of a reversible hydrogen electrode, i.e., the electrode in which hydrogen gas is bubbled through a solution of hydrogen ions of unit activity (1M) has been fixed as zero. This electrode is called ***standard hydrogen electrode (SHE)*** which may be represented as,

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$$Pt; H_{2(g)}(1atm), H^+_{(aq)} (c = 1M)$$

Other single electrode potentials measured with respect to SHE are called potentials on the hydrogen scale.

Suppose it is required to determine the electrode potential of zinc electrode, i.e., Zn/Zn^{++} electrode. For this, the zinc electrode is combined with the SHE to have a complete cell as,

$$Zn/Zn^{++}{}_{solution}; H^+_{(aq)} (c= 1M)/H_{2(g)} (1 atm); Pt$$

The emf of the cell is determined potentiometrically which is equal to the potential of the zinc electrode in the hydrogen scale since the potential of SHE is zero. However it is more convenient to use a better reference electrode like calomel electrode for measuring electrode potential.

Sign of Electrode Potential

According to the IUPAC convention, the electrode potential is given a *positive sign if the electrode reaction involves reduction* (i.e., taking up electrons from the electrode) when connected to the standard hydrogen electrode and a *negative sign if the electrode reaction involves oxidation* (i.e., liberation of electrons) when connected to the standard hydrogen electrode whose potential is taken as zero.

Thus, when copper electrode is connected with a standard hydrogen electrode, reduction takes place at the copper electrode.

The electrode reactions are:

Copper electrode : $Cu^{2+}_{(aq)} + 2e^- \rightleftharpoons Cu_{(s)}$ .... Reduction

Hydrogen electrode : $H_{2(g)} \rightleftharpoons 2H^+_{(aq)} + 2e^-$ .... Oxidation

Hence, according to the above convention, the potential of copper electrode is taken as *positive*. Thus, $E_{Cu^{2+}/Cu}$ is positive.

However, if zinc electrode is connected with the standard hydrogen electrode, oxidation takes place at the zinc electrode.

The electrode reactions are as follows:

Zinc electrode : $Zn(s) \rightleftharpoons Zn^{2+} + 2e^-_{(aq)}$ .... Oxidation

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Hydrogen electrode : $2H^+_{(aq)} + 2e^- \rightleftharpoons H_{2(g)}$ ... Reduction.

Hence the potential of the zinc electrode is taken as *negative*. Thus, $E_{(Zn^{2+}, Zn)}$ is negative.

E.M.F or Cell Potential

An electrochemical cell consists of two electrodes, i.e., two half-cells. When two half-cells are connected externally there is a potential difference between them. This difference in potential leads to a flow of current (i.e., electrons) from the electrode of lower reduction potential (higher oxidation potential) to the electrode of higher reduction potential (i.e., lower oxidation potential) through the external circuit. Thus, the motive force (i.e., driving force) for the flow of electric current is the potential difference and is called electromotive force (EMF).

The difference in potential between the two electrodes in a cell is called electromotive force or cell potential.

It is measured with a voltmeter and its unit is volt. It is given as,

EMF of cell = Reduction potential of cathode-Reduction potential of anode.

or EMF of cell= Oxidation potential of anode- Oxidation potential of cathode.

or EMF of cell = Oxidation potential of anode + reduction potential of cathode.

Standard electrode potential

When an electrode is in contact with its ions having unit activity (1M) at standard state (i.e., at 298K and 1 atm), it is called a standard electrode. The potential of a standard electrode is called *standard electrode potential* and is usually represented as E^0. Similarly, the EMF of a cell under standard conditions is called the standard EMF of the cell. The standard EMF of a cell can be calculated from the standard electrode potentials of the two electrodes.

i.e., Standard EMF=E^0 right -E^0 left

where E^0 right = reduction potential of cathode, i.e., right hand electrode and E^0 left = reduction potential of anode, i.e., left hand electrode.

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Example 3.3: The standard reduction potential of Cu is 0.34V and standard oxidation potential of Zn is 0.76V. Calculate the standard EMF of the cell,

$$Zn|\ Zn^{++}{}_{(1M)} \parallel Cu^{++}{}_{(1M)}\ |\ Cu$$

Solution: Standard reduction potential of Cu electrode =0.34V.

Standard oxidation potential of Zn electrode= 0.76V

$\therefore$ Standard reduction potential of Zn electrode =−0.76V

$\therefore$ Standard EMF $= E^0_{right} - E^0_{left} = E^0_{Cu} - E^0_{Zn} = 0.34V - (-076V)$

$= 0.34V + 0.76V = \mathbf{1.10V.}$

Self Study Problems

3.2. The standard electrode potential of Zn and Ag are - 0.80V and 0.76V respectively. Calculate the standard EMF of the cell,

$Zn|Zn^{++}||Ag^{+}|Ag$ **Ans :** 1.56V

Electro Chemical Series

A table in which electrodes are arranged in the increasing order of their standard electrode potentials is called electrochemical series or activity series. (Table 3.1)

Applications of electrochemical series

(i) Calculation of standard EMF of cells. From the electrochemical series, it is possible to calculate the standard potentials of the cells.

$$E^0_{cell} = E^0_{right} - E^0_{left}$$

(ii) To compare the reactivity of metals. The greater the value of E^0 of an electrode lesser will be its electron releasing capacity. Thus *any metal having lower reduction potential can displace a metal having higher reduction potential*. In other words, *any metal in the series can displace metals lying below it from their solutions*. For example, zinc displaces copper. Similarly, copper can displace silver.

(iii) To predict whether a metal can displace hydrogen from acids. Metals which lie above hydrogen in the electrochemcial series i.e., *metals having negative electrode potentials can liberate hydrogen from acids*. Thus, metals

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like Zn, Mg, etc., can displace hydrogen from acids while metals like Cu, Ag, etc., cannot liberate hydrogen.

(iv) To predict the feasibility of a redox reaction. In the given equation the substance undergoing oxidation will act as the anode and the substance undergoing reduction will act as the cathode. EMF of the cell is then calculated as, $E^0_{cathode} - E^0_{anode}$. If the EMF comes out to be positive the redox reaction is feasible. If the EMF comes out to be negative the redox reaction is not feasible.

Table 3.1. *Electrochemical series (Standard electrode potentials at 298 K)*

Electrode	Electrode reaction	E^0
Li, Li^+	$Li^+ + e^- \rightarrow Li$	-3.05
K, K^+	$K^{++} + e^- \rightarrow K$	-2.93
Ba, Ba^{2+}	$Ba^{2+} + 2e^- \rightarrow Ba$	-2.90
Ca, Ca^{2+}	$Ca^{2+} + 2e^- \rightarrow Ca$	-2.87
Na, Na^+	$Na^+ + e^- \rightarrow Na$	-2.71
Mg, Mg^{2+}	$Mg^{2+} + 2e^- \rightarrow Mg$	-2.37
Al, Al^{3+}	$Al^{3+} + 3e^- \rightarrow Al$	-1.66
Zn, Zn^{2+}	$Zn^{2+} + 2e^- \rightarrow Zn$	-0.76
Fe, Fe^{2+}	$Fe^{2+} + 2e^- \rightarrow Fe$	-0.44
Pb, $PbSO_4$, SO_4^{2-}	$PbSO_4 + 2e^- \rightarrow Pb + SO_4^{2-}$	-0.31
Co, Co^{2+}	$Co^{2+} + 2e^- \rightarrow Co$	-0.28
Ni, Ni^{2+}	$Ni^{2+} + 2e^- \rightarrow Ni$	-0.25
Sn, Sn^{2+}	$Sn^{2+} + 2e^- \rightarrow Sn$	-0.14
Pb, Pb^{2+}	$Pb^{2+} + 2e^- \rightarrow Pb$	-0.13
Pt, H_2; H^+	$2H^+ + 3e^- \rightarrow H_2$	+0.00
Cu; Cu^{2+}	$Cu^{2+} + 2e^- \rightarrow Cu$	+0.34
Ag; Ag^+	$Ag^+ + e^- \rightarrow Ag$	+0.80
Cl^-; Cl_2, Pt	$Cl_2 + 2e^- \rightarrow 2Cl^-$	+1.36
Au; Au^{3+}	$Au^{3+} + 2e^- \rightarrow Au$	+1.36
F^-; F_2, Pt	$F_2 + 2e^- \rightarrow 2F^-$	+2.87

Increasing (i) tendency for oxidation to occur ⟶

Increasing (i) tendency for reduction to occur ⟶

Example 3.4 Calculate the EMF of the cell, $Zn \mid Zn^{++} \| Cu^{++} \mid Cu$ at 298 K. Given $E^0_{Zn} = -0.76V$, $E^0_{Cu} = +0.34$ V. What is the cell reaction? Is the reaction

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spontaneous?

Solution: $E^0_{Cell} = E^0_{right} - E^0_{left} = 0.34 - (-0.76\ V) = 1.1\ V$

Here, E is positive, hence the cell reaction will be spontaneous. The cell reaction is $Zn + Cu^{++} \rightarrow Zn^{++} + Cu$

Example 3.5 Write the cell reaction for the cell, $Zn \mid Zn^{++}_{(1M)} \parallel Pb^{++}_{(1M)} \parallel Pb$. Calculate the standard EMF for the cell. Given $E^0Pb = -0.126$ V and $E^0_{Zn} = -0.763$ V. Is the reaction spontaneous?

Solution. Cell reaction. $Zn + Pb^{++} \rightleftharpoons Zn^{++} + Pb$

$E^0_{Cell} = E^0_{Pb} - E^0_{Zn} = -0.126\ V-(-0.763\ V) = 0.637\ V$

Since the E_{cell} is positive, the reaction will be spontaneous.

Self Study Problems

3.3 Calculate the EMF of the cell, $Zn \mid Zn^{++}_{(1M)} \parallel Cr^{3+}_{(1M)} \mid Cr$. Write the half cell reactions and the cell reactions. Is the reaction spontaneous?

Given $E^0_{Zn} = 0.76$ V, $E^0_{Cr} = -0.74$ V.
Ans. $E_{Cell} = 0.02$V. Reaction will be spontaneous.

3.4 Calculate the cell potentials and predict the feasibility of the cell reactions?

(i) $Cu \mid Cu^{2+}_{(1M)} \parallel Br^-_{(1M)} \mid Br_2$ (1 atm), Pt. Given $E^0_{Cu} = 0.34$ V; $E^0Br_2 = +1.08$ V

(ii) $Ni \mid Ni^{2+} \parallel Mg^{2+} \mid Mg$. Given $E^0_{Mg} = +2.37$V; $E^0_{Ni} = 0.25$ V.

Ans. (i) $E^0_{Cell} = 0.74$ V; (ii) $E^0_{Cell} = -2.12$ V

Electrical Energy in a Galvanic Cell

The electrical energy produced by a galvanic cell is equal to the product of the electromotive force (emf) and the amount of electricity which passes. For example, in the case of Daniell cell the emf is 1.1 volts. The cell reaction involves liberation of 2 electrons by zinc and taking up of 2 electrons by Cu^{++}, i.e., dissolution of 2 gram equivalents of zinc and deposition of 2 gram equivalent of copper. Thus, the amount of electricity produced is 2 faradays

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(Faraday's second law), i.e., 2 × 96500 coulombs. Thus, the amount of electrical energy produced for the complete cell reaction

= 2 × 96500 coulombs × 1.1 volts = 212300 volt. coulombs

= 212300 Joules.

Electrical Energy and Free Energy change of the Cell Reaction

Towards the close of the nineteenth century, it was pointed out by *Gibbs*, and independently by *Helmholtz*, that the electrical energy of a reversible cell originates from the decrease of free energy ($-\Delta G$) of the cell reaction.

Let us consider in a particular cell reaction n electrons are liberated at one electrode and n electrons are taken up at the other electrode. Then, the amount of electricity produced by the complete cell reaction is n faradays, i.e., nF.

Then, the amount of electrical energy produced by the cell

= emf × amount of electricity

= nFE, where E is the emf of the cell.

But the amount of electrical energy

= $-\Delta G$, i.e., decrease of free energy

$\therefore \quad -\Delta G = nFE$

The cell reaction can proceed spontaneously only if $\Delta G<0$ or , from the above equation, $E > 0$ (i.e., positive). Under these conditions, the cell can act as a source of electrical energy. However, a positive value of ΔG refers to a nonspontaneous change and E will be negative, the cell will not operate spontaneously. A zero value of ΔG or E shows that the cell is in a state of equilibrium. The criteria of spontaneity of a cell reaction in terms of ΔG or E are given as,

Magnitude of ΔG	*Magnitude of E*	*Reaction*
$\Delta G < 0$, i.e., negative	$E > 0$, i.e., positive	Spontaneous
$\Delta G > 0$, i.e., positive	$E < 0$, i.e., negative	Non-spontaneous
$\Delta G = 0$	$E = 0$	Equilibrium

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Electromotive force and Equilibrium constant of a cell Reaction

Let us consider the cell reaction in a reversible cell,

$$aA + bB \rightleftharpoons cC + dD$$

For this the decrease in free energy is, $-\Delta G = \Delta G^0 - RT\,ln Q$ ....(1), where Q is the reaction Quotient,

i.e., $Q = (a_C)^c (a_D)^d / (aA)^a (aB)^b$

But we have, $-\Delta G = nFE$ .....(2)

$\therefore nFE = -\Delta G^0 - Rt\,ln Q$ .....(3)

From Van't Hoff reaction isotherm, $-\Delta G^0 = RT ln K$ .....(4) where K is the equilibrium constant.

From Eqs. (3) and (4), $nFE = RT ln K - RT ln Q$ .....(5)

Substituting the expression for Q,

$$nFE = RT ln K - RT ln \frac{(aC)^c (aD)^d}{(aA)^a (aB)^b} \quad .....(6)$$

Replacing activities by concentrations and changing ln to $\log_{10}$,

$$E = \frac{2.303\,RT}{nF} \log K - \frac{2.303RT}{nF} \log \frac{[C]^c [D]^d}{[A]^a [B]^b} \quad .....(7)$$

$$\text{At } 25^0C, \quad \frac{2.303\,RT}{F} = \frac{2.303 \times 8.314\,JK^{-1}mol^{-1} \times 298K}{96500\,Cmol^{-1}}$$

$$= 0.0591 \text{ volt}$$

$$\therefore E = \frac{0.0591}{n} \log K - \frac{0.0591}{n} \log \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

Knowing E, and the concentrations of reactants and products the equilibrium constant K may be calculated.

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Standard emf and Equilibrium Constant

We have, $\Delta G^0 = -RT\,lnK$ and $\Delta G^0 = -nFE^0$

$$\therefore\ nFE^0 = RT\,lnK \quad \therefore\ lnK = \frac{nFE^0}{RT} \text{ or } \log K = \frac{nFE^0}{2.303\,RT}$$

Example 3.6. The emf of a standard cadmium cell is 1.01832 V at 25^0C. The temperature coefficient of the cell is -5.0×10^{-5} VK^{-1}. Calculate the ΔG, ΔH and ΔS of the cell reaction.

Solution. $\Delta G = -nFE = -2$ mol $\times$ 96500 C $mol^{-1} \times$ 1.01832 V

$$= \mathbf{-196.535\ kJ\ mol^{-1}}$$

$$\Delta S = nF\left(\frac{\partial E}{\partial T}\right)_P = 2 \text{ mol} \times 96500 \text{ coulombs mol}^{-1} \times -5 \times 10^{-5}\ VK^{-1}$$

$$= \mathbf{-9.65\ JK^{-1}\ mol^{-1}}$$

$\Delta G = \Delta H - T\Delta S$; $\therefore\ \Delta H = \Delta G + T\Delta S$

i.e., $\Delta H = -196.535$ kJ $mol^{-1} + (298 \times -9.65 \times 10^{-3}$ $kJK^{-1}mol^{-1})$

i.e., $\Delta H = -196.535$ kJ $mol^{-1} - 2.876$ kJ $mol^{-1} = \mathbf{-199.659\ kJ\ mol^{-1}}$

Example 3.7. The cell reaction of a galvanic cell is given as, $Cd_{(s)} + 2AgCl_{(s)} + aq \rightleftharpoons CdCl_2 . 5/2\ H_2O$ (sat) $+ 2\ Ag_{(s)}$. The emf of the cell is 0.6753 V at 25^0C and 0.6915 V at 0^0C. Calculate ΔG, ΔH and ΔS of the reaction at 25^0C.

Solution. $\Delta G = -nFE = -2$ mol $\times$ 96500 C $mol^{-1} \times$ 0.6753 V

$$= -130332.9\ J = -130.33\ kJ$$

$$\left(\frac{\partial E}{\partial T}\right)_P = -\left(\frac{0.6915V - 0.6753V}{298K - 273K}\right) = \frac{-0.0162V}{25K} = -6.48 \times 10^{-4}\ K^{-1}$$

$$E = -\frac{\Delta H}{2F} + T\left(\frac{\partial E}{\partial T}\right)_P$$

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$$\text{i.e. } 0.6753\ \text{V} = \frac{\Delta H}{2\ \text{mol} \times 96500\ \text{C mol}^{-1}} + 298\ \text{K} \times (-6.48 \times 10^{-4}\ \text{VK}^{-1})$$

$$\text{i.e., } 0.6753\ \text{V} = \frac{-\Delta H}{193000\text{C}} - 0.1931\ \text{V}$$

$$\therefore \frac{-\Delta H}{193000\text{C}} = 0.6753\ \text{V} + 0.1931\ \text{V} = \mathbf{0.8684\ V}$$

$$\therefore \Delta H = -193000\ \text{C} \times 0.8684\ \text{V} = -167601.2\ \text{V} = -167.6\ \text{kJ}$$

But $\Delta G = \Delta H - T\Delta S$

$\therefore T\Delta S = \Delta H - \Delta G$ or

$$\Delta S = \frac{\Delta H - \Delta G}{T} = \frac{-167.6\,\text{kJ} - (-130.33\,\text{kJ})}{298\,\text{K}} = \frac{37.27\,\text{kJ}}{298\,\text{K}}$$

$$= 0.125\ \text{kJK}^{-1} = \mathbf{125\ JK^{-1}}$$

Example 3.8. For the Daniell cell the standard free energy change of the cell reaction is –218.4 kJ. Calculate the standard emf.

Solution : $-\Delta G^0 = nFE^0$

$$\therefore E^0 = -\frac{\Delta G^0}{nF} = -\frac{-218.4\,\text{kJ}}{2 \times 96500\,\text{C}} = -\frac{-218400\,\text{J}}{2 \times 96500\,\text{C}} = \frac{218400\,\text{VC}}{2 \times 96500\,\text{C}}$$

$$= \mathbf{1.13\ V}$$

Self Study Problems

3.5. The standard emf of the Daniell cell is 1.1 V. Calculate the equilibrium constant at 298 K.

Hint: $\log K = \frac{nFE^0}{2.303RT}$ **Ans.** $K = 1.67\,9 \times 10^{37}$

3.6. For a galvanic cell, the cell reaction is $2Ag^+ + Zn \rightleftharpoons 2Ag + Zn^{++}$.

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Given that : $[Zn^{++}] = 0.1$ M, $[Ag^{+}]$ 10 M; emf at 25^0C = 1.62 volts. Calculate the equilibrium constant of the reaction.

Hint : $E = \frac{0.0591}{n} \log K - \frac{0.0591}{n} \log \frac{[C]^c[D]^d}{[A]^a[B]^b}$

Ans. $K = 8.13 \times 10^{52}$

3.7. The standard electrode potential for Hg_2^{2+}/Hg and Hg^{2+}/Hg are 0.799 V and 0.855 V respectively. Calculate the equilibrium constant of the reaction, $Hg^{2+} + Hg \rightleftharpoons Hg_2^{2+}$.

Ans. K= 78.38

3.8. For the cell, $Zn/Zn^{++}_{(a=0.1)}||Cu^{++}_{(a=0.1)}/Cu$, E^0_{cell} is 1.099 V. Compare the values of ΔG and ΔG^0 of this cell.

Hint : $\Delta G^0 = -nFE^0$ and $\Delta G = nFE$

Ans. $\Delta G^0 = 212.107$ kJ mol^{-1}

$\Delta G = 206.394$ kJ mol^{-1}

$E = 1.0694$ V

Nernst Equation

Nernst Equation for Electrode Potential

The potential of a single electrode or half-cell varies with the concentration of ions in the cell. In 1889 *Nernst* derived a mathematical relationship which enables us to calculate the electrode potential, E from the standard electrode potential, E^0, and the temperature of the cell. It also gives the relation between electrode potential and concentration. This equation is known as *Nernst equation.* It can be derived as follows. Consider a general redox reaction, $M^+ + ne^- \longrightarrow M$

The free energy change of a reaction is given as,

$$\Delta G = \Delta G^0 + RT \ln \frac{\text{Product of concentration of products}}{\text{Product of concentration of reactants.}}$$

Thus, for the electrode reaction given above,

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$$\Delta G = \Delta G^{o} + RT \ln \frac{[M]}{[M^{+}]} \quad ...(2)$$

where ΔG^{o} is the standard free energy change. But, the free energy change for a reversible electrode reaction is related to the electrode potential by,

$\Delta G = -nFE$ .....(3), where n is the number of moles of electrons transferred and F is the faraday constant. In the standard state, equation (3) can be written as,

$\Delta G^{o} = -nFE^{o}$ ....(4), where ΔG^{o} and E^{o} are the standard free energy change and standard electrode potential respectively. Thus, the electrical energy (nFE) is produced at the expense of the decrease of free energy of the reaction.

Substituting the values of ΔG and ΔG^{o} in equation (2), we get

$$-nFE = -nFE^{o} + RT \ln \frac{[M]}{[M^{n+}]} \quad ....(5)$$

$$\text{or } nFE = nFE^{o} + RT \ln \frac{[M^{n+}]}{[M]} \quad ....(6)$$

Dividing (6) by nF, we get

$$E = E^{0} + \frac{RT}{nF} \ln \frac{[M^{n+}]}{[M]} \quad .....(7)$$

$$\text{or } E = E^{0} + 2.303 \frac{RT}{nF} \log \frac{[M^{n}]}{[M]} \quad .....(8)$$

$$\text{Or, in general, } E = E^{0} + \frac{2.303RT}{nF} \log \frac{\text{concentration of oxidised state}}{\text{concentration of reduced state}}$$

For a pure metal [M] can be taken as unity. Therefore, equation (8) can be written as,

$$E = E^{0} + 2.303 \frac{RT}{nF} \log [M^{n+}] \quad .....(9)$$

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This equation is the ***Nernst equation*** for the electrode potential, where E^0 is the standard electrode potential. Substituting the values of R (8.314 JK^{-1} mol^{-1}) and Faraday constant (96500 C) at temperature 298 K, we get

$$E = E^0 + \frac{0.0591}{n}\log[M^{n+}] \quad ....(10)$$

Example 3.9. Write down the Nernst equation for the following electrodes at 298K (i) $Zn^{++}|Zn$ and (ii) $Cu^{++}|Cu$

Solution : (i)$Zn^{++}|Zn$; the electrode reaction is,

$$Zn^{++} + 2e^- \rightarrow Zn.$$

$$\therefore E = E^0 - \frac{2.303RT}{nF}\log\frac{[Zn]}{[Zn^{++}]}$$

Since the temperature is 298 K, $\frac{2.303RT}{F} = 0.059$; $n = 2; [Zn] = 1$

$$\therefore E = E^0 + \frac{0.059}{2}\log[Zn^{++}] \qquad \text{i.e., } E = E^0 + 0.0295 \log [Zn^{++}]$$

(ii) $Cu^{++}|Cu$; the electrode reaction is, $Cu^{++} + 2e^- \rightarrow Cu$

$$\therefore E = E^0 + 0.0295 \log [Cu^{++}]$$

Example 3.10. Calculate the electrode potential of $Zn^{++}_{(2M)}|Zn$ at 298 K. Given : $E^0_{Zn} = -0.762$ V.

Solution :The electrode reaction is, $Zn^{++} + 2e^- \rightarrow Zn$

$$\therefore E = E^0 + \frac{0.059}{2}\log [Zn^{++}], \text{ i.e., } E = E^0 + \frac{0.059}{2}\log 2$$

$$= -0.762 + 0.0295 \times 0.3010 = -0.762 + 0.008879 = -0.7531 \text{ V}$$

Example 3.11. A zinc rod is placed in a $ZnSO_4$ solution of concentration 0.1 M. If the salt is dissociated 95% at 298 K, calculate the potential of the electrode.

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Given: $E^0_{Zn^{++}/Zn} = -0.76\,V$

Solution: $[Zn^{++}] = C \times \alpha = 0.1 \times \frac{95}{100} = 0.095\ M$

The electrode reaction is,

$$Zn^{++} + 2e \rightleftharpoons Zn$$

$$\therefore E_{el} = E^0_{el} - \frac{RT}{nF} ln \frac{1}{[Zn^{++}]} = E^0_{el} + \frac{RT}{nF} ln[Zn^{++}]$$

At 298 K,

$$E_{el} = -0.76 + \frac{0.0591}{2} \log 0.05 = \mathbf{-0.79\ V}$$

Self Study Problems

3.9. At 298K what is the single electrode potential of a half cell for Zn electrode dipping in 0.01 M $ZnSO_4$ solution?

E^0_{Zn}= -0.763V **Ans:**-0.8222V

3.10. Given the electrode reaction for the calomel electrode as,

$$Hg_2Cl_{2(s)} + 2e^- \rightarrow 2Hg_{(s)} + 2Cl^-_{(0.1N)}$$

Calculate the electrode potential at 298 K, if the standard electrode potential is +0.28 V. **Ans.** 0.3391V

Nernst equation of Cell potential (E.M.F.)

The Nernst equation is applicable to cell potentials as well. It can be derived as follows.

Let us consider a general reaction as, $aA + bB \rightleftharpoons cC + dD$

The decrease in free energy of the reaction is given as,

$$\Delta G = \Delta G^o + RT \ln \frac{[C]^c\,[D]^d}{[A]^a\,[B]^b} \quad ....(1)$$

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But $\Delta G = -nFE$ and $\Delta G^o = -nFE^o$, therefore, eqn (1) becomes,

$$-nFE = -nFE^o + RT\ln \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$$\text{or } E = E^o - \frac{RT}{nF} \ln \frac{[C]^c [D]^d}{[A]^a [B]^b} \quad ....(3)$$

$$\text{or } E = E^o - \frac{2.303\ RT}{nF} \log \frac{[C]^c [D]^d}{[A]^a [B]^b} \quad ....(4)$$

This is the Nernst equation for cell potential. Substituting the values of R, and F at a temperature of 298 K, we get

$$E = E^o - \frac{0.0591}{n} \log \frac{[C]^c [D]^d}{[A]^a [B]^b} \quad ....(5)$$

Thus, at 298 K, we can write the Nernst equation as,

$E_{cell} = E^o_{cell} - \frac{0.0591}{n} \log K$, (6) where K is the equilibrium constant of the cell reaction.

For example, for the cell, $Zn|Zn^{++}(0.1M)||Cu^{++}(0.01M)|Cu$,

$$E_{cell} = E^0_{cell} + \frac{2.303RT}{nF} \log K$$

The electrode reactions are, $Zn \rightleftharpoons Zn^{++} + 2e^-$

$$Cu^{++} + 2e^- \rightleftharpoons Cu$$

$\therefore$ The cell reaction is, $Zn + Cu^{++} \rightleftharpoons Zn^{++} + Cu$

Thus, the equilibrium constant, $K = \frac{[Zn^{++}][Cu]}{[Zn][Cu^{++}]}$

But [Cu]=1; [Zn]=1; n=2; at 298 K, $\frac{2.303RT}{F} = 0.059$

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$$\therefore \text{At 298 K, } E_{cell} = E^0_{cell} - \frac{0.059}{2} \log \frac{[Zn^{++}]}{[Cu^{++}]}$$

$$\text{i.e., } E_{cell} = E^0_{cell} + \frac{0.059}{2} \log \frac{[Cu^{++}]}{[Zn^{++}]} = E^o_{cell} + \frac{0.059}{2} \log \frac{0.01}{0.1}$$

Example 3.12. Calculate the emf of the cell, Fe $|Fe^{++}(1M) \| Ni^{++}(1M)|$ Ni at 298 K. Given $E^0_{Fe} = -0.44$ V: $E^0_{Ni} = 0.25$V

Solution : The electrode reactions are,

$$Fe \rightarrow Fe^{++} + 2e^-$$

$$Ni^{++} + 2e^- \rightarrow Ni$$

$\therefore$ The cell reaction is, $Fe + Ni^{++} \rightleftharpoons Fe^{++} + Ni$

$$\therefore E = E^0 - \frac{0.059}{2} \log \frac{[Fe^{++}]}{[Ni^{++}]}$$

But, $[Fe^{++}] = 1M$ and $[Ni^{++}] = 1M$

$$\therefore E = E^0 - \frac{0.059}{2} \log 1, \text{ i.e., } E = E^0$$

But $E^0 = E^0_{Ni} - E^0_{Fe} = -0.25$ V $-(-0.44$ V$) = 0.19$ V $\quad \therefore$ **E=+0.19V**

Example 3.13. Calculate the emf of the cell, Zn| $Zn^{++}(0.001M||$ $g^+(0.1M)|$Ag at 298 K. $E^0_{Zn} = -0.76$ V; $E^0_{Ag} = +0.8$V

Solution : The electrode reactions are :

$$Zn \rightarrow Zn^{++} + 2e^-; \quad 2Ag^+ + 2e^- \rightarrow 2Ag$$

$\therefore$ The cell reaction is, $Zn + 2Ag^+ \rightarrow Zn^{++} + 2Ag$

$$\therefore E = E^0 - \frac{2.303RT}{nF} \log \frac{[Zn^{++}]}{[Ag^+]^2}$$

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Thus at 298 K, $E = E_0 - \frac{0.059}{2} \log \frac{[Zn^{++}]}{[Ag^{+}]^2}$

Here, $[Zn^{++}]=0.001$ M; $[Ag^{+}]=0.1$M;

$E^0_{cell}=E^0_{Ag}-E^0_{Zn}=0.8 - (-0.76)=0.8 + 0.76 = 1.56$ V

$$\therefore E_{cell} = 1.56 - 0.0295 \log \frac{0.001}{0.1 \times 0.1}$$

$$E_{cell} = 1.56 - 0.0295 \log \frac{10^{-3}}{10^{-2}} = 1.56 - 0.0295 \times \log 10^{-1}$$

i.e., $E_{cell}=1.56 + 0.0295 = \mathbf{1.5895\ V}$

Self Study Problems

3.11. Calculate the e.m.f. of the cell $Zn|Zn^{++}$ (0.1M)$||$$Cu^{++}$ (0.1M)$|$ Cu at 298 K, if E^0_{Zn} and E^0_{Cu} are -0.76 V and + 0.3370V respectively. **Ans :** 1.09 V

3.12. Calculate the emf of the cell, $Fe/Fe^{++}_{(a=1)}||\ Ni^{2+}_{(a=1)}/Ni$; $E^0_{Fe/Fe^{++}} = 0.44$ V and $E^0_{Ni/Ni^{++}} = 0.25$ V. **Ans.** 0.190 V

Reference Electrodes

Since a half-cell in an electrochemical cell can work only in combination with the other half-cell, it is not possible to determine the absolute electrode potential of an electrode. We can determine only the relative electrode potential. Thus, the electrode potential of an electrode can be measured only with the help of a reference electrode. Some commonly used reference electrodes are: standard hydrogen electrode, calomel electrode, quinhydrone electrode, etc.

Standard hydrogen electrode

A standard hydrogen electrode is set up by bubbling pure hydrogen gas at one atmosphere pressure through a solution of H^+ ions having 1M

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concentration (e.g., 1M HCl) at a temperature of 298 K. Since hydrogen gas is a non-conductor, a platinum strip (wire) is used for electric contact. The standard hydrogen electrode (S.H.E) or normal hydrogen electrode (N.H.E.) is represented as,

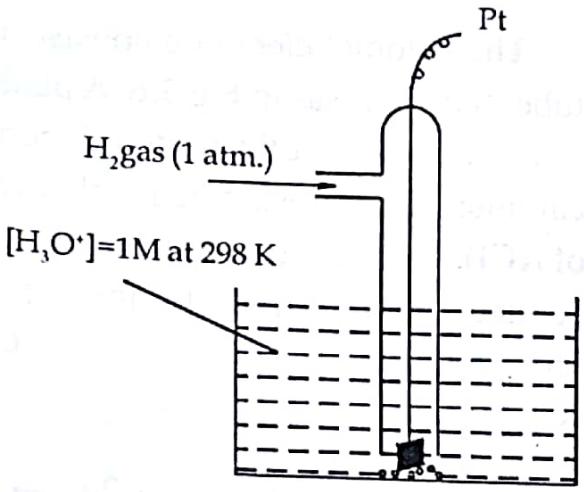

Fig.3.5 *Standard hydrogen electrode*

(Pt) H_2 (g, 1atm.) $|^{\bullet}H^+$(1M); when it acts as ***anode***

H^+ (1M) | H_2 (g, 1atm) (Pt); when it acts as ***cathode.***

The Pt metal is left unchanged and it merely serves as a site for the reaction. The following reactions occur in this half -cell depending upon whether it acts as an anode or as a cathode.

$$H_2(g) \rightarrow 2H^+ + 2e^- \qquad \text{(when it acts as anode)}$$

$$2H^+ + 2e^- \rightarrow H_2(g) \qquad \text{(when it acts as cathode)}$$

The electrode has a definite potential. But for the convenience of reference, the potential of S.H.E is arbitrarily fixed as zero. Hence when a S.H.E is combined with any other electrode, the EMF of the cell will be the single electrode potential of the electrode.

Calomel Electrode

S.H.E. is difficult to construct and maintain. Therefore any other electrode whose potential has been determined relative to the S.H.E. can be used as secondary reference electrode. An example is the calomel electrode.

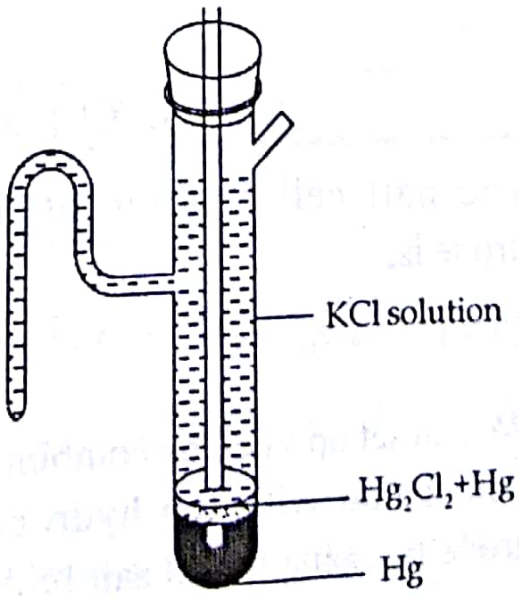

Fig.3.6 *Calomel Electrode*

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The calomel electrode consists of a wide glass tube with a narrow side tube. It is set up as in Fig.3.6. A platinum wire is dipping into liquid mercury covered with a solid paste of mercury and mercurous chloride (Hg_2Cl_2, calomel). The tube is filled with a 1M solution of KCl (or saturated solution of KCl). The side-tube containing KCl solution provides the salt bridge which connects the electrode to any other electrode. The calomel electrode is represented as, $Cl^-|Hg_2Cl_2$, Hg and the half cell reaction is,

$$Hg_2Cl_2 + 2e^- \rightarrow 2Hg + 2Cl^-$$

The single electrode potential of calomel electrode is determined by combining with a S.H.E. The potential of the calomel electrode depends on the concentration of KCl solution taken in the half cell. For 1M KCl solution, the electrode potential is 0.28V and for saturated KCl solution the potential is 0.242V. Knowing the single electrode potential, calomel electrode can be used as a reference electrode and it is called secondary ***reference electrode***. Calomel electrode has the advantages of retaining its constant voltage for a long period.

The Standard Silver-Silver Chloride electrode

In standard silver-silver chloride electrode system, silver wire (or strip) is covered with silver chloride. It is dipped in potassium chloride solution in which the concentration of Cl^- ion is 1 M. This electrode can be represented as,

$$Ag, AgCl/Cl^-(1M)$$

The half cell reaction of the electrode is,

$$AgCl + 1e^- \rightarrow Ag + Cl^-$$

We can set up a cell by combining this electrode with the hydrogen electrode by using a KCl salt bridge as,

$$Pt, H_2|H^+(1M)||Cl^-(1M)\,AgCl, Ag$$

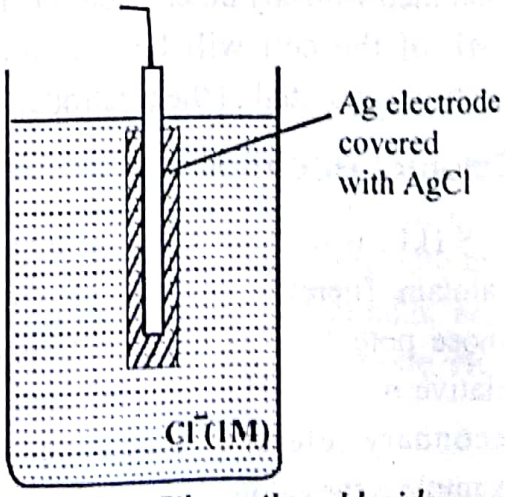

Fig.3.7 *Silver-silverchloride electrode*

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The emf of the cell is found to be 0.2224V. Thus the standard electrode potential for the silver-silver chloride electrode is 0.2224V.

Example 3.14 The emf of the cell, $Zn|Zn^{++}||KCl(std)|Hg_2Cl_2$. Hg is 1.10V. Calculate the electrode potential of $Zn|Zn^{++}$ electrode, if the electrode potential of calomel electrode is +0.242V.

Solution :

EMF of the cell=Reduction potential of cathode - Reduction potential of anode

i.e. $E_{cell}=E_{calomel}-E_{Zn}$

But, $E_{calomel}$ = 0.242 V and E_{cell} = 1.1 V

$\therefore$ 1.1 V=0.242 V - E_{Zn}

$\therefore E_{Zn}$=0.242V-1.1V=-0.858 V

Self Study Problems

3.13. Evaluate the standard electrode potential of $Zn|Zn^{++}$ electrode if the EMF of the cell, $Zn\,|Zn^{++}{}_{(a=1)}\,\|\,Ni^{++}{}_{(a=1)}\,|\,Ni$ is 0.51 V and $E^0{}_{Ni}$ is -0.25V

Ans: -0.76V

3.14 Evaluate the standard electrode potential of $Ni|Ni^{++}$ electrode if the emf of the cell $Ni\,|Ni^{++}{}_{(a=1)}\|\,KCl_{(std)}|\,Hg_2Cl_2$, Hg is 0.49V & E_{sce} is 0.24V

Ans: -0.25V

IMPORTANT POINTS FOR QUICK REVISION

Electrochemical change : Change involving production or consumption of electrical energy.

Electrolyte : Substances which allow passage of electricity through them in molten state or in aqueous solution and undergo decomposition.

Ohm's law : I = V/R

Resistance : Measure of obstruction to the flow of electricity.

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Specific resistance : Resistance of a conductor of unit length and unit area of cross section. $\rho = R \times \frac{a}{l}$

Conductance : Reciprocal of resistance, $C = 1/R$.

Specific conductance: Conductance of a conductor of unit length and unit area of cross section. $k = C \times \frac{l}{a}$.

Galvanic cell : Device in which chemical energy is converted into electrical energy.

Electrode potential : Tendency of an electrode to lose or gain electrons when it is in contact with its own ions in solution.

EMF : Difference between the electrode potentials of two electrodes when no current is drawn from the cell. $EMF = E_{cathode} - E_{anode}$.

Nernst equation for electrode potential :

$$E = E^0 + \frac{2.303\,RT}{nF} \log \frac{[M^{n+}]}{M}$$

Nernst Equation for EMF : $E = E^0_{cell} + \frac{0.0591}{n} \log \frac{[Rectants]}{[Products]}$

Electrolysis : Decomposition of an electrolyte by the passage of electric current.

Faraday : Quantity of electricity required to liberate or deposit one gram equivalent of a substance.

MODEL QUESTIONS

Part A [Each questions carries 2 marks]

1.. One gram equivalent of every substance is deposited by one Faraday. Explain.

2. State Faraday's laws of electrolysis. What is one Faraday?

3. What do you understand from galvanic cells? How do they differ from electrolytic cell?

4. What is meant by a standard electrode and standard electrode potential?

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5. What is a calomel electrode? Give the reaction taking place at the calomel electrode.
6. What are reference electrodes? Give their significance.
7. How will you set up a calomel electrode?
8. What are electrode potential and cell potential?
9. What is a calomel electrode? What are its advantages as a reference electrode?
10. State and explain Nernst equation for emf of a cell.
11. What is a Galvanic cell ? What are its cell reactions?
12. What is N.H.E.? How is it constructed? Represent it in the conventional manner?
13. How standard hydrogen electrode is constructed? What is its potential?
14. What is the difference between EMF and potential difference?
15. What do you understand by the terms (a) standard hydrogen electrode (b) electrode potential.
16. Write down Debye-Huckel-Onsager equation and explain the terms.

Part B [Each question carries 4 marks]

- Write a brief note on calomel electrode
- Write Nernst equation for cell EMF and explain the terms.
- Explain the setting and working of
(i) Standard hydrogen electrode (ii) Calomel electrode
The standard reduction potentials of Zn and Cu electrodes determined by using the reference electrode, standard hydrogen electrode (SHE) are -0.76 V and 0.34V respectively. In these determinations, SHE acts as anode in one case and as cathode in other case.
(a) In which case SHE acts as anode? What is the electrode reaction of SHE in this case?
(b) In which case SHE acts as cathode? What is the electrode reaction of SHe here?
(c) Write the cell reaction in both the cases.
(d) What is the standard potential of the cell obtained by the combination of Zn and Cu electrodes?

ns. (a) With copper electrode SHE acts as anode.

$$\frac{1}{2}H_2 \rightarrow H^+ + 1e^- \text{ (oxidation of } H_2)$$

(b) With zinc electrode SHE acts as cathode.

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$$H^+ + 1e^- \rightarrow \frac{1}{2}H_2 \text{ (reduction of } H^+)$$

(c) $Zn + 2H^+ \rightarrow Zn^{++} + H_2$

$Cu^{++} + H_2 \rightarrow Cu + 2H^+$

(d) $E^0_{cell} = E^0_{cathode} - E^0_{anode} = E^0_{Cu} - E^0_{Zn} = 0.34 - (-0.76) = 1.1$ volt

5. A zinc electrode may be combined with a copper electrode to form a galvanic cell.
(a) Identify the anode and cathode. Write the cell notation.
(b) Give the half cell reactions and cell reaction
(c) If zinc is replaced by aluminum what happens to the voltage?

Ans. (a) Zinc-anode, copper-cathode

$Zn \mid Zn^{++} \| Cu^{++} \mid Cu$

(b) $Zn \rightarrow Zn^{++} + 2e$ (anode half cell reaction)

$Cu^{++} + 2e \rightarrow Cu$ (cathode half cell reaction)

$Zn + Cu^{++} \rightarrow Zn^{++} + Cu$ (overall cell reaction)

(c) In the electrochemical series aluminium is above zinc. Hence aluminium produces a high voltage.

6. The standard reduction potentials of some electrodes are :
$Mg^+, Mg = -2.37$; $Zn^{2+}, Zn = -0.76$,
$H^+, H_2, Pt = 0.0$; $Ag^+, Ag = +0.80$; $F, F_2, Pt = +2.87$
(a) Which of these is most easily reduced?
(b) Which of these metals can liberate hydrogen from HCl?
(c) Which of the following reactions are feasible?

(i) $Zn^{2+} + 2Ag \rightarrow Zn + 2Ag^+$

(ii) $Zn + Cu^{++} \rightarrow Zn^{++} + Cu$

Ans. (a) F_2, because of high reduction potential
(b) Mg and Zn
(c) (i) is not feasible as Ag^+ is more readily reduced than Zn^{++}
(ii) feasible.

Part C [Each question carries 10 marks]

1. Give an account of the Debye-Huckel theory of strong electrolytes. Explain clearly what is meant by asymmetry effect and electrophoretic effect.

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B. PHOTOCHEMISTRY

Photochemistry, in its broadest sense, embraces any chemical effect associated with the absorption or emission of any kind of radiation. The term radiation includes the whole electromagnetic spectrum ranging from high frequency (i.e., low wavelength) γ-rays through X-rays, ultraviolet rays, visible and infrared rays to the lowest frequency (i.e., highest wavelength) radio waves. However, *in the restricted sense photochemistry is the science of the chemical effects of radiations whose wavelengths lie in the visible and ultraviolet region that is in the wavelength range from 800 nm to 200 nm.* The chemical effects produced by the high energy radiations like X-rays, γ-rays, etc. (i.e., the ionizing radiations) comprise the subject matter of *radiation chemistry.*

In ordinary thermal reactions, the molecules are activated by random intermolecular collisions. In photochemical reactions, on the other hand, molecules are activated by the absorption of light photons with a particular amount of energy. Combination of hydrogen and chlorine to give hydrochloric acid, and of hydrogen and bromine to give hydrogen bromide are well known examples of photochemical reactions brought about by sunlight. In photochemical reactions, the first step which is produced directly by the absorption of light is called primary process and the reactions which occur as secondary effects of light absorption are called secondary processes.

Difference between Thermal and Photochemical Reactions

As already mentioned, in thermal reactions effective collisions between the reacting molecules provide the energy of activation for the reaction to occur, whereas in photochemical reactions absorption of light photons (visible or ultraviolet) provide the required energy.

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All ordinary reactions are accompanied by a decrease in free energy. However, certain photochemical reactions are accompanied by an increase of free energy. Ozonisation of oxygen, photosynthesis, etc., are examples of such photochemical reactions. In these, the energy of the absorbed radiation is converted into free energy of the products.

In a photochemical reaction only a single selected species, i.e., atom or molecule can be promoted to an excited state independent of the other species present in the reaction system. Thermal reactions, on the other hand, do not permit such selectivity. Thermal excitation increases the translational, rotational and vibrational energies of all the molecules in random manner resulting in the excitation of all the molecules to almost the same extent. Thus, exposure of a mixture of hydrogen and bromine to radiations of wavelengths 450 nm – 500 nm, cases the excitation of only the bromine molecules whereas heat would cause the excitation of the molecules of both bromine and hydrogen.

The photochemical reactions may occur even at room temperature whereas thermal reactions occur at elevated temperatures. From this it is evident that the rate of latter reactions depends on the temperature while the rate of photochemical reactions is independent of temperature. But the rate of photochemical reactions depends on the intensity of the light radiations used. In certain photochemical reactions the rate of reaction is found to vary with temperature but this is due to the temperature dependence of the thermal reactions which subsequently takeplace after the absorption of light radiation (secondary process).

Light Absorption by Solutions : Lambert-Beer Law

The absorption of light in the visible and near U.V. regions by a solution is governed by a law called *Lambert-Beer Law.*

When a beam of monochromatic light of suitable frequency is passed through a solution, a part of it is absorbed. As a result, the intensity of the emerging light will be considerably reduced. Let I_0 be the intensity of the incident light and I_t the intensity of the transmitted light. Then the intensity of the absorbed light I_a is given as,

$$I_a = I_0 - I_t$$

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The intensity of radiation can be defined as the number of photons that pass across a unit area in unit time. According to Lambert Beer's Law:

> *"For a monochromatic radiation travelling through a solution, the rate of decrease of intensity of radiation with thickness of the solution is proportional to the intensity of the incident radiation as well as concentration of the solution."*

Mathematically, $-\frac{dI}{dx} = \alpha cI$ or $\frac{dI}{I} = -\alpha c\, dx$ .....(i)

where dI is the change in intensity produced by the absorption of radiation, c is the molar concentration of the solution, dx, is the thickness of the solution and α is the proportionality constant. The minus sign is used because there is a decrease in intensity. Integrating equation (i) between the limits $I = I_0$ at $x = 0$ and $I = I$ at $x = b$.

$$\int_{I_0}^{I} \frac{dI}{I} = -\alpha c \int_{0}^{b} dx \quad .....(ii)$$

$$\ln \frac{I}{I_0} = \alpha bc \quad .....(iii) \quad \text{or} \quad -2.303 \log \frac{I}{I_0} = \alpha bc \quad .....(iv)$$

Equation (iv) is the *Lambert Beer Law*. Thus, the law may also be states as:

> *"the intensity of a beam of monochromatic radiations decreases exponentially with increase in the thickness x and the concentration c of the absorbing medium."*

$$\therefore \log \frac{I}{I_0} = \frac{\alpha}{2.303} bc = -\varepsilon bc \quad .....(v)$$

$$\text{or} \quad \log \frac{I_0}{I} = \varepsilon bc \quad .....(vi)$$

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where $\varepsilon = \frac{\alpha}{2.303}$ is called *molar absorption coefficient* or *molar extinction coefficient.* The quantity $\log \frac{I_0}{I}$ is called *absorbance* A.

i.e., $$\log \frac{I_0}{I} = A \quad \text{.....(vii)}$$

From (vi) and (vii), $A = \varepsilon bc$ ..... (viii)

From this, it is evident that absorbance is directly proportional to the concentration c and the path length b. The transmittance T is defined as,

$$T = \frac{I}{I_0} \qquad \therefore A = -\log T \quad \text{or} \quad T = 10^{-A} = 10^{-\varepsilon bc}$$

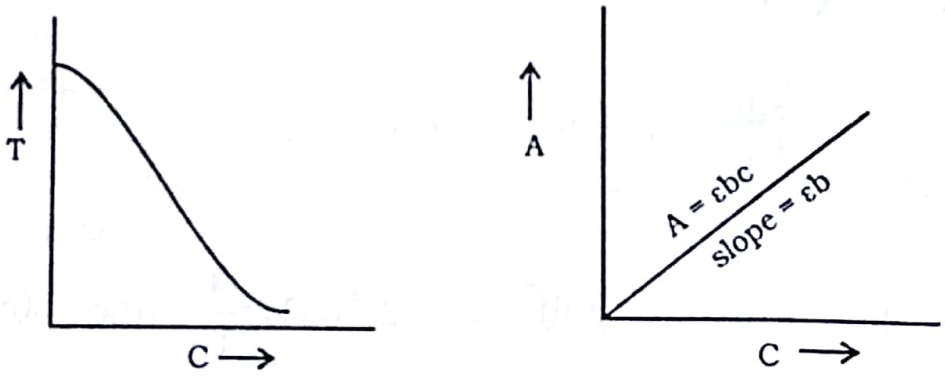

Fig.3.8 *Plots of T and A versus C*

From the plots of T and A versus c [Fig.3.8], it is evident that the plot of A versus c is a straight line passing through the origin. So this plot is more convenient than the plot of T versus c for interpretation. So modern spectrophotometers are calibrated to read absorbance though it is the transmitted light which is actually measured.

Limitations of Lambert Beer's Law

1. If the radiation is not monochromatic the law is not obeyed.
2. The law governs the absorption behaviour of dilute solutions only. When the concentration is high, the ions of the solute (if it is an electrolyte) will be close enough to disturb the charge distribution of the neighbours.

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The interionic attraction may alter the ability of the solute to absorb a given wavelength of the incident radiation to a large extent. As a result, the relationship between A and C will no longer be linear. Thus, at higher concentrations, there are large deviations from linearity as shown in the fig. 3.9.

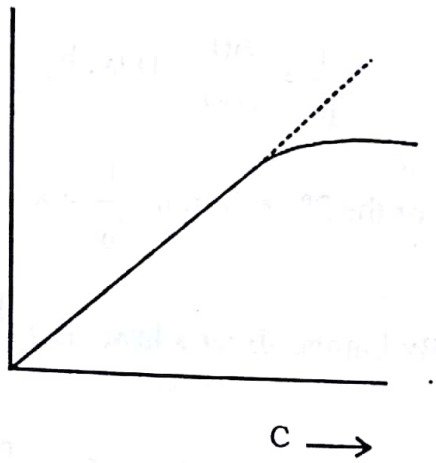

Fig.3.9 *Deviation from Lambert-Beer law.*

With increase of temperature the absorption band will shift towards longer wavelength (bathochromic shift). So temperature of the system should not be allowed to alter to a large extent.

EXAMPLE

3.15 When a monochromatic radiation is incident on a solution of 0.05 molar concentration of an absorbing substance, the intensity of the radiation is reduced to one fourth of the initial value after passing through 10 cm length of the solution. Calculate the molar extinction co-efficient of the substance.

Solution : $\frac{I}{I_0} = \frac{1}{4}$ $\quad \therefore \quad \frac{I_0}{I} = 4$, b = 10 cm

$$\log \frac{I_0}{I} = \varepsilon bc$$

i.e., $\log 4 = \varepsilon \times 10 \text{ cm} \times 0.05 \text{ mol dm}^{-3}$

$$\therefore \varepsilon = \frac{\log 4}{10 \text{ cm} \times 0.05 \text{ mol dm}^{-3}} = \frac{0.6021}{10 \text{ cm} \times 0.05 \text{ mol dm}^{-3}}$$

$$= 1.2042 \text{ dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$$

3.16 A 1% solution of a dye transmitted 60 per cent of the blue light in a cell of thickness 1cm. Calculate the percentage of light absorbed by a 2% solution of the same substance in the same cell.

Solution : For the 1% solution,

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$$\frac{I}{I_0} = \frac{60}{100} = 0.6\,,\ b = 1\ cm,\ c = 1\%$$

For the 2% solution, $\frac{I}{I_0} = x$, b = 1 cm and c = 2%

By Lamber-Beer's law, $\log \frac{I}{I_0} = \frac{-\alpha bc}{2.303}$

For 1% solution, $\log 0.6 = \frac{-\alpha \times 1.0 \times 1.0}{2.303}$ .....(1)

For 2% solution, $\log x = \frac{-\alpha \times 1.0 \times 2.0}{2.303}$ .....(2)

Dividing (2) by (1), $\frac{\log x}{\log 0.6} = 2.0$

$\therefore \log x = 2.0 \times \log 0.6 = \log (0.6)^2 = \log 0.36$

$\therefore$ The fraction of light absorbed by the 2% solution, x = 0.36. That is light absorbed is 36%.

SELF STUDY PROBLEMS

3.15. The percentage transmittance of 5×10^{-4} M solution of a substance is 19.2% in a 1 cm cell. Calculate the absorbance A and molar absorption co-efficient ε.

Ans. A = 0.7166; $\varepsilon = 1.433 \times 10^3\ dm^3\ mol^{-1}\ cm^{-1}$.

Hint : $A = \log \frac{I_0}{I}$ Given $T = \frac{I}{I_0} = \frac{19.2}{100}$; $\varepsilon = \frac{A}{bc}$

3.16. A substance when dissolved in water at 10^{-3} M concentration absorbs 10 per cent of an incident radiation in a path of 1 cm length. Calculate the concentration of the solution which can absorb 90 percent of the same radiation.

[Hint: $C_2 = 0.0218\ mol\ dm^{-3}$]

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THE LAWS OF PHOTOCHEMISTRY

There are two basic laws which govern the photochemical reactions. They are : (i) *The Grothus -Draper Law* and (ii) *The Stark-Einstein Law.*

The Grothus-Draper Law

When light falls on a cell containing a reaction mixture, some light is absorbed and the remaining is transmitted. From this it is obvious that, only the absorbed component of light is capable of producing the reaction. The transmitted light is chemically ineffective. In the 19th century, *Grothus* and *Draper* studied a number of photochemical reactions and enunciated a generalisation known as *Grothus-Draper law.* The law may be stated as

> *"only that light radiation which is absorbed by a system can bring about a photochemical change"*

However, it does not mean that the absorption of light must necessarily be followed by a reaction. When the conditions are not favourable for the molecules to react, the light energy remains unused. It may be re-emitted as heat or light.

Grothus-Draper law is purely quantitative in nature. It gives no idea of the relation between the absorbed light and the molecules undergoing change.

Stark-Einstein law of photochemical equivalence

Stark and *Einstein* studied the quantitative aspect of photochemical reactions by applying quantum theory. They noted that each molecule taking part in the reaction absorbs only a single quantum of light. The molecule that absorbs one photon-equivalent energy is activated and enters into reaction. *Stark* and *Einstein* thus proposed a basic law of photochemistry. The *Stark Einstein law* of photochemical equivalence states that:

> *In a photochemical reaction, each molecule of the reacting substance absorbs a single photon of radiation, causing the reaction and is activated to form the products.*

Thus, when a molecule A absorbs a photon of radiation, it is got activated (A*). The activated molecule is then converted to the product (B) as,

$$A + h\nu \rightarrow A^{*} \text{ (activated molecule)}$$

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$$A^* \rightarrow B \quad \text{(product)}$$

$$A + h\nu \rightarrow B \quad \text{(overall reaction)}$$

Energy per quantum = hν. Thus for one molecule to undergo a chemical change, the radiant energy required is hν. Therefore, for one mole to undergo chemical change, the number of quanta required is N (Avogadro number). Thus, the quantity of radiant energy required for one mole to undergo chemical change is Nhν. The energy absorbed per mole (E) of the reacting substance is called *one einstein*. The magnitude of einstein is given as

$$E = Nh\nu = Nh\frac{c}{\lambda}$$

$$= \frac{6.02\times10^{23}\ \text{mol}^{-1}\times6.62\times10^{-34}\times\text{Js}\times3\times10^{8}\ \text{ms}^{-1}}{\lambda}$$

$$= \frac{11.955}{\lambda}\times10^{-2}\ \text{mol}^{-1}$$

In this equation λ is in metres. Expressing it in Angstrom units ($1\text{Å} = 10^{-10}$ m,) we have,

$$E = \frac{11.955}{\lambda}\times10^{-2}\times10^{10}\ \text{J mol}^{-1}$$

$$= \frac{11.955}{\lambda}10^{8}\ \text{J mol}^{-1} = \frac{11.955}{\lambda}\times10^{5}\ \text{kJ mol}^{-1}$$

$$= \frac{2.857}{\lambda}\times10^{5}\ \text{kcal mol}^{-1}$$

Quantum yield or Quantum efficiency

The number of molecules reacted or formed per photon absorbed is called quantum yield or quantum efficiency. It is denoted by φ and may be written as:

$$\varphi = \frac{\text{Number of molecules reacted}}{\text{Number of photons absorbed}}$$

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$$= \frac{\text{Number of moles reacted}}{\text{Number of einsteins absorbed}}$$

or $$\varphi = \frac{\text{Number of molecules of product formed}}{\text{Number of photons absorbed}}$$

$$= \frac{\text{Number of moles of product formed}}{\text{Number of einsteins absorbed}}$$

For a reaction that obeys strictly the *Stark-Einstein law*, one molecule of reactant decomposes per photon or one molecule of product forms per photon (or one mole of reactant decomposes per einstein or one mole of product forms per einstein), the quantum yield $\varphi = 1$. This is, however, very rare. When two or more molecules are decomposed per photon, $\varphi > 1$ and the reaction has a high quantum yield. If the number of molecules decomposed is less than one per photon, the reaction has low quantum yield.

Causes of high quantum yield

(i) Reactions subsequent to the primary reaction

One photon absorbed in a primary reaction dissociates one molecule. But the resulting excited atoms may start a subsequent secondary reaction in which a further molecule is dissociated.

$$AB + h\nu \rightarrow A + B \text{ (primary reaction)}$$

$$AB + B \rightarrow A_2 + B \text{ (secondary reaction)}$$

Thus, one photon decomposes two molecules and $\varphi = 2$.

The decomposition of HI by absorption of wavelength less than 4000 Å is an example of this type of reaction.

(ii) A reaction chain forms many molecules per photon

When a reaction involves two or more reactants, a molecule of one of them absorbs a photon and dissociates in the primary reaction chain.

$$A + A + h\nu \rightarrow 2A \quad \text{.....(i) (primary reaction)}$$

$$A + B - B \rightarrow A - B + B \quad \text{.....(ii)}$$

(secondary reaction chain)

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$B + A - A \rightarrow A - B + A$ .....(iii)

The steps (ii) and (iii) will continue. As a result, the number of molecules of AB formed per photon will be very large and hence ϕ will be very large.

Hydrogen-chlorine reaction is a well known example of a photochemical chain reaction.

Causes of low quantum yield

(i) Deactivation of molecules

The excited molecules formed in the primary reaction may be deactivated by fluorescence or by collision with some inert molecules.

$$A + h\nu \rightarrow A^* \text{ (activation)}$$

$$A^* \rightarrow A + h\nu' \text{ (fluorescence)}$$

(ii) Recombination of dissociated fragments

During the primary reaction, the reactant molecules may dissociate to give smaller fragments. These fragments may recombine to give back the reactant.

$$A - B + h\nu \rightarrow A + B \text{ (primary reaction)}$$

$$A + B \rightarrow A - B$$

Thus, the secondary reactions involving the fragments to form the product will not occur.

The dimerisation of anthracene is an example of a reaction having low quantum yield. When anthracene dissolved in benzene is exposed to UV light, it is converted to dianthracene as follows,

$2C_{14}H_{10} + h\nu \rightarrow C_{28}H_{20}$.

From this it is obvious that the quantum yield should be 2 but the actual quantum yield is only 0.5. The low quantum yield is due to fluorescence accompanied by the reaction which deactivates the excited anthracene molecules. In addition to this, the reaction is reversible. Thus the conversion of the product back to the reactant till a state of equilibrium is attained.

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$$2C_{14}H_{10} \underset{\text{Thermal}}{\overset{h\nu}{\rightleftharpoons}} C_{28}H_{20}$$

This further lowers the quantum yield.

Determination of Quantum Yields

Quantum yields of photochemical reactions can be determined by the determination of (i) the number of moles of the light absorbing substance which react in a given time and (ii) the number of einsteins of light of required wavelength which are absorbed by the same substance in the same time.

(i) Determination of the number of moles of the substance reacting in a given time

Usual analytical techniques employed in chemical kinetics can be used for the determination of the number of moles of the substance reacting in a given time. The reaction cell is usually made of glass unless violet or ultraviolet radiation is to be used. In that case the cell is made of quartz. A parallel beam of incident radiation is allowed to fall at right angles to it. The extent of a photochemical reaction depends upon the intensity of light used. It has been calculated that one candle power of light falling on one square centimetre of surface per second corresponds to 2×10^{14} quanta. Thus, 2×10^{14} molecules or 3.32×10^{-10} moles of light absorbing substance would react per square centimetre of surface per second. It is a very small quantity. So light of high intensity is to be used.

In photochemical experiments, it is desirable to use monochromatic light, i.e., light of a single wavelength. This can be done by using discharge tubes. Mercury vapour lamp is used as the common source of light for this purpose. Metal filament lamps like tungsten lamp, iron and carbon arcs, etc. can also be used as the source of light. It is to be noted that if we use polychromatic light, some other reactions may also occur simultaneously.

(ii) Determination of the number of einsteins of light absorbed

The number of einsteins absorbed can be determined by using a thermopile consisting of a number of thermocouples joined in series. One set of junctions of the thermocouples are blackened such that all radiations falling on it are absorbed. The other set is protected from radiation and maintained at a fixed temperature. When light falls on one set of junctions of the thermocouples,

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the heat radiation associated with the light generates a thermoelectric current measuring the EMF developed by the thermoelectric effect, the energy of the incident radiation can be calculated. The measurements are done before and after passing light through the reaction cell. The amount of energy absorbed (i.e., number of einsteins absorbed) by the substance is obtained from the difference in the two measurements.

Another method of using the energy of radiation is by using and ***actinometer*** in which a standard photochemical reaction is employed to estimate the energy. ***Uranyl oxalate actinometer*** is a commonly used actinometer which consists of a dilute solution of oxalic acid mixed with uranyl sulphate. Uranyl sulphate sensitises the decomposition of oxalic acid on exposure to violet or ultraviolet light the wave length range of which is 250 to 435 nm. At the end of the experiment, oxalic acid is titrated against potassium permanganate to determine the extent of decomposition. The amount of decomposition is assumed to be proportional to the product of intensity of light of a given wavelength and the time of exposure. The apparatus is first standardised with respect to the radiations of different wavelengths. Thus, the amount of energy of light radiation of a given wavelength absorbed in a photochemical reaction can be evaluated. The measurements are done before and after passing light through the cell.

EXAMPLE 3.17

Calculate the energy per photon and per einstein (i.e., per mole) of the light with wavelength 500 nm.

Solution : Energy per photon $= \frac{ch}{\lambda}$

$$= \frac{3\times10^{8}\,ms^{-1}\times6.62\times10^{-34}\,Js}{5\times10^{-7}\,m}$$

Energy per einstein, $E = \frac{N.ch}{\lambda}$

$$= \frac{6.02\times10^{23}\,mol^{-1}\times3\times10^{8}\,ms^{-1}\times6.62\times10^{-34}\,Js}{5\times10^{-7}\,m}$$

$$= 2.391\times10^{5}\,J\,mol^{-1}$$

3.18. A radiation of 253 nm incident on HI results in the decomposition of

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1.85×10^{-2} mole per 1000 cals of radiant energy. Determine the quantum efficiency.

[$N = 6.02 \times 10^{23}$ mol^{-1}; $c = 3 \times 10^{8}$ ms^{-1}; $h = 6.62 \times 10^{-34}$ Js]

Solution : $\lambda = 253$ nm $= 2.53 \times 10^{-7}$ m

HI decomposed by 1000 cals or 1000×4.184 J

$= 4184$ J of radiation $= 1.85 \times 10^{-2}$ mol

Energy per einstein, $E = \frac{Nch}{\lambda}$

$$= \frac{6.02 \times 10^{23}\,\text{mol}^{-1} \times 3 \times 10^{8}\,\text{ms}^{-1} \times 6.62 \times 10^{-34}\,\text{Js}}{2.53 \times 10^{-7}\,\text{m}}$$

$= 4.728 \times 10^{5}$ J mol^{-1}

Number of einsteins in the given 4184 J of energy

$$= \frac{4184}{4.728 \times 10^{5}} = 0.0885$$

$\therefore$ Quantum efficiency, $\phi = \frac{\text{Number of moles reached}}{\text{Number of einsteins absorbed}}$

$$= \frac{1.85 \times 10^{-2}}{0.0885}\ 0.209 \text{ or } \mathbf{20.9\ \%}$$

SELF STUDY PROBLEMS

3.17. Calculate the energy per mole of light having wavelengths of (a) 85 nm and (b) 300 nm.

(**Hint :** $E = \frac{Nhc}{\lambda}$; **Ans.** (a) 1.407×10^{6} J mol^{-1}

(b) 3.986×10^{5} J mol^{-1})

3.18 Radiation of wavelength 2500Å was passed through a cell containing 10 ml of a solution which was 0.05 molar in oxalic acid and 0.01 molar in uranyl sulphate. After absorption of 80 joules of radiation energy, the concentration of oxalic acid was reduced to 0.04 molar. Calculate the quantum yield for the photochemical decomposition of oxalic acid at the given wavelength. **Ans.** $\phi = 0.60$

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3.19 When a substance A was exposed to light, 0.002 mole of it reacted in 20 minutes and 4 seconds. In the same time A absorbed 2.0×10^6 photons of light per second. Calculate the quantum yield. **Ans.** 4.99×10^{11}

3.20 On irradiation with light of wavelength 3020 Å propionaldehyde is decomposed to carbon monoxide as, $CH_3CH_2CHO + h\nu \rightarrow CH_3 - CH_3 + CO$. The quantum yield of the reaction is 0.54. Propionaldehyde absorbed 3.78×10^{-9} einstein. Calculate the amount of CO formed.

Ans. 2.04×10^{-9} moles

Rate Equation for the Decomposition of Hydrogen iodide

The decomposition of hydrogen iodide is brought about by the absorption of light of wavelength less than 4000 Å. The following mechanism has been suggested for this photochemical decomposition.

(i) $HI + h\nu \xrightarrow{k_1} H + I$

(ii) $H + HI \xrightarrow{k_2} H_2 + I$

(iii) $I + I \xrightarrow{k_3} I_2$

Thus, the rate law for the decomposition of HI is given as,

$$-\frac{d}{dt}[HI] = k_1 I_a + k_2[H]\,[HI] \ldots\ldots(1)$$

where I_a is the intensity of the absorbed radiation. By applying steady state approximation to [H], we have

$$\frac{d[H]}{dt} = k_1 I_a - k_2[H][HI] = 0$$

$$\therefore\ k_2[H][HI] = k_1 I_a \ \ldots\ldots(2)$$

Substituting (2) in (1),

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$$-\frac{d[HI]}{dt} = 2k_1 I_a \quad .....(3)$$

The quantum yield of the reaction,

$$\phi = \frac{\text{Rate of disappearance of HI}}{\text{Rate of absorption of light}} = \frac{2k_1 I_a}{k_1 I_a} = 2$$

In this case ϕ decreases as the reaction proceeds. This is due to the fact that as iodine accumulates, the thermal reaction, $H_2 + I_2 \xrightarrow{k_4} HI + I$ becomes significant. By including this reaction also in the mechanism, the steady state approximation for [H] becomes,

$$d[H]/dt = k_1 I_a - k_2[H][HI] - k_4[H][I_2] = 0 \quad .....(4)$$

$$\therefore\ [H] = \frac{k_1 I_a}{k_2[HI] + k_4[I_2]} \quad .....(5)$$

Substituting (5) in (1),

$$-\frac{d[HI]}{dt} = k_1 I_a + k_2[HI]\frac{k_1 I_a}{k_2[HI] + k_4[I_2]}$$

$$= k_1 I_a\left[1 + \frac{1}{1 + \{k_4[I_2]/k_2[HI]\}}\right] \quad .....(6)$$

$$\therefore\ -\frac{d[HI]/dt}{I_a} = k_1 + \frac{k_1}{1 + \{k_4[I_2]/k_2[HI]\}} \quad .....(7)$$

When the reaction proceeds, $[I_2]$ increases and hence the quantum yield decreases.

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H_2–Cl_2 Reaction - Qualitative treatment

This is a typical example of a photochemical chain reaction. A mixture of hydrogen and chlorine is exposed to light of wavelength less than 4000 Å. The hydrogen and chlorine react rapidly to form hydrogen chloride. In the primary step, a molecule of chlorine absorbs a photon and dissociates into two Cl atoms. This is followed by the secondary reactions. Thus,

$$Cl_2 + h\nu \rightarrow 2Cl \quad \text{.....(1) Primary Reaction}$$

$$\left.\begin{aligned} Cl + H_2 &\rightarrow HCl + H \text{.....(2)} \\ H + Cl_2 &\rightarrow HCl + Cl \text{.....(3)} \end{aligned}\right\} \text{Secondary Reactions}$$

The Cl atom is used up in step (2) but it is regenerated in step (3). Thus the steps (2) and (3) constitute a self propagating chain reaction. This results in the formation of two molecules of HCl in each cycle. Thus one photon of light absorbed in step (1) forms a large number of HCl molecules by the repetition of steps (2) and (3). The reaction terminates by the recombination of Cl atoms at the walls of the vessel where they lose the excess energy.

$$Cl + Cl \xrightarrow{\text{walls}} Cl_2$$

In this reaction, the number of HCl molecules formed per photon of light is very high. The quantum yield of the reaction is very high and it varies from 10^5 to 10^6. The rate of this reaction is found to be directly proportional to the intensity of the absorbed radiation.

Photochemical Process

Primary Processes in Photochemical Reactions

In a photochemical process, the first step is the absorption of a quantum of radiation by the reacting molecule. This step is called the primary process in photochemistry. The energy absorbed from the photon in the visible or UV region raises the molecules to an excited state. The excited molecule may then behave in different ways. There are four distinct possibilities for the excitation of the molecule. [Fig.3.10].

1. The electronic transition from a stable ground state to a stable excited state (Fig.3.10a). There will be no direct dissociation of the molecule.

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As a result, the spectrum consists of discontinuous bands with a fine structure of closely packed lines.

2. When the molecule is raised to a higher quantum state above the maximum stability level, the energy acquired is more than the binding energy, it would undergo dissociation (Fig.3.10 b) in its first oscillation and the spectra will show continuum.
3. The energy may raise the molecule to an unstable electronic state and the potential energy curve does not show any minimum (Fig.3.10c). The molecules breaks up into atoms or radicals immediately on absorbing the photon. The dissociated fragments are produced with different kinetic energy and a continuous spectrum is obtained without any fine structure.
4. The initial electronic transition is from a stable to a higher stable state. Sometimes the higher state is intersected by the potential energy surface of an unstable state. When the molecule vibrates in the excited state, the molecule may shift from the stable state to the unstable state (Fig.3.10 d). During a shift of this type the molecule would dissociate to produce atoms or radicals. At lower levels of vibrations the spectrum would show fine structure which is followed by a continuum. This type of phenomenon is called *predissociation.*

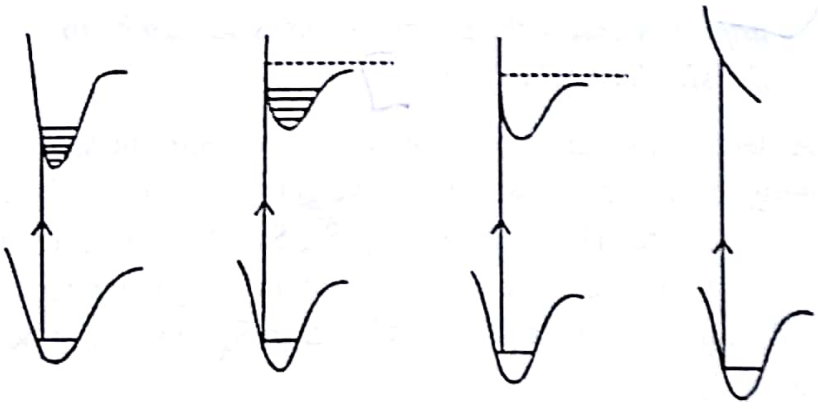

Fig.3.10 *Primary photochemical processes*

Secondary Photochemical Processes - Excited states

Let us consider the transitions between electronic energy levels only. In order to understand the nature of electronic transitions, it is essential to know

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the concept of spin multiplicity. Most molecules have an even number of electrons and thus in the ground state, all the electrons are spin paired (↑↓). The quantity 2S+1, where S is the total electron spin, is called *spin multiplicity* of a state. Thus, when the spins are paired (Fig.3.11a), the upward orientation of the electron spin is cancelled by the downward orientation.

That is, $s_1 = +½$; $s_2 = -½$ $\therefore S = s_1 + s_2 = +½ - ½ = 0$

$\therefore$ Spin multiplicity = $2S+1 = (2\times0)+1 = 1$. Now the molecule is said to be in the ***singlet ground state***, usually denoted by S_0. Thus:

A molecule with electrons paired and with opposite spins is said to be in singlet ground state

When one of the paired electrons jumps to a higher energy level, (excited state) by absorption of a photon, the spin orientations of the two single electrons may be either parallel, ↑↑ (Fig. 3.11b)] or anti parallel, ↑↓ [Fig.3.11c)]. If the spins are antiparallel, we get, $S = s_1+s_2 = +½-½ = 0$, $\therefore 2S+1 = (2\times0)+1=1$ That is, the spin multiplicity is one. Such a molecule is said to be in the ***singlet excited state.*** Thus:

An excited molecule with two of its electrons unpaired and with opposite spins is said to be in the ***singlet excited state.***

Since the electron can jump to any of the higher electronic states depending upon the energy of the photon absorbed, we get a series of excited states, S_n where n = 1, 2, 3... etc. Thus we have S_1, S_2, S_3, etc. singlet excited states which are called first single excited state, second single excited state, third single excited state, etc. [*Note:* The symbol S in S_n should not be confused with total spin, S].

On the other hand, if the spins are parallel, then

$S = s_1 + s_2 = +½+½ = 1$, $\therefore 2S+1 = (2\times1)+1=3$

That is, the spin multiplicity is three. Such a molecule is said to be in the ***triplet excited state***. Thus:

An excited molecule with two of its electrons are not paired and with parallel spins is said to be in the ***triplet excited state.***

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As described above, since the electrons can jump to any of the higher electronic states, we get a series of excited states, T_n, where n = 1, 2, 3, ...etc. Thus, we have excited triplet states T_1, T_2, T_3.... etc., which are called first excited triplet state, second excited triplet state, third excited triplet state, etc., respectively.

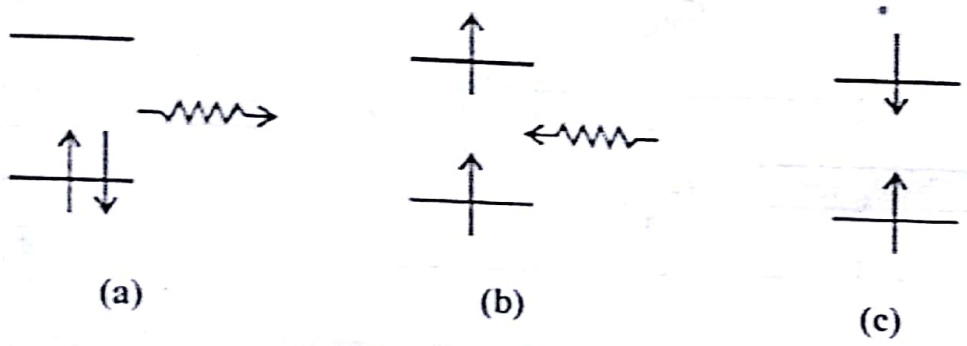

Fig.3.11 spin orientation on the absorption of a light photon

Quantum mechanically it has been shown that a singlet excited state has higher energy than the corresponding triplet excited state. Thus, the energy sequence is given as,

$$E_{S_1} > E_{T_1};\ E_{S_2} > E_{T_2};\ E_{S_3} > E_{T_3} \text{ and so on.}$$

When a molecule in the S_0 state absorbs a light photon, it is get excited and its paired electrons become unpaired. Excitation generally leads to singlet excited state S_1, S_2, S_3, etc., rather than to triplet excited states T_1, T_2, T_3 etc. Transition from $S_0 \rightarrow T_1$ is highly forbidden as it involves a change in spin multiplicity. After the molecule is excited to some vibrational level of an excited state, it loses the vibrational energy instantaneously until it reaches the zeroth vibrational level of that electronic state. In other words, when a molecule in the ground state absorbs energy it may leads to allowed transitions like $S_0 \rightarrow S_1$, $S_1 \rightarrow S_2$, $S_2 \rightarrow S_3$, etc. Subsequently the molecule loses energy while reaching the ground state. The various processes by means of which an excited molecule lose energy while reaching the ground state are best explained with the help of the ***Jablonski diagrams*** as shown in the Fig. 3.12. In these diagrams, solid line represents *radiative processes;* i.e., a molecule discards its excitation energy as a photon. In other words, such processes involve emission of radiation. The wavy horizontal line denotes *nonradiative processes,* which do not involve the emission of any radiation.

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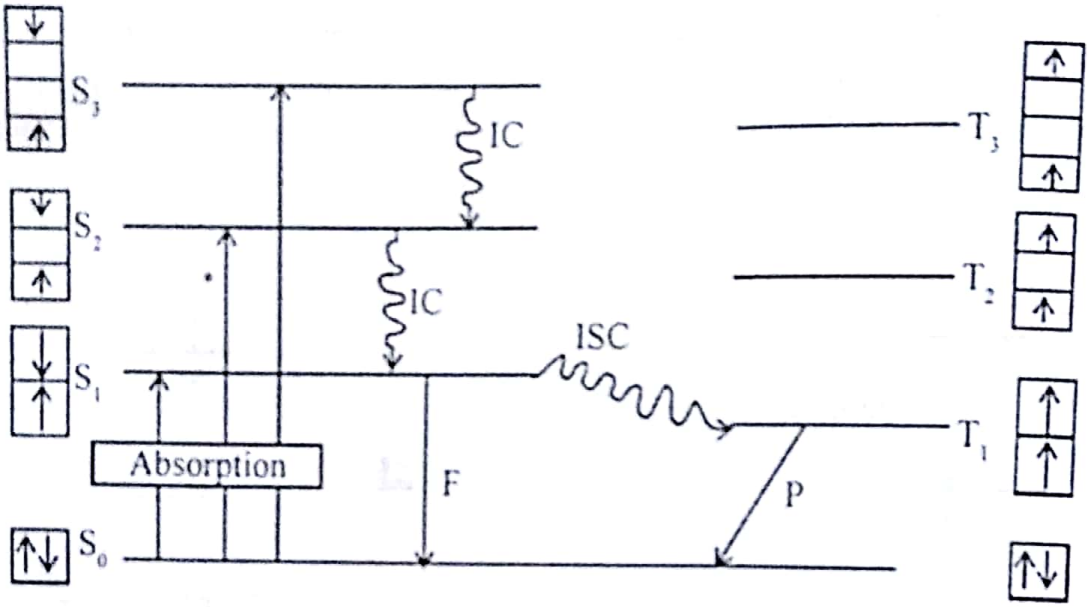

Fig. 3.12 The Jablonski diagram showing various photophysical processes

To sum up : The activated molecule returns from the excited state to the ground state by dissipating its energy through the following types of processes.

(i) Internal Conversion (IC)

An excited molecule may undergo very rapid radiationless transition from a higher electronic state to a lower electronic state of same multiplicity, i.e., S_3 to S_2, S_2 to S_1 etc. The energy of the activated molecule is dissipated in the form of heat through molecular collisions. The process is called internal conversion (IC). The rate constants for conversion of S_3 to S_2 and S_2 to S_1 are in the range of 10^{-11} to 10^{-13} S^{-1}.

(ii) Inter System Crossing (ISC)

It is the *radiationless transition between states of different spin multiplicities.* The excited molecule could cross over to the first triplet state through $S_1 \rightarrow T_1$ transition. Because of the spin interchange in inter system crossing, the rate constant for this process is 10^{-2} to 10^{-6} as fast as internal conversion.

(iii) Fluorescence (F)

The excited molecule may undergo the transition $S_1 \rightarrow S_0$. Such transitions are accompanied by the emission of radiation. Spectroscopically, the transition

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from S_1 to S_0 state is an allowed transition and occurs in 10^{-8} second. The emission of radiation in a transition between the states of same multiplicity ($S_1 \rightarrow S_0$) is called ***fluorescence.***

Organic dyes like eosin, fluorescein, compounds like chlorophyll , etc., exhibit fluorescence in the visible and UV region. The radiation emitted in fluorescence has a lower frequency than that of the incident radiation. This is because of the fact that the absorbing molecules initially in their lower vibrational energy state may acquire higher vibrational energy states. As atoms do not possess vibrational energy and in returning to their initial state from the excited state, they emit radiation of exactly the same frequency which they absorbed. This phenomenon is called *resonance fluorescence.* Thus, mercury vapours at low pressure with atoms in their normal state, when exposed to radiation of wavelength 253.7 nm, are excited and subsequently return to their normal state emitting radiation of same frequency which they absorbed. In this example, if the mercury vapours are mixed with the vapours of silver, thalium, lead or zinc which do not absorb radiation at 253.7 nm and then exposed to radiation a part of the excitation energy from mercury atoms is transferred to the atoms of silver, thalium, lead, zinc, etc. As a result, these atoms are raised to higher energy states. When the excited atoms return to their lower energy states they emit radiation. This is termed *sensitized fluorescence.*

When a photochemically excited atom collides with another atom the intensity of the fluorescent radiation is diminished. This is termed *quenching* of fluorescence and is due to the transfer of energy from the exited atom to the colliding atom.

(iv) Phosphorescence

Phosphorescence is another process of deactivation of the excited state. When the excited molecule at T_1 level undergoes transition to S_0 level by emitting light (transition between different states of multiplicity) is called *phosphorescence (D).* It is generally shown by solids. The substance which shows phosphorescence is called *phosphorescent substance*. Phosphorescence is mainly caused by ultraviolet and visible light. Zinc sulphide, alkaline earth sulphides, etc., are examples of substances exhibiting phosphorescence. Fluorescein in boric acid shows phosphorescence in the blue region at 5700 Å wavelength.

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Explanation. In phosphorescence, as in the case of fluorescence, a molecule absorbs light radiation and gets excited. On returning to the ground state, it emits light energy of longer wavelength. In doing so the excited molecule passes from one series of electronic states to another and gets trapped. This shows the emission of light which persists even after the removal of light source. Thus phosphorescence may be considered as *delayed fluorescence.*

$$A + h\nu \rightarrow A^{*} \xrightarrow{\text{slow}} h\nu'$$

(v) Chemiluminescence

Some chemical reactions are accompanied by emission of visible light at ordinary temperature. For example, when a beam of atomic hydrogen falls on the surface of liquid mercury, blue radiations are emitted. The spectrum of these consists of resonance line 253.7 nm. *The emission of light, at ordinary temperature, as a result of chemical reaction is calied* ***chemiluminescence.*** It may be regarded as the reverse of photochemical reaction because here chemical reaction brings about emission of radiation while in photochemical reaction radiation (photon) brings about chemical reaction.

Explanation. A part of the heat of reaction excites some of the molecules and these excited molecules while reverting to the ground state emit radiation usually in the visible region. The radiation emitted is also called *'cold light'* as the emission occurs at ordinary temperatures.

Some examples of chemiluminescence are : (i) The glow of fire flies due to the aerial oxidation of *luciferon* (a protein) in the presence of enzyme *luciferase* (ii) Oxidation of 5-aminophthalic acid hydrazine in alkaline solution by hydrogen peroxide emitting bright green light. (iii) Glow of yellow phosphorous in air at ordinary temperatures.

Photosensitization

Some chemical reaction occur not by the absorption of light radiation by one of the reactants but by a third substance which transfers the absorbed energy to the reactants. A species which can both absorb and transfer radiant energy for activation of the reactant molecule is called a photosensitizer and the process is called *photosensitization*. The reaction so caused is called a

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photosensitized reaction. Commonly used photosensitizers are the atomic sensitizers such as mercury, cadmium and zinc and the molecular photosensitizers like benzophenone and sulphur dioxide.

An example of photosensitization by mercury atoms is the synthesis of formaldehyde from carbon monoxide and hydrogen. When a mixture of carbon monoxide and hydrogen containing traces of mercury vapours are exposed to ultraviolet light of 253.7 nm, the mercury atoms are excited. These excited atoms of mercury transfers their energy by collisions to hydrogen molecules to dissociate them into hydrogen atoms. This initiates the reaction. The overall reaction may be represented as,

$$Hg + h\nu \rightarrow Hg^{*} \quad \text{(Primary absorption)}$$

$$Hg^{*} + H_2 \rightarrow Hg + H + H \quad \text{(Energy transfer)}$$

$$\left.\begin{aligned} &H + CO + HCO \\ &HCO + H_2 \rightarrow HCHO + H \\ &HCO + HCO \rightarrow HCHO + CO \end{aligned}\right\} \text{(Reaction)}$$

Some glyoxal, CHO–CHO, is also formed by dimerisation of formyl radicals, HCO.

Another example is the reaction between hydrogen and oxygen photosensitized by mercury vapour to give hydrogen peroxide. The reaction may be represented as,

$$Hg + h\nu \rightarrow Hg^{*} \quad \text{(Primary absorption)}$$

$$Hg^{*} + H_2 \rightarrow 2H^{\bullet} + Hg \quad \text{(Energy transfer)}$$

$$\left.\begin{aligned} &H^{\bullet} + O_2 \rightarrow HO_2^{\bullet} \\ &HO_2^{\bullet} + HO_2^{\bullet} \rightarrow H_2O_2 + O_2 \end{aligned}\right\} \text{(Reaction)}$$

Hydrogen peroxide may decompose to form water, H_2O.

The most familiar instance of photosensitization is the action of chlorophyll in permitting carbon dioxide and water to react in presence of sunlight to produce carbohydrates.

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$$xCO_2 + xH_2O + nh\nu \rightarrow (CH_2O)_x + xO_2$$

Chlorophyll absorbs strongly in the region between 600 and 700 nm. This region is most effective for photosynthetic reactions.

Decomposition of H_2, NH_3, H_2O, C_2H_4, etc. are sensitized by mercury vapours.

Photopolymerization

Photopolymerization is a light induced reaction that converts a liquid monomer into a solid polymer.

Photopolymerization processes make use of liquid, radiation curable resins or photopolymers as their primary materials. *Photopolymers are polymers which change their properties when exposed to light, often in the UV or visible region.* Upon irradiation, these materials undergo a chemical reaction to become solid. This reaction is called *photopolymerization.*

The changes in the properties of a photopolymer on exposure to light are often manifested structurally. For example, hardening of the material occurs as a result of cross linking when exposed to light. An example of it is shown in Fig. 3.13 depicting a mixture of monomers, oligomers and photoinitators which conform into a hardened polymeric material through a process called *curing.*

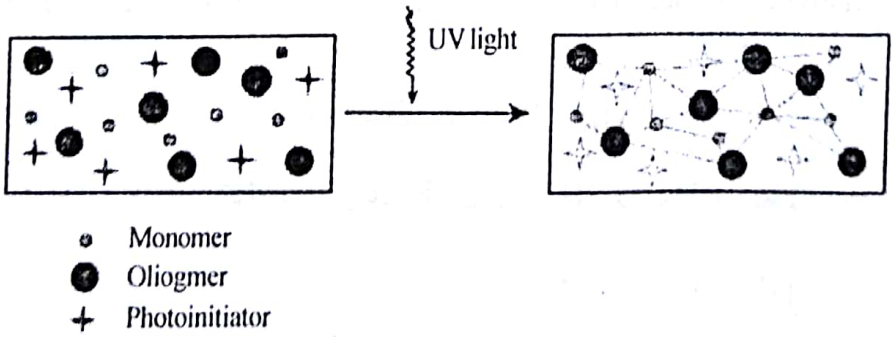

Fig. 3.13

Typically a photopolymer consists of a mixture of multifunctional monomers and oligomers in order to achieve the desired properties. Therefore, a wide variety of monomers and oligomers have been developed which can polymerize in the presence of light. Photopolymers undergo a process called curing, where oligomers are cross linked on exposure to light to form a network polymer. One of the advantages of photo curing is that it can be done selectively

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using high energy light sources. However, most systems are not readily activated by light, and in this case a photoinitiator is needed. ***Photoinitiators*** *are compounds that on exposure to light decompose into reactive species that activate polymerization of specific functional groups on the oligomers.* An example of a mixture that undergoes cross linking when exposed to light is shown below. The mixture consists of monomeric styrene and oligo acrylates.

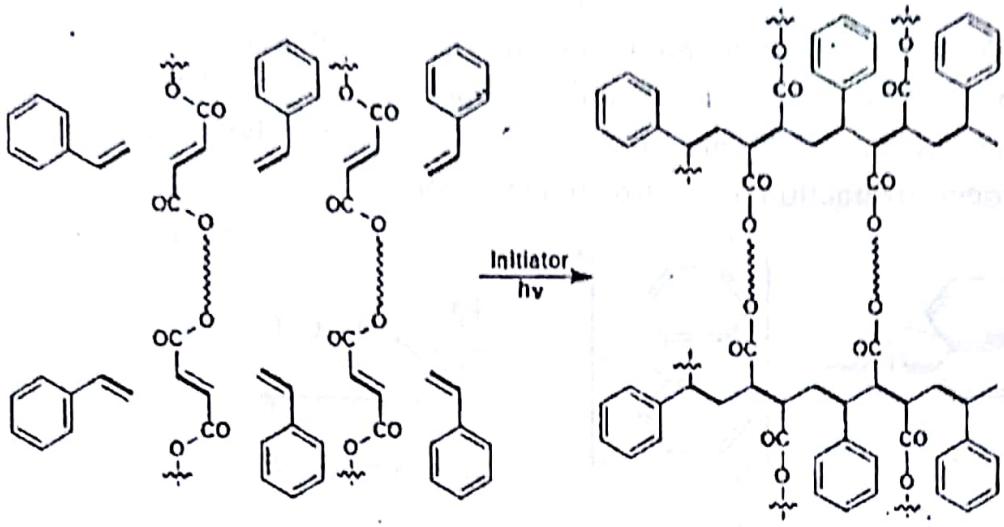

Fig. 3.14 *Monomeric styrene and olygomeric acrylates giving cross linked polymer by photopolymerization.*

There are two types of photoinitiation - free radical and ionic. The general process involves doping a batch of polymer with a small quantity of photoinitiator, followed by exposure to light radiation which results in a highly cross linked product.

A. Photoinitiated Free radical Polymerisation: When polymerizations are initiated by light and both the initiating species and the growing chain ends are radicals the process is called *radical photopolymerization*. In most of the photo induced polymerization, initiators are used to generate radicals. Photoinitiators are of two types: *Unimolecular photoinitiators and bimolecular photoinitiators.*

Unimolecular photoinitiators undergo homolytic bond cleavage on absorption of light. Benzoin and its derivatives are the most widely used photoinitiators for radical polymerization of vinyl monomers.

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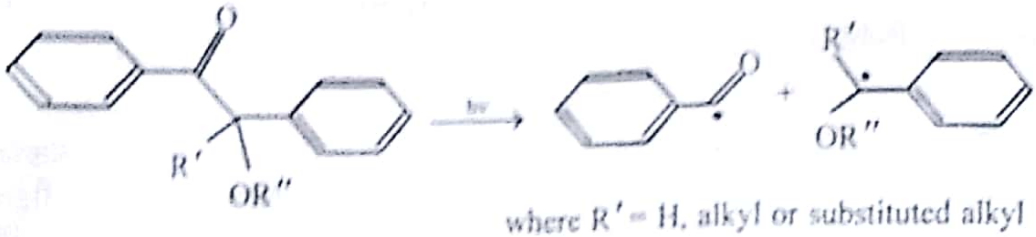
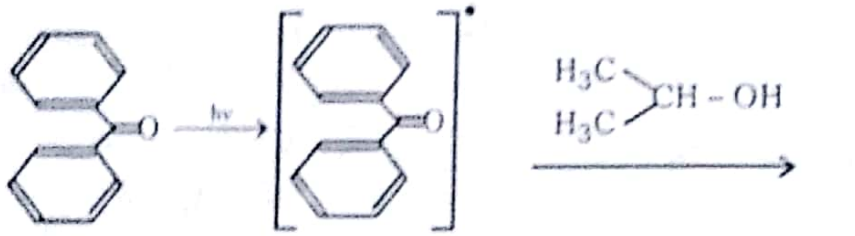
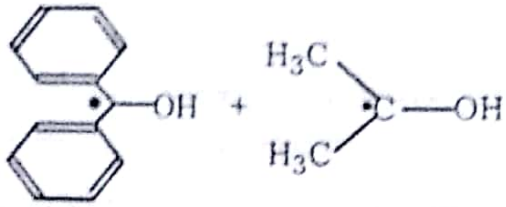


where R′ = H, alkyl or substituted alkyl
R″ = H, alkyl or substituted alkyl

Bimolecular photoinitiators consist of a photoinitiator like benzophenone and a coinitiator like alcohol or amine. Radicals are generated in a bimolecualr process by the reduction of photoexcited aromatic carbonyl compound by hydrogen abstraction or electron transfer reactions.





B. Photoinitiated Cationic Polymerization

Cationic photopolymerization begins with the photoexcitation of the initiator. Aryl onium salts like iodonium and sulphonium salts can be used as photoinitiators. These thermally and hygroscopically stable initiators, on photolysis undergo irreversible photofragmentation to produce cation radicals and Bronsted acids. Reactive species thus formed photochemically with onium salts initiate the cationic polymerization of suitable monomers as given below.

$$On^{+} \xrightarrow{h\nu} \dot{O}n + H^{+}$$

$$\dot{O}n^{+} + M \rightarrow \text{Polymer}$$

$$H^{+} + M \rightarrow \text{Polymer}$$

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MODEL QUESTIONS

Part A (Each question carries 2 marks)

1. Explain fluorescence with an example.
2. Explain photosensitisation with one example.
3. Distinguish between a photochemical reaction from a thermal reaction.
4. Distinguish fluorescence from phosphorescence.
5. State and explain Lambert-Bee Law.
6. State and explain Stark Einstein law.
7. State and explain Grothus-Draper law.
8. Explain the term quantum efficiency.
9. Define quantum yield of a reaction.
10. How do photochemical reactions differ from thermal reactions.
11. Discuss the use of photosensitizers.

Part B (Each Question Carries 4 marks)

1. Explain briefly fluorescence and chemiluminescence.
2. Calculate the values of frequency, quantum energy, and einstein for 500 nm radiation.
3. How would you explain very high and very low quantum efficiencies of some photochemical reaction?
4. Derive the Lambert-Beer Law.
5. Brief about primary and secondary photochemical processes.
6. How is quantum yield of a photochemical reaction determined experimentally?
7. How would you explain very high and very low quantum efficiencies of some photochemical reactions?
8. What is an actinometer? Describe how a uranyl oxalate actinometer may be used.
9. Explain briefly fluorescence and chemilumin essence.
10. Differentiate between fluorescence and phosphoroscence.
11. Explain the causes of high and low quantum efficiency of photochemical reactions.

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Part C (Each Question Carries 10 Marks)

1. (i) Distinguish between photochemical and thermal reactions.

   (ii) Derive the Lambert-Beer Law.

   (iii) How would you explain very high and very low quantum efficiencies of some photochemical reactions?

2. State an explain the term 'quantum yield'. How do you account for the fact that the quantum yield of the photochemical reaction, $H_{2(g)}+Br_{2(g)} \rightarrow 2HBr_{(g)}$ is low (=0.01) while that of the reaction, $H_{2(g)}+Cl_{2(g)} \rightarrow 2HCl_{(g)}$ is very large ($=10^5$)?Ir

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SYLLABUS

Chemistry 1 (Common to all branches)

Module 1

Chemical Bonding **(14 hours)**

Review of orbital concept and electronic configuration, electrovalency and ionic bond formation, ionic compounds and their properties, lattice energy, solvation enthalpy and solubility of ionic compounds, covalent bond, covalency, orbital theory of covalency - sigma and pi bonds - formation of covalent compounds and their properties Hybridization and geometry of covalent molecules - VSEPR theory - polar and non-polar covalent bonds, polarization of covalent bond - polarizing power, polarisability of ions and Fajan's rule, dipole moment, percentage ionic character from dipole moment, dipole moment and structure of molecules - co-ordinate covalent compounds and their characteristics, molecular orbital theory for H_2, N_2, O_2 and CO, metallic bond - free electron, valence bond and band theories, weak chemical bonds - inter and intra molecular hydrogen bond - van der Waals forces.

Module 2

Thermodynamic Parameters **(8 hours)**

Stoichiometry-mole concept, significance of balanced chemical equation - simple calculations - Conditions for occurrence of chemical reactions - enthalpy, entropy and free changes - spontaneity - Thermochemistry-heats of reactions - (formation, combustion, neutralization) - specific heats - variation of enthalpy change with temperature - Kirchhoff' relation (integrated form) - bond enthalpy and bond order - Problems based on the above.

Kinetics **(8 hours)**

Review of molecularity and order of a reaction, rate law expression and rate constant, - first, second, third and zero order reactions, pseudo-first order reactions (pseudo-unimolecular reactions)-complex reactions - equilibrium and steady state approximations - mechanism of these reactions - effect of temperature on reaction rates - Arrhenius equation and its significance-Michaelis Menden kinetics - enzyme catalysis.

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Module 3

Electrochemistry **(8 hours)**

Electrolytes- strong and weak, dilution law, Deby-Huckel theory, Faraday's laws, origin of potential, single electrode potential, electrochemical series, electrochemical cells, Nernst equation and its application, reference electrodes - SHE, Ag/AgCl, Calomel.

Photochemistry

Photochemistry, laws of photochemistry - Stark-Einstein law, Beer-Lamberts law, quantum efficiency-determination, photochemical processes - Jablonsky diagram, internal conversion, inter-system crossing, fluorescence, phosphorescence, chemiluminescence and photo sensitization - photopolymerization.

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